

Round33 research addendum: boundary decay, locality and convergence of the named constructions, certificates for the limit, and applications to related problems

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25 September 2026

Round33 scientific addendum: BA1–BD2

Abstract

This addendum records the Round33 cycle: eight prospectively contracted investigations in three research sub-rounds and an applications stage, all accepted within their stated model scope. They are the paired BA1, BA2, BB1, BB2 and BD1, the statement investigation BC1 and the single-direction investigations BC2 and BD2. BA1, BA2, BB1 and BB2 concern the zero-selected patterned family of the SU(2) Kogut–Susskind form at fixed spacing and the original coupling cap $|\tau| \leq 10^{-8}$, and BC1 the same family at the cap $\tau = \pm 10^{-8}$ only, with the two named construction families of Round32: F1, AQ1’s centered whole-star boxes, and F2, II’s all-contained-face boxes with padding, both on $\Lambda_N = [-N, N]^3$, $N \geq 2$. BA1 shows that replacing one box of the named constructions by another moves the AM2 creation coefficients on supports meeting the Wilson cover $R = \{0, e_z\}$ by at most Kq^{N-1} in the anchored norm, where N is the smaller box size, in each on-site cutoff space and uniformly in the cutoff: $K = 49/111790368 \approx 4.383 \times 10^{-7}$ at $q = 1/64$, and $K = 49/2018304 \approx 2.428 \times 10^{-5}$ at the per-step ball ratio $q = 37888|\tau| = 148/390625$, for every comparison of the contract and both signs. It is proved by two routes, a weighted-norm contraction and analyticity on a complex-coupling disc. This is decay of coefficients at a rate in N , not decay of the reduced density. BA2 shows that the finite-box Heisenberg evolutions of F1 and F2 of every $A \in B(\mathcal{H}_R)$ differ in norm by at most $K_{\text{cmp}}(5N+1)N^{-3}\|A\|$, with $K_{\text{cmp}} = 3969/155720800000000 \approx 2.549 \times 10^{-11}$, for $|\theta| \leq 8$ in $\theta = \alpha t/\hbar$. The evolutions of each family are Cauchy in N at the rate $1/(N-1)$, with constants about 5.947×10^{-11} (F1) and 9.452×10^{-11} (F2), and the F2 limit dynamics equals the AQ1 limit dynamics. This closes two obligations left open by Round32, boundary independence of the dynamics on compact windows and the dynamics of the padded family, within that scope. Both BA2 constants are quadratic in τ . The BA2 statements concern the algebraic dynamics of the named constructions on a compact window. They assert no equality of GNS dynamics or of correlation functions of different states and nothing uniform in time. BB1 turns coefficient decay into locality of the reduced densities. For each of five comparisons of the named constructions, among them the fixed- N change of boundary prescription and two volumes of one prescription, the reduced densities of the finite-box ground vectors on R differ in trace norm by at most Cq^{N-1} with $q = 1/64$ and $C \approx 8.905 \times 10^{-7}$. On every finite region $Y \subset \Lambda_N$ they differ by at most $c_{\text{site}}|Y|e^{|Y|/10^8}q^{d_Y}$ with $c_{\text{site}} \approx 8.768 \times 10^{-7}$. The bounds hold in each on-site cutoff space and, at fixed N , for the untruncated ground vectors. They are proved by a hard-core polymer expansion with a Kotecký–Preiss condition, whose cited criterion was checked after the freeze against a committed excerpt, and by an iterated product-ordering split. BB1 is static and per comparison. BB2 uses BB1’s frozen targets as hypotheses, which the BB1 gate discharges item by item. The reduced densities of F1 and F2 then converge as whole sequences, without compactness, on every finite region: the supremum over $M > N$ of the difference on R is at most $C'q^{N-1}$ with $C' = 4/984375 \approx 4.06 \times 10^{-6}$. They converge to one common limit, the limit of the named constructions. That limit equals every AQ1 and every F2 subsequential limit, inherits exactly their admitted properties, and is invariant under coarse translations. Its correlation functions on $|\theta| \leq 8$ are the limits of the finite-box ones. The correlation bound carries an explicit rate only for $5 \leq N \leq 14000$ and is vacuous from $N = 14419$. BC1 restates two Round32 certificates for that one state with every constant unchanged: the node certificate

for the centered Euclidean Wilson correlation at $s = 1$, whose interval still contains the free reference $e^{-3}/4$, and the enclosure $[\tau/144 - K_2^+ \tau^2, \tau/144 + K_2^+ \tau^2]$ of the static Wilson mean. The whole-sequence bound then certifies the sign of the static Wilson mean of the untruncated padded boxes for every $N \geq N_{\text{sign}} = 4$ at $\tau = \pm 10^{-8}$; nothing is claimed below the cap, the boxes with $N = 2, 3$ stay open, and no finite-box node is claimed. BC1 also splits the Round32 obligation “uniqueness of the limit”: the coincidence of all subsequential limits of the two named families is closed within the BB2 scope, and uniqueness of any ground state stays open. BC2 concerns the uniform Kogut–Susskind SU(2) model at fixed spacing and strong bare coupling, in which every face carries the coupling, handled by Round32’s route B. It re-derives the chain of BA1, BB1 and BB2 with the route-B constants (per-site sum $29|\tau|$, 52 first-order faces and 5 interaction groups per site): coefficient decay with $K_B = 13/27941256 \approx 4.653 \times 10^{-7}$, locality with $C_B \approx 9.452 \times 10^{-7}$ and $c_{\text{site},B} \approx 9.306 \times 10^{-7}$, and whole-sequence convergence of that model’s one named construction for every $N \geq 2$. The limit is also reached along every exhausting sequence of the same prescription, is identified with every AQ1-type subsequential state of that construction, is mirrored pointwise between the two signs of τ and is invariant under coarse translations. At $\tau = +10^{-8}$ and $s = 1$ only, it satisfies the Round32 node certificate of that model with unchanged constants. None of these investigations establishes uniqueness of any ground state, a statement about states outside the named constructions or about other boundary prescriptions, equality of GNS dynamics of different states, route-B dynamics, an interaction shift of the Euclidean correlation, an estimate uniform in the lattice spacing or a continuum statement. The applications stage records what transfers and what does not. BD1 carries the Round32 link-flip lemma and first-order parity theorem to other gauge groups on one-plaquette and box models under a frozen convention. The flip transfers to SU(2), SU(4), U(1) and Z2, whose centres contain an element acting as -1 on the Wilson representation, and the parity to SU(2), SU(4), SU(5), U(1) and Z2, whose third Haar moment of the Wilson variable vanishes. SU(3) and SO(3) are obstructions with exact nonzero coefficients ($1/589824$ and $1/331776$ at second order), and SU(5) is a flip obstruction with the fourth-order coefficient $1/(2^{30}3^{10})$. For SU(2) in the zero-selected family at $\kappa = 0$ the Wilson-loop mean is odd or even in τ with the parity of the area of any spanning surface, in every open centered whole-star box and, through BB2, for the limit; no parity is claimed for F2 boxes. BD2 turns to new SU(2) observables and a new dimension. On the two-plaquette graph with independent couplings it gives exact coefficients through total order 4 and certifies the 1×2 loop positive at $l_1 = l_2 \in \{\pm 10^{-1}, \pm 10^{-2}, \pm 10^{-3}\}$, with $D = 8$ relative width at most 1.79645×10^{-17} . In the zero-selected family the 1×2 Wilson mean vanishes at first order and is bounded by $K_2' \tau^2$, with only a formal second-order coefficient, and the electric energy on the cover lies in the certified band $[\frac{3}{2}L^2, 98|\tau|]$. In 2+1 dimensions only the AM2 constants are recounted, with no theorem. The addendum closes with the process history, the reviewed state of the 21 obligations, what the applications did not transfer, and a ranked roadmap of unexecuted goals. The limit throughout is the limit of the named constructions, nothing is uniform in the lattice spacing, and investigation counts are never a fraction of the four-dimensional continuum Yang–Mills problem, which remains open.

Status and reading guide

This is a separate scientific addendum; Draft03 [2] and the Round31 and Round32 addenda [3, 4] are unchanged. The Round33 cycle consists of eight prospectively contracted investigations in four sub-rounds of two. Three are research sub-rounds: locality (BA1, BA2), convergence of the named constructions (BB1, BB2) and consequences for the admitted certificates (BC1, BC2). The fourth is an applications stage (BD1, BD2). It applies admitted Round32 and Round33 equations to related problems and records an explicit obstruction where an equation does not transfer. An expert panel fixed the goals, directions and contract rules in two deliberation loops before any production. The panel consists of three methodological lenses (Newton/Tesla, Jung/Pauli and Penrose/Feynman), an advisor and a skeptic. After each sub-round one research assistant per lens runs the tests named in the lens updates, and the advisor may reorder, replace or retitle the next pair. Assistant work counts as zero investigations.

This is the complete edition. It records all four sub-rounds, and all eight investigations are gated and accepted within their stated model scope. It is authored from the eight reviewed gates, and its

builder refuses to run before all eight exist and verifies every gate binding.

BA1, BA2, BB1, BB2 and BC1 concern one model, the zero-selected patterned family of Round32, and the two named construction families of Section 1.2. BC2 concerns the uniform model handled by route B and its one named construction (Section 7.2). BD1 concerns one-plaquette and box models of seven gauge groups and, for $SU(2)$, the zero-selected family; BD2 concerns the Round11 two-plaquette graph, the zero-selected family and a declared 2+1-dimensional model. BA1, BA2, BB1 and BB2 are paired: a forward and a reverse producer proved the same frozen targets by different routes, the reverse under premise isolation. The skeptic derived the constants in a pre-comparison package, committed before both producer commits for BA1, BA2 and BB1 and between the two for BB2, and replayed and mutated both producers after the comparison. BC1 is a statement investigation with one forward producer; the skeptic’s pre-comparison replay was committed 3 min 32 s after the forward commit, with its values final before the forward checker was last written. BC2 is a single-direction investigation with one forward producer; the skeptic’s pre-comparison replay from the contract was committed 3 min 44 s before the forward commit. BD1 is paired, with a characters route and a Weyl-integration route, and BD2 is single-direction; for both, the skeptic’s pre-comparison package was committed after the producer commits, with its values final before them. BA1 carries the sub-label `boundary_decay_rate_only` with the secondary label `static_not_dynamic`: it is a statement about creation coefficients, not about states or dynamics. BA2 carries `dynamics_on_compact_windows`: it is a statement about the algebraic Heisenberg dynamics on the window $|\theta| \leq 8$, not about the states. BB1 carries `boundary_decay_rate_only` with `static_not_dynamic`: it is a statement about the reduced densities of finite-box ground vectors, one comparison at a time, and forms no limit object. BB2 carries `convergence_of_named_constructions` with `common_limit_of_named_constructions`: it is a statement about the limit of the named constructions only. BC1 carries `certificate_restated_for_limit` with `reference_unresolved` for the node and `static_not_dynamic` for the static mean: it restates admitted certificates for that one limit and adds a finite-box corollary. BC2 carries `convergence_of_named_constructions` with `certificate_restated_for_limit`, `reference_unresolved` and `static_not_dynamic`: it concerns the one named construction of the route-B model, its limit and the restated node, and no route-B real-time dynamics. BD1 carries `transfer_to_named_model` with `obstruction_recorded`, and BD2 carries `sign_certified_finite_graph` with `transfer_to_named_model`, `obstruction_recorded` and `static_not_dynamic`: they are statements about named models and a finite graph.

The headline results, each within the scope its gate records, are these.

- Boundary decay of the AM2 creation coefficients (BA1): $\sum_{I \ni u} \|c_I^{\text{box } 1} - c_I^{\text{box } 2}\| \leq K q^{N-1}$ for $u \in R$, with $K \approx 4.383 \times 10^{-7}$ at $q = 1/64$ (margin about 1.14, recorded as thin) and $K \approx 2.428 \times 10^{-5}$ at $q = 37888|\tau|$, for every comparison of two named boxes, both signs and every on-site cutoff; the crude tier misses the headline target and is retained.
- Comparison of the dynamics (BA2): $\|T_\theta^{F2,N}(A) - T_\theta^{F1,N}(A)\| \leq 2.549 \times 10^{-11}(5N+1)N^{-3}$ for $\|A\| \leq 1$ and $|\theta| \leq 8$, with the first order in τ vanishing exactly.
- Cauchy estimates and the F2 limit dynamics (BA2): rate $1/(N-1)$ in N for both families, whole-sequence convergence of the evolutions, the F2 limit dynamics identified with AQ1’s, and AQ1 §§4–5 rerun for every F2 subsequential limit state. This closes the Round32 obligations O5 and O6 within that scope.
- Locality of the reduced densities (BB1): $\|\rho_R^{\text{box } 1} - \rho_R^{\text{box } 2}\|_1 \leq C q^{N-1}$ with $C \approx 8.905 \times 10^{-7}$ at $q = 1/64$ (margin about 4.49), and a region form with $c_{\text{site}} \approx 8.768 \times 10^{-7}$ (margin about 2.28, the binding one), for every comparison, both signs, each cutoff space and the untruncated vectors at fixed N ; the crude tier fails and is retained.
- Convergence of the named constructions (BB2), with every BB1 hypothesis discharged: whole-sequence Cauchy bounds with $C' = 4/984375$ and $c'_{\text{site}} = 2/984375$ (margin 315/128), one common limit identified with every AQ1 and F2 subsequential limit, coarse translation invariance

with $\|\rho_R^{\Lambda_N+v} - \rho_R^{\Lambda_N}\|_1 \leq C_h q^{N-|v|_\infty-1}$, and convergence of the correlation functions on $|\theta| \leq 8$ with $C_{\text{dyn}} \approx 1.2532 \times 10^{-10}$, at a rate K_5/N on $5 \leq N \leq 14000$ only. This closes the Round32 obligations O2, O3 and O4 for the named constructions within that scope.

- Certificates for the limit (BC1), at $\tau = 10^{-8}$ with $\tau = -10^{-8}$ a mirror replay: the node certificate $|C(1) - d| \leq R$ with $R \approx 1.832 \times 10^{-7}$ and the free reference $e^{-3}/4$ inside the interval, and the static enclosure of $\omega_\infty(W)$ in about $[6.91090 \times 10^{-11}, 6.97799 \times 10^{-11}]$, both restated for the limit of the named constructions with unchanged constants; the sign of the static Wilson mean of the untruncated padded boxes certified for every $N \geq 4$ at $\tau = \pm 10^{-8}$ only, with the boxes $N = 2, 3$ open and no finite-box node; and a reviewed obligations table of 21 rows, in which O1 is split into O1a, closed within the BB2 scope, and O1b, open.
- Convergence for the route-B uniform model (BC2): every-site coefficient decay with $K_B = 13/27941256$ (margin about 2.149) on the disc $|z| \leq 64|\tau|$, admissible under $J' = 29|\tau|$ while the route-A extremes fail the self-map by the factor $29/28$; locality by the iterated split with $C_B \approx 9.452 \times 10^{-7}$ and $c_{\text{site},B} \approx 9.306 \times 10^{-7}$ (margin about 2.149); whole-sequence convergence by one direct comparison for every $N \geq 2$, exhaustion within the prescription, the pointwise sign mirror, identification before inheritance and coarse translation invariance; and the AX2 node at $\tau = +10^{-8}$, $s = 1$ restated for the limit, with the free reference still inside the interval.
- Centre symmetry across gauge groups (BD1): exact Haar moments and first-order Wilson coefficients for SU(2), SU(3), SU(4), SU(5), U(1), Z2 and SO(3) (1/144, 1/1152, 1/2880, 1/5760, 1/96, 1/48, 1/864), equal on two routes and the skeptic's three in 141 cells; the flip lemma transferred to SU(2), SU(4), U(1) and Z2 and the parity theorem to SU(2), SU(4), SU(5), U(1) and Z2, box by box inside a Kato radius; SU(3), SO(3) and SU(5) recorded as obstructions with exact coefficients; and the SU(2) area parity at $\kappa = 0$ in every open centered whole-star box and for the limit.
- New observables and a new dimension (BD2): exact two-coupling tables through total order 4 on the two-plaquette graph with the one-link flip parities, and the 1x2 loop certified positive at $l_1 = l_2$ with $|l| \in \{10^{-1}, 10^{-2}, 10^{-3}\}$ at $D = 8$ relative width at most 1.79645×10^{-17} against 10^{-16} ; the $\mathbb{Z}^3$ 1x2 Wilson mean zero at first order, bounded by $K'_2 \tau^2 \approx 1.34 \times 10^{-12}$, even for F1 and the limit, with the formal coefficient 7/124416 labelled; the electric band $[\frac{3}{2}L^2, 98|\tau|]$ for every box and the limit; and the 2+1D AM2 constants $G_3 = 9e^{3/32}$, $G'_3 = 118e^{3/32}$ with the cap about 1.5807×10^{-3} , no theorem claimed.

The four-dimensional continuum Yang–Mills problem remains open, and no investigation count is a fraction of it.

Section 1 fixes the model dictionary, the two families, the Round32 checkpoint and the frozen Round33 plan. Section 2 derives the coefficient decay, Section 3 the comparison of the dynamics, Section 4 the locality of the reduced densities, Section 5 the convergence of the named constructions, Section 6 the certificates for their limit, Section 7 the convergence for the route-B uniform model and Section 8 the applications, ending with Table 29 of what transferred and what did not. Section 9 records the process history, Section 10 what the record does and does not establish, with the ranked roadmap, and Appendix A the contribution and evidence ledger. Each investigation opens with its gate's accepted statement, quoted verbatim in the gate's ASCII notation as the reviewed scope, and closes with the limitations the gate recorded. Exact rational arithmetic decides admission. Decimals summarize exact rationals, and labelled values are recorded but not bound.

The human project author is Hruday N M (BUNZEEY). *AI-assistance disclosure*: the advisor, the skeptic, the expert-panel lenses (Newton/Tesla, Jung/Pauli and Penrose/Feynman as documented methodological lenses), the research assistants and the forward and reverse producers are model-agent roles with shared mathematical ancestry. Their agreement is correlated model-agent work, not external human peer review or independent physical observation. Historical figures supply documented methodological lenses; none participated in or endorsed this work. No historical, occult, mystical or governmental source supplies a premise, and the Jung lens's reading ledger records custody, not confirmation. HNM contribution aliases identify project records; scientific priority is

unverified. Investigation counts are never a fraction of the continuum problem.

BA2 applies the Lieb–Robinson bound and the limit-dynamics theorem of Nachtergaele and Sims ([arXiv:1410.8174v1](#)) [11], quoted from a committed excerpt. Its rerun for the padded family follows AQ1 §§4–5, whose strategy AQ1 credits to Gauvin, [arXiv:2503.15539v3](#), Supplement A.10 [14]. BA1 applies standard tools to the admitted AM2 fixed point: weighted norms that pay a fixed loss per interaction, Banach’s fixed-point theorem, Weierstrass’s theorem, the maximum principle and Cauchy’s estimates. The BB1 polymer route cites the Kotecký–Preiss criterion (1986) [13] in the form of Ueltschi ([arXiv:math-ph/0304003v3](#)) [12], checked after the freeze against a committed excerpt. BD1 applies Peter–Weyl theory, Schur’s lemma, the Pieri rule, Clebsch–Gordan series, Schur–Weyl duality with the hook-length formula, the Weyl integration and character formulas and Kato’s analytic perturbation theory; centre symmetries of lattice gauge theories are a known kind of argument, and no priority is claimed for either criterion. BD2 applies the Weyl, Temple, Eckart and Davis–Kahan bounds and the Fuchs–van de Graaf inequality to the Round11 graph reduction and the admitted AZ2 certificate code. The attribution sentence of the Round32 addendum [4] for the AM2 creation expansion and the AQ construction is inherited unchanged. Scientific priority is unverified, and no independence from those sources is claimed.

1 Model dictionary, Round32 checkpoint and Round33 plan

This section fixes the objects that every Round33 investigation names. It restates the Round32 model dictionary without change ([Round32 addendum](#), Section 2), adds the two named construction families as Round32’s AY1 admitted them, records where Round32 stopped and lists the frozen Round33 plan. Nothing in this section is a new result.

1.1 Model dictionary

The theory is the $SU(2)$ Kogut–Susskind form [15] on the spatial lattice $\mathbb{Z}^3$ at fixed spacing a . A coarse site $b \in \mathbb{Z}^3$ owns the three positive links from each fine tail $(4b_x + r, 2b_y + q, b_z)$, $0 \leq r < 4$, $0 \leq q < 2$: exactly 24 untruncated $L^2(SU(2))$ link factors. One coarse step is therefore $(4a, 2a, a)$ in fine units. Both original endpoint gauge actions are retained. Write $C_e = -\sum_a X_{ea}^2$ for the link Casimir, with generators $i\sigma_a/2$, and $W_f = \frac{1}{2} \text{Tr } U_f$ for the real fundamental Wilson plaquette, so $\|W_f\| = 1$.

An elementary face in directions $a < c$ at p has positive link tails at p , $p + e_a$, $p + e_c$ and p , so its coarse owner set is $M_f = \{\pi(p), \pi(p + e_a), \pi(p + e_c)\}$, with two or three elements. Grouping by anchor gives 24 face classes per coarse site, with relative supports inside the star $S = \{0, e_x, e_y, e_z\}$ ([I1 dictionary](#)). Three classes lie inside one factor and form the selected strip. Throughout Round33 their coefficient triple is exactly $(0, 0, 0)$, an admissible choice inside the closed AQ coefficient box with the Haar product as reference. It is neither the uniform model nor a replacement of a nonzero selected vacuum by Haar. The other 21 classes cross a block boundary. These *omitted* classes (Table 1) are the only interaction.

Two counts read off the table recur below. The classes whose support avoids e_x , e_y or e_z number 17, 13 and 5. Those avoiding two directions number 10 (avoiding e_x, e_y), 3 (avoiding e_x, e_z) and 1 (avoiding e_y, e_z), and no class avoids all three. By translation covariance a coarse factor lies in the owner sets of exactly $21 + 4 + 8 + 16 = 49$ omitted faces, grouped in 15 owner sets. The faces meeting the cover R defined below number 82; 16 of them contain both sites of R , and ten have owner set exactly R .

In normalized units $\delta = \alpha/8$ the omitted faces are grouped by anchor into whole stars:

$$\phi_b = -\frac{\tau}{3} \sum_{f \in O_b} W_f, \quad |O_b| = 21, \quad \|\phi_b\| = 7|\tau|, \quad h_b = 8 \sum_{e \text{ owned by } b} C_e. \quad (1)$$

Table 1: The 21 omitted face classes of the I1 dictionary, grouped by relative owner set K ; q is the fine y phase and r the fine x phase.

Relative support K	Orientation and phases	Directions in K	Classes
$\{0, e_x\}$	$xy: r = 3, q = 0$	x	1
$\{0, e_y\}$	$xy: r = 0, 1, 2, q = 1$	y	3
$\{0, e_z\}$	$xz: r = 0, 1, 2, q = 0, 1; yz: r = 0, \dots, 3, q = 0$	z	10
$\{0, e_x, e_y\}$	$xy: r = 3, q = 1$	x, y	1
$\{0, e_x, e_z\}$	$xz: r = 3, q = 0, 1$	x, z	2
$\{0, e_y, e_z\}$	$yz: r = 0, \dots, 3, q = 1$	y, z	4
Total	14 two-site and 7 three-site classes		21

At the zero triple the on-site operator h_b is an unbounded pure Casimir sum with compact resolvent, constant (Haar) vacuum projection P_b , zero ground scalar and $h_b \geq 6(I - P_b)$. The physical Hamiltonian is δ times the normalized one; in the energy unit α each star is $\phi_b/8$. A site u lies in the four stars anchored at $u - S$, incoming anchors included, so the per-site interaction sum is

$$J = \max_u \sum_{b: u \in b+S} \|\phi_b\| \leq 4 \cdot 7|\tau| = 28|\tau| \leq J_0 = \frac{7}{25000000} \quad (|\tau| \leq 10^{-8}). \quad (2)$$

The outgoing count $7|\tau|$ is a rejected control (`missing_incoming_stars`) in both sub-round-1 contracts.

For the original xz Wilson plaquette W at the origin, the complete-factor cover is $R = \{0, e_z\}$: 48 stored links and 36 original endpoints. Its seven incident whole-star anchors are $R - S = \{0, -e_x, -e_y, -e_z, e_z, e_z - e_x, e_z - e_y\}$. The on-site product reference on R is the Haar vector Ω_R with projection P_R .

AM2 creation expansion. AM2 (Round29) proves, for every finite complete-factor volume and both signs of $|\tau| \leq 10^{-8}$, a simple ground state and a centered gap at least $\alpha/16$ ([AM2 gate](#)). For nonempty I a creation vector $c_I \in \bigotimes_{x \in I} Q_x \mathcal{H}_x$, $Q_x = I - P_x$, defines $\hat{c}_I = |c_I\rangle\langle\Omega_I| \otimes 1$. These operators commute, $\hat{c}_I \hat{c}_J = 0$ when $I \cap J \neq \emptyset$, and with $C = \sum_I \hat{c}_I$ the ground vector is $\psi = e^{-C} \Omega_0$. In the anchored norm $\|c\|_a = \max_u \sum_{I \ni u} \|c_I\|$ the coefficients are the unique fixed point in the ball of radius R_a of

$$c = \sum_{k=0}^8 \frac{L_k(c, \dots, c)}{k!}, \quad (L_k)_M = H_M^{-1} P_M \text{ad}_{C_1} \cdots \text{ad}_{C_k}(V) \Omega_0, \quad \|c\|_a \leq R_a = \frac{1}{64}, \quad (3)$$

where the commutator nest terminates at order eight. The all-order majorant and its evaluated constants are

$$G(t) = 16e^{8t}(1 + 10t), \quad G(R_a) < \frac{148}{7}, \quad G'(R_a) < 352, \quad (4)$$

$$J_0 G(R_a) < \frac{37}{6250000} < R_a, \quad 2J_0 G'(R_a) < 1,$$

from the directed enclosure $e^{1/8} < 8/7$. The Round33 gate records write $R = 1/64$ for the anchored radius; this edition writes R_a so that R always denotes the cover. The exact first-order collection is AV1's $c_M^{(1)} = -(\tau/72) \sum_{M_f=M} W_f \Omega_0$, with $\|c^{(1)}\|_a \leq t_1 = 49|\tau|/144$; the AV1 self-consistent remainder gives $\|c\|_a \leq t_1/(1 - 352J)$ ([AV1 gate](#)).

On-site cutoff. Let $Q_L = \bigotimes_x 1_{[0,L]}(h_x)$. AM2 holds in every cutoff space Q_L with rank-independent constants; the compressions keep supports and do not increase J , and for $L \geq 24$ every first-order face vector $W_f \Omega_0$ is kept exactly. AM2 §6 removes the cutoff for the eigenvalues and the gap, and AV1 removes it for the ground vector. Neither defines untruncated creation coefficients; BA1 states its coefficient bounds in each Q_L , uniformly in L , and asserts nothing about untruncated coefficients.

States, dynamics and clock. AQ1 (Round29) supplies, by diagonal extraction along centered whole-star boxes, locally normal stationary subsequential states, with trace-norm convergence of the local densities on every finite region; no uniqueness, whole-sequence convergence or rate in N is assumed. AQ1 §3 places the whole-star model in the Nachtergaele–Sims setting and obtains the limit dynamics T_θ of local observables. The clocks are

$$G_\tau = H_\tau/\alpha, \quad s = \frac{\alpha t_E}{\hbar}, \quad \theta = \frac{\alpha t}{\hbar}, \quad T_\theta^{F,N}(A) = e^{i(\theta/8)\hat{H}_N^F} A e^{-i(\theta/8)\hat{H}_N^F}, \quad (5)$$

where $\hat{H}_N^F$ is the normalized box operator of a family F below. The normalized time $u = \theta/8$ is used only internally, with this conversion, so the BA2 window $|\theta| \leq 8$ is $|u| \leq 1$. Ground scalars cancel in the conjugation. The positive energy α , action $\hbar$ and spacing a stay fixed, so θ and τ are dimensionless; a window in u labelled θ is an eightfold clock error and a rejected control.

1.2 The two named construction families

Both families use the centered coarse cubes $\Lambda_N = [-N, N]^3$ with $N \geq 2$, exactly as Round32’s AY1 admitted them (AY1 gate):

$$\begin{aligned} \text{F1 (AQ1): } \hat{H}_N^{F1} &= \sum_{b \in \Lambda_N} h_b + \sum_{b+S \subset \Lambda_N} \phi_b, \\ \text{F2 (I1 §6): } \hat{H}_N^{F2} &= \sum_{b \in \Lambda_N} h_b + \sum_{f: M_f \subset \Lambda_N} \left(-\frac{\tau}{3}\right) W_f, \quad \hat{H}_N^{F2, \text{pad}} = \hat{H}_N^{F2} \otimes 1 + 1 \otimes \sum_{x \in B_+ \setminus \Lambda_N} h_x. \end{aligned} \quad (6)$$

F1 is I1’s empty-boundary rule: an anchor group is retained only when its whole star lies in the box. F2 retains every omitted face whose owner set lies in the box, grouped at its anchor. I1 §6 pads the box to $B_+ = \Lambda_N + S$ and places the on-site operator alone on the $3(2N+1)^2$ padding sites. On the centered cubes F1 retains $21(2N)^3$ faces and F2 retains $28N(2N+1)(3N+1)$, so F2 keeps

$$|E_N| = 28N(2N+1)(3N+1) - 21(2N)^3 = 28N(5N+1) \quad (7)$$

extra faces, where $E_N = F2(\Lambda_N) \setminus F1(\Lambda_N)$. They sit at the $(2N+1)^3 - (2N)^3 - 1$ anchors with some $b_i = N$; the corner (N, N, N) carries none (Table 2). The face sets are totally ordered, $F1(\Lambda_N) \subset F2(\Lambda_N) \subset F1(\Lambda_{N+1})$. For every $N \geq 2$ both families retain the same 82 faces meeting R , because the owner sets of the faces meeting R lie in $[-1, 1]^2 \times [-1, 2] \subset \Lambda_2$.

Table 2: Retained omitted faces of the two families on the centered cubes. The Λ_2 and Λ_3 rows are enumerated identically in AY1, BA1 and BA2; in the Λ_4 row the extra faces and partial groups are enumerated in BA1 and the totals follow from the all-size formulas. The extra faces sit at the anchors of the partial groups.

Box	F1 faces	F2 faces	Extra faces $28N(5N+1)$	Partial groups	Padding sites $3(2N+1)^2$
Λ_2	1344	1960	616	60	75
Λ_3	4536	5880	1344	126	147
Λ_4	10752	13104	2352	216	243

Other prescriptions are not covered. The literal vertex boxes of I1 §7 clip the on-site operator and form another model; periodic and orthant boxes are not among the named families either. Two coarse metrics occur, and neither is a fine-site metric. BA1 and BB1 use the coarse ℓ^∞ metric, in which every star has diameter 1; BA2 uses the coarse ℓ^1 metric of AQ1 §3, in which every star has diameter 2; BB2 uses the first for states and the second for dynamics. They satisfy $d_\infty \leq d_1 \leq 3d_\infty$, and no constant mixes them.

1.3 Round32 checkpoint

Round32 closed with ten investigations accepted within scope ([Round32 addendum](#)). Its fourth sub-round compared the two families statically. AY1 proved that every pair of subsequential limits of F1 or F2 at the same coupling satisfies $\|\rho_R - \rho'_R\|_1 \leq 2D \approx 2.72249 \times 10^{-8}$ on the cover, with D the AV1 forward tier-(ii) bound. The limits share the first-order density $\rho_R^{(1)}$ of the ten faces with owner set R , and beyond it they differ by at most $2K'_2\tau^2 \approx 2.6835 \times 10^{-12}$, with the R -local constant $K'_2 \approx 13417.53$. AY2 certified a static two-sided distance of every limit from the Haar product on R , between about 4.3786×10^{-10} and 4.4055×10^{-10} , and listed six obligations that the static bounds leave open ([AY2 gate](#)). Table 3 repeats them with their standing in this edition.

Table 3: The six AY2 obligations and their standing in this edition, with O1 split into O1a and O1b as in the BC1 review. “Open” means that no Round33 gate has closed the obligation. The reviewed table of the BC1 gate, with 21 rows, is Table 18.

AY2 obligation	Standing in this edition
O1a: all subsequential limits of F1 and F2 coincide on every finite region (from O1, uniqueness of the limit)	Closed within the BB2 scope, as the BC1 review records it (Section 6.8): every AQ1 and every F2 subsequential limit equals the limit of the named constructions on every finite region (BB2 items 2–3). The scope is the named constructions F1 and F2 only, at fixed spacing and both signs of τ at the cap.
O1b: uniqueness of any ground state, states outside the named constructions and other boundary conditions (the general part of O1)	Open. No Round33 gate closes it, and the plan fixes the gate field <code>uniqueness_of_ground_state_claimed</code> to false for every investigation. No separate Dobrushin or HTW loop is scheduled (the Jung lens’s narrowed dissent is recorded); the BC1 table names these as candidate routes.
O2: whole-sequence convergence	Closed for the named constructions within the BB2 scope: the reduced densities of F1 and F2 converge as whole sequences on every finite region, without compactness, to one common limit that equals every AQ1 and F2 subsequential limit (Section 5), from the BB1 locality bounds (Section 4).
O3: translation invariance	Closed for the named constructions within the BB2 scope: the limit is invariant under coarse translations; non-coarse fine translations are not symmetries of the model and are rejected.
O4: a rate in N	Closed for the named constructions within the BB2 scope: the rate q^{N-1} , $q = 1/64$, per coarse step at fixed spacing for the reduced densities; the correlation functions converge with an explicit rate only on $5 \leq N \leq 14000$.
O5: boundary independence of dynamics on compact time windows	Closed within the BA2 scope: an algebraic comparison of the F1 and F2 finite-box evolutions on $ \theta \leq 8$ (Section 3).
O6: dynamics of the padded family	Closed within the BA2 scope: the F2 limit dynamics is AQ1’s, and AQ1 §§4–5 are rerun for every F2 subsequential limit state (Section 3.8).

The Round32 roadmap ranked five unexecuted goals: uniqueness of the local restriction, boundary independence of dynamics on compact windows, whole-sequence convergence with a rate in N , one estimate uniform along a named trajectory in the lattice spacing, and a sharper R -local second-order constant, the last triggered only ([Round32 roadmap](#)). Its missing premise for whole-sequence convergence was a decay factor for the AM2 fixed point under a change of the box boundary. The Round33 brief proposed that factor as the first candidate direction: each step of a chain of stars might cost a factor of order $JG(R_a)$, which the advisor’s selection note writes as the ball ratio $64JG(R_a) = 37888|\tau|$. BA1 tests that proposal for coefficients.

1.4 Round33 plan

On 24 September 2026, after the Round32 merge, the user asked to continue with the next three rounds and then to apply the admitted solutions and new equations to their related problems, if any ([brief](#)). As in Round32, a “round” is read as a sub-round of two investigations with the advisor, the skeptic and the expert panel, followed by one research assistant per lens. Round33 runs three research sub-rounds and an applications stage.

Two deliberation loops preceded production. In loop 1 each lens wrote a memo and a source ledger, and the skeptic wrote a prospective triage and a control catalogue; the advisor’s plan v1 fixed four sub-rounds, adopted the rate $37888|\tau|$ as the candidate per-step factor and did not schedule a Dobrushin/HTW loop, recording the Jung lens’s dissent. In loop 2 the three lenses signed off, and the skeptic signed off the architecture with thirteen required changes (P1–P13), vetoing the BA1 and BA2 freezes until its blocking edits were applied ([deliberation](#), [loop 2](#)). Among them: the BA1 floor target moved from 1/16 to 1/12, because 1/16 lacked the freeze-time margin of 2; the BA2 reverse route was replaced, since a triangle inequality through the two limits cannot meet the comparison target; the within-family Cauchy targets were quantified over all $M > N$; and the Nachtergaele–Sims excerpt was committed as a source. The panel previews that justify the frozen margins are recorded in that planning file only; they are not producer premises and are not results. Plan v2 was frozen before BA1 and BA2 ([plan record](#)). Its history records plan sha256 5310d5af...0cdc4b at the BA1/BA2 freeze, a panel update after sub-round 1 that kept every goal, the BB1/BB2 freeze after a skeptic pre-freeze review, and the BC1/BC2 freeze on 25 September after the skeptic’s pre-freeze review of the BC drafts (30 blocking and 28 non-blocking edits applied), and the BD1/BD2 freeze later that day after the review of the BD drafts (34 blocking and 21 non-blocking edits applied, with the BD2 width target tightened from $1/10^{10}$ to $1/10^{16}$). All eight investigations are gated. [Table 4](#) lists the plan.

Direction rules. Derivation-bearing investigations are paired, with genuinely different forward and reverse routes named in the contract. Formula-instantiating certificates use one direction plus an independent skeptic replay written from the contract alone, and statements use a statement plus skeptical review. Same-author passes are never independent review. For paired investigations the reverse producer receives only `AGENTS.md`, the contract and its declared shared premises, and the freezing tool enforces that inventory. After each sub-round the advisor may reorder, replace or retitle the next pair; the change is recorded in the next selection note and in the plan history. The applications stage records, for each admitted equation it applies, either a labelled transfer to a named model with its own exact checks or the obstruction that prevents the transfer.

Contract rules of plan v2. Every Round33 contract carries the Round32 pre-registration block (model identifier, selected triple, coupling and signs, state provenance, observable, nodes, itemized error terms, target, outcome types, sub-labels, claim exclusions, control ids, a mandatory sentence template quoted once as one unbroken span, and the required gate fields), and the Round32 closing-panel rules R1–R10, of which the contract freezer enforces R1–R5 and R8 and the gate recorder R6 and R7. Plan v2 adds the following rules.

- *One tier and one route per constant.* Every constant carries exactly one tier and exactly one route from the closed lists of [Table 5](#).
- *Margins.* The (rate, constant) pair of a decay target is frozen before production with a margin of at least 2 against the recorded previews. The margin is a freeze-time property of the target; acceptance asks only that the proved constant be at most the frozen target.
- *Scaling brackets per constant.* Each constant’s $\tau \rightarrow \tau/100$ bracket is declared before production and admits every valid form of its τ order: $[95, 105]$ for linear and $[9500, 10500]$ for quadratic constants, unless exact variants justify a narrower bracket.
- *No global Lipschitz decay.* The global Lipschitz constant of a contraction is never a decay factor.

Table 4: The Round33 plan (v2), all eight investigations gated. Direction: P = paired forward/reverse, S = single direction plus an independent skeptic replay, T = statement plus skeptic.

Sub-round, goal	Investigations	Dir.	Status in this edition
1, locality	BA1: boundary decay of the AM2 creation coefficients (weighted norm / analyticity).	P	gated: accepted within scope; boundary decay rate only; static, not dynamic
	BA2: boundary comparison of the Heisenberg dynamics on compact windows.	P	gated: accepted within scope; dynamics on compact windows
2, convergence	BB1: marginal locality, from coefficient decay to the reduced density (polymer / iterated split).	P	gated: accepted within scope; boundary decay rate only; static, not dynamic
	BB2: whole-sequence convergence, common limit and coarse translation invariance (nested telescoping / general-volume unions).	P	gated: accepted within scope; convergence and common limit of the named constructions
3, consequences	BC1: node and sign certificates restated for the limit; updated obligations.	T	gated: accepted within scope; certificates restated for the limit of the named constructions
	BC2: route-B uniform model, convergence and its node.	S	gated: accepted within scope (run because BB1 and BB2 were accepted); convergence of the route-B named construction, one family
4, applications	BD1: center-symmetry transfer of the parity theorem and the link-flip lemma across gauge groups.	P	gated: accepted within scope; transfer to named models, obstructions recorded
	BD2: new observables and dimensions in $SU(2)$ with the finite graphs.	S	gated: accepted within scope; sign certified on the finite graph, transfers to named models, obstructions recorded

- *Committed source excerpts.* An external theorem is quoted from a committed source excerpt with its PDF hash, never from a report that cites it.
- *Cauchy quantifier.* A within-family Cauchy target is quantified over all $M > N$; an N to $N + 1$ bound alone is not a Cauchy estimate.
- *Bookkeeping.* Every contract names its metric, weights, window and clock in its parameters and gives new control semantics for each new control id; `selected_after` never names the plan, and each freeze records the plan’s sha256 in the plan history.

Vocabulary and gate fields. Tiers, routes and sub-labels come from the closed lists of Table 5; the sub-labels are the five of Round32 and seven new ones.

Table 5: The closed Round33 vocabulary of plan v2 for tiers, routes and sub-labels.

List	Allowed values
Tiers	<code>crude_majorant</code> , <code>exact_first_order</code> , <code>polynomial_lieb_robinson</code> , <code>exponential_lieb_robinson</code> , <code>first_order_distance_from_product</code>
Routes	<code>weighted_norm</code> , <code>analytic_disc</code> , <code>polymer_kp</code> , <code>iterated_split</code> , <code>duhamel_inner_f1</code> , <code>duhamel_inner_f2</code>
Sub-labels from Round32	<code>reference_unresolved</code> , <code>sign_certified_finite_graph</code> , <code>sign_certified_below_cap</code> , <code>static_not_dynamic</code> , <code>uniform_local_closeness_not_uniqueness</code>
New sub-labels	<code>boundary_decay_rate_only</code> , <code>convergence_of_named_constructions</code> , <code>common_limit_of_named_constructions</code> , <code>dynamics_on_compact_windows</code> , <code>certificate_restated_for_limit</code> , <code>transfer_to_named_model</code> , <code>obstruction_recorded</code>

The gate fields form a closed vocabulary as well. `uniqueness_of_ground_state_claimed`, `rate_in_a_claimed` and `continuum_claim` are always false; `whole_sequence_claimed` may be true only with a nonempty scope naming the construction, and `translation_invariance_claimed` only with a scope naming coarse translations and the construction. The field `rate_in_N_claimed` is kept separate from `rate_in_a_claimed`. The field `dynamics_level` takes the value `algebraic_heisenberg_compact_window` for BA2 or `correlation_functions_compact_window` for BB2. P1 retires phrasings that presuppose a limit object, among them any name for a single infinite-volume ground state or a single limit and the usual statistical-mechanics names for the infinite-volume limit and for the decay length of correlations; an unqualified “rate” and an unqualified uniformity in the lattice spacing are never used. The admitted object would be the limit of the named constructions, and every rate is a rate in N at fixed spacing. A `limited` or `insufficient` verdict is a retained outcome, and investigation counts are never a fraction of the continuum problem.

2 BA1: boundary decay of the AM2 creation coefficients

Contribution HNM-C-BA1. The BA1 contract, recorded under the title “Hruday boundary decay of the AM2 creation coefficients for the named constructions”, was frozen before production on 24 September 2026 as investigation 1 of 8. It asks how far a change of the box boundary can move the AM2 creation coefficients on supports meeting the cover R , for the two named families F1 and F2 of Section 1.2. Two routes were assigned ([selection note](#)). The forward producer uses a diameter-weighted anchored norm that pays a fixed loss once per interaction term, together with a difference weight that grows with distance from the boundary source (route `weighted_norm`). The reverse producer, under premise isolation, continues the AM2 fixed point to complex couplings, counts the order at which a distant term can enter, and applies Cauchy’s estimate in maximum-modulus form (route `analytic_disc`). The routes share the AM2 majorant and differ in the decay mechanism. Both producers received the same 27 inputs: `AGENTS.md`, the contract and 25 shared premises, among them the AM2, AQ1, AV1, AW1, AY1 and AY2 records and the I1 dictionary.

The contract froze two (rate, constant) pairs. The headline pair is $q = 1/64$ with target $K \leq 1/2000000$ at the `exact_first_order` tier, where $q = 1/64$ is a chosen ratio realized by the weight $w = 64$ or the disc radius $64|\tau|$, not the ball radius R_a used as a per-step ratio. The floor pair is the per-step ball ratio $q_{\min} = 37888|\tau| = 148/390625$ at the cap, with target $K \leq 1/12$. The `crude_majorant` tier at $q = 1/64$ is reported, not targeted. The scaling brackets are $[95, 105]$ for the headline constant, $[99/100, 101/100]$ for the floor constant and exactly 100 for $q_{\min}$. The itemized error terms are `weighted_contraction_loss`, `boundary_source_terms`, `order_versus_distance_count`, `exact_first_order_remainder`, `cutoff_uniformity` and `arithmetic`, and the contract lists 37 controls.

2.1 Reviewed scope

The [BA1 gate](#) records the verdict `accepted_within_scope`, with sub-label `boundary_decay_rate_only` and secondary sub-label `static_not_dynamic`. Its accepted statement, quoted verbatim below in the gate’s ASCII notation, is the scope of every claim in this section. In it $R=1/64$ is the anchored radius R_a , while $R=\{0, e_z\}$ is the cover. The BA1 gate was recorded at 23:11 UTC, three minutes after the BA2 gate; the investigation numbers follow the plan, not the order of the gates.

```
In the AM2/AQ1 zero-selected patterned family (model AQ_patterned_zero_selected: SU(2)
Kogut-Susskind form on  $Z^3$  at fixed spacing, coarse 24-link factors, selected triple
exactly (0,0,0) with Haar product reference, 21 omitted faces per anchor entering as
 $-(\tau/3)W_f$  in normalized units  $\delta\alpha=\alpha/8$ ; both signs  $|\tau|\leq 10^{-8}$ ; AM2 creation
expansion with  $J\leq 28|\tau|$ ,  $R=1/64$ ,  $G(t)=16e^{\{8t\}(1+10t)}$ ; cover  $R=\{0, e_z\}$ ; coarse
1-infinity metric with star diameter  $d_X=1$ ), with the named construction families  $F1 =$ 
AQ1 centered whole-star boxes  $\Lambda_N=[-N, N]^3$  and  $F2 = I1$  section 6
all-contained-face boxes with padding on the same  $\Lambda_N$ ,  $N$  at least 2: For the
zero-selected patterned family at the same coupling  $|\tau|\leq 10^{-8}$ , and for the named
construction families  $F1$  (centered whole-star boxes) and  $F2$  (all-contained-face boxes
with padding) on centered coarse cubes, the AM2 creation coefficients in each on-site
cutoff space, on supports meeting the cover  $R$ , differ between two boxes of the named
constructions by at most  $K q^{(N-1)}$  in the anchored norm, uniformly in the cutoff, where
 $N$  is the smaller box size; this is decay of the coefficients of the named constructions
at a rate in  $N$ , not decay of the reduced density, not a statement about untruncated
creation coefficients, not uniqueness of any ground state, and not a statement uniform
in the lattice spacing  $a$ . Constants (exact, both signs, every comparison of the
contract:  $F1$  on  $\Lambda_N$  versus  $\Lambda_{(N+1)}$ ,  $F2$  on  $\Lambda_N$  versus  $\Lambda_{(N+1)}$ ,  $F1$ 
versus  $F2$  on the same  $\Lambda_N$ , any two centered boxes of size at least  $N$  and any two
complete-factor volumes of one prescription containing  $\Lambda_N$ ): headline pair  $q=1/64$ 
with  $K=49/111790368$  (about  $4.3832e-7$ ; exact_first_order tier, analytic_disc route on
the disc of radius  $64|\tau|$ ; margin about 1.1407 against the frozen  $1/2000000$ ), also
certified by the weighted_norm route with the smaller  $K=2734375/12204185915601$  (about
 $2.2405e-7$ , labelled); floor pair  $q=148/390625=37888|\tau|$  with  $K=49/2018304$  (about
 $2.4278e-5$ ; exact_first_order tier, analytic_disc route on the disc of radius
 $\tau_{\text{star}}=1/37888$ ), also certified by the weighted_norm route with
 $K=708203125/43148545682688$  (about  $1.6413e-5$ , labelled). The crude_majorant tier at
 $q=1/64$  ( $296/390625$  analytic_disc,  $9472/24454143$  weighted_norm) misses  $1/2000000$  and is
retained. The  $\tau$  to  $\tau/100$  ratios are  $4340004/43129$  for the headline  $K$ , exactly 1 for
the floor  $K$  and exactly 100 for  $q_{\min}$ . The rate is in  $N$ , per coarse 1-infinity step at
fixed spacing; every interaction term present in one box and not the other lies at
coarse 1-infinity distance at least  $N-1$  from  $R$ , and the bound uses the contract form
 $\text{ceil}(d/\text{diam})$  of the order-versus-distance count. For each family the whole sequence of
coefficients on supports meeting  $R$  is Cauchy in each on-site cutoff space; this is a
coefficient statement only, with no state, reduced density, common limit or untruncated
coefficient asserted.
```

The rest of this section derives that statement. The limitations recorded by the gate close the section (Section 2.12).

2.2 Metric, weights and the diameter convention

Distances are coarse ℓ^∞ distances on factor sites, $d(p, p') = \max_i |p_i - p'_i|$, and $\text{diam } I$ is the ℓ^∞ diameter of a support. Since $S - S \subset \{0, \pm e_i, \pm(e_i - e_j)\}$, every whole star has ℓ^∞ diameter exactly 1 and ℓ^1 diameter exactly 2, every face and every clipped F2 group has ℓ^∞ diameter at most 1, and each site lies in at most four interaction terms. Both producers verified these values by enumeration on Λ_2 , Λ_3 and Λ_4 (64, 216 and 512 stars). The forward norms are

$$\begin{aligned} \|c\|_w &= \max_u \sum_{I \ni u} w^{\text{diam } I} \|c_I\|, & 1 \leq w \leq w_{\max} = \frac{390625}{148}, \\ \|\delta\|_{B, \beta} &= \max_u \sum_{I \ni u} e^{\beta \rho_B(I)} \|\delta_I\|, & \rho_B(I) = \max_{p \in I} d(p, B), \end{aligned} \quad (8)$$

with $1 \leq e^\beta \leq w$. Here B is the source set of a comparison (Section 2.4). The difference weight grows with distance from B , that is towards R . The headline uses $w = e^\beta = 64$ and the floor $w = e^\beta = w_{\max}$. The loss $w^{d_X} = w$ is charged once per interaction term and never per creation. The reverse route replaces the weight by a disc of complex couplings of radius ρ , with $q = |\tau|/\rho$: $\rho = 64|\tau|$ for the headline and $\rho = \tau_* = 1/37888$ for the floor.

The convention matters. In the ℓ^1 reading each star costs w^2 , the self-map needs $w^2 \leq 390625/148$, that is $w \leq 51.37\dots$, and the rate per ℓ^1 step is at least about 1.9465×10^{-2} . The ratio $q = 1/64$ therefore does not close in ℓ^1 ; the verified ℓ^∞ diameter 1 carries the headline rate. The distances that matter below are the same in both metrics, so the ℓ^1 reading would give only the slower rate. A cardinality weight $e^{\mu|I|}$ would pay $e^{4\mu}$ per term and give a rate of at least $(148/390625)^{1/4} \approx 0.1395$; it is labelled and not used.

2.3 The order-versus-distance lemma

Scale the interaction by a coupling z . The fixed point is then analytic in z near zero (Section 2.6), and its Taylor coefficients obey $c_1 = L_0^{(1)}$ and $c_n = \sum_{k \geq 1} \frac{1}{k!} \sum_{n_1 + \dots + n_k = n-1} L_k^{(1)}(c_{n_1}, \dots, c_{n_k})$. Expanding each interaction slot writes c_n as a sum over families of n interaction pieces $Y_1, \dots, Y_n$ (faces, stars or clipped groups). Two facts of the AM2 estimate drive the lemma: a creation disjoint from the piece in the interaction slot commutes with it and kills the word, and the output support satisfies $\mathcal{N} \setminus X \subset M \subset \mathcal{N} \cup X$, with $\mathcal{N} = \bigcup_j I_j$. By induction on n , a nonzero contribution to $c_{n,M}$ comes from a family that is connected in its intersection graph and whose union contains M . The diameters then add through the interaction:

$$\text{diam } M \leq d_X + \sum_j \text{diam } I_j. \quad (9)$$

The sum cannot be replaced by a maximum. For $X = \{0, e_x\}$, $I_1 = \{-e_x, 0\}$ and $I_2 = \{e_x, 2e_x\}$ the output $M = \mathcal{N} \setminus X = \{-e_x, 2e_x\}$ is disconnected, with $\text{diam } M = 3 = d_X + 1 + 1$, while $d_X + \max_j \text{diam } I_j = 2$. Both producers use this fixture as a damaging control, and the reverse checker verifies (9) exhaustively on 3033 configurations.

Lemma (order versus distance). Let M meet R , and let a contribution to $c_{n,M}$ involve a piece Y at distance $d \geq 1$ from R , every piece having diameter at most diam . Then

$$n \geq 1 + \left\lceil \frac{d}{\text{diam}} \right\rceil \geq \left\lceil \frac{d}{\text{diam}} \right\rceil. \quad (10)$$

Proof. A point of $M \cap R$ lies in some piece Z_0 . Connectivity gives a chain $Z_0, Z_1, \dots, Z_m = Y$ of distinct pieces with consecutive members intersecting, and stepping through the intersections gives $d \leq m \text{ diam}$. Since $d \geq 1$, $Y \neq Z_0$, so the family has at least $m + 1$ pieces. $\square$

Two interactions that differ only in terms at distance at least d from R therefore give coefficients on supports meeting R that agree through order $\lceil d/\text{diam} \rceil$ in the coupling. The contract states the weaker form $\lceil d/\text{diam} \rceil$; the sharp form, with the $+1$ because the terminal piece need not be crossed, is proved by both producers. Every bound constant uses the contract form, for a reason given in Section 2.7. The proof uses the ℓ^∞ convention with diameter 1.

The contract asked for the order-versus-distance tables in both conventions on Λ_2 , Λ_3 and Λ_4 (Tables 6 and 7). Each cell gives the number of interaction terms at distance d from R and the least tree order at which such a term can enter, found by breadth-first search on the enumerated intersection graph. Every cell is at least the bound $1 + \lceil d/\text{diam} \rceil$. In ℓ^∞ the bound is attained for every $d \leq N - 1$, along the chain of stars anchored at $(0, 0, k)$; the larger values at $d = N$ come from the box walls. The seven terms at distance 0 are the seven incident anchors.

Table 6: Order versus distance in the coarse ℓ^∞ convention (diameter 1, used in the proof). Entries: terms at distance d / least tree order.

d	$1 + \lceil d \rceil$	F1 Λ_2	F1 Λ_3	F1 Λ_4	F2 Λ_2	F2 Λ_3	F2 Λ_4
0	1	7/1	7/1	7/1	7/1	7/1	7/1
1	2	47/2	62/2	62/2	62/2	62/2	62/2
2	3	10/4	131/3	166/3	55/3	166/3	166/3
3	4	–	16/6	255/4	–	107/4	318/4
4	5	–	–	22/8	–	–	175/5

Table 7: Order versus distance in the coarse ℓ^1 convention (diameter 2, a labelled alternative not used in the proof).

d	$1 + \lceil d/2 \rceil$	F1 Λ_2	F1 Λ_3	F1 Λ_4	F2 Λ_2	F2 Λ_3	F2 Λ_4
0	1	7/1	7/1	7/1	7/1	7/1	7/1
1	2	19/2	22/2	22/2	22/2	22/2	22/2
2	2	20/2	42/2	45/2	37/2	45/2	45/2
3	3	12/3	51/3	73/3	35/3	68/3	76/3
4	3	5/4	45/3	90/3	19/3	74/3	107/3
5	4	1/6	29/4	92/4	4/4	64/4	121/4
6	4	–	14/5	78/4	–	40/4	119/4
7	5	–	5/6	54/5	–	18/5	101/5
8	5	–	1/9	31/6	–	4/6	69/5
9	6	–	–	14/7	–	–	39/6
10	6	–	–	5/8	–	–	18/7
11	7	–	–	1/12	–	–	4/8

The reverse producer adds an exact sharpness fixture, a finite model that transfers to nothing: a qubit chain on sites $0, \dots, 4$ with $H_0 = \sum_x n_x$, bonds $X_i X_{i+1}$ and $R = \{0\}$, whose creation coefficients are obtained by exact cumulant inversion of the Rayleigh–Schrödinger ground vector. Removing the bond $(d, d+1)$ first changes a Taylor coefficient on a support containing 0 at order exactly $1 + d$. The witnesses are $c_{\{0,2\}}$ at order 2 for $d = 1$ (coefficient $-1/2$ with the bond, 0 without it), $c_{\{0,3\}}$ at order 3 for $d = 2$ ($5/8$ against 0) and $c_{\{0,4\}}$ at order 4 for $d = 3$ ($-7/8$ against 0). An exponent read from the far endpoint of the removed bond would claim vanishing one order too far.

2.4 Boundary sources and exact distances

In every comparison the smaller box’s interaction is contained in the larger one’s, and the source is the set of terms present in one box and not the other, each charged once. Table 8 gives the sources

of the three named comparisons. The counts are enumerated for $N = 2, 3, 4$ and have all-size forms: $(2N + 2)^3 - (2N)^3 = 8(3N^2 + 3N + 1)$ new stars for F1, $56(9N^2 + 14N + 6)$ new faces for F2 (an all-size form that the historical assistant’s independent enumeration reproduces), and $28N(5N + 1)$ extra faces for F1 versus F2.

Table 8: Boundary sources of the BA1 comparisons (ℓ^∞ distances; the ℓ^1 values are the same).

Comparison	New terms, charged once	Count at $N = 2, 3, 4$	Least distance from $e_z / 0$
F1 Λ_N vs Λ_{N+1}	whole stars inside Λ_{N+1} , not inside Λ_N ; each meets the shell $\Lambda_{N+1} \setminus \Lambda_N$	152, 296, 488 stars	$N - 1 / N$ (shell alone: $N / N + 1$)
F2 Λ_N vs Λ_{N+1}	faces with $M_f \subset \Lambda_{N+1}$, $M_f \not\subset \Lambda_N$, at 60, 126, 216 gaining anchors; at most 45 new faces per site	3920, 7224, 11536 faces	$N - 1 / N$
F1 vs F2 on Λ_N	the extra faces E_N ; 17, 13, 5 per anchor with one coordinate N , 10, 3, 1 with two, none at the corner	616, 1344, 2352 faces	$N - 1 / N$

The exact distances are

$$d(e_z, \text{shell}) = N, \quad d(0, \text{shell}) = N + 1, \quad d(e_z, \text{outer layer}) = N - 1, \quad d(0, \text{outer layer}) = N, \quad (11)$$

where the outer layer of Λ_N is $\{\max_i |p_i| = N\}$. A shell point has $|p_i| = N + 1$ for some i ; the only way to come within N of e_z is $p_z = N + 1$, attained at $(0, 0, N + 1)$. The same argument with N in place of $N + 1$ gives the layer distances, attained at $(0, 0, N)$. The two producers read the nested source set differently, and both readings give the same bound. The forward producer takes the shell, at distance N from e_z , and charges the source loss w once, since every new term meets the shell. The reverse producer uses the distance $N - 1$ of the new terms themselves: the new F1 star $(0, 0, N) + S$ contains $(0, 0, N) \in \Lambda_N$. The sites met by the new F1 stars are the shell together with the whole outer layer of Λ_N , with 98, 218 and 386 outer-layer sites at $N = 2, 3, 4$. The $28N(5N + 1)$ new F1 faces lying inside Λ_N are exactly the extra F2 faces $F2(\Lambda_N) \setminus F1(\Lambda_N)$. Each extra face keeps the coordinate that equals N at all its sites, so it lies in the positive outer layer $\{\max_i p_i = N\}$.

The reverse producer proves the distance bound for every comparison at once. *Common-core lemma*: for $N \geq 2$, every omitted face at ℓ^∞ distance at most $N - 2$ from R is retained by both families on every finite complete-factor volume containing Λ_N . Such a face has a point q within $N - 2$ of 0 or of e_z , so $q \in \Lambda_{N-1}$; its anchor $b = q - s$, $s \in S$, has $b_i \in [-N, N - 1]$, so $b + S \subset \Lambda_N$. The enumeration against all faces of F2 on Λ_5 agrees with it (82, 1044 and 3910 faces within distance $N - 2$ for $N = 2, 3, 4$). Hence every source term of every comparison lies at distance at least $N - 1$ from R , and the three named comparisons attain $N - 1$.

The F2 padding carries no interaction. The padded fixed point is therefore supported in Λ_N , and coefficients on supports meeting the $3(2N + 1)^2$ padding sites vanish; both producers work on Λ_N directly. In each comparison the larger interaction, split into its old and new terms with faces charged once, has per-site sum at most $28|\tau|$ (the enumerated maximum is 84 face units of $|\tau|/3$), so the AM2 constants apply to it. The fixed point does not depend on the decomposition, because each L_k is linear in V .

2.5 The weighted-norm route

Weighted multilinear estimate. For collections $c_1, \dots, c_k$ in one box, with interaction terms of at most $p = 4$ sites and diameter at most 1,

$$\|L_k(c_1, \dots, c_k)\|_w \leq w J L_k^{\text{num}} \prod_j \|c_j\|_w, \quad L_k^{\text{num}} = 16 \cdot 8^k (1 + \frac{5k}{4}), \quad (12)$$

and $\sum_k L_k^{\text{num}} t^k / k! = G(t)$. The proof follows AM2's itemized steps. Support localization is unchanged. Diameter subadditivity (9) gives $w^{\text{diam } M} \leq w^{d_X} \prod_j w^{\text{diam } I_j}$ for $w \geq 1$. The root in X contributes $16 \cdot 8^k w J$, and the root in a creation I_l contributes $16 \cdot 8^k (5k/4) w J$, the denominator $|I_l|$ cancelling against $\sum_{X \cap I_l \neq \emptyset} w^{d_X} \|V_X\| \leq w J |I_l|$. The loss $w^{d_X} \leq w$ enters once per interaction term, independent of k and of the number of creations. In the mixed form one slot holds a difference measured in $\|\cdot\|_{B,\beta}$, and the one-point quantity ρ_B satisfies $\rho_B(M) \leq \rho_B(I_j^*) + d_X + \sum_{j \neq j^*} \text{diam } I_j$; the loss is then e^β . For a source term X meeting B the loss is $e^{\beta \rho_B(X)} \leq e^\beta$.

Self-map, contraction and identification. Summing (12) over k and telescoping the k -linear terms give, for $\|c\|_w, \|c'\|_w \leq R_a$,

$$\|\Phi(c)\|_w \leq J w G(\|c\|_w), \quad \|\Phi(c) - \Phi(c')\|_w \leq \Gamma \|c - c'\|_w, \quad \Gamma = J w G'(R_a), \quad (13)$$

re-evaluated as exact rationals at the declared weights. The self-map $J_0 w G(R_a) \leq R_a$ holds exactly when $w \leq w_{\max} = 390625/148$. At $w = 64$ the self-map value is $148/390625$ and $\Gamma = 2464/390625 \approx 6.30784 \times 10^{-3}$. At $w_{\max}$ the rational self-map bound equals R_a exactly, the true $G(R_a) \leq 20.9632 \dots$ making it strict, and $\Gamma = 77/296 \approx 0.260135$. For $w \geq 1$ the weighted ball lies inside AM2's anchored ball, so Banach's theorem and AM2's uniqueness in its ball identify the weighted fixed point with AM2's: $\|c\|_w \leq R_a$ for every $w \in [1, w_{\max}]$, both families, every box, every Q_L and both signs.

Tiers. Every omitted owner set has ℓ^∞ diameter exactly 1, so $\|c^{(1)}\|_w = w \|c^{(1)}\|_a \leq w t_1$. Convexity of G gives $\|c - c^{(1)}\|_w \leq J w (G(t_w) - 16) \leq \Gamma t_w$, hence the `exact_first_order` tier

$$t_w := \|c\|_w \leq T_w = \frac{w t_1}{1 - J w G'(R_a)}, \quad (14)$$

$$T_{64} = \frac{49}{223580736} \approx 2.19160 \times 10^{-7}, \quad T_{w_{\max}} = \frac{49}{4036608} \approx 1.21389 \times 10^{-5},$$

the contract's self-consistent inequality re-derived in the weighted norm. The `crude_majorant` tier is $t_w \leq J w G(R_a)$: $148/390625$ at $w = 64$ and $1/64$ at $w_{\max}$. An exact first-order source combined with the crude t_w would be labelled crude, and the checker rejects every mislabel.

Difference theorem. Fix a comparison, a cutoff L , a sign and $|\tau| \leq 10^{-8}$. Let c be the smaller box's fixed point, zero-extended by ι , c' the larger box's, and $\delta = c' - \iota c$. Then, for every $u \in R$ and $1 \leq e^\beta \leq w \leq w_{\max}$,

$$\sum_{I \ni u} \|c'_I - c_I\| \leq e^{-\beta d(u,B)} \frac{S_B}{1 - \Gamma_\beta}, \quad \Gamma_\beta = e^\beta J G'(R_a), \quad S_B \leq e^\beta t_1 + \Gamma_\beta T_w. \quad (15)$$

Proof. The map is local: $\Phi_{\Lambda'}(\iota c) = \iota \Phi_\Lambda(c) + \Psi_{\text{new}}(\iota c)$, where Ψ_{new} is the AM2 series with V replaced by the new terms, because $\text{ad}_{\iota C}^k(\iota V_\Lambda) \Omega$ factors as box 1's value tensored with the outside vacuum. Hence $\delta = [\Phi_{\Lambda'}(c') - \Phi_{\Lambda'}(\iota c)] + \Psi_{\text{new}}(\iota c)$. The mixed estimate contracts the bracket by Γ_β . The source's first-order part is exactly $-(\tau/72) \sum_{\text{new } f} W_f \Omega_0$, with at most 49 new faces per site and $\rho_B(M_f) \leq 1$, so it costs at most $e^\beta t_1$; the remainder costs at most $\Gamma_\beta T_w$ by the source form. The volume is finite, so the difference is absorbed, and $\rho_B(I) \geq d(u, B)$ for $I \ni u$ extracts the decay. $\square$

The decay comes only from the weight $e^{-\beta d(u,B)}$; the Lipschitz factor enters only as $1/(1 - \Gamma)$. At $e^\beta = w$ the source bound is $S_B \leq T_w$, so the constant multiplying $q^{d(e_z, B)}$, $q = 1/w$, is

$$K_{\text{own}} = \frac{T_w}{1 - \Gamma} = \frac{w t_1}{(1 - \Gamma)^2}, \quad K_{12} = K_{\text{own}}, \quad K_{\text{nest}} = q K_{\text{own}}, \quad K_{\text{gen}} = K_{12} + \frac{K_{\text{nest}}}{1 - q}, \quad (16)$$

for F1 versus F2 at exponent $N - 1$, for a nested comparison (the shell lies at distance N), and for any two centered boxes by telescoping. At the headline, $K_{12} = 19140625/86785322066496 \approx$

2.20551×10^{-7} , $K_{\text{nest}} = 19140625/5554260612255744 \approx 3.44612 \times 10^{-9}$ and $K_{\text{gen}} = \frac{64}{63}K_{12} = 2734375/12204185915601 \approx 2.24052 \times 10^{-7}$, below $1/2000000$ with margin about 2.2316. At the floor the corresponding values are $1813/110502144$, $67081/10791225000000$ and $K_{\text{gen}} = 708203125/43148545682688 \approx 1.64131 \times 10^{-5}$, with margin about 5077 against $1/12$. The crude tier gives $9472/24454143 \approx 3.873 \times 10^{-4}$ at the headline, which fails $1/2000000$ and is retained, and $14453125/684115704 \approx 2.1127 \times 10^{-2}$ at the floor, which meets $1/12$ and is reported only.

The global contraction bound $\|c^{N+1} - c^N\|_a \leq J_0 G(R_a)/(1 - J_0 G'(R_a)) = 37/6249384$ is the same for every N ; it has no decay and is never used as a per-shell factor. Its denominator uses the Lipschitz constant $J_0 G'(R_a) < 77/781250$ of the fixed-point map; the contract's control text called $2J_0 G'(R_a)$, which is AM2's exclusion constant, the Lipschitz constant, and both producers corrected that wording independently.

2.6 The analytic-disc route

Complexified map. Write the interaction at unit coupling as $V_1 = \sum_X V_{X,1}$, $\|V_{X,1}\| \leq 7$, and set $V(z) = zV_1$. Then

$$F_z(c) = \sum_{k=0}^8 \frac{L_k^{(z)}(c, \dots, c)}{k!}, \quad L_k^{(z)} = z L_k^{(1)}, \quad \|L_k^{(z)}(c_1, \dots, c_k)\|_a \leq 28|z| 16 \cdot 8^k (1 + \frac{5k}{4}) \prod_j \|c_j\|_a. \quad (17)$$

The reverse producer re-derives AM2's estimate step by step and records where z enters: only through $\|V_X(z)\| = |z| \|V_{X,1}\|$. No step uses self-adjointness of V or reality of the coupling, and the coefficient equation does not involve the ground energy, so nothing else is complexified. For $|z| \leq \rho$ this gives the self-map bound $28\rho G(R_a)$ and the Lipschitz constant $\kappa(\rho) = 28\rho G'(R_a)$. The self-map holds up to $\tau_* = R_a/(28 \cdot 148/7) = 1/37888$. At the headline radius $\rho = 64|\tau| = 1/1562500$ the values are below $148/390625$ and $2464/390625$; at $\rho = \tau_*$ they are at most $1/64$ and $77/296$. A radius $2\tau_*$, and the radius $4\tau_*$ that an outgoing-only count $7|z|$ would allow, are rejected controls.

Analyticity. For $\rho \leq \tau_*$, every finite complete-factor volume of F1 or F2 and every L , F_z has exactly one fixed point $c(z)$ in the ball. The map $z \mapsto c(z)$ is continuous on $|z| \leq \rho$ and holomorphic on $|z| < \rho$, and at real $z = \tau$ it is the AM2 fixed point. The iterates $c^{(m+1)} = F_z(c^{(m)})$, $c^{(0)} = 0$, are polynomials in z that converge uniformly on the closed disc at rate $\kappa^m R_a$; Weierstrass's theorem gives holomorphy, contraction gives uniqueness in the ball, and AM2's uniqueness identifies the real value. On the circle, with $t(w) = \|c(w)\|_a$ and $t_1(\rho) = 49\rho/144$,

$$t(w) \leq T(\rho) = \frac{t_1(\rho)}{1 - 28\rho G'(R_a)} \quad (\text{exact first order}), \quad t(w) \leq 28\rho G(R_a) \quad (\text{crude majorant}), \quad (18)$$

and with $J = 28|\tau|$ and $w = \rho/|\tau|$, $T(\rho)$ is exactly the weighted T_w of (14).

Cauchy's estimate. Let Λ^A, Λ^B be finite complete-factor volumes containing Λ_N , with prescriptions in $\{F1, F2\}$, and $0 < |\tau| < \rho \leq \tau_*$. Then, for $u \in R$,

$$\sum_{I \ni u} \|c_I^A(\tau) - c_I^B(\tau)\| \leq 2t(\rho) \left(\frac{|\tau|}{\rho}\right)^{n_0} \leq 2t(\rho) \left(\frac{|\tau|}{\rho}\right)^{N-1}, \quad n_0 \geq N. \quad (19)$$

Proof. Embed both volumes in their union with unperturbed on-site factors on the complement; the fixed points keep their supports and values. The difference $g(z)$ is holomorphic on the open disc and continuous on the closed one. Terms of the Taylor recursion whose faces are all common to both volumes coincide, and every other term contains a source face at distance at least $N - 1$ (common-core lemma), so by (10) $g_n = 0$ for $n < n_0$ with $n_0 \geq N$. The maximum principle applied

to $g(z)/z^{n_0}$ through norming functionals, with $r \uparrow \rho$, gives the factor $(|\tau|/\rho)^{n_0}$, and the circle bound is $t^A + t^B \leq 2t(\rho)$. $\square$

No Lipschitz constant enters as a decay factor, and no source norm is charged: the source enters only through its distance, as the vanishing order. The constants are $K = 2T(\rho)$ at the exact tier and $K = 2 \cdot 28\rho G(R_a)$ at the crude tier. The general comparison is made directly through the union. Telescoping at exponent $N - 1$ would give $889/1006113312 \approx 8.836 \times 10^{-7}$ and a two-step union $49/55895184 \approx 8.766 \times 10^{-7}$, both above the headline target. The headline ratio $q = 1/64$ equals R_a numerically only because the frozen disc radius is $64|\tau|$.

What is not complexified. Only the creation coefficients are continued in z . The reduced density needs the normalization, the continued $\|\psi\|^2$, and that can vanish inside the disc: on two qubits $\psi(z) = \Omega - 2z|11\rangle$ has the entire coefficient $2z$, while its normalization $1 + 4z^2$ vanishes at $z = i/2$. No zero-free region is proved, and extending analyticity to the reduced density is a rejected control.

2.7 The gated constants

Following the skeptic’s recommendation and the Round32 precedent, the gate binds for each pair the larger valid constant, the reverse one, which both routes certify because the forward route proves a smaller constant for the same statement. At the cap $|\tau| = 10^{-8}$, both signs:

$$q = \frac{1}{64} : \quad K = 2T(64|\tau|) = \frac{49}{111790368} \approx 4.38320 \times 10^{-7} \leq \frac{1}{2000000}, \quad (20)$$

$$q = \frac{148}{390625} = 37888|\tau| : \quad K = 2T(\tau_*) = \frac{49}{2018304} \approx 2.42778 \times 10^{-5} \leq \frac{1}{12}. \quad (21)$$

Table 9 lists every constant with its tier and route. Read with the exponent, the headline bound is about 6.849×10^{-9} at $N = 2$, 1.070×10^{-10} at $N = 3$ and 1.672×10^{-12} at $N = 4$.

Table 9: BA1 constants at $|\tau| = 10^{-8}$, both signs; decimals are summaries of exact rationals. Only the rows marked “bound” are the gate’s admitted values.

Pair	Tier / route	Exact K	Status	$\approx K$
$q = 1/64$	exact / analytic disc, $\rho = 64 \tau $	$49/111790368$	bound; margin 1.1407	4.3832×10^{-7}
$q = 1/64$	exact / weighted, $w = 64$	$2734375/12204185915601$	labelled; margin 2.2316	2.2405×10^{-7}
$q = 1/64$	crude / analytic disc	$296/390625$	fails; retained	7.578×10^{-4}
$q = 1/64$	crude / weighted	$9472/24454143$	fails; retained	3.873×10^{-4}
$q = q_{\min}$	exact / analytic disc, $\rho = \tau_*$	$49/2018304$	bound; margin 3432.49	2.4278×10^{-5}
$q = q_{\min}$	exact / weighted, $w = w_{\max}$	$708203125/43148545682688$	labelled; margin 5077	1.6413×10^{-5}
$q = q_{\min}$	crude / analytic disc	$1/32$	meets 1/12; reported	3.125×10^{-2}
$q = q_{\min}$	crude / weighted	$14453125/684115704$	meets 1/12; reported	2.113×10^{-2}

Both signs, smaller couplings and scaling. Every formula depends on $|\tau|$ only, so the $\tau = -10^{-8}$ value replays the same exact rational; it is a replay, not a second confirmation. The headline K is increasing in $|\tau|$ at fixed $q = 1/64$, so the cap value bounds every smaller coupling. The floor constant does not depend on τ : at $w = 1/q_{\min}$ both $Jw = 7/9472$ and $wt_1 = 49/5455872$

are invariant, while $q = 37888|\tau| \leq 148/390625$ scales with $|\tau|$. The preregistered $\tau \rightarrow \tau/100$ test, evaluated from the same exact formula without intermediate rounding, gives

$$\frac{K_{\text{head}}(\tau)}{K_{\text{head}}(\tau/100)} = \frac{4340004}{43129} \approx 100.628, \quad \frac{K_{\text{floor}}(\tau)}{K_{\text{floor}}(\tau/100)} = 1, \quad \frac{q_{\text{min}}(\tau)}{q_{\text{min}}(\tau/100)} = 100, \quad (22)$$

inside $[95, 105]$, $[99/100, 101/100]$ and “exactly 100”. The labelled forward constants give 101.261 and $3255196/3253975 \approx 1.000375$; the latter differs from 1 only through the telescoping factor $w_{\text{max}}/(w_{\text{max}} - 1)$.

Why the contract form of the exponent. The sharp count $1 + \lceil d/\text{diam} \rceil$ gives exponent N at e_z . Written against q^{N-1} , a floor constant that uses it absorbs one factor $q_{\text{min}} = 37888|\tau|$ and becomes linear in τ , with ratio exactly 100, outside the prefrozen floor bracket. Every sharp form is therefore labelled, and every bound constant uses exponent $N - 1$.

2.8 The thin headline margin and the labelled refinements

The bound headline constant is only about 1.1407 below its target. The freeze-time margin rule was checked against the forward preview only, and a looser reverse majorant, or telescoping on the analytic route, would have missed the target. Acceptance asks only that the proved constant be at most the frozen target, so the thin margin is recorded, not blocking, and nothing was retuned after the constants were seen. The producers recorded refinements that are valid but not bound, each in the tier `exact_first_order`:

- the reverse sharp exponent, $49/7154583552 \approx 6.85 \times 10^{-9}$ at the headline and $1813/197100000000$ at the floor;
- the reverse first-order cancellation on R , $3773/1364628515625 \approx 2.76 \times 10^{-9}$: the exact first-order parts on supports through $u \in R$ coincide, so only the remainder is charged on the circle. It is quadratic in τ (ratio about 10062.8), outside the prefrozen linear headline bracket;
- the forward loss-one F1-versus-F2 value $57270955/16662781836767232 \approx 3.44 \times 10^{-9}$: the extra faces lie in the outer layer, where $\rho_B = 0$, so no source loss is needed;
- the reverse coefficientwise Cauchy form $14/31441041$, which is weaker than the maximum-modulus form.

A later loop that needs more room would have to prefreeze one of these as its own pair, with its exponent and bracket.

2.9 Coefficient decay is not decay of the reduced density

The ground vector $\psi = e^{-C}\Omega_0$ depends on every coefficient, and its R -marginal couples all supports through the normalization and through the straddling contractions of AV1. A bound on coefficients meeting R does not by itself control $\rho_{N,R}$; that step is BB1. Both producers exhibit the exact fixture (AV1 type, a finite model): qubit sites r, o, o' with $R = \{r\}$ and creations $c_{\{r,o\}} = 1/3$ (straddling, the only coefficient meeting R), $c_{\{o\}} = b$ and a decoupled $c_{\{o'\}}$. Changing b from $1/2$ to 0 leaves every coefficient meeting R unchanged, yet

$$\rho_r = \frac{1}{49} \begin{pmatrix} 45 & 6 \\ 6 & 4 \end{pmatrix} \longrightarrow \begin{pmatrix} 9/10 & 0 \\ 0 & 1/10 \end{pmatrix}, \quad \|\Delta\rho_r\|_1^2 = \frac{3681}{60025}. \quad (23)$$

The coefficient Cauchy property is a whole-sequence statement about coefficients in each cutoff space, obtained from the bound itself and not from compactness. The gate fields keep `whole_sequence_claimed`, `common_limit_claimed` and `state_decay_claimed` false, because they concern states; `rate_in_N_claimed` and `coefficient_cauchy_claimed` are true, the latter with the scope “AM2 creation coefficients of F1 and F2 on supports meeting R , in each on-site cutoff space”.

2.10 Uniformity in the cutoff, traps and error ledger

Every estimate is dimension-free and holds in each Q_L with constants independent of L . The compressed interactions keep their supports and do not increase J . The compression $Q_L c^{(1)}$ keeps or annihilates each face vector: $W_f \Omega_0$ lies in a product of on-site eigenspaces with site energies at most 18, so every face vector is kept for $L \geq 24$, and for smaller L the first-order counts only decrease. No cutoff removal is applied to coefficients.

The contract required exact fixtures for three traps; each is a damaging control in both checkers, and all are finite models that transfer to nothing.

- *A weight growing from R fails.* On a chain with a local averaging map and a unit source at the far end, the weight growing towards R certifies decay (for example $|\Delta_0| \leq 1/6$ against the exact $1/195$ in the reverse fixture), while the reversed weight gives $128/3$, above the unweighted bound 2; a non-submultiplicative weight 2^{r^2} and a min-distance weight are also rejected.
- *A global Lipschitz bound does not decay.* The nonlocal map $(A\delta)_i = \frac{1}{2}\delta_n$ has Lipschitz constant $1/2$ and fixed-point difference $\delta_0 = 1$ for every n ; the mean-field map on six sites keeps $c_0 = 1/6$ for every position of the source.
- *A per-creation loss charge fails.* A first-order coefficient on a two-site support has diameter $1 = d_X$ with no creation, so a loss charged per creation misses it. In the forward qubit fixture ($V = \sigma^x \otimes \sigma^x$ on $X = R$, $L_0 = \frac{1}{2}|11\rangle$) the weighted norm 32 at $w = 64$ exceeds the per-creation bound 16, while the per-interaction bound is 1024.

Table 10: The preregistered BA1 error terms, added linearly (weighted-norm values; the reverse reading is in the text).

Term	Headline, $q = 1/64$	Floor, $q = q_{\min}$
<code>weighted_contraction_loss</code>	$\Gamma = 2464/390625$; amplification $(1 - \Gamma)^{-2} \approx 1.012736$	$\Gamma = 77/296$; $(1 - \Gamma)^{-2} \approx 1.826818$
<code>boundary_source_terms</code>	$49/225000000 +$ $3773/2729257031250 = T_{64}$	$49/5455872 + 3773/1194835968 =$ $T_{w_{\max}}$
<code>order_versus_distance_count</code>	not a numeric cost: exponents $N - 1, N, N + 1$ enter as integers	same
<code>exact_first_order_remainder</code>	$\Gamma T_{64} = 3773/2729257031250$	$\Gamma T_{w_{\max}} = 3773/1194835968$
<code>cutoff_uniformity</code>	not a numeric cost: dimension-free in each Q_L	same
<code>arithmetic</code>	not a numeric cost: exact rationals; only the directed enclosure $e^{1/8} < 8/7$	same

On the reverse route the contraction loss is the circle factor $\rho/|\tau|$ inside $t(\rho)$ (64 at the headline, $1/q$ at the floor), the source enters only through its distance, and the remainder $28\rho G'(R_a)T(\rho)$ equals the forward values above. Deterministic terms add linearly; a root sum of squares and a division by $\sqrt{N}$ or by the square root of a face count are rejected controls.

2.11 Controls, replays and review

Each producer implements all 37 contract controls as damaging mutations that must raise explicit exceptions. The forward checker runs 51 exact checks with 96 rejected mutations and the reverse checker 61 exact checks with 132. Both checkers verify the contract hash before any evaluation, read their targets and brackets from the contract, pin every admitted gate they parse and record their own hash before the full evaluation. All four producer replays, forward and reverse under normal and optimized Python, reproduce the frozen outputs byte for byte, and each closure (31 files) verifies file by file. Both input inventories equal `AGENTS.md`, the contract and the 25 shared premises; the reverse inventory holds no Round33 skeptic, lens, forward, deliberation, plan or brief file.

The skeptic’s pre-comparison package, 101 exact checks, was committed before either producer’s commit. Its values equal the reverse headline, floor and crude constants exactly. The forward values differ from its values only because the forward evaluates G' at R_a rather than at T_w and telescopes the general comparison (factor 64/63); both are valid upper bounds, and the forward F1-versus-F2 value equals an exact variant the skeptic had listed. The post-comparison review adds 20 exact checks, 81 source-edit runs on temporary copies (two unmutated baselines, 74 validator weakenings, one per control per producer, and five input edits, every one aborting as expected) and 10 damaged packets rejected by the review’s own value validator. The producers wrote their final files after the skeptic’s commit, so isolation rests on disclosure and content, not on timing. Both producers disclose protocol and code-style reads outside their inputs; none carries premise weight. The review found no blocking issue; its non-blocking findings are the thin margin, the two readings of the nested source set, the reverse report’s “positive outer layer” where the literal set is the full outer layer, the forward union wording, and the Lipschitz wording of the contract.

2.12 Limitations recorded by the gate

The BA1 gate records the following limitations verbatim.

- Only the zero-selected patterned family (selected triple (0,0,0), Haar reference), the cover $R=\{0,e_z\}$, fixed spacing and $|\tau|\leq 10^{-8}$, with F1 and F2 on centered cubes Λ_N , N at least 2, and complete-factor volumes of one prescription containing Λ_N ; nothing transfers to nonzero selected triples, the uniform route-B model, literal vertex boxes, weak coupling or the continuum; every constant is uniform in N at fixed spacing and in the cutoff, never in a .
- Coefficients only: the statements concern AM2 creation coefficients in each on-site cutoff space Q_L with constants independent of L ; they are not decay of the reduced density or of any state (BB1/BB2), not a statement about untruncated creation coefficients, and the analyticity in the coupling is not extended to the reduced density (no zero-free region of the complexified normalization is proved).
- The coefficient Cauchy property gives whole-sequence convergence of coefficients per cutoff space only; `whole_sequence_claimed` and `common_limit_claimed` concern states and stay false; no uniqueness of any ground state is asserted.
- Thin headline margin: the bound headline $K=49/111790368$ is about 1.1407 below $1/2000000$ (the freeze-time margin rule was checked against the forward preview only); the labelled refinements (reverse sharp exponent $49/7154583552$, reverse first-order cancellation $3773/1364628515625$ which is quadratic in τ , forward loss-one F1-versus-F2 value $57270955/16662781836767232$) are recorded and not bound.
- Exponent convention: the bound uses the contract form $\text{ceil}(d/\text{diam})$ (exponent $N-1$ at e_z for every comparison); the sharp count $1+\text{ceil}(d/\text{diam})$ gives exponent N , but a floor constant written that way is linear in τ (ratio exactly 100) and falls outside the prefrozen floor bracket, so every sharp form is labelled.
- General comparisons: the reverse compares any two volumes directly (constant K); telescoping at exponent $N-1$ would give $889/1006113312$ and a two-step union $49/55895184$, both above $1/2000000$; the forward telescoped constant $2734375/12204185915601$ and its union constant are valid labelled values; the face sets are totally ordered ($F1_N$ inside $F2_N$ inside $F1_{(N+1)}$), so one nested comparison suffices for any pair of centered boxes.
- Contract wording read as follows: the source set of a nested comparison is the shell, and the literal set of sites met by the new terms adds the outer layer of Λ_N (98, 218, 386 sites at $N=2,3,4$) at distance $N-1$ from e_z , which gives the same bound once the source loss is charged; the fixed-point map has Lipschitz constant $J_{OG'}(R)$ below $77/781250$, while $2J_{OG'}(R)$ is the AM2 exclusion constant; `gate_fields_topic_specific` is read as the 12 frozen gate fields; the floor difference weight is $e^{\beta w}=390625/148$.
- Inherited without re-proof: the AM2 multilinear majorant, fixed point, uniqueness in the ball and cutoff facts; the AV1 exact first-order coefficient and self-consistent remainder; the AY1 item-by-item F2 premises; the I1 dictionary; Banach, Weierstrass and the maximum principle are standard and not machine-checked.
- Independence is limited to routes, enumerations and code: the contract and selection note name both mechanisms, the skeptic triage proposed both routes and the frozen

- parameters, all agents are correlated model agents, and isolation is verified for repository inputs only.
- Upper bounds only; the `-tau` values replay the same `|tau|` formula; scientific priority is unverified.

Full derivations and reviewed implementations: [BA1 forward](#), [BA1 reverse](#), [independent model-agent review](#) and the [frozen contract](#). BA1 is a statement about coefficients; Section 3 turns to the dynamics.

3 BA2: boundary comparison of the Heisenberg dynamics on compact windows

Contribution HNM-C-BA2. The BA2 contract, recorded under the title “Hruday boundary comparison of the Heisenberg dynamics of the named constructions on compact windows”, was frozen with BA1 before production as investigation 2 of 8. It takes up the AY2 obligations O5 and O6 (Table 3), whose route Round32 had named but not written: a Duhamel and Lieb–Robinson comparison over the extra F2 faces with the AQ1 Nachtergaele–Sims constants ([selection note](#)). The model is that of BA1; the observables are $A \in B(\mathcal{H}_R)$ on the cover; the clock is $\theta = \alpha t/\hbar$ with the window $|\theta| \leq 8$; the metric is the coarse ℓ^1 metric of AQ1 §3; the decay function is the polynomial $F(r) = (1+r)^{-4}$, tier `polynomial_lieb_robinson`. An exponential instance was allowed as a labelled extra without a target and was not executed.

Both directions are paired. The forward producer runs Duhamel’s formula over the extra faces with the F1 evolution inside the integral and the whole-star constants (route `duhamel_inner_f1`). The reverse producer, under premise isolation, puts the F2 evolution inside and uses the owner-set constants (route `duhamel_inner_f2`), and compares each family with its own Nachtergaele–Sims limit for the Cauchy estimates. Loop 2 of the deliberation replaced an earlier reverse route: the triangle inequality through the two limits gives only a bound of order $1/N$ and cannot meet the comparison target, so it may appear only as a labelled weaker bound. Both producers received `AGENTS.md`, the contract and 27 shared premises, among them the committed Nachtergaele–Sims excerpt, the AQ1 report, gate and review, the Round29 source dictionary and primary bindings, and the AY1 and AY2 records.

The frozen targets, each with the freeze-time margin of 2, are

$$\begin{aligned} \left\| T_{\theta}^{F2,N}(A) - T_{\theta}^{F1,N}(A) \right\| &\leq 6 \times 10^{-11} (5N+1)N^{-3} \|A\|, \\ \sup_{M>N} \left\| T_{\theta}^{F,M}(A) - T_{\theta}^{F,N}(A) \right\| &\leq \frac{2.5 \times 10^{-10}}{N-1} \|A\|, \end{aligned} \tag{24}$$

for $|\theta| \leq 8$, $N \geq 2$ and $F \in \{F1, F2\}$, together with whole-sequence convergence of the F2 evolutions, identification of their limit with the AQ1 limit dynamics T_{θ} , and a rerun of AQ1 §§4–5 for every F2 subsequential limit state. Both coefficients are quadratic in τ , with the bracket $[9500, 10500]$. The itemized error terms are `lieb_robinson_tail`, `duhamel_boundary_sum`, `interaction_picture_onsite`, `inner_family_constants` and `arithmetic`, and the contract lists 35 controls.

3.1 Reviewed scope

The [BA2 gate](#) records the verdict `accepted_within_scope` with sub-label `dynamics_on_compact_windows` and no secondary sub-label. Its gate fields set `dynamics_level` to `algebraic_heisenberg_compact_window`; `whole_sequence_claimed`, `dynamics_limit_identified_claimed` and `rate_in_N_claimed` are true, the first with the scope “finite-box Heisenberg evolutions of F1 and F2 of A in $B(\mathcal{H}_R)$, in norm, uniformly for $|\theta|$ at most 8; dynamics only, no state”; and `gns_dynamics_equality_claimed`, `uniform_in_time_claimed`, `common_limit_claimed`, `state_convergence`

claimed, translation_invariance_claimed, uniqueness_of_ground_state_claimed, rate_in_a_claimed, continuum_claim, weak_coupling_claim and scientific_priority_verified are false. Both producers read common_limit_claimed=false as “no common limit state”; the identification of the dynamics is carried by the separate true field. The accepted statement, quoted verbatim below in the gate’s ASCII notation, is the scope of every claim in this section.

In the AM2/AQ1 zero-selected patterned family (model AQ_patterned_zero_selected: SU(2) Kogut-Susskind form on Z^3 at fixed spacing, coarse 24-link factors, selected triple exactly (0,0,0), 21 omitted faces per anchor entering as $-(\tau/3)W_f$ in normalized units $\delta\alpha=\alpha/8$; both signs $|\tau|\leq 10^{-8}$, the $-\tau$ value a replay of the same $|\tau|$ formula; cover $R=\{0,e_z\}$ of the original xz Wilson loop, 48 links, 36 endpoints; observables A in $B(H_R)$; common clock $\theta=\alpha t/\hbar$ with the window $|\theta|\leq 8$, i.e. $U=1$ in the internal normalized time $u=\theta/8$), with the two named construction families $F1$ = AQ1 centered whole-star boxes and $F2$ = I1 section 6 all-contained-face boxes with padding on the same $\Lambda_N=[-N,N]^3$, N at least 2, with the Lieb-Robinson bound of Nachtergaele-Sims arXiv:1410.8174v1 Theorem 3.1, eqs. (51)–(52), quoted from the committed excerpt (PDF sha256 501af040512f2bdb62344ecbd093664e79ad1861338eabbce2c9791fca4e2fba), the unbounded on-site Casimirs h_b in the on-site slot of (44) and handled by the interaction picture, and the AQ1 constants $F(r)=(1+r)^{-4}$ on the coarse l1 metric, $C\leq 224$, $||\Phi||_F\leq 2268|\tau|$ for the whole-star interaction Φ ($J=28|\tau|$, four incident stars) and $||\Phi'||_F\leq 1323|\tau|$ for the owner-set interaction Φ' (49 faces per site), substituted as upper bounds by monotonicity: (1) $H^{\{F2, \text{pad}\}}_N$ minus $H^{\{F1\}}_N$ is the sum of the $28N(5N+1)$ extra faces (enumerated on Λ_2 to Λ_5 ; per anchor 17, 13 and 5 faces when exactly one coordinate equals N , 10, 3 and 1 when two do, none at the corner), each charged once with norm $|\tau|/3$, every owner on the outer layer at coarse l1 distance at least $N-1$ from e_z and at least N from 0 (both attained), plus the padding on-site terms, which factor out of the evolution exactly and are never charged. (2) Define $b(N) = K_{\text{cmp}} (5N+1) N^{-3}$ with $K_{\text{cmp}} = 3969/155720800000000$ (about $2.54879e-11$; forward route: Duhamel with the inner $F1$ evolution, whole-star Φ , tier polynomial_lieb_robinson, route duhamel_inner_f1; margin about 2.354 against the frozen 6×10^{-11}). For the zero-selected patterned family at the same coupling $|\tau|\leq 10^{-8}$ and every A in $B(H_R)$ with $||A||\leq 1$ on the fixed cover R , the finite-box Heisenberg evolutions $T^{\{F1,N\}}_\theta$ and $T^{\{F2,N\}}_\theta$ of the named construction families $F1$ and $F2$ on the same centered coarse cube Λ_N satisfy $||T^{\{F2,N\}}_\theta(A) - T^{\{F1,N\}}_\theta(A)|| \leq b(N)$ for $|\theta|\leq 8$ in the common clock $\theta=\alpha t/\hbar$; this is an algebraic comparison of the dynamics of the named constructions on a compact time window at a rate in N , not equality of GNS dynamics or of correlation functions of different states, not a uniform-in-time statement, and not a statement uniform in the lattice spacing a . The reverse route (Duhamel with the inner $F2$ evolution, owner-set Φ' , route duhamel_inner_f2) proves the same inequality with the smaller labelled second-route constant $K'_{\text{cmp}} = 452554889222515527/3048140234375000000000000000$ (about $1.48469e-11$, margin about 4.041); both constants are quadratic in τ (the first order vanishes because no extra face meets R), with exact $\tau/100$ ratios $1953058850/194651$ and $38176688920663121234473000000/3810205403940325763421973$ inside the preregistered bracket $[9500,10500]$. (3) For $F = F1$ and $F = F2$, sup over M greater than N of $||T^{\{F,M\}}_\theta(A) - T^{\{F,N\}}_\theta(A)|| \leq K_F/(N-1) ||A||$ for $|\theta|\leq 8$ and N at least 2, at the polynomial rate in N at fixed spacing: $F1$ with $K_{F1} = 9261/155720800000000$ (about $5.94718e-11$, forward, faces charged once, inner $F1$) and the second route $12128856933354903/1189670410156250000000000000$ (about $1.01951e-10$, reverse, whole stars, inner $F1$ on Λ_M); $F2$ with $K_{F2} = 117747/1245766400000000$ (about $9.45177e-11$, forward, one $F1$ -inner Duhamel plus the comparison) and the second route $1055961408185869563/3048140234375000000000000000$ (about $3.46428e-11$, reverse, owner-set Φ' , faces charged once); every value is at most the frozen 2.5×10^{-10} (margins about 4.204, 2.452, 2.645 and 7.217); a bound for N to $N+1$ alone is not used as a Cauchy estimate. (4) Hence the $F1$ and $F2$ finite-box evolutions of every A in $B(H_R)$ converge as whole sequences in norm, uniformly for $|\theta|\leq 8$, and the $F2$ limit dynamics equals the AQ1 limit dynamics T_θ : $||T^{\{F2,N\}}_\theta(A) - T_\theta(A)|| \leq \min(K_{F1}, \text{second-route } K_{F2})/(N-1) ||A||$, and on every local algebra the same Duhamel sum with R replaced by a finite region tends to zero on every compact window, so the two automorphism groups agree on the quasi-local algebra. (5) AQ1 sections 4–5 are rerun for every subsequential $F2$ limit state with $F2$'s own diagonal extraction (AY1): stationarity under T_θ , strong continuity of the GNS evolution and a nonnegative

generator, with the physical cyclic space reducing it. Not admitted: equality of GNS dynamics or of correlation functions of different states, a common limit state or convergence of states, uniqueness of every infinite-volume ground state, translation invariance, any uniform-in-time statement, exponential decay in N with the polynomial Lieb–Robinson function, any rate or estimate in the lattice spacing a , a continuum or weak-coupling statement, and scientific priority.

The rest of this section derives that statement, and the limitations recorded by the gate close it (Section 3.11).

3.2 The Lieb–Robinson theorem as the committed excerpt gives it

Plan rule P4 requires an external theorem to be quoted from a committed source excerpt, not from a report that cites it. The AQ1 report places the model in the theorem but does not state the inequality. The advisor therefore committed [an excerpt](#) of B. Nachtergaele and R. Sims, “On the dynamics of lattice systems with unbounded on-site terms in the Hamiltonian”, [arXiv:1410.8174v1](#) [11], Section 3 (pages 7–8) and Section 4 (page 11). The excerpt binds the PDF (203467 bytes, 14 pages) by the sha256

501af040512f2bdb62344ecbd093664e79ad1861338eabbce2c9791fca4e2fba,

identical to the Round29 primary binding ns-2014. Its Part A transcribes the displayed statements and was checked by the advisor against the rendered page images; its Part B is the verbatim `pdftotext` extraction, in which norm bars and superscripts are lost; where they differ, the rendered page is authoritative. The excerpt file itself has sha256 6a28f4cd...c6921, pinned in the forward checker. The theorem is quoted below as Part A gives it, set in mathematical notation. In the quotation τ_t^Λ is the paper’s symbol for the dynamics and t its time; neither is the coupling τ or the physical time of this edition.

Setting (Section 3). Γ is a countable set with a metric d . If Γ is infinite, $F : [0, \infty) \rightarrow (0, \infty)$ is non-increasing and

$$\|F\| = \sup_{x \in \Gamma} \sum_{y \in \Gamma} F(d(x, y)) < \infty \quad (\text{uniform integrability}); \quad (40)$$

$$C = \sup_{x, y \in \Gamma} \sum_{z \in \Gamma} \frac{F(d(x, z))F(d(z, y))}{F(d(x, y))} < \infty \quad (\text{convolution condition}). \quad (41)$$

To each $x \in \Gamma$ a separable complex Hilbert space $\mathcal{H}_x$ is associated; for finite $\Lambda \subset \Gamma$, $\mathcal{H}_\Lambda = \bigotimes_{x \in \Lambda} \mathcal{H}_x$ and $\mathcal{A}_\Lambda = \mathcal{B}(\mathcal{H}_\Lambda)$. For each x there is a self-adjoint operator H_x with dense domain $D_x \subset \mathcal{H}_x$; the bounded interaction Φ maps finite $X \subset \Gamma$ to $\Phi(X)^* = \Phi(X) \in \mathcal{A}_X$, and

$$H_\Lambda = \sum_{x \in \Lambda} H_x + \sum_{X \subset \Lambda} \Phi(X), \quad (44)$$

essentially self-adjoint on the dense domain $D_\Lambda = \text{span}\{\bigotimes_{x \in \Lambda} \psi_x \mid \psi_x \in D_x\}$ (45); $\tau_t^\Lambda(A) = e^{itH_\Lambda} A e^{-itH_\Lambda}$ for $A \in \mathcal{A}_\Lambda$ (46).

$$\|\Phi\| = \sup_{x, y \in \Gamma} \frac{1}{F(d(x, y))} \sum_{X \subset \Gamma: x, y \in X} \|\Phi(X)\| < \infty \text{ defines the space } \mathcal{B}_F(\Gamma). \quad (48)$$

$$S_\Lambda(X) = \{Z \subset \Lambda : Z \cap X \neq \emptyset \text{ and } Z \cap (\Lambda \setminus X) \neq \emptyset\}, \quad (49)$$

the surface of X in Λ , and $S(X) = S_\Gamma(X)$.

$$\partial_\Phi X = \{x \in X : \exists Z \in S(X) \text{ with } x \in Z \text{ and } \Phi(Z) \neq 0\}, \text{ the } \Phi\text{-boundary of } X. \quad (50)$$

Theorem 3.1. Let Γ and F be as indicated above. Fix a collection of local Hamiltonians $\{H_x\}_{x \in \Gamma}$ and an interaction $\Phi \in \mathcal{B}_F(\Gamma)$. Let $X, Y \subset \Gamma$ be finite disjoint sets. For any finite $\Lambda \subset \Gamma$ with $X \cup Y \subset \Lambda$ and any $A \in \mathcal{A}_X$ and $B \in \mathcal{A}_Y$, the bound

$$\|[\tau_t^\Lambda(A), B]\| \leq \frac{2\|A\|\|B\|}{C} (e^{2\|\Phi\|C|t|} - 1) D(X, Y) \quad (51)$$

holds for all $t \in \mathbb{R}$, where the quantity $D(X, Y)$ is given by

$$D(X, Y) = \min \left\{ \sum_{x \in X} \sum_{y \in \partial_\Phi Y} F(d(x, y)), \sum_{x \in \partial_\Phi X} \sum_{y \in Y} F(d(x, y)) \right\}. \quad (52)$$

Theorem 4.1. Let Γ and F be as described in Section 3. Fix a collection of on-site Hamiltonians $\{H_x\}_{x \in \Gamma}$ and an interaction $\Phi \in \mathcal{B}_F(\Gamma)$. For each $t \in \mathbb{R}$ and $A \in \mathcal{A}_\Gamma^{\text{loc}}$, the norm limit

$$\tau_t(A) = \lim_{\Lambda \rightarrow \Gamma} \tau_t^\Lambda(A) \quad (77)$$

exists and the convergence is uniform for t in compact sets. The limit may be taken along any increasing sequence of finite sets Λ which tend to Γ , and the result is independent of the particular sequence. This limiting dynamics $\tau_t(\cdot)$ can be uniquely extended to a one-parameter group of $*$ -automorphisms on $\mathcal{A}_\Gamma$.

The excerpt adds two notes of its own. The proof of Theorem 3.1 (steps 1–4) uses an interaction-picture dynamics (57) generated from $H_0 = \sum_{x \in \Lambda} H_x + \sum_{Z \subset X} \Phi(Z)$ (54), which is how the unbounded on-site terms are handled; and the proof of Theorem 4.1 uses $U_\Lambda(t, s) = e^{itH_\Lambda^{\text{loc}}} e^{-i(t-s)H_\Lambda} e^{-isH_\Lambda^{\text{loc}}}$ (78) with $H_\Lambda^{\text{loc}} = \sum_{x \in \Lambda} H_x$ (79). The section title of Theorem 4.1 is omitted here, as in both producer reports. Equation (47), Stone’s theorem for $t \mapsto e^{-itH_\Lambda}$, is available only in Part B and is used only for strong continuity and domain invariance. Both producers quote every statement sentence and equation of these passages verbatim; the forward report drops the bold marker of one heading and replaces the section title. Both checkers require the quotations verbatim, and the forward controls reject a paraphrase, a dropped velocity factor 2 or prefactor $1/C$, the minimum in (52) replaced by a maximum and a quotation taken from the AQ1 report. The theorems are cited, not machine-checked.

3.3 Placement in the model

Table 11 places the model. The unbounded Casimirs sit in the on-site slot of (44) and enter no constant; only bounded operators are ever subtracted between two generators below.

Table 11: Placement of the model in the Nachtergaele–Sims setting.

Source object	This model
Γ , metric d	coarse $\mathbb{Z}^3$ with the ℓ^1 metric
F , (40), (41)	$F(r) = (1+r)^{-4}$; $4r^2 + 2$ sites at distance $r \geq 1$; $\ F\ \leq 1 + 6 \sum_{k \geq 2} k^{-2} \leq 7$; $C \leq 32 \ F\ \leq 224$
H_x on D_x	$h_x = 8 \sum_{e \in x} C_e$, unbounded, self-adjoint, compact resolvent, on the separable $L^2(\text{SU}(2)^{24})$
whole-star Φ , (48)	$\Phi(b + S) = \phi_b$; ℓ^1 diameter 2, four stars per site, so $\ \Phi\ _F \leq 81J = 2268 \tau $; F1 on Λ_N is its native restriction (44)
owner-set Φ' , (48)	$\Phi'(M) = -(\tau/3) \sum_{M_f=M} W_f$; 49 faces per site, so $J' \leq 49 \tau /3$ and $\ \Phi'\ _F \leq 81J' = 1323 \tau $; F2 on unpadded Λ_N is its native restriction
time t in (46)	the normalized time $u = \theta/8$

The convolution bound comes from splitting the sum at the midpoint: for any z , $d(x, z) \geq d/2$ or $d(z, y) \geq d/2$, and $F(d/2) \leq 16F(d)$. The factor $(2/C)(e^{2nC|t|} - 1) = 4n|t| \varphi(2nC|t|)$, with $\varphi(y) = (e^y - 1)/y$ increasing, is increasing in C and in $n = \|\Phi\|_F$, and so is its time integral; the admitted upper bounds may therefore be substituted. The exact suprema in (48), found by pair enumeration, are $567|\tau|$ for Φ and $108|\tau|$ for Φ' , factors 4 and 12.25 below the admitted values; they are recorded and used nowhere. Placing an on-site operator inside the bounded interaction would make $\|\Phi\|_F$, and the velocity $2\|\Phi\|_F C$, grow with every on-site cutoff (at least $448L$ for a ladder of height L); both checkers reject it.

3.4 The boundary source and the padding

On the same box the two padded and unpadded generators differ by

$$\hat{H}_N^{F2,\text{pad}} - \hat{H}_N^{F1} \otimes 1 = \sum_{f \in E_N} \left(-\frac{\tau}{3}\right) W_f + \sum_{x \in B_+ \setminus \Lambda_N} h_x, \quad |E_N| = 28N(5N + 1), \quad (25)$$

read on $\mathcal{H}_{\Lambda_N} \otimes \mathcal{H}_{\text{pad}}$.

The extra faces. A face (b, K) , with D_K the set of directions in K , lies in F2 but not in F1 on Λ_N exactly when $b \in \Lambda_N$, $b_j < N$ for every $j \in D_K$, and $b_i = N$ for some $i \notin D_K$. Counting by anchor, the $(2N)^2$ anchors on each of the three planes with exactly one coordinate N carry 17, 13 and 5 faces, the $2N$ anchors on each of the three edges carry 10, 3 and 1, and the corner carries none: $4N^2(17 + 13 + 5) + 2N(10 + 3 + 1) = 140N^2 + 28N = 28N(5N + 1)$. Counting by class instead, each of the 14 one-direction classes contributes $2N(4N + 1)$ and each of the 7 two-direction classes $4N^2$, with $28N(11N + 2)$ face–owner incidences. The enumerations give 616, 1344 and 2352 extra faces at $N = 2, 3, 4$, and the class identities are checked up to $N = 199$. An owner $y = b + k$ of an extra face keeps $y_i = b_i = N$, so every owner lies on the outer layer, at ℓ^1 distance at least $N - 1$ from e_z and at least N from 0; both minima are attained at the owner $(0, 0, N)$, and eleven faces reach distance $N - 1$. No extra face meets R , so both families keep the same 82 faces meeting R . Each face is one pair (anchor, class), charged once with norm $|\tau|/3$.

The padding factors out. The padding term acts on its own tensor factor and commutes with $\hat{H}_N^{F2}$, with every face and with $B(\mathcal{H}_R)$. The closure of a sum of self-adjoint operators on separate factors generates the product of their groups, so

$$T_\theta^{F2,\text{pad},N}(A \otimes 1) = T_\theta^{F2,N}(A) \otimes 1 \quad (A \in B(\mathcal{H}_{\Lambda_N})), \quad (26)$$

and the Duhamel source on $\mathcal{H}_{\Lambda_N}$ is the bounded sum over E_N alone. The padding has $3(2N + 1)^2$ sites (75 at $N = 2$, 147 at $N = 3$). Charging the padding terms as a source would give a norm at least the cutoff L while their exact contribution is zero; both checkers reject it, and both verify (26) on exact Gaussian-rational fixtures.

Fine units. A coarse step is $(4a, 2a, a)$. The link tails of the extra faces are at least $N - 1$ fine steps from those of R along z , the plaquette corners come within $N - 2$ fine steps of the 36 endpoints of R (they touch at $N = 2$), and no link is shared. Only link sets matter for commutation, and no constant uses the fine metric.

3.5 Duhamel’s formula with a bounded difference

Let $H_1 = H_{\text{loc}} + V_1$ and $H_2 = H_{\text{loc}} + V_2$ on one finite box, with V_1, V_2 bounded and self-adjoint, so $D(H_1) = D(H_2) = D(H_{\text{loc}})$. Write $\tau_u^{(j)}(B) = e^{iuH_j} B e^{-iuH_j}$. Then, for bounded B and real u ,

$$\begin{aligned} \tau_u^{(2)}(B) - \tau_u^{(1)}(B) &= i \int_0^u \tau_s^{(2)}([V_2 - V_1, \tau_{u-s}^{(1)}(B)]) ds, \\ \|\tau_u^{(2)}(B) - \tau_u^{(1)}(B)\| &\leq \int_0^{|u|} \|[V_2 - V_1, \tau_r^{(1)}(B)]\| dr. \end{aligned} \quad (27)$$

The cocycle $W(s) = e^{isH_2} e^{-isH_1}$ satisfies $W(s)\psi - \psi = i \int_0^s e^{irH_2} (V_2 - V_1) e^{-irH_1} \psi dr$ on the common domain and then on every vector, so it is strongly differentiable; the product rule on vectors gives the identity. Only vector derivatives of bounded families are used, never a norm derivative in time. The evolution inside the commutator, $\tau^{(1)}$, is the *inner* family, and its Nachtergaele–Sims constants

are the ones that enter. Wherever on-site terms of a larger box or the padding appear, they are placed in the inner Hamiltonian, where they factor out; the unbounded terms never enter a source. For each face the bound (51), with $X = R$, $Y = M_f$ and the first entry of the minimum in (52), gives

$$\| [W_f, \tau_r^{(1)}(A)] \| \leq \frac{2 \|A\|}{C} (e^{v|r|} - 1) \sum_{x \in R} \sum_{y \in M_f} F(d(x, y)), \quad v = 2 \|\Phi_{\text{inner}}\|_F C, \quad (28)$$

since $\partial_\Phi M_f \subset M_f$ and $R \cap M_f = \emptyset$.

Why the first order vanishes. With the on-site evolution alone, $e^{irH_{\text{loc}}} A e^{-irH_{\text{loc}}}$ stays in $B(\mathcal{H}_R)$, and no extra face acts on a link owned in R , so the first-order Dyson term of the difference vanishes identically. In the bound one factor $|\tau|$ is the face coefficient and the other is the Lieb–Robinson growth $e^{vr} - 1 = O(|\tau|r)$. Every constant below is therefore quadratic in τ ; a linear label, as from the trivial commutator bound, is a rejected control.

3.6 The comparison of the two families

Forward route (inner F1, whole stars). Apply (27) on Λ_N with inner $\hat{H}_N^{F1}$ and outer $\hat{H}_N^{F2}$, add (28) linearly over the extra faces and integrate over $|u| \leq U$:

$$\begin{aligned} b(N) &= \frac{|\tau|}{3} \Lambda(U) S_b(N), \quad S_b(N) = \sum_{f \in E_N} \sum_{x \in R} \sum_{y \in M_f} F(d(x, y)), \\ \Lambda(U) &= \frac{2}{Cv} (e^{vU} - 1 - vU) \leq \frac{2 \|\Phi\|_F U^2}{1 - vU/3}, \end{aligned} \quad (29)$$

using $e^x - 1 - x \leq x^2/(2(1 - x/3))$ for $0 \leq x < 3$. Every extra face has at most three owners, each contributing $F(d(e_z, y)) + F(d(0, y)) \leq N^{-4} + (N + 1)^{-4} \leq 2N^{-4}$, so $S_b(N) \leq 168(5N + 1)N^{-3}$ and, at $U = 1$,

$$b(N) \leq K_{\text{cmp}}(5N + 1)N^{-3}, \quad K_{\text{cmp}} = \frac{254016 \tau^2}{1 - 338688|\tau|} = \frac{3969}{155720800000000} \approx 2.54879 \times 10^{-11}. \quad (30)$$

At the cap $v = 0.01016064$ exactly and $\Lambda(1) \leq 567/12457664 \approx 4.5514 \times 10^{-5}$. The margin against 6×10^{-11} is about 2.354. For example, $b(2) \leq \frac{11}{8} K_{\text{cmp}} \approx 3.5046 \times 10^{-11}$ against the target 8.25×10^{-11} , $b(3) \leq \frac{16}{27} K_{\text{cmp}} \approx 1.5104 \times 10^{-11}$ and $b(10) \leq 0.051 K_{\text{cmp}} \approx 1.2999 \times 10^{-12}$. The bound decays like N^{-2} , because the source is a surface of about N^2 faces at distance about N and $F \sim N^{-4}$.

Reverse route (inner F2, owner sets). The reverse producer puts the F2 evolution inside, which is the native restriction of Φ' , and bounds the time integral by $2 \|\Phi'\|_F U^2 E(vU)$ with $E(y) = 2(e^y - 1 - y)/y^2 \leq E_{\text{up}}(y) = 1 + y/3 + y^2/(12(1 - y/5))$, from $(j + 2)! \geq 24 \cdot 5^{j-2}$. With $\sum_f \|v_f\| D(R, M_f) \leq (|\tau|/3) 28N(5N + 1) 3(N^{-4} + (N + 1)^{-4}) \leq 56|\tau|(5N + 1)N^{-3}$,

$$K'_{\text{cmp}} = 148176 \tau^2 U^2 E_{\text{up}}(592704|\tau|U) = \frac{452554889222515527}{3048140234375000000000000000} \approx 1.48469 \times 10^{-11}, \quad (31)$$

margin about 4.041, where $E_{\text{up}}(9261/1562500) = 48866741088707/48770243750000 \approx 1.001979$. The skeptic's own closed form $9261/623765200000000$ and its sharp exponential enclosure bracket this value: sharp enclosure $\leq$ reverse $\leq$ closed form, the three differing only in how $e^x - 1 - x$ is bounded beyond second order. The gate labels K'_{cmp} as the reverse's own formula.

Constants swapped between families. The constants of the inner family, and only those, enter. On Λ_2 the F1 box is not the native restriction of Φ' (616 faces missing) and the F2 box is not the native restriction of Φ (60 partial star groups), so an F1-inner bound quoted with $1323|\tau|$, or an F2-inner bound with $2268|\tau|$, is a rejected control.

Labelled values. The enumerated boundary sum at $N = 2$ is

$$S_b(2) = \frac{616396498531}{82978560000} \approx 7.428,$$

against the all-size value 231 in (29); the all-size face sum exceeds the enumerated one by factors of 20 to 31 at $N = 2$ to 5. The enumerated sums give sharper values at small N only (about 1.127×10^{-12} forward and 6.565×10^{-13} reverse at $N = 2$); the headline is the all-size bound. The triangle inequality through the common limit gives only $2K_{F1}/(N-1)$ (forward) or $(c_1 + c_2)/(N-1)$ (reverse), which misses the comparison target at every $N \geq 2$; it is retained as a labelled weaker bound.

3.7 Within-family Cauchy estimates

Let E'_N be the set of faces whose anchor star is not inside Λ_N . Every point of such a star has $\|y\|_\infty \geq N$, so every owner of such a face lies at ℓ^1 distance at least N from 0 and at least $N-1$ from e_z , and no such face meets R . For every $M > N$ the differences $F1(\Lambda_M) \setminus F1(\Lambda_N)$, $F2(\Lambda_M) \setminus F1(\Lambda_N)$ and $F1(\Lambda_M) \setminus F2(\Lambda_N)$ lie in E'_N . With at most 49 faces through a site, $4r^2 + 2 \leq 4(1+r)^2$ and $\sum_{k>m} k^{-2} \leq 1/m$, the tail over all such faces is

$$S_c(N) = \sum_{f \in E'_N} \sum_{x \in R} \sum_{y \in M_f} F(d(x, y)) \leq 49 \left(\frac{4}{N} + \frac{4}{N-1} \right) \leq \frac{392}{N-1}, \quad (32)$$

uniformly in M . Each Duhamel pair lives on Λ_M (or its padding), with the on-site terms of the larger box in the inner Hamiltonian, where they factor out.

Forward. For F1 the inner Hamiltonian is $\sum_{x \in \Lambda_M} h_x$ plus the F1 interaction of Λ_N , the native restriction of the truncated interaction Φ_N , with $\|\Phi_N\|_F \leq \|\Phi\|_F$. The source is the new stars, split into their faces, each charged once. For F2 the forward producer goes through F1: one Duhamel pair between F1 on Λ_N and padded F2 on Λ_M , plus the comparison at N , using $\max_{N \geq 2} (5N + 1)(N-1)N^{-3} = 11/8$. This gives

$$\sup_{M > N} \left\| T_\theta^{F,M}(A) - T_\theta^{F,N}(A) \right\| \leq \frac{K_F}{N-1} \|A\|, \quad (33)$$

$$K_{F1} = \frac{592704 \tau^2}{1 - 338688|\tau|} = \frac{9261}{155720800000000}, \quad K_{F2} = K_{F1} + \frac{11}{8} K_{\text{cmp}} = \frac{117747}{1245766400000000},$$

that is, $K_{F1} \approx 5.94718 \times 10^{-11}$ (margin 4.204) and $K_{F2} = 941976 \tau^2 / (1 - 338688|\tau|) \approx 9.45177 \times 10^{-11}$ (margin 2.645). Since $F2(\Lambda_N) \subset F1(\Lambda_M)$ for $M > N$, a direct F1-inner pair also gives $\left\| T_\theta^{F2,N}(A) - T_\theta(A) \right\| \leq K_{F1}/(N-1) \|A\|$.

Reverse. Each family is compared with itself, with the larger box Λ_M inside the integral, so each uses its own interaction: Φ with whole stars for F1 and Φ' with faces charged once for F2. Exchanging sums bounds the source by $J_F(4/N + 4/(N-1)) \leq 8J_F/(N-1)$, with $J_{F1} = 28|\tau|$ and $J_{F2} = 49|\tau|/3$, uniformly in M :

$$c_1 = 1016064 \tau^2 E_{\text{up}}(1016064|\tau|) = \frac{12128856933354903}{118967041015625000000000000}, \quad (34)$$

$$c_2 = 345744 \tau^2 E_{\text{up}}(592704|\tau|) = \frac{1055961408185869563}{304814023437500000000000000000},$$

that is, $c_1 \approx 1.01951 \times 10^{-10}$ (margin 2.452) and $c_2 \approx 3.46428 \times 10^{-11}$ (margin 7.217). Charging whole anchor groups instead of faces would recharge old faces: from Λ_2 to Λ_3 it would count 616 old faces again, while the owner-set indexing charges only the 3920 new ones.

The rate is $1/N$, and it is honest. With the polynomial F the tail over $\{\|y\|_\infty \geq N\}$ is of order $1/N$. Exact lower witnesses show that no faster rate follows from these constants: the reverse checker proves $mS(m) \geq 1/2$ for the half-shell sums $S(m)$, $m = 2, \dots, 256$, and the forward checker a slab lower bound that contradicts any tail $C_0 q^N$ with $C_0 = 1$, $q = 1/2$ from $N = 9$ on. The comparison decays faster, like N^{-2} , only because its source is one surface layer. A bound for N to $N + 1$ alone is not a Cauchy estimate, since increments of order $1/N$ sum like the divergent harmonic series; every estimate above holds for all $M > N$ at once. No exponential decay in N is claimed. Every rate is a rate in N at fixed spacing, per coarse step $(4a, 2a, a)$, and nothing is uniform in the lattice spacing.

Gated values. Table 12 collects the constants. The gate binds the forward route as the headline, because it is the contract’s designated forward route, its constants are closed rationals equal to the skeptic’s independent closed forms (with K_{F2} an exact composition of two of them), and its margins are at least 2.35. The reverse constants are a labelled second route; the smaller of the two is a valid bound for each quantity.

Table 12: BA2 constants at $|\tau| = 10^{-8}$, $|\theta| \leq 8$, both signs. The $\tau \rightarrow \tau/100$ ratios lie in the preregistered [9500, 10500].

Quantity	Route (inner family, interaction)	$\approx$ value	Margin	$\tau/100$ ratio
Comparison K	forward headline, <code>duhamel_inner_f1</code> , Φ	2.54879×10^{-11}	2.354	10033.644
	reverse, <code>duhamel_inner_f2</code> , Φ'	1.48469×10^{-11}	4.041	10019.588
F1 Cauchy	forward headline, inner F1, faces once	5.94718×10^{-11}	4.204	10033.644
	reverse, inner F1 on Λ_M , whole stars	1.01951×10^{-10}	2.452	10033.615
F2 Cauchy	forward headline, through F1	9.45177×10^{-11}	2.645	10033.644
	reverse, inner F2 on Λ_M , faces once	3.46428×10^{-11}	7.217	10019.588

The exact ratios are 1953058850/194651 for all three forward constants, which share the form $c\tau^2/(1-338688|\tau|)$, and 38176688920663121234473000000/3810205403940325763421973 (comparison and F2) and 13564609990853609831750000/1351916508459800973113 (F1) for the reverse ones. The contract’s control semantics carry a stale bracket [9900, 10100]; every ratio happens to lie inside it too, and the preregistered [9500, 10500] governs. Every constant is increasing in $|\tau|$ and in U , so the cap values bound every smaller coupling and every $|\theta| \leq 8$, and the $-\tau$ evaluations replay the same $|\tau|$ formula.

3.8 Whole-sequence convergence and the F2 limit dynamics

Because the Cauchy bounds hold for all $M > N$, both whole sequences $(T_\theta^{F1,N}(A))_N$ and $(T_\theta^{F2,N}(A))_N$ converge in norm, uniformly for $|\theta| \leq 8$. For F1 the limit is AQ1’s T_θ (Theorem 4.1 for Φ). For F2, Theorem 4.1 applied to Φ' gives a limit dynamics of its own, and the comparison identifies it with T_θ : $\|T_\theta^{F2,N}(A) - T_\theta^{F1,N}(A)\| \leq b(N) \rightarrow 0$. Letting $M \rightarrow \infty$ in the Cauchy bounds,

$$\|T_\theta(A) - T_\theta^{F1,N}(A)\| \leq \frac{K_{F1}}{N-1} \|A\|, \quad \|T_\theta^{F2,N}(A) - T_\theta(A)\| \leq \frac{\min(K_{F1}, c_2)}{N-1} \|A\|, \quad (35)$$

where the second minimum is the gate’s “ $\min(K_{F1}, \text{second-route } K_{F2})$ ”. For a general local observable $A \in B(\mathcal{H}_X)$, $X \subset \Lambda_n$, the same Duhamel sum with R replaced by X and distance $N - n$

gives, for example on the forward route, $\|T_u^{F2,N}(A) - T_u(A)\| \leq (|\tau|/3)\Lambda(U) \cdot 49|X| \cdot 4/(N-n) \cdot \|A\| \rightarrow 0$ on every compact window $|u| \leq U$; this is qualitative, and the frozen targets concern $|\theta| \leq 8$ only. The two automorphism groups therefore agree on the local algebra and, being isometric, on the quasi-local algebra: the F2 limit dynamics is the AQ1 limit dynamics. This closes obligation O5 within the stated scope.

AQ1 §§4–5 for the F2 limit states (O6). AY1 admits for F2 its own local trace-norm compactness and diagonal extraction. Let $\omega^{(2)}$ be a subsequential limit along F2’s own subsequence N_k , the local trace-norm limit of the simple F2 box ground states. Both producers rerun AQ1 §§4–5 for every such state, each one separately.

1. *Stationarity.* Each box ground state is invariant under its own evolution. For local A and fixed n , $T_u^{F2,n}(A)$ is local, so local trace-norm convergence passes $k \rightarrow \infty$ and the Cauchy bound controls the remainder: $|\omega^{(2)}(T_\theta(A)) - \omega^{(2)}(A)| \leq 2c_2/(L-1)$ for every L (reverse), or $\leq 2\varepsilon_n$ with $\varepsilon_n = \sup_{|u| \leq 1} \|T_u(A) - T_u^{F2,n}(A)\|$ (forward). Hence $\omega^{(2)} \circ T_\theta = \omega^{(2)}$, then on the quasi-local algebra by density and for every θ by the group property.
2. *GNS unitaries and strong continuity.* Stationarity makes $U_\theta \pi(A)\Omega = \pi(T_\theta(A))\Omega$ a unitary group with $U_\theta \Omega = \Omega$. The map $\theta \mapsto \omega^{(2)}(A^* T_\theta^{F2,n}(A))$ is continuous through the normal density on Λ_n , and the uniform approximation passes continuity to $\omega^{(2)}(A^* T_\theta(A))$, so $\|(U_\theta - 1)\pi(A)\Omega\|^2 \rightarrow 0$. Density of local cyclic vectors and Stone’s theorem give a self-adjoint generator.
3. *Nonnegative generator.* For local A the finite correlation $\omega_N(A^* T_\theta^{F2,N}(A))$ is the Fourier transform of a positive measure on $[0, \infty)$, because the F2 box operator minus its ground energy is nonnegative (simple ground state and gap at least $1/2$ in normalized units, AY1 via AM2). These correlations converge for every θ and are bounded by $\|A\|^2$, so nonnegative smooth tests supported in $(-\infty, 0)$ pass their zero integrals to the limit spectral measure; the negative spectral projection annihilates every local cyclic vector and vanishes. Stationarity alone would not suffice: for $H = \text{diag}(0, 1)$ the invariant excited vector has a correlation of negative frequency.
4. *Physical sector.* Every retained F2 term is an on-site Casimir or a Wilson loop and the box ground states are simple, so $\omega^{(2)}$ is gauge invariant, T_θ preserves the gauge-invariant algebra, and the physical cyclic space reduces the unitary group and its generator.

Only strong continuity of the GNS group is claimed; norm continuity in time on all of $B(\mathcal{H})$ fails, as AQ1’s shift fixture shows (a difference of norm 2). This closes obligation O6 within the stated scope. The F2 limit states remain subsequential, and nothing identifies them with each other or with F1 limit states.

3.9 Meaning

BA2 concerns the algebraic Heisenberg dynamics of the named constructions: the finite-box evolutions of bounded observables on the cover, in operator norm, on the compact window $|\theta| \leq 8$, at a rate in N at fixed spacing, and their common norm limit. It does not assert that the GNS dynamics of different states agree, or that correlation functions of different states agree; that needs a proved common state, which is BB2’s subject. Exact fixtures in both checkers show one automorphism group with two invariant vector states whose σ_x correlations differ, $(3 \pm 4i)/5$ in the forward and $\pm i$ in the reverse fixture. The bounds grow like $U^2/(1 - vU/3)$ with the window, exactly like $(2/(Cv))(e^{vU} - 1 - vU)$, and nothing is claimed uniformly in time. The rates are N^{-2} for the comparison and $1/N$ for the Cauchy estimates, with the polynomial F ; no exponential decay in N and no estimate in the lattice spacing a is claimed.

3.10 Error ledger, controls, replays and review

The preregistered terms are itemized in both reports and added linearly. `lieb_robinson_tail` is $S_b(N) \leq 168(5N + 1)N^{-3}$ for the comparison and $S_c(N) \leq 392/(N - 1)$ for the Cauchy estimates (the reverse writes $F(N) + F(N - 1) \leq 2N^{-4}$ per owner and $8J_F/(N - 1)$). `duhamel_boundary_sum` is the face norm $|\tau|/3$ times the time integral times the F -sum. `interaction_picture_onsite` is zero and not applicable: the on-site terms are identical in every Duhamel pair, and the padding and larger-box on-site terms sit in the inner Hamiltonian. `inner_family_constants` names $C \leq 224$ and $\|\Phi\|_F \leq 2268|\tau|$ (inner F1) or $\|\Phi'\|_F \leq 1323|\tau|$ (inner F2). `arithmetic` is zero: exact rationals, with $e^x - 1 - x$ replaced by the directed rational bounds above. No term is divided by $\sqrt{N}$ or by the square root of a face count.

Each producer implements all 35 contract controls as damaging mutations. The forward checker runs 44 exact checks with 116 rejections and the reverse checker 59 exact checks with 149 rejections; four controls without frozen semantics were implemented from their earlier-round meaning. The skeptic's pre-comparison package (95 exact checks, 38 controls, 101 rejected mutations) was committed before both producer commits. The post-comparison review runs 113 checks: unmutated copies reproduce both frozen outputs byte for byte; 70 control weakenings, 35 per producer, each abort with the intended damaging mutation; 19 must-abort source edits all abort at the intended check; and five silent edits pass their producer checkers and are caught only by the review's exact value validator. That last finding is why the gate binds the exact rationals from the review. All four producer replays reproduce the frozen outputs byte for byte, both closures (33 files each) verify, and both producers hold exactly the 29 declared inputs, the reverse none of the skeptic, lens, deliberation, plan or forward files. The producers wrote their final files after the skeptic's commit, so isolation rests on disclosure, content and computation, not on commit order.

The review found no blocking issue. Its non-blocking findings are recorded in the gate: the stale bracket; four controls without frozen semantics; the observable identifier, which names N to $N + 1$ while the target and the proof concern all $M > N$; the notation $b(N)$ and $B(N)$, and the evolutions written with the letter of the coupling; no N_0 or clock field in the parameters; the route labels living only in the plan; the section title of Theorem 4.1, cited and not used as project wording; the exact suprema of (48); the forward F2 constant, built through F1 and 2.73 times the reverse's own-family value; the reverse comparison constant, its own directed formula between the skeptic's sharp enclosure and closed form; and the commit order.

3.11 Limitations recorded by the gate

The BA2 gate records the following limitations verbatim.

- Only the zero-selected patterned family (selected triple $(0,0,0)$), the cover $R=\{0,e_z\}$, fixed spacing and $|\tau| \leq 10^{-8}$, with the two named families F1 and F2 on centered cubes Λ_N , N at least 2; nothing transfers to literal vertex boxes, periodic or orthant boxes, nonzero selected triples, the uniform route-B model, weak coupling or the continuum
- Algebraic Heisenberg dynamics of the named constructions on a compact window only: no equality of GNS dynamics, no equality of correlation functions of different states, no common limit state and no convergence of states is established (F1 and F2 limit states remain subsequential and are not identified with each other); those need a proved common state (BB2)
- The frozen constants hold for $|\theta| \leq 8$ ($U=1$); the bounds grow like $U^2/(1-vU/3)$ and nothing is claimed uniformly in time; outside the window only qualitative convergence on compact windows is used (for the 06 rerun)
- Every rate is a rate in N at fixed spacing and strong bare coupling (a coarse step is $(4a, 2a, a)$): order N^{-2} for the comparison and $1/N$ for the Cauchy estimates with the polynomial $F(r)=(1+r)^{-4}$; no exponential decay in N is claimed, the optional exponential instance was not executed, and no estimate is uniform in the lattice spacing a

- All constants are upper bounds: the admitted $||\Phi||_F \leq 2268|\tau|$ is four times the exact (48) supremum $567|\tau|$ and $||\Phi'||_F \leq 1323|\tau|$ is 12.25 times $108|\tau|$ (recorded, not used); the all-size face sum $168(5N+1)N^{-3}$ is 20 to 31 times the enumerated sum at $N=2$ to 5
- Inherited without re-proof: Nachtergaele-Sims Theorems 3.1 and 4.1 as transcribed in the committed excerpt (not machine-checked); AQ1's $C \leq 224$ and $||\Phi||_F$ bounds (arithmetic re-verified); AM2's simple ground state and gap for F2 boxes and F2 compactness and diagonal extraction via AY1; the I1 dictionary; Stone's theorem, Fourier inversion and the tensor-product exponential of commuting self-adjoint operators on separate factors
- Independence is limited: the contract, the selection note and the AY2 obligations table name the route and the constants; the routes differ in the inner family, the interaction indexing, the constants, the exponential remainder bound, the enumeration and the code; all agents are correlated model agents; both producers wrote their final files after the skeptic pre-comparison commit, so isolation is shown by disclosure and content, not by commit order
- The producer checkers do not pin every labelled value: five silent source edits (labelled refinements, a retained weaker bound, preview strings) pass the producer checkers and are caught only by the skeptic value validator; the gate binds the exact rationals above
- The $-\tau$ evaluations replay the same $|\tau|$ formula and are not a second confirmation
- Scientific priority is unverified; the Lieb-Robinson and limit-dynamics theorems are credited to Nachtergaele-Sims and the 06 strategy to AQ1 (Gauvin arXiv:2503.15539v3 Supplement A.10)

Full derivations and reviewed implementations: [BA2 forward](#), [BA2 reverse](#), [independent model-agent review](#), the [committed source excerpt](#) and the [frozen contract](#). With BA1 and BA2 gated, sub-round 1 is complete; the panel kept every goal for sub-round 2 ([panel update 1](#)).

4 BB1: locality of the reduced densities

Contribution HNM-C-BB1. BA1 bounds how far a change of box moves the AM2 creation coefficients on supports meeting the cover; it says nothing about the reduced density, which depends on every coefficient through the normalization and the straddling supports (Section 2.9). The BB1 contract, recorded under the title “Hruday marginal locality: from coefficient decay to the reduced densities of the named constructions”, was frozen after the BA1 gate and before production, on 24 September 2026, as investigation 3 of 8 and the first of sub-round 2 ([selection note](#)). It asks for the step from coefficients to reduced densities: a bound on $\|\rho_Y^{\text{box } 1} - \rho_Y^{\text{box } 2}\|_1$ on the cover and on finite regions Y , for every comparison of the two named families, in each on-site cutoff space and, at fixed N , for the untruncated ground vectors. Both directions are paired. The forward producer writes the reduced density as a hard-core polymer gas and verifies a Kotecký–Preiss condition (route `polymer_kp`). The reverse producer, under premise isolation, applies the AV1 product-ordering split recursively with per-site charging (route `iterated_split`). Both received `AGENTS.md`, the contract and 33 shared premises, among them the BA1 and BA2 gates and reviews.

The contract froze the rate $q = 1/64$ and two targets in the `exact_first_order` tier: $C \leq 1/250000$ in the R form and $c_{\text{site}} \leq 1/500000$ in the region form

$$\|\rho_R^{\text{box } 1} - \rho_R^{\text{box } 2}\|_1 \leq C q^{N-1}, \quad \|\rho_Y^{\text{box } 1} - \rho_Y^{\text{box } 2}\|_1 \leq c_{\text{site}} |Y| e^{|Y|/10^8} q^{d_Y}, \quad d_Y = N - \max_{y \in Y} \|y\|_\infty, \quad (36)$$

for every finite complete-factor region $Y \subset \Lambda_N$, where N is the smaller box size and $d_R = N - 1$. A labelled secondary pair at $q_2 = 151552|\tau|$ has targets $1/20000$ and $1/40000$; the crude tier is reported, never targeted; and the per-step ball ratio of BA1 is not a target for densities. The scaling brackets are $[95, 105]$ for C and c_{site} , $[99/100, 101/100]$ for the secondary constants and exactly 100 for q_2 . The itemized error terms are `coefficient_input_every_site`, `straddling_supports`, `normalization`, `polymer_or_split_remainder`, `cutoff_removal_at_fixed_N` and `arithmetic`, and the contract lists 37 controls. The metric is BA1's coarse ℓ^∞ metric, with star diameter 1.

4.1 Reviewed scope

The [BB1 gate](#) records the verdict `accepted_within_scope`, with sub-label `boundary_decay_rate_only` and secondary sub-label `static_not_dynamic`. Its gate fields set `state_decay_claimed` and `rate_in_N_claimed` to true, the first with the scope “reduced densities of F1/F2 finite-box ground vectors on R and on regions Y, per comparison”; `whole_sequence_claimed`, `common_limit_claimed`, `translation_invariance_claimed`, `uniqueness_of_ground_state_claimed`, `rate_in_a_claimed`, `continuum_claim`, `weak_coupling_claim` and `scientific_priority_verified` are false. The accepted statement, quoted verbatim below in the gate’s ASCII notation, is the scope of every claim in this section. In it $R=1/64$ is the anchored radius R_a , and the exact rationals are the gate’s bound values; where a long numeral breaks across lines in the quotation, no character is added or removed. The gate was recorded on 25 September 2026 at 03:01 UTC.

```
In the AM2/AQ1 zero-selected patterned family (model AQ_patterned_zero_selected: SU(2)
Kogut-Susskind form on  $Z^3$  at fixed spacing, coarse 24-link factors, selected triple
exactly (0,0,0) with Haar product reference, 21 omitted faces per anchor entering as
 $-(\tau/3)W_f$  in normalized units  $\delta=\alpha/8$ ; both signs  $|\tau|\leq 10^{-8}$ ; AM2 creation
expansion with  $J\leq 28|\tau|$ ,  $R=1/64$ ,  $G(t)=16e^{\{8t\}}(1+10t)$ ; cover  $R=\{0,e_z\}$ ; coarse
1-infinity metric with star diameter 1; trace norm on  $B(H_Y)$ ), with the named
construction families  $F1 = AQ1$  centered whole-star boxes  $\Lambda_N=[-N,N]^3$  and  $F2 = I1$ 
section 6 all-contained-face boxes with padding on the same  $\Lambda_N$ ,  $N$  at least 2, and
with the BA1 gate bound value  $K=49/111790368$  used at every site of the union volume
through the every-site form (b), proved in both packets: For the zero-selected
patterned family at the same coupling  $|\tau|\leq 10^{-8}$ , and for every comparison of the
named construction families  $F1$  and  $F2$  on centered coarse cubes, the reduced densities
of the finite-box ground vectors on the cover  $R$  differ by at most  $C q^{(N-1)}$  in trace
norm, in each on-site cutoff space and, at fixed  $N$ , for the untruncated ground vectors,
where  $N$  is the smaller box size; this is locality of the reduced densities of the named
constructions at a rate in  $N$ , not uniqueness of any ground state, not a statement about
other boundary conditions, and not a statement uniform in the lattice spacing  $a$ .
Constants (exact; both signs; each of the five comparisons of the contract, namely  $c1$ 
 $F1$  on  $\Lambda_N$  versus  $\Lambda_{(N+1)}$ ,  $c2$   $F2$  on  $\Lambda_N$  versus  $\Lambda_{(N+1)}$ ,  $c3$   $F1$  versus
 $F2$  on the same  $\Lambda_N$ ,  $c4$  any two centered boxes of size at least  $N$  compared directly,
and  $c5$  two finite complete-factor volumes of one prescription containing  $\Lambda_N$ 
compared directly; in each on-site cutoff space with constants independent of the
cutoff, and at fixed  $N$  for the untruncated ground vectors):  $q=1/64$  and  $C=5784307935$ 
011820953163614046687121178140973664974888824331236872077433489356435905949478780525
21565633072717772804866626438590065415931413647325/649548567473042337664409810664855
266791889949984121342174530956201606073245694809426490159178862195173312829634280741
896750749402941566759659762614272 (about  $8.9051e-07$ ; exact_first_order tier, polymer_kp
route; margin about 4.49 against the frozen  $1/250000$ ); the iterated_split route proves
the smaller  $C=1649304957714574761387109521705122525/18522658383323707613583283296586$ 
87644475392 (about  $8.9043e-07$ , labelled). Region form: for every finite complete-factor
region  $Y$  inside  $\Lambda_N$ ,  $||\rho^{\text{box}1}_Y - \rho^{\text{box}2}_Y||_1 \leq c_{\text{site}} |Y| e^{\{|Y|/10^8\}} q^{\{d_Y\}}$ 
with  $d_Y=N-\max\{y \text{ in } Y\}|y|_{\text{inf}}$  and  $c_{\text{site}}=20704240478148137003187223317052643618088$ 
2031154261216585642344563461649618003876497888095865804915933/2361309485145316408070
35481765249880343283808663306771542081002582898728697162432799219796273340474330185728
(about  $8.7681e-07$ ; polymer_kp; margin about 2.28 against the frozen  $1/500000$ );
iterated_split proves the smaller  $c_{\text{site}}=25373922426378073252109377257001885/28941$ 
653723943293146223880150916994444928 (about  $8.7673e-07$ , labelled). Labelled secondary
pair at  $q_2=151552|\tau|$  with the re-instantiated input  $K_2=49/10202112$  (disc radius
 $1/151552$ , proved in both packets):  $C_2=254713191786858884385903518855136871787/2628$ 
8652445972361083565698627981857600000000 (about  $9.6891e-06$ ) and  $c_{\text{site},2}=651079047$ 
656055039494458366725211/67298950261689244373928188487633555456 (about  $9.6744e-06$ ),
iterated_split, meeting  $1/20000$  and  $1/40000$ ; polymer_kp proves the smaller values  $C_2$ 
about  $9.6440e-06$  and  $c_{\text{site},2}$  about  $9.6294e-06$  (labelled). The crude_majorant tier at
 $q=1/64$  ( $C$  about  $2.0529e-03$  polymer_kp and  $1.8006e-03$  iterated_split) misses  $1/250000$ 
and is retained. The  $\tau$  to  $\tau/100$  ratios of the bound constants are about 100.647876
for  $C$  and 100.647875 for  $c_{\text{site}}$ , about 1.001500 and 1.000000 for the secondary pair,
and exactly 100 for  $q_2$ . The rate is per coarse 1-infinity step in  $N$  at fixed spacing,
with constants uniform in  $N$  and in the cutoff; no limit object, whole-sequence
convergence, common limit or translation statement is formed, and no dynamics is
asserted.
```

The rest of this section derives that statement, and the limitations recorded by the gate close it (Section 4.7).

4.2 The decomposition and the every-site input

The AV1-type decomposition. Fix a comparison and a cutoff Q_L , and let Λ be the union of the two volumes, on which both finite-box ground vectors live with the smaller box’s collection zero-extended (the AM2 fixed point of a volume stays supported in it). For a region Y let A be the supports meeting Y , $T_c = \prod_{I \in A} (1 - \hat{c}_I) = 1 + D_c$, and $\varphi_{\text{out}} = \prod_{I \notin A} (1 - \hat{c}_I) \Omega$. Every creation not in A acts as the identity on $\mathcal{H}_Y$, so $\psi(c) = T_c(\Omega_Y \otimes \varphi_{\text{out}})$. Let $O = \bigcup_{I \in A} (I \setminus Y)$ be the straddle region and ω_O the normalized marginal of φ_{out} on O . Then, exactly,

$$N_c(X) = \text{Tr}_O[T_c(P_Y \otimes X)T_c^*], \quad \rho_Y(c) = \frac{N_c(\omega_O)}{\text{Tr } N_c(\omega_O)}, \quad \text{Tr } N_c(\omega) = 1 + \text{Tr}[D_c(P_Y \otimes \omega)D_c^*] \geq 1, \quad (37)$$

for every state ω on $\mathcal{H}_O$, pure or mixed. The last identity is AV1’s orthogonality: every term of D_c contains a creation excited at a site of Y , so $(P_Y \otimes 1)D_c = 0$ and the cross terms vanish. Two Lipschitz bounds follow, and both producers name their constants. In the outside state, $\|\rho_c(\omega) - \rho_c(\omega')\|_1 \leq 2(2\varepsilon_Y + \varepsilon_Y^2) \|\omega - \omega'\|_1$ with $\varepsilon_Y \leq \prod_{y \in Y} (1 + \sum_{I \ni y} \|c_I\|) - 1$; for $Y = R$ the constant is about 2.72×10^{-8} . It does not depend on N and is never used as a decay factor. In the coefficients, the forward proves $\|\rho_Y(c) - \rho_Y(c')\|_1 \leq 2(1+t)^{|Y|-1} \sum_{I \cap Y \neq \emptyset} \|c_I - c'_I\|$ from a derivative along the segment, with $t = 49/14398580736$ the AV1 self-consistent anchored bound. The reverse proves, for a change at a single support J and *every* region Y ,

$$\|\rho_Y(c) - \rho_Y(c')\|_1 \leq 2 \|c_J - c'_J\| \quad (\eta = 0), \quad (38)$$

by splitting at J itself: $\psi(c) - \psi(c') = -\hat{\delta}_J \psi_{\neg J}$ with $\psi_{\neg J} = (P_J \otimes 1)\psi(c)$, and the trace distance of two normalized rank-one projections is at most twice the distance of the vectors over the larger norm. Summed over far supports, (38) would grow with the shell, so it is used only for supports meeting Y .

Neither Lipschitz bound can simply be iterated. Iterating the outside-state bound on the straddle region, then on its straddle region, and so on, multiplies factors of order $|O_k|t_0$ with regions of growing size; with $|O_k| = (2k+1)^3$ that factor exceeds one at depth 264. Comparing the two global ground vectors directly fails too: in the global-fidelity fixture the one-site marginals stay a fixed distance apart while the global overlap collapses with the volume. Both are rejected controls. Two further fixtures fix what must be charged. Coefficient decay is not marginal decay (the fixture (23)); and on the chain $r-o_1-o_2$ with supports $\{r, o_1\}$, $\{o_1, o_2\}$ and $\{o_2\}$ of amplitudes a , b and c , the off-diagonal element of the unnormalized R -marginal is exactly $-abc$: a change two supports away moves the marginal at the order ab of the two straddling amplitudes, with no first-order term.

The every-site input, form (b). The BA1 gate admits the coefficient bound for $u \in R$ only. Both routes need it at every site of the union volume, and both prove it in full as a BB1 lemma (forward Theorem 8.1, reverse Theorem B6); the contract forbids citing it as an admitted BA1 statement. For every site u ,

$$\sum_{I \ni u} \|c_I^A(\tau) - c_I^B(\tau)\| \leq K q^{(N-\|u\|_\infty)_+}, \quad K = 2T(\rho), \quad T(\rho) = \frac{49\rho/144}{1 - 28\rho G'(R_a)}, \quad q = \frac{|\tau|}{\rho}. \quad (39)$$

The proof is BA1’s analytic-disc argument (Section 2.6) with R replaced by $\{u\}$. The circle bound $2T(\rho)$ is uniform in u , because the anchored norm is a maximum over sites. The vanishing order at u comes from the order-versus-distance lemma and from the *common-core lemma at u*: every omitted face with a site within distance $N - \|u\|_\infty - 1$ of u lies in a star $b + S \subset \Lambda_N$, and so is retained by both prescriptions on every volume containing Λ_N . Such a face has a site q' with $\|q'\|_\infty \leq N - 1$,

and its anchor $b = q' - s$, $s \in S$, has $b_i \in [-N, N - 1]$. Every source term of every comparison therefore lies at distance at least $N - \|u\|_\infty$ from u . Sites of the union volume outside Λ_N take the exponent 0, from the per-box circle bound at the real point. The instances are the headline radius $\rho = 64|\tau|$, which gives exactly the BA1 gate's bound value $K = 49/111790368$ at $q = 1/64$; the secondary radius $\rho = 1/151552 = \tau_*/4$, which gives $K_2 = 49/10202112$ at $q_2 = 151552|\tau|$ and is re-instantiated and labelled, since the BA1 gate binds only $64|\tau|$ and τ_* ; and the crude input $296/390625$. Both producers enumerate the lemma at every site of Λ_N , the reverse for $N = 2, 3, 4$ (125, 343 and 729 sites) and the forward for $N = 2, 3$, for the named comparisons and for general volumes, with the distance $N - \|u\|_\infty$ attained. The historical assistant's enumeration covers all five comparisons at $N = 2, 3, 4$.

4.3 Forward route: hard-core polymers and the Kotecký–Preiss condition

Polymers, Y-clusters and vacuum probabilities. The product expands as $\psi(c) = \sum_{\mathcal{F}} (-1)^{|\mathcal{F}|} c_{\mathcal{F}}$ over families $\mathcal{F}$ of pairwise disjoint supports, with $\|c_{\mathcal{F}}\| = \prod_{I \in \mathcal{F}} \|c_I\|$. Because each c_I is excited at every site of I , two families pair to zero against $A \otimes 1$, $A \in B(\mathcal{H}_Y)$, unless their excitation sets agree outside Y . A *polymer* $\gamma = (\mathcal{F}, \mathcal{F}')$ is a pair of nonempty families with the same excitation set whose bipartite overlap graph is connected; its activity is $w(\gamma) = (-1)^{|\mathcal{F}|+|\mathcal{F}'|} \langle c_{\mathcal{F}}, c_{\mathcal{F}'} \rangle$, and polymers are compatible when their supports are disjoint (hard core). A *Y-cluster* η is a Y -matched pair all of whose members are connected to Y , and $\rho_\eta \in B(\mathcal{H}_Y)$ is its signed partial trace. Every Y -matched pair splits uniquely into one Y -cluster and compatible polymers away from it, so, with $Z(\Lambda') = \sum_{\Gamma} \prod_{\gamma \in \Gamma} w(\gamma)$,

$$\begin{aligned} \langle \psi, (A \otimes 1) \psi \rangle &= \sum_{\eta} \text{Tr}(\rho_{\eta} A) Z(\Lambda \setminus \text{supp } \eta), & \rho_Y &= \sum_{\eta} \rho_{\eta} \pi(\text{supp } \eta), \\ \pi(S) &= \frac{Z(\Lambda \setminus S)}{Z(\Lambda)} = \frac{\langle \psi, P_S \psi \rangle}{\langle \psi, \psi \rangle} \in (0, 1]. \end{aligned} \quad (40)$$

The outside state enters only through the vacuum probabilities of the Y -cluster supports, each at most 1. This replaces the factor $e^{a|\gamma|}$ of a rooted cluster sum and keeps every later bound free of an exponential in the size of the differentiated support. The activities obey $|w(\gamma)| \leq \prod_{K \in \gamma} \|c_K\|$ and $\|\rho_{\eta}\|_1 \leq \prod_{K \in \eta} \|c_K\|$. An exploration tree that adds, at each step, the member of the other side through the least unbalanced site bounds the combinatorics: with per-site sum $\sigma \leq 1$, the polymers through a site cost at most σ^2 , and the Y -clusters cost at most $N_Y = (1 + t_a)^{2|Y|}$, with a factor w^{-d} for a member at distance d from Y . These counts are brute-forced on a five-site chain (233 polymers, and 275 and 856 Y -clusters), and the polymer identity is checked exactly on four qutrit sites with all 15 supports.

The mixed-weight contraction lemma. The Kotecký–Preiss hypothesis needs a cardinality factor, which BA1's diameter weight does not carry. The forward therefore proves a new lemma in full. With $W(I) = w^{\text{diam } I} e^{b|I|}$ and $\|c\|_{w,b} = \max_u \sum_{I \ni u} W(I) \|c_I\|$,

$$\|L_k(c_1, \dots, c_k)\|_{w,b} \leq w e^{4b} J L_k^{\text{num}} \prod_j \|c_j\|_{w,b}. \quad (41)$$

The proof is BA1's weighted estimate with one addition. Besides $\text{diam } M \leq \text{diam } X + \sum_j \text{diam } I_j$, the output also satisfies $|M| \leq |X| + \sum_j |I_j|$ with $|X| \leq 4$, so $W(M) \leq w e^{4b} \prod_j W(I_j)$: the loss is charged once per interaction term. It is attained at order zero, where a four-site piece gives $M = X$. The checker verifies the weight inequality over 19432 configurations at three weight pairs and rejects a loss $w e^{3b}$, a loss per creation, the maximum-instead-of-sum shortcut, $e^b = 2$ and $w = 42/q$. With $\hat{w} = w e^{4b}$ and $\Gamma = J\hat{w}G'(R_a)$, the self-map and contraction hold when $\hat{w} \leq w_{\max} = 1/(37888|\tau|)$, and AM2's uniqueness identifies the fixed point. The two tiers are $\bar{\tau} = w e^{3b} t_1/(1 - \Gamma)$ (exact first order) and $\bar{\tau} = J\hat{w}G(R_a)$ (crude).

The proof weights were declared before any constant, by one rule for both pairs: the creation weight $w = 3/q$, the cluster distance weight $v = e^\mu = 2/q$, the cardinality factor $e^b = 1001/1000$ and the Kotecký–Preiss parameter $a = \bar{\tau}^2$. The far-site lattice sum needs $qv > 1$ ($qv = 2$), the reach sum needs $v/w < 1$ ($v/w = 2/3$), admissibility needs $\hat{w} \leq w_{\max}$, and the criterion needs $a \leq 2b$. At the headline, $w = 192$, $v = 128$,

$$\begin{aligned}\hat{w} &= \frac{3012018012003}{15625000000} \approx 192.769, & \Gamma &= \frac{231925386924231}{12207031250000000} \approx 1.89993 \times 10^{-2}, \\ \bar{\tau} &= \frac{6143393381125}{9196881302842190592} \approx 6.67987 \times 10^{-7},\end{aligned}\tag{42}$$

and $a = \bar{\tau}^2 \approx 4.46206 \times 10^{-13}$. At the secondary pair $w = 1171875/592$, $v = 390625/296$, $\hat{w} \approx 1987.45$, $\Gamma \approx 0.195883$, $\bar{\tau} \approx 8.40187 \times 10^{-6}$ and $a \approx 7.05915 \times 10^{-11}$. The lattice sums $S_x = \sum_{y \in \mathbb{Z}^3} x^{d(y,0)}$ are $S_{2/3} = 725$ and $S_{1/2} = 147$ at both pairs.

The Kotecký–Preiss condition. With $a(\gamma) = a|\text{supp } \gamma|$ and the distance weight $d(\gamma) = \mu \sum_{K \in \gamma} \text{diam } K$, the per-site bound $\sum_{\gamma \ni x} |w(\gamma)| e^{a(\gamma)+d(\gamma)} \leq \bar{\tau}^2 \leq a$ follows from the activity bound, $a|\text{supp } \gamma| = (a/2) \sum_{K \in \gamma} |K|$ (each side of a polymer is a disjoint union), the exploration count and the mixed norm. Summing over the sites of a polymer, or of a test set of sites, gives the hypothesis $\sum_{\gamma' \not\sim \gamma} |w(\gamma')| e^{a(\gamma')+d(\gamma')} \leq a(\gamma)$. The criterion then gives, for the Ursell-weighted cluster sums Φ^T , the site decay $\sum_{X \not\sim z, X \not\sim s} |\Phi^T(X)| \leq a v^{-d(z,s)}$, and for $S \supset Y$ and a polymer γ away from S the truncated vacuum probabilities satisfy

$$|\pi_{\Lambda \setminus S}(\gamma) - \pi_\Lambda(\gamma)| \leq \sum_{z \in \text{supp } \gamma} \sum_{s \in S} a v^{-d(z,s)},\tag{43}$$

with no factor $e^{a|\gamma|}$, because both probabilities are real, positive and at most 1. The checker verifies (43) exactly on a six-site chain.

The cited theorem and its post-hoc check. The frozen forward packet states the criterion as a cited theorem: Kotecký and Preiss (Comm. Math. Phys. 103 (1986) 491–498) [13] in the decay-function form of Ueltschi (Moscow Math. J. 4 (2004) 511–522) [12], transcribed as known to the producer, with the primary sources not re-inspected. The skeptic found that the packet’s own lemmas bound polymer sums, which is the hypothesis side, not the Ursell-weighted cluster sums behind the site decay and (43). Plan rule P4, that an external theorem is quoted from a committed excerpt, had not been carried into the BB1 contract (defect D2), so the producer had no excerpt to quote. The skeptic ruled this a limitation, not a blocking issue, since every bound value is at least the value of the self-contained reverse route. After that ruling, and after the freeze, the advisor committed an excerpt of D. Ueltschi, “Cluster expansions and correlation functions”, [arXiv:math-ph/0304003v3](https://arxiv.org/abs/math-ph/0304003v3) (commit cca0584, 02:51 UTC on 25 September) as a repair source. It is not a premise of either frozen packet. Its PDF is bound by the sha256

2cc3e036af5b2a46afe95acf0b5863fe5c7d31554debb10a2a5cade4122bdfc

(191795 bytes, 11 pages); the excerpt file itself has sha256 17b0b3ff...0eed66. As in the Nachtergaele–Sims excerpt, Part A transcribes the displayed statements and was checked by the advisor against the rendered pages, and Part B is the raw `pdftotext` text. Ueltschi states that his criterion (3) is the same as in Kotecký–Preiss when polymers are subsets of a lattice; the original paper is not committed. The statements used are quoted below as Part A gives them, set in mathematical notation.

Setting (Section 2). $(\mathbb{A}, \mathcal{A}, \mu)$ is a measure space; μ is a complex measure with $|\mu|(\mathbb{A}) < \infty$, $|\mu|$ its total variation. ζ is a complex measurable symmetric function on $\mathbb{A} \times \mathbb{A}$.

$$Z = \sum_{n \geq 0} \frac{1}{n!} \int d\mu(A_1) \dots \int d\mu(A_n) \prod_{1 \leq i < j \leq n} (1 + \zeta(A_i, A_j)); \text{ the term } n = 0 \text{ is } 1. \tag{1}$$

$\mathcal{G}_n$ is the set of (unoriented) graphs with n vertices and $\mathcal{C}_n \subset \mathcal{G}_n$ the connected ones.

$$\varphi(A_1, \dots, A_n) = 1 \text{ if } n = 1, \text{ and } \frac{1}{n!} \sum_{G \in \mathcal{C}_n} \prod_{(i,j) \in G} \zeta(A_i, A_j) \text{ if } n \geq 2 \quad (2)$$

(product over the edges of G). A sequence $(A_1, \dots, A_n)$ is a cluster if the graph with an edge between i and j whenever $\zeta(A_i, A_j) \neq 0$ is connected.

Theorem 1 (Cluster expansion). Assume that $|1 + \zeta(A, A')| \leq 1$ for all $A, A' \in \mathbb{A}$, and that there exists a nonnegative function a on $\mathbb{A}$ such that for all $A \in \mathbb{A}$,

$$\int d|\mu|(A') |\zeta(A, A')| e^{a(A')} \leq a(A), \quad (3)$$

and $\int d|\mu|(A) e^{a(A)} < \infty$. Then $Z = \exp\{\sum_{n \geq 1} \int d\mu(A_1) \dots \int d\mu(A_n) \varphi(A_1, \dots, A_n)\}$. Combined sum and integrals converge absolutely. Furthermore, for all $A_1 \in \mathbb{A}$,

$$1 + \sum_{n \geq 2} n \int d|\mu|(A_2) \dots \int d|\mu|(A_n) |\varphi(A_1, \dots, A_n)| \leq e^{a(A_1)}. \quad (4)$$

Weights (Section 3). $b \geq 0$ is a function on $\mathbb{A}$ and $c \geq 0$ a symmetric function on $\mathbb{A} \times \mathbb{A}$ (both may vanish identically). $\mu_b(A) = \mu(A) e^{b(A)}$; $\zeta_c(A, A') = \zeta(A, A') e^{c(A, A')}$; $c(A_1, \dots, A_n) = \min_{G \in \mathcal{C}_n} \sum_{(i,j) \in G} c(A_i, A_j)$ if $n \geq 3$; $\varphi_c(A_1, \dots, A_n) = \varphi(A_1, \dots, A_n) e^{c(A_1, \dots, A_n)}$.

Theorem 3 (Decay of correlations). Assume that $|1 + \zeta_c(A, A')| \leq 1$ for all $A, A' \in \mathbb{A}$, and that there exists a nonnegative function a on $\mathbb{A}$ such that

$$\int d|\mu_b|(A') |\zeta_c(A, A')| e^{a(A')} \leq a(A) \quad \text{for all } A \in \mathbb{A}. \quad (16)$$

Then for all $m \geq 1$ and all $A_1, \dots, A_m \in \mathbb{A}$,

$$\begin{aligned} & \sum_{n \geq m} \frac{n!}{(n-m)!} \int d|\mu_b|(A_{m+1}) \dots \int d|\mu_b|(A_n) |\varphi_c(A_1, \dots, A_n)| \\ & \leq \exp\left\{ \frac{1}{m\gamma} [(1+\gamma)^m - 1] \sum_{i=1}^m a(A_i) \right\} \prod_{1 \leq i < j \leq m} (1 + |\zeta_c(A_i, A_j)|), \end{aligned}$$

where

$$\gamma = \sup_n \sup_{A_0, \dots, A_n} \left| \prod_{i=1}^n (1 + \zeta(A_0, A_i)) - 1 \right| \quad (17)$$

(clearly $0 \leq \gamma \leq 2$). The case $m = 1$ is

$$1 + \sum_{n \geq 2} n \int d|\mu_b|(A_2) \dots \int d|\mu_b|(A_n) |\varphi_c(A_1, \dots, A_n)| \leq e^{a(A_1)}, \quad (18)$$

and multiplying (18) by $|\zeta_c(A, A_1)|$ and integrating over A_1 , from (16),

$$\sum_{n \geq 1} \int d|\mu_b|(A_1) \dots \int d|\mu_b|(A_n) \left(\sum_{i=1}^n |\zeta_c(A, A_i)| \right) |\varphi_c(A_1, \dots, A_n)| \leq a(A) \quad \text{for all } A \in \mathbb{A}. \quad (19)$$

The paper states that (18) and (19) “can be proved exactly the same way as (4) and (5)”.

Section 4.2, lattice polymer models: a polymer is a connected subset of $\mathbb{Z}^d$; μ is the counting measure times a weight $w(A)$; polymers interact through non-intersection, $\zeta(A, A') = -1$ if $A \cap A' \neq \emptyset$ and 0 otherwise (the hard-core case, where $|1 + \zeta| \leq 1$ holds).

The skeptic’s [repair record](#) compares the forward’s statement and its use with this text, with the dictionary hard-core $\zeta \in \{0, -1\}$, $c \equiv 0$, $b = d$ (the distance weight) and $a(\gamma) = a|\text{supp } \gamma|$. Ueltschi’s (16) is then exactly the forward’s hypothesis. Theorem 1 gives the cluster expansion of every $Z(\Lambda')$, since restricting the finite polymer set only removes terms. (19) gives the forward’s conclusion with

the same weight $e^{d(X)}$, counted with multiplicity, and with Ueltschi’s count $\sum_i |\zeta(A, A_i)| \geq 1$ in place of the forward’s indicator, which only strengthens it. The test sets enter as atoms of μ -mass zero with $a(S) = a|S|$, which changes no integral; that is the forward’s own parenthetical. The site decay and (43) add only the forward’s elementary steps. The record finds no missing factor and no different hypothesis form: the cited statement and its use are covered. The limitation stays, because the check is post hoc and outside the frozen packet. The skeptic’s conditional alternative, a **limited** verdict with the reverse constants bound if P4 were held binding on the frozen producer, is retained in its review as the advisor asked and is not recommended.

Marginal locality from a real-parameter derivative. Put $c_\lambda = c + \lambda(c' - c)$ with $\lambda \in [0, 1]$ *real*. Every coefficient of $\psi(c_\lambda)$ is a real polynomial in λ and $Z_\lambda(\Lambda) \geq 1$, so $\lambda \mapsto \rho_Y(c_\lambda)$ is smooth, and no complex λ is used. A zero-free region is not needed and none is proved; the two-qubit vector $\Omega - 2z|11\rangle$, whose normalization $1 + 4z^2$ vanishes at $z = i/2$, shows why analyticity of the reduced density is never claimed. The derivative splits into the changes on supports meeting Y , charged by the coefficient Lipschitz bound, and the changes on the other supports. The latter act through the Y -cluster derivatives (the straddling term) and through the derivatives of the vacuum probabilities (the normalization term), which (43) controls. The result is

$$\|\rho_Y(c) - \rho_Y(c')\|_1 \leq 2(1+t)^{|Y|-1} \sum_{I \cap Y \neq \emptyset} \|\delta_I\| + N_Y \kappa_0 \sum_{x \notin Y} \left(\sum_{y \in Y} v^{-d(x,y)} \right) \sum_{I \ni x} \|\delta_I\|, \quad (44)$$

with $\kappa_0 = 4\bar{\tau} + 2A(t + \bar{\tau}(S_{v/w} - 1))$ and $A = 2\bar{\tau}^2 + a(1 + 2\bar{\tau}^2 S_{v/w})$; at the headline $\kappa_0 \approx 2.67195 \times 10^{-6}$. In the contract’s lemma form this is $\kappa(I) \leq \kappa_0(Y)(w')^{-d_\infty(I,Y)} p(|I|)$ with $\kappa_0(Y) = N_Y |Y| \kappa_0$, $w' = v$ and $p(s) = s$.

4.4 Reverse route: the iterated split with per-site charging

Weights. The reverse uses only a diameter weight: $W = 1024$ at the headline and $W_2 = 1/(37888|\tau|)$ at the secondary pair, with the site ratio $\lambda = 2/W = 1/512$ and lemma weight $w' = W/2 = 512$. At the cap $W = 1024$ gives the weighted self-map value $2368/390625 < 1/64$ and the contraction constant $39424/390625 \approx 0.1009$. The lattice sums need $\lambda < q$, so $W = 64$ is rejected. For a comparison put $m_I = \max(\|c_I^A\|, \|c_I^B\|)$, with per-site sums $t_0 \leq 2T(|\tau|) = 49/7199290368$ and $t_W \leq 2T(W|\tau|) = 49/6321618 \approx 7.75 \times 10^{-6}$. The reverse also proves the mixed-weight lemma for completeness, with the loss $w e^{4b}$ and a fixture showing that $w e^b$ is insufficient, but does not use it.

Single-support telescoping. Order the supports of the union volume and change them one at a time from c^A to c^B . Every intermediate collection lies in the class $\mathcal{C}(m)$ of collections with $\|c_I\| \leq m_I$, and

$$\|\rho_Y(c^A) - \rho_Y(c^B)\|_1 \leq \sum_k \kappa(I_k) \|c_{I_k}^A - c_{I_k}^B\|, \quad \kappa(J) = \sup \frac{\|\rho_Y(c) - \rho_Y(c')\|_1}{\|c_J - c'_J\|}, \quad (45)$$

the supremum over pairs in $\mathcal{C}(m)$ differing only on J . By (38), $\kappa(J) \leq 2$ for every J ; that bound is used for supports meeting Y , with $\eta = 0$.

The split at the changed support. For J missing Y , apply the AV1 product-ordering split at J : $\psi(c) = \psi_J + \chi(c)$ with $\psi_J = \Omega_J \otimes \varphi$, φ common to c and c' , and $(P_J \otimes 1)\chi = 0$. Because J is traced out and χ is excited on J , the cross terms vanish exactly. This gives an exact identity for $\rho_Y(c) - \rho_Y(c')$ with two channels, and

$$\|\rho_Y(c) - \rho_Y(c')\|_1 \leq 2m_J \|\delta\| \|\rho_Y(\varphi) - \rho_Y(c')\|_1 + \frac{2}{n^2} \|\Gamma(\delta) - \text{Tr } \Gamma(\delta) \rho_Y(c')\|_1, \quad (46)$$

with $n = \|\varphi\|$. The first term is the normalization channel, carrying the weight $\|c_J\|^2$ of the J -excited component; the second is the straddling channel. No intermediate normalization is ever divided by, since orthogonality at J gives $\|\psi(c)\|^2 \geq n^2$.

Only coverings survive. Expand the creations meeting J into families. Since $\delta = c_J - c'_J$ is excited at every site of J , the contraction with δ kills every family that does not cover J , and a covering family contributes with the product of its members' amplitudes. The spectator families cancel exactly, so no factor $\prod_x (1 + s_x)$ over a region ever appears. The same holds at the next level. Contracting the outside vector against a covering region S keeps only the coverings of S by supports meeting S , which gives a recursion on records $\Gamma[U, S, g]$ with a strictly growing removed set U . The recursion terminates in every finite volume. Its depth is not bounded uniformly, and no constant depends on it. Both identities are verified exactly on a five-site fixture with two excited levels per site.

Per-site charging and the closure. With the site potential $h(x) = \sum_{y \in Y} \lambda^{d(x,y)}$ and $H(I) = \sum_{x \in I} h(x)$, three per-site facts carry the estimate:

- a support through x meeting Y has diameter at least $d(x, Y)$, so $\sum_{I \ni x, I \cap Y \neq \emptyset} m_I \leq t_W h(x)$;
- $\sum_{I \ni x} m_I H(I) \leq 8t_W h(x)$, from $|I| \leq (\text{diam } I + 1)^3 \leq 8 \cdot 2^{\text{diam } I}$, so the size of a support is charged once, at the site where the chain enters it, never as a product of growing region sizes;
- the coverings of a region T cost at most t_0 in total.

The per-level factor is therefore $8t_W = 196/3160809 \approx 6.2009 \times 10^{-5}$ at every depth. Writing $r = \max_{I \cap Y = \emptyset} \kappa(I)/H(I)$, the chain bound gives $\kappa(J) \leq (c_1 + c_2 r)H(J)$, hence

$$r \leq \beta^* = \frac{c_1}{1 - c_2}, \quad c_1 = 4t_0 t_W + \frac{4t_W + 4t_0 t_W / (1 - t_0)}{1 - 8t_W}, \quad c_2 = 16t_0 t_W + \frac{16t_0 t_W}{(1 - t_0)(1 - 8t_W)}, \quad (47)$$

with $c_1 \approx 3.10066 \times 10^{-5}$ and $c_2 \approx 1.69 \times 10^{-12}$ at the headline, so that

$$\beta^* = \frac{8027378053355287230305861131}{258892194754590065811608028259796} \approx 3.10066 \times 10^{-5}.$$

In the contract's lemma form, $\eta = 0$, $\kappa(I) \leq \beta^* H(I) \leq \kappa_0(w')^{-d_\infty(I,Y)} |I|$ with $\kappa_0 = \beta^* |Y|$, $w' = 512$ and $p(n) = n$. The lemma is checked end to end on a seven-site qubit chain with 23 supports, and each of its 18 single far changes stays within $\beta^* H(J) \|\delta_J\|$.

4.5 The density bounds for every comparison

Both routes insert the every-site input (39) into their lemma and sum over $\mathbb{Z}^3$ with $q^{N - \|x\|_\infty} \leq q^{N - \|y\|_\infty} q^{-d(x,y)}$. The forward obtains

$$C = K(1 + q) \left[2(1 + t) + (1 + t)^4 \kappa_0 (S_{1/(qv)} - 1) \right], \quad c_{\text{site}} = K \left[2 + \kappa_0 (S_{1/(qv)} - 1) \right], \quad (48)$$

where $(1 + t)^{2|Y|} \leq e^{|Y|/10^8}$ because $2t + t^2 \approx 6.806 \times 10^{-9}$ is below 10^{-8} . The reverse obtains $c_{\text{site}} = K(2 + \beta^* S_\lambda)$ with $S_\lambda = 2169/343$ at $\lambda/q = 1/8$, and $C = c_{\text{site}}(1 + q)$, in the stronger form $c_{\text{site}} \sum_{y \in Y} q^{N - \|y\|_\infty}$ without the exponential factor. The factor $1 + q$ in both is the site 0 charged at q^N and e_z at q^{N-1} .

The five comparisons. The density lemma sees a comparison only through the per-site coefficient differences and the norms of the two fixed points. The every-site input covers any two volumes of either prescription containing Λ_N , and the norms are bounded for every such volume, so one pair of constants holds for all five comparisons of the contract:

- c1: F1 on Λ_N against F1 on Λ_{N+1} ;
- c2: F2 on Λ_N against F2 on Λ_{N+1} ;
- c3: F1 against F2 on the same Λ_N , the fixed- N change of boundary prescription, named as its own item: its source is the $28N(5N+1)$ extra faces (616, 1344 and 2352 at $N = 2, 3, 4$) on the outer layer, at distance $N-1$ from e_z and N from 0, enumerated by all three packets;
- c4: any two centered boxes of size at least N , of either family, compared directly and not by telescoping;
- c5: two finite complete-factor volumes of one prescription, both containing Λ_N , compared directly: the lemma runs once on their union, which serves only as the embedding space. The route through the union volume, each volume against the union and then the triangle inequality, costs a factor 2 and is labelled only ($2C \approx 1.781 \times 10^{-6}$).

Every source site in the enumeration audits has $\|p\|_\infty \geq N$. The $-\tau$ values replay the same $|\tau|$ formulas.

Cutoff removal at fixed N . Every constant is independent of the on-site cutoff, so the bounds hold in each Q_L . At fixed N and for fixed volumes, AM2 §6 (and, for F2, the AY1 itemization) gives a simple untruncated ground state with gap at least $1/2$ in normalized units, and AV1’s Eckart-type bound makes the cutoff ground vectors converge to it; the reduced densities converge in trace norm, and the bound passes through the closed ball. The order is fixed: a bound uniform in L at fixed N , then $L \rightarrow \infty$ at fixed N , then the reading as a function of N . The fixture $(1 - 2^{-L})^{(2N+1)^3}$, whose iterated limits are 1 and 0, shows why the two limits are never exchanged.

Constants. Table 13 lists the constants. For each quantity the gate binds the larger valid constant and labels the other route’s value. The polymer_kp values are the larger at the headline and the iterated_split values at the secondary pair, so every bound value is at least the value that the self-contained reverse route proves.

Table 13: BB1 constants at $|\tau| = 10^{-8}$, both signs, every comparison, each cutoff space and the untruncated vectors at fixed N . Decimals summarize exact rationals; only the rows marked “bound” are the gate’s admitted values.

Quantity	Route	$\approx$ value	Margin	Status
$C, q = 1/64$	polymer_kp	8.90512×10^{-7}	4.4918	bound; target 1/250000
	iterated_split	8.90426×10^{-7}	4.4922	labelled
$c_{\text{site}}, q = 1/64$	polymer_kp	8.76812×10^{-7}	2.2810	bound; target 1/500000; the binding margin
	iterated_split	8.76727×10^{-7}	2.2812	labelled
$C_2, q_2 = 151552 \tau $	iterated_split	9.68909×10^{-6}	5.1604	bound (labelled pair); target 1/20000
	polymer_kp	9.64401×10^{-6}	5.1846	labelled
$c_{\text{site},2}$	iterated_split	9.67443×10^{-6}	2.5841	bound (labelled pair); target 1/40000
	polymer_kp	9.62942×10^{-6}	2.5962	labelled
crude $C, q = 1/64$	polymer_kp	2.05285×10^{-3}	–	fails 1/250000; retained
	iterated_split	1.80057×10^{-3}	–	fails 1/250000; retained
crude c_{site}	iterated_split	1.77287×10^{-3}	–	fails 1/500000; retained

The exact bound rationals of the headline pair, with numerators and denominators of about 150 digits, are those of the gate statement quoted above. The labelled iterated_split values are

$$\begin{aligned}
C &= \frac{1649304957714574761387109521705122525}{1852265838332370761358328329658687644475392}, \\
c_{\text{site}} &= \frac{25373922426378073252109377257001885}{28941653723943293146223880150916994444928},
\end{aligned} \tag{49}$$

and the bound values of the labelled secondary pair are

$$\begin{aligned} C_2 &= \frac{254713191786858884385903518855136871787}{26288652445972361083565698627981857600000000}, \\ c_{\text{site},2} &= \frac{651079047656055039494458366725211}{6729895026168924437392818848763355456}. \end{aligned} \quad (50)$$

The crude `polymer_kp` estimate has no region constant in the frozen form, because its anchored norm gives $(1+t)^2 - 1 \approx 1.18 \times 10^{-5}$, above 10^{-8} . The reverse’s crude region constant is in the frozen form, because its near term carries no exponential factor, and it fails its target. The skeptic’s pre-comparison review had said that no crude region constant can be written in the frozen form, and it corrected itself on this point. Read with the exponent, the bound $C q^{N-1}$ is about 1.391×10^{-8} at $N = 2$, 2.174×10^{-10} at $N = 3$ and 3.397×10^{-12} at $N = 4$.

Scaling. Evaluated from the same exact formulas at τ and $\tau/100$, the ratios of the bound constants are about 100.647876 for C and 100.647875 for c_{site} , inside $[95, 105]$ (the reverse values give about 100.638). The secondary ratios are about 1.001500 and 1.000000, inside $[99/100, 101/100]$, since K_2 does not depend on τ and only the factor $1 + q_2$ remains; and q_2 scales by exactly 100.

The input governs the constant. The near term $2K(1+q) = 3185/3577291776 \approx 8.90338 \times 10^{-7}$ is 99.98% of C in all three computations: the forward’s, the reverse’s and the skeptic’s own pre-comparison value 8.903452×10^{-7} . The three differ only in how the far sites are charged, about 10^{-4} of C (Table 14). The forward pays the far cost twice, through the straddling families and through the normalization, which is why its constant is the larger and is bound. The three derivations agree to within 2 parts in 10^4 on C and c_{site} . The region margin, about 2.28, is the binding one. A later loop that needs more room would have to improve the coefficient input, not the density lemma.

Table 14: The preregistered BB1 error terms as contributions to the headline C , added linearly.

Term	polymer_kp	iterated_split
<code>coefficient_input_every_site</code>	$2K(1+q) = 3185/3577291776 \approx 8.90338 \times 10^{-7}$	the same
<code>straddling_supports</code>	$\approx 8.683 \times 10^{-11}$ (Y-cluster derivatives)	$\approx 8.729 \times 10^{-11}$ (covering chains reaching Y)
<code>normalization</code>	$\approx 8.683 \times 10^{-11}$ (vacuum-probability derivatives and η_R)	$\approx 5.94 \times 10^{-19}$ (the $\ c_J\ ^2$ channel)
<code>polymer_or_split_remainder</code>	$\approx 2.81 \times 10^{-20}$ (cluster sums in the truncated probabilities)	$\approx 5.94 \times 10^{-19}$ (removal terms) plus about 1.5×10^{-22} (closure)
<code>cutoff_removal_at_fixed_N</code>	not a numeric cost: an exact limit at fixed N	the same
<code>arithmetic</code>	not a numeric cost: exact rationals, directed enclosures only	the same

4.6 Readings, defects and review

Contract readings. The review reconciles the contract’s wording with both packets as follows.

- *D1.* The semantics of `normalization_couples_supports` say that a support strictly containing R has a nonzero first-order R -marginal. It does not. At first order only supports contained in R move ρ_R , namely the ten faces with owner set exactly R . Every straddling support has zero first-order R -marginal, including the 6 of the 72 straddling faces that strictly contain R , and differs from zero only at second order. All three packets exhibit the three kinds, and all three found the defect.

- *R10, non-centred F2 volumes.* The AY1 itemization H1–H5 of the AM2 premises is gated for F2 on Λ_N . Both producers apply it to every F2 volume containing Λ_N (on-site operator, grouping at the anchor inside the star, $|X_b| \leq 4$, per-site sum through $b \in u - S$, cutoff compression), because no item uses the cube shape. The skeptic checked this against the AY1 report and accepts it as a local reading. It is not a gated statement, and it affects the F2 branch of c5.
- *Union sites and lemma constants.* Sites of the union volume outside Λ_N use the exponent $(N - \|u\|_\infty)_+ = 0$. The lemma constant κ_0 carries the factor $|Y|$. The reverse split weight is read as $W \leq 1/(37888|\tau|)$. Tier, route and BA1 input are three separate labels.
- *D2, the excerpt rule.* Plan rule P4 was not carried into the BB1 contract; the repair is described in Section 4.3.
- *D3.* The reverse exports comparison names truncated to 80 characters. Each prefix is unique.

Controls, replays and review. Each producer implements all 37 contract controls as damaging mutations. The forward checker runs 51 exact checks with 107 rejected mutations, the reverse checker 78 exact checks with 132. The skeptic’s pre-comparison package, 88 exact checks with all 37 controls as 106 damaging mutations, was committed before both producer commits. The post-comparison review runs 21 exact checks, among them the check of the Kotecký–Preiss citation against the committed excerpt. It adds 82 source-edit runs on temporary copies (two unmutated baselines, 74 validator weakenings, one per control per producer, and six input edits, each aborting as required) and 15 damaged packets rejected by its own value validator. All four producer replays reproduce the frozen outputs byte for byte under normal and optimized Python. Both closures (39 files each) verify, and both input inventories equal `AGENTS.md`, the contract and the 33 shared premises. Both producers wrote their reports through a shell heredoc after the file tool refused, and saw other agents’ file names or commit subjects without opening them. The forward disclosed this in its packet; the reverse disclosed it only in its hand-back, which the gate records. Isolation rests on disclosure and content. The reverse’s far-site mechanism (covering chains, β^* , $W = 1024$) and its far term differ from the forward’s, and the forward’s weights differ from the skeptic’s feasibility weights. The reverse read only §§1–8 of the AV1 reverse report; the skeptic found that everything it uses lies in those parts. The review found no blocking issue.

Meaning. BB1 is a static statement about the reduced densities of the finite-box ground vectors of the named constructions, one comparison at a time, at a rate in N per coarse ℓ^∞ step at fixed spacing. It forms no limit object. Whole-sequence convergence, a common limit and translation statements are BB2’s subject, and BB2’s contract uses BB1’s frozen targets as hypotheses. BB1 asserts no dynamics, no uniqueness of any ground state, nothing about boundary conditions outside F1 and F2, no analyticity of the reduced density in the coupling, and no estimate in the lattice spacing.

4.7 Limitations recorded by the gate

The BB1 gate records the following limitations verbatim.

- Scope: only the zero-selected patterned family (selected triple (0,0,0), Haar reference), fixed spacing, $|\tau| \leq 10^{-8}$, the named families F1 and F2 on centered cubes Λ_N with N at least 2, finite complete-factor volumes of one prescription containing Λ_N , the cover $R = \{0, e_z\}$ and finite complete-factor regions inside Λ_N ; nothing transfers to other boundary conditions, nonzero selected triples, literal vertex boxes, weak coupling or the continuum; the constants are uniform in N at fixed spacing and in the cutoff, never in the lattice spacing.
- Static statement about reduced densities of finite-box ground vectors per comparison: no dynamics, no limit object, no whole-sequence convergence or common limit (BB2), no translation statement, no uniqueness of any ground state and no analyticity of the reduced density in the coupling (no zero-free region is proved; the forward uses a real-parameter derivative and the reverse single-support telescoping).

- The `polymer_kp` route uses the Kotecky–Preiss criterion (cited as Kotecky–Preiss 1986 in the form of Ueltschi 2004) as an external theorem that the frozen packet transcribes without a committed source excerpt (the plan rule that an external theorem is quoted from a committed excerpt with its PDF hash was not carried into the BB1 contract); its exploration-tree lemmas bound polymer sums, not the Ursell-weighted cluster sums behind Corollary 6.3 and Lemma 6.4. The citation was checked post hoc against the Ueltschi excerpt committed after the freeze (`research/round33/sources/ueltschi-math-ph-0304003v3.md`, `arXiv:math-ph/0304003v3`, PDF sha256 2cc3e036af5b2a46afe95acf0b5863fe5c7d31554debb10a2a5cade4122bdbfc; repair record `bb1-kp-repair.md`): with hard-core zeta in $\{0, -1\}$, $c=0$, b equal to the forward distance weight d and $a(\gamma)=a|\text{supp } \gamma|$, Theorem 3 (16) is the forward hypothesis, (19) gives its conclusion with the same weight $e^{-d(X)}$, Theorem 1 gives the cluster expansion of every $Z(\Lambda_N)$, and the test sets enter as mass-zero atoms; no factor is missing, so the cited statement and its use are covered. The excerpt is not a premise of the frozen packet, the original Kotecky–Preiss paper is neither committed nor needed, and every bound value is at least the value proved by the self-contained `iterated_split` route.
- F2 volumes other than centred cubes (fifth comparison): the AY1 itemization H1–H5 is gated for F2 on Λ_N ; both producers apply it to every F2 volume containing Λ_N as a local reading (on-site operator, per-anchor grouping inside the star, $|X_b| \leq 4$, per-site sum through b in u -S, cutoff compression), reviewed here as correct but not a gated statement.
- Inherited without re-proof: the AM2 multilinear majorant, fixed point, uniqueness in the ball and section 6 facts; the AV1 product-ordering split, first-order coefficient (49 faces per site) and cutoff-vector removal F20–F23; the AY1 F2 itemization; the I1 dictionary; Banach, Weierstrass and the maximum principle are standard and not machine-checked. The every-site coefficient input (form (b)) and the secondary input $K_2=49/10202112$ are BB1 lemmas proved in both packets, not admitted BA1 statements.
- Upper bounds only; the $-\tau$ values replay the same $|\tau|$ formula. The near term $2K(1+q)=3185/3577291776$ is almost the whole headline constant (the far-site terms are about 10^{-4} of C), so the constants are governed by the BA1 input; the region margin (about 2.28) is the binding one. The forward, reverse and skeptic derivations agree to within 2 parts in 10^4 on C and c_{site} .
- Contract wording read as follows: at first order only supports contained in R move ρ_R , and every straddling support, including the 6 of the 72 straddling faces that strictly contain R , has zero first-order R -marginal (D1); the lemma constant κ_0 carries the factor $|Y|$; union-volume sites outside Λ_N use the exponent $(N-|u|_{\infty})_+=0$; the reverse split weight is read as $W \leq 1/(37888|\tau|)$; tier, route and BA1 input are three separate labels.
- Independence is limited to derivations, constants and code: the contract and selection note name both mechanisms and the frozen targets, the skeptic triage and pre-freeze review shaped the parameters, all agents are correlated model agents, isolation is verified for repository inputs only, and the reverse packet does not itself record its heredoc report write or the file names it saw.
- Scientific priority is unverified.

Full derivations and reviewed implementations: [BB1 forward](#), [BB1 reverse](#), [independent model-agent review](#), the [Kotecký–Preiss repair record](#), the [committed Ueltschi excerpt](#) and the [frozen contract](#).

5 BB2: convergence of the named constructions

Contribution HNM-C-BB2. BB1 bounds the difference of the reduced densities of two boxes one comparison at a time and forms no limit. The BB2 contract, recorded under the title “Hruday whole-sequence convergence of the named constructions, their common limit and its coarse translation invariance”, turns those comparisons into statements about a limit. It was selected after the BA2 gate and frozen together with BB1 on 24 September 2026, before either was produced, as investigation 4 of 8 ([selection note](#)). Because BB1 was still in production, BB2 uses BB1’s *frozen targets* as explicit hypotheses, and its gate was recorded after BB1’s, on 25 September at 03:31 UTC. Both directions are paired, and they differ in how the BB1 comparisons are assembled, recorded in

an assembly field rather than as a route label. The forward producer telescopes nested comparisons (`nested_telescoping`). The reverse producer, under premise isolation, compares any two boxes directly through their union (`union_comparison`). Both received `AGENTS.md`, the contract and 36 shared premises, among them the BB1 contract, the BA1 and BA2 gates, the AQ1 and AQ2 records and the Nachtergaele–Sims excerpt; neither read a BB1 producer, skeptic or gate file.

The contract names five items: a whole-sequence Cauchy estimate over all $M > N$ with the limit obtained without compactness, the common limit of F1 and F2, identification with every AQ1 and F2 subsequential limit, coarse translation invariance, and convergence of the correlation functions on the window $|\theta| \leq 8$. Its targets are $C' \leq 1/100000$ and $c'_{\text{site}} \leq 1/200000$ for the state constants, evaluated at the hypothesis values, and $C_{\text{dyn}} \leq 1/2000000000$ for the dynamics constant. A labelled secondary constant has target $1/8000$. The scaling brackets are “exactly 1” for C' and c'_{site} at the hypothesis values, $[9500, 10500]$ for C_{dyn} , $[99/100, 101/100]$ for the secondary constants, exactly 100 for q_2 , and no single bracket for the item-5 sum. The index $r_N = \lfloor (N-1)/2 \rfloor$ of item 5 is frozen. The itemized error terms are `bb1_hypothesis_constants`, `cauchy_telescoping_or_union`, `identification_with_subsequential_limits`, `translation_general_volume`, `dynamics_constant_from_ba2`, `region_form_on_Lambda_rN` and `arithmetic`, and the contract lists 34 controls.

5.1 Reviewed scope

The `BB2 gate` records the verdict `accepted_within_scope`, with sub-label `convergence_of_named_constructions` and secondary sub-label `common_limit_of_named_constructions`. Its gate fields set `whole_sequence_claimed`, `common_limit_claimed`, `state_convergence_claimed`, `translation_invariance_claimed` and `rate_in_N_claimed` to true. The whole-sequence scope is “reduced densities of the named constructions F1 and F2 on every finite region”, the translation scope is “coarse translations; the limit of the named constructions”, and `dynamics_level` is `correlation_functions_compact_window`. The fields `uniqueness_of_ground_state_claimed`, `gns_dynamics_equality_claimed`, `uniform_in_time_claimed`, `rate_in_a_claimed`, `continuum_claim`, `weak_coupling_claim` and `scientific_priority_verified` are false. The claimed rate in N is the state rate q^{N-1} of item 1; the correlation rate holds only on a certified range (Section 5.6). The accepted statement, quoted verbatim below in the gate’s ASCII notation, is the scope of every claim in this section.

```
In the AM2/AQ1 zero-selected patterned family (model AQ_patterned_zero_selected: SU(2)
Kogut-Susskind form on Z^3 at fixed spacing, coarse 24-link factors, selected triple
exactly (0,0,0) with Haar product reference, 21 omitted faces per anchor entering as
-(tau/3)W_f in normalized units delta=alpha/8; both signs |tau|<=10^-8; AM2 creation
expansion in each on-site cutoff space; cover R={0,e_z}; coarse l-infinity metric with
star diameter 1 for states and l1 on the coarse factor lattice with F(r)=(1+r)^-4 for
dynamics; trace norm on B(H_Y); clock theta=alpha t/hbar), with the named construction
families F1 = AQ1 centered whole-star boxes Lambda_N=[-N,N]^3 and F2 = I1 section 6
all-contained-face boxes with padding on the same Lambda_N, N at least 2, with the BA2
gate dynamics constants, and with the BB1 hypotheses discharged by the BB1 gate (sha256
18141fea672e5bae09024a5fddc56ea7102aae56ecca16fbb30d67a1354a3827,
accepted_within_scope), which admits for each of the five comparisons c1-c5 of the BB1
contract, at both signs, in each on-site cutoff space with constants independent of the
cutoff and at fixed N for the untruncated ground vectors, the R form C q^(N-1) with
q=1/64 and C about 8.9051e-07 (polymer_kp bound value; iterated_split 8.9043e-07), at
most C_h=1/250000, and the region form c_site |Y| e^{|Y|/10^8} q^{d_Y} with c_site
about 8.7681e-07 (iterated_split 8.7673e-07), at most c_h=1/500000, so that every
hypothesis used by either route is discharged: For the zero-selected patterned family
at the same coupling |tau|<=10^-8, under the BB1 locality bounds at their frozen
targets as hypotheses (discharged only when the BB1 gate admits constants at most those
targets), the reduced densities of the named construction families F1 and F2 on
centered coarse cubes converge as whole sequences, at a rate in N, to one common limit
on every finite region; this limit coincides with every AQ1 subsequential limit and
every F2 subsequential limit, is invariant under coarse translations, and has as
```

correlation functions on $|\theta| \leq 8$ the limits of the finite-box correlation functions; this is convergence of the named constructions, not uniqueness of any ground state, not a statement about states outside the named constructions, and not a statement uniform in the lattice spacing a . Constants (exact; both signs; F_1 and F_2 ; in each on-site cutoff space uniformly in the cutoff and, at fixed N , for the untruncated ground vectors; evaluated at the hypothesis values as the contract target specifies, and valid under the discharge because each is nondecreasing in the BB1 constants). Item 1: sup over M greater than N of $\|\rho^{\{F,M\}}_{R-\rho^{\{F,N\}}_R\|_1 \leq C' q^{(N-1)}$ with $C'=4/984375$ (about $4.0635e-06$; exact_first_order, nested_telescoping, hypothesis source bbl_frozen_targets, BB1 route polymer_kp; margin $315/128$ against $1/100000$), and for every finite complete-factor region Y inside Λ_N , sup over M greater than N of $\|\rho^{\{F,M\}}_{Y-\rho^{\{F,N\}}_Y\|_1 \leq c'_{\text{site}} |Y| e^{-\{|Y|/10^8\}} q^{d_Y}$ with $d_Y = N - \max\{y \text{ in } Y\} |y|_{\text{inf}}$ and $c'_{\text{site}} = 2/984375$ (about $2.0317e-06$; margin $315/128$ against $1/200000$); the union_comparison route proves $C'=1/250000$ and $c'_{\text{site}}=1/500000$ (labelled); re-evaluated at the admitted BB1 values (labelled, not bound), C' is about $9.0465e-07$ and c'_{site} about $8.9073e-07$ (nested_telescoping), about $8.9051e-07$ and $8.7681e-07$ (union_comparison). The limit on every finite region follows from this Cauchy bound and trace-class completeness, without compactness. Item 2: the F_1 and F_2 limits coincide on every finite region (BB1 comparison c3, the fixed- N comparison of the two prescriptions). Item 3: the limit equals every AQ1 subsequential limit (F_1) and every F_2 subsequential limit on every finite region and inherits exactly the admitted properties of those states: AQ1 stationarity under T_θ , strong continuity of the GNS evolution and a nonnegative generator (for F_2 limits through BA2's rerun of AQ1 sections 4-5), and the AQ2 statement for the actual AQ1 centered subsequential state, H_θ at least $(\alpha/16)(I-P_\Omega)$ with a simple vacuum on the invariant-local cyclic completion, the full-GNS strengthening only as qualified in the AQ2 gate. Item 4: for every coarse translation v (fine translation $(4v_x, 2v_y, v_z)$, preserving residues and face classes) and N at least $|v|_{\text{inf}}+2$, $\|\rho^{\{\Lambda_N+v\}}_{R-\rho^{\{\Lambda_N\}}_R\|_1 \leq C_h q^{(N-|v|_{\text{inf}}-1)}$ with $C_h=1/250000$ (the direct BB1 comparison c5 of two one-prescription volumes containing $\Lambda_N - |v|_{\text{inf}}$; re-evaluated at the admitted C about $8.9051e-07$, labelled; a two-step comparison through the union is labelled only), and the limit is invariant under coarse translations on every finite region (region form of c5); non-coarse translations are rejected. Item 5: for N at least 5, $r_N = \text{floor}((N-1)/2)$, $|\theta|$ at most 8 and A in $B(H_R)$, $|c^{\{F,N\}}_A(\theta) - c^{\text{inf}}_A(\theta)| \leq \|A\|^2 [C_{\text{dyn}}/(r_N-1) + c'_{\text{site}} |\Lambda_{r_N}| e^{\{|\Lambda_{r_N}\|/10^8\}} q^{(N-r_N)} + 2C' q^{(N-1)}]$ with complex-mean centering, $c^{\{F,N\}}_A(\theta) = \omega^{\{F,N\}}(A * T_\theta(A)) - \omega^{\{F,N\}}(A)$ and $c^{\text{inf}}_A(\theta) = \omega_{\text{inf}}(A * T_\theta(A)) - \omega_{\text{inf}}(A)$, and $C_{\text{dyn}} = 78057/622883200000000$ (about $1.2532e-10$; polynomial_lieb_robinson, duhamel_inner_f1, $2K_{F1} + K_{\text{cmp}}/4$ from the BA2 gate values, one constant for both families; margin about 3.99 against $1/2000000000$); the reverse per-family values $9261/77860400000000$ (F_1 , duhamel_inner_f1, about $1.1894e-10$) and $1055961408185869563/152407011718750000000000000000$ (F_2 , duhamel_inner_f2, about $6.9286e-11$) are labelled. This inequality holds for every N at least 5. It gives an explicit rate in N only on the certified range $5 \leq N \leq 14000$, where N $|c^{\{F,N\}}_A(\theta) - c^{\text{inf}}_A(\theta)| \leq K_5 \|A\|^2$ with K_5 about $6.10e-09$ (about $5.96e-09$ and $5.46e-09$ with the reverse F_1 and F_2 constants); no $O(1/N)$ rate is asserted beyond that range. The bracket stays below 2 up to $N=14418$ and is at least 2 at $N=14419$ and at every larger N , so the bound is vacuous from $N=14419$ (certified here at the hypothesis values and at the admitted values alike; the historical assistant-2 found the same first exceedance independently at the hypothesis values). The finite-box correlation functions converge to $c^{\text{inf}}_A(\theta)$ along the whole sequence of N , for every A in $B(H_R)$ and $|\theta|$ at most 8, without a rate outside the certified range (the inequality with any fixed r in place of r_N gives a limsup of at most $C_{\text{dyn}}/(r-1) \|A\|^2$). Labelled secondary at $q_2=151552|\tau|$ (592/390625 at the cap): $C'_2=625/12481056$ (about $5.0076e-05$, target $1/8000$) and $c'_{\text{site},2}=625/24962112$ (about $2.5038e-05$, no target), discharged by the BB1 secondary pair (C_2 about $9.689e-06$ at most $1/20000$ and $c_{\text{site},2}$ about $9.674e-06$ at most $1/40000$, iterated_split bound values). The τ to $\tau/100$ ratios are exactly 1 for C' and c'_{site} at the hypothesis values (non-discriminating), $1953058850/194651$ (about 10033.64) for C_{dyn} and the reverse F_1 value, about 10019.59 for the reverse F_2 value, $1085053/1083425$ (about 1.0015) for C'_2 , and exactly 100 for q_2 . Rates are per coarse step in N at fixed spacing.

The rest of this section derives that statement, and the limitations recorded by the gate close it (Section 5.9).

5.2 Hypotheses and their discharge

The hypotheses. For every BB1 comparison, both signs, each on-site cutoff space and, at fixed N , the untruncated ground vectors, with N the smaller box size,

$$\begin{aligned} \left\| \rho_R^{\text{box } 1} - \rho_R^{\text{box } 2} \right\|_1 &\leq C_h q^{N-1}, & \left\| \rho_Y^{\text{box } 1} - \rho_Y^{\text{box } 2} \right\|_1 &\leq c_h |Y| e^{|Y|/10^8} q^{d_Y}, \\ C_h &= \frac{1}{250000}, & c_h &= \frac{1}{500000}, \end{aligned} \quad (51)$$

with $q = 1/64$, and at the labelled secondary pair $q_2 = 151552|\tau|$, $C_{2h} = 1/20000$ and $c_{2h} = 1/40000$. These are BB1’s frozen targets. Both producers mark every BB1-dependent statement `conditional_on_bb1_targets` and write each BB2 state constant as an explicit function of (C_h, c_h) , nondecreasing in each and evaluated at the hypothesis values. A constant proved at the hypothesis values then stays valid for any admitted BB1 constants that are at most those values.

The protocol. The skeptic pre-registered the discharge in its pre-comparison checker, committed before it read either producer or the BB1 gate. An item becomes unconditional only when, at both signs, some route proves it from comparisons admitted with a bound dominated pointwise by the hypothesis form and a constant at most the hypothesis value. The skeptic’s admission record transcribes the BB1 gate into 30 anchored entries: 10 bound values, 10 iterated_split values and 10 secondary cells. Each entry must be a verbatim substring of the gate’s text, including its constant, bound shape, comparison label, signs and regimes. The record is cross-checked against the BB1 review’s table in 40 cells (five comparisons, two forms, two cutoff regimes, two signs). The BB1 gate admits the following values (Section 4.5).

Table 15: BB1 admitted values used for the discharge; each has exactly the hypothesis shape, for every comparison, both signs and both cutoff regimes.

BB1 constant	Admitted value	Route	Hypothesis	Margin
C	$\approx 8.9051 \times 10^{-7}$	polymer_kp	$C_h = 1/250000$	4.49
c_{site}	$\approx 8.7681 \times 10^{-7}$	polymer_kp	$c_h = 1/500000$	2.28
C_2 (labelled secondary)	$\approx 9.689 \times 10^{-6}$	iterated_split	$C_{2h} = 1/20000$	5.16
$c_{\text{site},2}$ (labelled secondary)	$\approx 9.674 \times 10^{-6}$	iterated_split	$c_{2h} = 1/40000$	2.58

Item by item. Item 1 needs c_1 and c_2 (forward) or c_4 (reverse) in both forms and both families; item 2 needs item 1 and c_3 ; item 3 needs item 2; item 4 needs c_5 in both forms and item 1; item 5 needs items 1 and 2 (the R form, and the region form on Λ_{r_N}) together with BA2. Every one is discharged in full at both signs, and both routes are available for every family and form. The producers’ own maps of what they use are discharged cell by cell: the forward’s 104 primary and 24 secondary cells, and the reverse’s 68 primary and 24 secondary cells. The reverse’s item-5 map lists only item 1, but its F2 part also needs c_3 ; that reading was run as well and its 12 cells are discharged. Two labelled runs support the discharge. The secondary pair discharges every item at $q_2 = 592/390625$. The self-contained iterated_split values of BB1 discharge every item too, so the discharge does not depend on the post-hoc Kotecký–Preiss check of BB1’s polymer route. Eighteen discharge controls, run in memory on edited copies of the gate and the record, behave as required. Eight downgrade the verdict (for example, dropping c_5 drops item 4). One is accepted, because c_4 alone may be dropped: the forward route does not need it. Nine are refused outright.

5.3 Item 1: whole-sequence Cauchy estimates and the limit

Forward: nested telescoping. For $F \in \{F1, F2\}$, $N \geq 2$ and every $M > N$, write $\rho_Y^{F,M} - \rho_Y^{F,N}$ as the sum of the nested steps $\rho_Y^{F,k+1} - \rho_Y^{F,k}$, $k = N, \dots, M-1$. Each step is the BB1 comparison

c1 or c2 with smaller box Λ_k , and for $Y \subset \Lambda_N$ its exponent is $d_Y + (k - N)$. Summing the full geometric tail gives

$$\begin{aligned} \sup_{M>N} \left\| \rho_R^{F,M} - \rho_R^{F,N} \right\|_1 &\leq C' q^{N-1}, \\ \sup_{M>N} \left\| \rho_Y^{F,M} - \rho_Y^{F,N} \right\|_1 &\leq c'_{\text{site}} |Y| e^{|Y|/10^8} q^{d_Y}, \end{aligned} \quad (52)$$

with

$$C' = \frac{C_h}{1-q} = \frac{64}{63} C_h = \frac{4}{984375} \approx 4.0635 \times 10^{-6}, \quad c'_{\text{site}} = \frac{64}{63} c_h = \frac{2}{984375} \approx 2.0317 \times 10^{-6}. \quad (53)$$

Each meets its target with margin $315/128 = 2.4609375$. The bound is the limit of the partial sums and is not attained, and nothing of the tail is dropped.

Reverse: one direct comparison. The reverse compares every M with N directly. For two volumes V_1, V_2 of one prescription containing Λ_n , the triangle inequality through the union $U = V_1 \cup V_2$ costs the assembly factor $[V_1 \neq U] + [V_2 \neq U]$. For nested centered cubes the union is the larger cube, so the factor is 1 and $C' = C_h = 1/250000$, $c'_{\text{site}} = c_h = 1/500000$, each with margin $5/2$. The gate binds the forward values, the larger ones, which are valid under either route's discharge since $(64/63)C_h \geq C_h$; the reverse values are labelled. Re-evaluated at the admitted BB1 constants, C' is about 9.0465×10^{-7} and c'_{site} about 8.9073×10^{-7} on the forward assembly, and about 8.9051×10^{-7} and 8.7681×10^{-7} on the reverse. These re-evaluated values are labelled, not bound.

The limit without compactness. For a fixed finite region Y the whole sequence $(\rho_Y^{F,N})_N$ is Cauchy in the Banach space of trace-class operators on $\mathcal{H}_Y$, so it converges in trace norm; no compactness, subsequence or diagonal extraction is used. Positivity and unit trace are closed conditions. The partial trace is a trace-norm contraction, so the limits on nested regions are compatible. They define a locally normal state on the local algebra, which extends by norm continuity to the quasi-local algebra, the norm completion of the local algebra in the Nachtergaele–Sims setting. Letting $M \rightarrow \infty$ in (52) gives $\left\| \rho_R^{F,N} - \rho_R^{F,\infty} \right\|_1 \leq C' q^{N-1}$ and the region analogue. The order of limits is fixed. The bound holds in each cutoff space uniformly in L . Then $L \rightarrow \infty$ at fixed N and M , by AV1's cutoff-vector removal (and the AY1 itemization for F2), which the closed trace-norm ball survives. Only then is the supremum over M taken and N sent to infinity. Two exact fixtures mark what is not enough. The sequence $\text{diag}(\frac{1}{2} \pm (-1)^N \delta)$ lies in a one-state ball in every box yet has two subsequential limits. A zigzag walk with steps $1/(n+1)$ satisfies every N -to- $N+1$ bound and keeps oscillating.

5.4 Items 2 and 3: the common limit and its identification

Item 2. For every finite region Y and N large enough, the fixed- N comparison c3 of the two prescriptions on the same Λ_N , with item 1 twice, gives

$$\left\| \rho_Y^{F1,\infty} - \rho_Y^{F2,\infty} \right\|_1 \leq (2c'_{\text{site}} + c_h) |Y| e^{|Y|/10^8} q^{d_Y}, \quad (54)$$

and $(2C' + C_h)q^{N-1}$ on R . The left side does not depend on N , so the two limits coincide on every finite region. This common state, ω_∞ , is *the limit of the named constructions*: $\omega_\infty(A) = \lim_N \omega^{F1,N}(A) = \lim_N \omega^{F2,N}(A)$ for every quasi-local A . Nothing is said about other boundary conditions or other states.

Item 3. An AQ1 subsequential limit is a state whose local densities are trace-norm limits of the F1 densities along some subsequence N_k . Every subsequence of a convergent sequence has the same limit, so every AQ1 subsequential limit equals ω_∞ on every finite region, hence on the quasi-local

algebra. The same holds for every F2 subsequential limit of AY1’s own extraction, by item 2. The set of AQ1 subsequential limits, nonempty by AQ1’s compactness, is therefore the single state ω_∞ , whichever subsequence was chosen. Only after this identification does ω_∞ inherit properties, and in its own GNS representation, which is unitarily equivalent to AQ1’s because the states are equal. It inherits exactly the admitted ones:

- from AQ1: a compatible, locally normal, gauge-invariant state, stationary under T_θ , with a strongly continuous GNS evolution and a nonnegative self-adjoint physical generator reduced by the physical cyclic space; for F2 limits the same list comes through BA2’s rerun of AQ1 §§4–5, which now concerns the same state and the same dynamics;
- from AQ2, for the actual AQ1 centered subsequential state: the complete original-endpoint gauge-fixed space equals the invariant-local cyclic completion and reduces the generator, and on it $H_{\text{phys}} \geq (\alpha/16)(I - P_\Omega)$ as a quadratic-form inequality, with a simple vacuum on this completion;
- the full-GNS strengthening of that inequality only as qualified in the AQ2 gate, which uses AM2’s separately reviewed full-Hilbert finite gap.

Not inherited: uniqueness of any ground state, anything about states outside the named constructions, equality of GNS dynamics of different states, an unqualified full-GNS gap, and translation invariance, which AQ1 and AQ2 exclude and which is proved separately as item 4. The reverse also lists the AQ2 Wilson-variance witness of the same state; the gate’s list does not include it.

5.5 Item 4: coarse translations

Covariance. A coarse translation v is the fine translation $t_v = (4v_x, 2v_y, v_z)$. It maps each fine tail $(4i + r, 2j + q, k)$ to $(4(i + v_x) + r, 2(j + v_y) + q, k + v_z)$, so the links owned by b go to the links owned by $b + v$ with the same residues, every face goes to a face of the same class, and owner sets are translated. The site relabelling T_v therefore satisfies $T_v \hat{H}_V^P T_v^* = \hat{H}_{V+v}^P$ for both prescriptions, in each cutoff space; the ground states are simple, so $\rho_{Y+v}^{V+v} = T_v \rho_Y^V T_v^*$.

The bound. For $N \geq |v|_\infty + 2$ the volumes $\Lambda_N + v$ and Λ_N both contain $\Lambda_{N-|v|_\infty}$, but for $v \neq 0$ they are not nested and their union is not a centered cube. The direct BB1 comparison c5 of two one-prescription volumes containing $\Lambda_{N-|v|_\infty}$, used by both producers, gives

$$\left\| \rho_R^{\Lambda_N+v} - \rho_R^{\Lambda_N} \right\|_1 \leq C_h q^{N-\|v\|_\infty-1}, \quad C_h = \frac{1}{250000}, \quad (55)$$

with the hypothesis constant itself. Re-evaluated at the admitted C , it is about 8.9051×10^{-7} (labelled). A two-step comparison through the union, with factor 2, is labelled only; the skeptic notes that it is valid but looser than the available $C_h(1 + q^{|v|})$. The region form of c5 gives $|\omega^{P,N}(\alpha_v(A)) - \omega^{P,N}(A)| \leq c_h |Y| e^{|Y|/10^8} q^{N-|v|_\infty - \max_{y \in Y} \|y\|_\infty} \|A\| \rightarrow 0$ for $A \in B(\mathcal{H}_Y)$, so with item 1 the limit is invariant under every coarse translation, on every finite region and on the quasi-local algebra. Nested cubes alone could not give this: a covariant one-dimensional domain-wall fixture has constant nested marginals and a translated marginal at trace distance 2 for every N .

Non-coarse shifts are rejected, face by face. Of the 64 fine translations of the period window $[0, 8) \times [0, 4) \times [0, 2)$, exactly 8 preserve the selected set, every face class and the owner map, namely $(4v_x, 2v_y, v_z)$ with $v \in \{0, 1\}^3$. The witness $t = (1, 0, 0)$ maps the selected xy face at $r = 2, q = 0$ to the class $r = 3, q = 0$, which is omitted with owner set $\{0, e_x\}$. Similarly $(0, 1, 0)$ maps the selected row to an omitted one, and $(2, 0, 0)$ maps $r = 1$ to the omitted $r = 3$. Nothing is claimed for such shifts. The check must be made face by face. A per-anchor histogram of owner types is the same dictionary at every anchor, so a histogram-level check accepts all 64 translations, including the 56 non-coarse ones. The skeptic’s pitfall fixture and the historical assistant’s enumeration both show

this, and neither producer uses such a check. The forward checks role, class and a constant owner shift face by face on a fine window. The reverse checks all 64 translations.

5.6 Item 5: correlation functions on the window

The bound. For $A \in B(\mathcal{H}_R)$ put

$$c_A^{F,N}(\theta) = \omega^{F,N}(A^* T_\theta^{F,N}(A)) - |\omega^{F,N}(A)|^2, \quad c_A^\infty(\theta) = \omega_\infty(A^* T_\theta(A)) - |\omega_\infty(A)|^2, \quad (56)$$

centered with the complex mean; centering with $\omega(A)^2$ is rejected, since at $\theta = 0$ it is not the variance. $T_\theta(A)$ lies in the quasi-local algebra by Nachtergaele–Sims Theorem 4.1 (Section 3.2). With the local approximant $X_r = T_\theta^{F1,r}(A) \in B(\mathcal{H}_{\Lambda_r})$ and $r = r_N$, the difference splits into four terms:

- the finite-box evolution against X_r , bounded by the BA2 Cauchy constant $K_{F1}/(r-1)$, plus for F2 the BA2 comparison $b(N)$;
- the state difference on the region Λ_r that the approximant reaches, not only on R , bounded by the region form of item 1;
- X_r against the limit dynamics, bounded by $K_{F1}/(r-1)$;
- the means, bounded by $2C'q^{N-1}$.

The new lemma $(5N+1)(r_N-1) \leq N^3/4$ for $N \geq 5$ turns $b(N) = K_{\text{cmp}}(5N+1)N^{-3}$ into $(K_{\text{cmp}}/4)/(r_N-1)$. It follows from $2(r_N-1) \leq N-3$ and

$$N^3 - 2(5N+1)(N-3) = N^3 - 10N^2 + 28N + 6 = N(N-5)^2 + 3N + 6 > 0, \quad (57)$$

an identity the skeptic used to re-prove it; the lemma was also checked exactly for $N < 4000$. The result, for $N \geq 5$, $|\theta| \leq 8$, both families and both signs, is

$$|c_A^{F,N}(\theta) - c_A^\infty(\theta)| \leq \|A\|^2 \left[\frac{C_{\text{dyn}}}{r_N-1} + c'_{\text{site}} |\Lambda_{r_N}| e^{|\Lambda_{r_N}|/10^8} q^{N-r_N} + 2C'q^{N-1} \right], \quad (58)$$

$$C_{\text{dyn}} = 2K_{F1} + \frac{K_{\text{cmp}}}{4} = \frac{78057}{622883200000000},$$

with $C_{\text{dyn}} \approx 1.2532 \times 10^{-10}$ (tier `polynomial_lieb_robinson`, route `duhamel_inner_f1`, one constant for both families, margin about 3.99). The three constants are separate, each checked against its own target. The reverse gives one constant per family, $2K$ from the BA2 gate, and these are labelled: $C_{\text{dyn}}[F1] = 9261/77860400000000 \approx 1.1894 \times 10^{-10}$ (`duhamel_inner_f1`, margin about 4.20) and

$$C_{\text{dyn}}[F2] = \frac{1055961408185869563}{152407011718750000000000000000} \approx 6.9286 \times 10^{-11}$$

(`duhamel_inner_f2`, margin about 7.22). At $N = 5$ the forward bracket is about 1.0946×10^{-9} , with the region term dominant; at $N = 10$ it is about 4.179×10^{-11} , with the dynamics term dominant.

The rate, and where it stops. The inequality holds for every $N \geq 5$, but its rate is explicit only on a certified range. On $5 \leq N \leq 14000$, $N \leq 10(r_N-1)$ and the region term is at most $c'_{\text{site}}g(N)$ with $g(x) = x^4 e^{x^3/10^8} 8^{-(x+1)}$. The function $\ln g$ has a convex derivative that is negative at 5, so g decreases and then increases on the range, and $g(14000) \leq g(5)$ exactly. Hence

$$N |c_A^{F,N}(\theta) - c_A^\infty(\theta)| \leq K_5 \|A\|^2 \quad (5 \leq N \leq 14000), \quad (59)$$

$$K_5 = 10C_{\text{dyn}} + c'_{\text{site}} g(5) + 10C'q^4 \approx 6.10 \times 10^{-9},$$

about 5.96×10^{-9} and 5.46×10^{-9} with the reverse F1 and F2 constants. The reverse's own block certificate, about 1.415×10^{-8} , is valid but looser. Beyond the range the frozen region factor

takes over. $|\Lambda_{r_N}|$ is of order N^3 inside $e^{|\Lambda_{r_N}|/10^8}$ and eventually beats q^{N-r_N} . The skeptic certifies with logarithmic enclosures that the bracket stays below 2 up to $N = 14418$ (the tightest point is $N = 14417$). It is at least 2 at $N = 14419$ and at every larger N , for the forward and both reverse constant sets, at the hypothesis and at the admitted values. Since each centered correlation has modulus at most $\|A\|^2$, the bound is vacuous from $N = 14419$. The historical assistant found the same first exceedance by its own scan, and the reverse certified vacuity at $N = 14421$. The finite-box correlation functions still converge to $c_A^\infty(\theta)$ along the whole sequence, uniformly for $|\theta| \leq 8$, but without a rate outside the certified range. The inequality with any fixed $r \geq 2$ in place of r_N gives $\limsup_N |c_A^{F,N} - c_A^\infty| \leq C_{\text{dyn}} \|A\|^2 / (r - 1)$, and r is arbitrary. This does not re-choose r_N for any quantitative statement.

The rejected overclaim and defect D12. The forward report states that the rate is $O(1/N)$ while the state terms decay geometrically, the cube volume growing only like N^3 . That paragraph is false from $N = 14419$ and is rejected; it enters no gate text. The forward checker evaluates item 5 only for $N \leq 59$, and its enclosure $e^x \leq 1/(1-x)$ cannot be applied from $N = 465$. The forward’s inequality and constants are correct, and its frozen files are retained as evidence. The reverse producer found the problem (its wording finding W1). The contract’s required item 5 asks for a rate in N of $O(1/N)$, which the frozen r_N and region form allow only on a certified range. The skeptic’s own pre-comparison derivation repeated the unqualified statement, and it records the defect against itself as D12. The gate text therefore claims no correlation rate beyond the certified range. The skeptic’s planning advice notes that a region form without the exponential, such as BB1’s iterated_split form $c_{\text{site}} \sum_{y \in Y} q^{N-\|y\|_\infty}$, would give a rate for every N ; that is a candidate goal, not a result.

What item 5 does not say. Only the correlation numbers converge. $\omega^{F,N}$ and ω_∞ are different states, and one automorphism group can carry two invariant states with different correlations (the fixture with $(3 \pm 4i)/5$), so no equality of GNS dynamics of different states is asserted. The BA2 constants hold for $|\theta| \leq 8$ and grow with the window, so nothing is uniform in time.

5.7 Scaling, the secondary pair and the error ledger

At the hypothesis values C' and c'_{site} do not depend on τ : their ratio under $\tau \rightarrow \tau/100$ is exactly 1, which meets the bracket but cannot discriminate. The linear τ -scaling of the underlying BB1 constants, about 100.648, is recorded by BB1. The dynamics constant has the structure of the BA2 constants, $C_{\text{dyn}} = 1248912 \tau^2 / (1 - 338688|\tau|)$, with ratio $1953058850/194651 \approx 10033.64$ (the reverse F1 value shares it, and the reverse F2 value gives about 10019.59), inside $[9500, 10500]$. At the labelled secondary pair $q_2 = 151552|\tau| = 592/390625$, the forward gives $C'_2 = 625/12481056 \approx 5.0076 \times 10^{-5}$, below $1/8000$ with margin 2.4962112, and $c'_{\text{site},2} = 625/24962112 \approx 2.5038 \times 10^{-5}$, which has no target. Its ratio is $1085053/1083425 \approx 1.0015$, and q_2 scales by exactly 100. The BB1 secondary pair discharges them.

5.8 Evidence and review

Each producer implements all 34 contract controls as damaging mutations: the forward checker runs 52 exact checks with 122 rejected mutations, the reverse 59 exact checks with 162. The skeptic’s pre-comparison package (81 checks, 38 controls, 142 rejected mutations) was committed after the forward commit and before the reverse commit, so the forward could not have read it. The post-comparison checker runs 11 checks. The review adds 68 validator weakenings, one per control per producer, and 13 must-abort source edits, all caught. It also records 6 silent edits: three are caught only by the review’s value validator and three by the reverse checker itself. The two unmutated copies reproduce the frozen results. The 18 discharge controls are described above.

Table 16: The preregistered BB2 error terms, added linearly.

Term	Entry
bb1_hypothesis_constants	$C_h = 1/250000$ and $c_h = 1/500000$ at $q = 1/64$ (secondary $1/20000$ and $1/40000$ at q_2), discharged by the BB1 gate
cauchy_teslaping_or_union	forward factor $1/(1 - q) = 64/63$ from the full geometric tail; reverse factor 1 for nested cubes (2 for non-nested pairs, labelled)
identification_with_subsequential_limits	0: exact, since every subsequence of a convergent sequence has the same limit
translation_general_volume	$C_h q^{N- v _\infty-1}$ on R and the region form of c5 on Y , factor 1 by the direct comparison
dynamics_constant_from_ba2	$C_{\text{dyn}} = 2K_{\text{F1}} + K_{\text{cmp}}/4$ from the BA2 gate values, with the lemma (57)
region_form_on_Lambda_rN	$c'_{\text{site}} \Lambda_{rN} e^{ \Lambda_{rN} /10^8} q^{N-rN}$ with $ \Lambda_r = (2r + 1)^3$; the source of the certified range
arithmetic	0: exact rationals and directed enclosures of the exponential

All four producer replays reproduce the frozen outputs byte for byte under normal and optimized Python. Both closures (42 files each) verify, and each producer holds exactly its 38 declared inputs.

Both producers disclosed name-only exposures to untracked file names of other agents after their first freeze; they opened none of them, and the state constants are fixed by the contract. One of the skeptic’s replay outputs was written outside the checkout by an empty shell variable. It is identical to the others and is disclosed. The review found no blocking issue for the loop. Its recorded findings include:

- the rejected forward rate wording, and the forward’s conditional gate-field value, which the Jung assistant found outside the closed vocabulary;
- the reverse’s item-5 map, which omits c3 for its F2 part, and its labelled union fallback for item 4;
- the non-discriminating “exactly 1” brackets;
- the contract readings of the union route (D2), the composite C_{dyn} (D3, D10), the untargeted $c'_{\text{site},2}$ (D5), inherited control texts (D6), the BB1 definition of d_Y (D7), the missing frame for limited outcomes (D9) and the scope of the item-3 inheritance (D4, D11).

Meaning. BB2 establishes convergence of the named constructions. The reduced densities of F1 and F2 converge as whole sequences, at the rate q^{N-1} per coarse ℓ^∞ step at fixed spacing, to one common limit on every finite region. That limit is every AQ1 and every F2 subsequential limit and is invariant under coarse translations, and its correlation functions on $|\theta| \leq 8$ are the limits of the finite-box ones. It is not uniqueness of any ground state and says nothing about states outside the named constructions. It is not equality of GNS dynamics of different states, and nothing in it is uniform in time or in the lattice spacing.

5.9 Limitations recorded by the gate

The BB2 gate records the following limitations verbatim.

- Scope: only the zero-selected patterned family (selected triple (0,0,0), Haar reference), fixed spacing, $|\tau| \leq 10^{-8}$, the named families F1 and F2 on centered cubes Λ_N with N at least 2 and the limit of the named constructions, the cover R and finite complete-factor regions, coarse translations and the window $|\theta| \leq 8$; nothing transfers to other boundary conditions, states outside the named

- constructions, nonzero selected triples, literal vertex boxes, weak coupling or the continuum; the constants are uniform in N at fixed spacing and in the cutoff, never in the lattice spacing.
- Convergence of the named constructions only: no uniqueness of any ground state and no statement about any other infinite-volume state; the limit is identified only with the AQ1 and F2 subsequential limits and inherits only the listed admitted properties (the full-GNS strengthening only as qualified in the AQ2 gate); no equality of GNS dynamics of different states and no uniform-in-time statement.
 - Discharge: both routes prove items 1-5 from the BB1 frozen targets $C_h=1/250000$ and $c_h=1/500000$ at $q=1/64$ as hypotheses. The BB1 gate (sha256 18141fea672e5bae09024a5fddc56ea7102aae56ecca16fbb30d67a1354a3827, commit eef57b4) admits C about $8.9051e-07$ and c_{site} about $8.7681e-07$ for all five comparisons, both forms, both cutoff regimes and both signs. The skeptic discharge (the protocol committed in `bb2_check.py` before either producer or the gate was read, applied to the anchored admission record `bb2-bb1-admission.json` and cross-checked with the BB1 review table in 40 cells) finds every item full and every cell of both producers' hypothesis maps discharged. The bound constants are evaluated at the hypothesis values; the values re-evaluated at the admitted constants are labelled only.
 - The discharge carries the BB1 limitations: the `polymer_kp` bound values rest on the Kotecky-Preiss criterion, whose citation was checked post hoc against the committed Ueltschi excerpt and found covered; the self-contained `iterated_split` values discharge every item as well (labelled run). F2 volumes other than centred cubes in `c5` (item 4) rest on the AY1 itemization as a reviewed local reading, not a gated statement.
 - Item-5 rate (reverse W1; skeptic defect D12): contract required item 5 says the rate in N is $O(1/N)$, but with r_N and the region form frozen the region term on $\Lambda_{\{r_N\}}$ carries $e^{-|\Lambda_{\{r_N\}}|/10^8}$ with $|\Lambda_{\{r_N\}}|$ about N^3 , which eventually outgrows $q^{(N-r_N)}$. The rate holds only on the certified range $5 \leq N \leq 14000$ and the bound is vacuous from $N=14419$, so that clause of the requirement is met on the certified range only. The skeptic's own pre-comparison derivation (section 8) stated the rate without the range, and its pre-freeze and pre-comparison reviews missed the defect; the reverse producer found it.
 - Rejected forward overclaim: the forward report's rate paragraph (the rate in N is $O(1/N)$ while the state terms decay geometrically in N , the cube volume growing only like N^3) and the matching results text are rejected, because the region term is at least 2 from $N=14419$ on; the forward checker evaluates item 5 only for $N \leq 59$, and its enclosure $e^x \leq 1/(1-x)$ is unusable from $N=465$. The forward's item-5 inequality and constants are correct; the frozen forward files are retained as evidence and the rejected wording enters no gate text.
 - Constants are upper bounds only; the minus-sign values replay the same $|\tau|$ formula. C' and c'_{site} at the hypothesis values do not depend on τ (ratio exactly 1, non-discriminating); the linear τ -scaling of the BB1 constants is recorded by BB1 (about 100.648). The dynamics constant is BA2's, and the correlation bound holds per fixed local A on $|\theta| \leq 8$ only.
 - Contract wording read as follows: `union_comparison` is one direct application of `c4` or `c5` (a two-step comparison through a third volume is labelled, and the reverse's labelled item-4 fallback $2C_h$ is valid but looser than the available $C_h(1+q^{|v|})$; C_{dyn} is bound as the forward's single-route constant with the reverse per-family values labelled (D3, D10); $c'_{\text{site},2}$ has no target (D5); d_Y is the BB1 definition (D7); no frame for limited outcomes (D9, moot at this verdict); several inherited control texts were written for BA2 or BB1 (D6); the item-3 inheritance is read within the admitted scope (D4, D11).
 - Independence is limited to derivations, constants and code: all agents are correlated model agents; the frozen targets, the brackets and the conditional semantics came from the skeptic's BB pre-freeze review; the skeptic pre-comparison package was committed after the forward commit and before the reverse commit; isolation is verified for repository inputs only, and both producers disclosed name-only exposures to other agents' file names.
 - Scientific priority is unverified.

Full derivations and reviewed implementations: [BB2 forward](#), [BB2 reverse](#), [independent model-agent review](#), the [BB1 admission record](#) and the [frozen contract](#). With BB1 and BB2 gated, sub-round 2 is complete.

6 BC1: certificates for the limit

Contribution HNM-C-BC1. Two Round32 certificates were stated before any limit object existed. AV2 certified the centered Euclidean Wilson correlation at the node $s = 1$ for every AQ1 subsequential state, each chosen state separately. AW2 enclosed the static Wilson mean for every state in the whole set of AQ1 subsequential limits, and also in every centered whole-star box. BB2 has since admitted that this set is one state, the limit of the named constructions (Section 5.4). The BC1 contract, recorded under the title “Node and sign certificates restated for the limit of the named constructions; a finite-box sign corollary for the padded family; updated obligations”, restates both certificates for that state, turns the whole-sequence rate of BB2 into a sign certificate for the padded boxes, states what the common GNS representation carries, and updates the AY2 obligations table with the Round33 gates. It was selected after the BB2 gate as investigation 5 of 8, opening sub-round 3 (selection note). It was frozen together with BC2 on 25 September 2026 at 04:21:44 UTC, before either was produced and after the skeptic’s pre-freeze review of the two BC drafts (30 blocking and 28 non-blocking edits, all applied before the freeze). Its gate is recorded under the title “Hruday node and sign certificates for the limit of the named constructions”.

BC1 is a statement investigation (direction T of Table 4). One forward producer restates the admitted certificates with unchanged constants, proves the finite-box corollary by exact arithmetic from admitted constants and writes the obligations table. The skeptic replays the restatement and the corollary from the contract and the gates alone. The forward producer received AGENTS.md, the contract and 38 shared premises: the AQ1 and AQ2 gates and reports; the AV1, AV2, AW1, AW2, AY1 and AY2 gates with the relevant Round32 producer reports and reviews; the BA1, BA2, BB1 and BB2 gates and contracts; both BB2 producer reports; three Round33 reviews; the selection note; and the method files. It read nothing of BC2 and no skeptic, advisor or expert file outside those premises. The contract names seven items: the two restatements, the finite-box sign corollary, a determination of the finite-box node from the admitted record, the common GNS item, the obligations table, and the mandatory sentence with the gate fields and an exact checker. Its target is exact equality of every restated value with its gate value, and N_{sign} computed exactly. The contract calls this a feasibility and format check, not a blind discriminating threshold, because every value follows from admitted constants. The parameters fix $N_{\text{min}} = 2$ and the single node $s = 1$, and declare no weights, since BC1 derives no new decay estimate. The itemized error terms are `identification_premise_bb2`, `restated_constants_exactness`, `finite_box_whole_sequence_term`, `second_order_remainder_K2plus` and `arithmetic`, and the contract lists 25 controls.

6.1 Reviewed scope

The BC1 gate records the verdict `accepted_within_scope`, with sub-label `certificate_restated_for_limit` and secondary sub-labels `reference_unresolved` (the node) and `static_not_dynamic` (the static mean). Its gate fields set `node_certificate_restated_for_limit`, `sign_certificate_restated_for_limit`, `finite_box_sign_claimed`, `dynamics_limit_identified_claimed` and `rate_in_N_claimed` to true. The finite-box sign scope is “F2 boxes Λ_N , untruncated ground vectors at fixed N , N at least N_{sign} , both signs of τ , through the BB2 whole-sequence bound at the BB2 gate bound value; F1 boxes as certified by AW2”. Nine fields are false: `finite_box_node_claimed` and `correlation_shift_resolved` for the two certificates; `uniqueness_of_ground_state_claimed` and `gns_dynamics_equality_claimed` for the limit; `uniform_in_time_claimed` and `rate_in_a_claimed` for the rates; and `continuum_claim`, `weak_coupling_claim` and `scientific_priority_verified`. The field `dynamics_limit_identified_claimed` restates BA2 item (4) and adds no dynamics claim. The claimed rate in N is the density rate q^{N-1} for every $N \geq 2$; the correlation rate stays on the range certified in BB2. The accepted statement, quoted verbatim below in the gate’s ASCII notation, is the scope of every claim in this section.

54

representation (AQ1 for F1, BA2 item 4 for F2); its correlation functions on $|\theta\rangle$ at most 8 are the limits of the finite-box correlation functions of either family (BB2 item 5), with a rate in N ($K5/N$) only on $5 \leq N \leq 14000$ and convergence without a rate beyond; this concerns one state and is not equality of GNS dynamics of different states. Obligations (the reviewed table of 21 rows): AY2 row 01 is closed only in the narrow form that all subsequential limits of F1 and F2 coincide on every finite region (one limit of the named constructions, BB2 items 2–3, row 01a); uniqueness of any ground state, states outside the named constructions and other boundary conditions are not proved and remain open (rows 01b and N1); rows 02–06 are closed within the scopes of the BB2 and BA2 gates; rows N1–N14 are open. Labelled information, not claimed: $L/(C' q^3)$ is about 4.4584 and the widened exclusion margin at $N=4$, $(L-C' q^3)/(K_2 + \tau^2 + C' q^3)$, is about 3.3851.

The rest of this section derives that statement, and the limitations recorded by the gate close it (Section 6.10).

6.2 The identification premise

Three admitted statements are used, all read from hash-pinned gates. For $F \in \{F1, F2\}$, both signs, every $N \geq 2$ and every $M > N$, BB2 item 1 gives

$$\left\| \rho_R^{F,M} - \rho_R^{F,N} \right\|_1 \leq C' q^{N-1}, \quad C' = \frac{4}{984375}, \quad q = \frac{1}{64}, \quad (60)$$

in each on-site cutoff space and, at fixed N , for the untruncated ground vectors (Section 5.3). BB2 items 2–3 give that the F1 and F2 limits coincide on every finite region and that, in the gate’s words, “this limit coincides with every AQ1 subsequential limit and every F2 subsequential limit”. BA2 item (4) gives that the F2 limit dynamics equals the AQ1 limit dynamics T_θ (Section 3.8).

The identification. At each sign of τ , let $S(\tau)$ be the set of AQ1 subsequential limits. It is nonempty by the AQ1 gate. By BB2 item 3 every member equals ω_∞ on every finite region, and two states that agree on the local algebra agree on its norm closure. Hence

$$S(\tau) = \{\omega_\infty\}, \quad (61)$$

and the set of F2 subsequential limits is the same singleton. This is the forward’s statement HNM-BC1-F01. Every statement that a Round32 gate admits for every AQ1 subsequential state, or for the whole set of them, therefore holds for ω_∞ with the same constants. The GNS triple of ω_∞ is that of the one AQ1 subsequential state up to the canonical unitary equivalence, its GNS evolution implements the same automorphism group T_θ , and its generator G is nonnegative (AQ1 §5, inherited through BB2 item 3). The AV2 window lemma applies to ω_∞ word for word.

Order and what it does not license. The identification precedes every inheritance (control `limit_identified_with_aq1_limits`). A restatement drawn from uniform local closeness, from one BB1 comparison or from a subsequence is rejected. AV1 and AY1 place any two subsequential limits within $2D$ of each other on R , and within $2K'_2 \tau^2$ beyond the common first-order density; closeness of that kind does not make them one state. The identification changes no constant and adds nothing about $C(s)$ or $\omega(W)$ beyond the Round32 gates. It says nothing about states outside the named constructions or other boundary conditions, and nothing about finite boxes except through the explicit bound (60) (Section 6.5).

6.3 The node certificate restated

Let $(\pi_\infty, \mathcal{H}_\infty, \Omega_\infty)$ be the GNS triple of ω_∞ and $G = H/\alpha$ its generator in the Euclidean clock $s = \alpha t_E / \hbar$. Put

$$\chi = (\pi_\infty(W) - \omega_\infty(W))\Omega_\infty, \quad C(s) = (\chi, e^{-sG}\chi). \quad (62)$$

At the single preregistered node $s = 1$ and $\tau = +10^{-8}$, the AV2 certificate holds for ω_∞ with its constants unchanged (HNM-BC1-F02):

$$\begin{aligned} |C(1) - d| &\leq R, \\ d &= \frac{497870683678639429793424156500617766317}{4 \cdot 10^{40}} \approx 1.24467670920 \times 10^{-2}, \\ R &= \frac{1831967503411879425147166810021607}{10^{40}} \approx 1.83196750341 \times 10^{-7}, \end{aligned} \quad (63)$$

so that $C(1) \in [d - R, d + R]$ with

$$\begin{aligned} d - R &= \frac{497863355808625782275723567833377679889}{4 \cdot 10^{40}} \approx 1.24465838952 \times 10^{-2}, \\ d + R &= \frac{99575602309730615462224949033571570549}{8 \cdot 10^{39}} \approx 1.24469502887 \times 10^{-2}. \end{aligned} \quad (64)$$

The proof is the identification (61). The AV2 gate admits the certificate for every AQ1 subsequential state, each chosen state separately, and ω_∞ is that state. Nothing is re-derived. The checker parses d , R and the interval ends from the gate, requires $d \mp R$ to equal the ends exactly, and requires the gate's accepted statement and decision to carry the same d and R .

What the constants are. The radius bounds the AV2 point error $2(D + D^2) + 49|\tau|s/\pi$ plus the arithmetic half-width $1/(4 \cdot 10^{40})$ at $s = 1$. Here $D \approx 1.36124528702 \times 10^{-8}$ is the AV1 tier-(ii) state bound, and the window is the frozen C^2 window with $M_0 = 2$ and $M_1 = 4s/\pi$. Three exact consistency replays, not sources of any value, were run by the forward and reproduced by the skeptic with its own Machin enclosure of π and series for e^3 :

- R is a valid directed bound of the gate's formula, and exceeds $2(D + D^2) + 49|\tau|/\pi + 1/(4 \cdot 10^{40})$ by about 2.65×10^{-41} , less than 10^{-40} ;
- R is the common outward bound of the two AV2 producers, not a producer radius; the AV2 forward exact radius lies below R by less than 10^{-40} , and restating that smaller radius in place of R is rejected (control `certificate_values_unchanged`);
- d is the midpoint of an enclosure of $e^{-3}/4$, with $|d - e^{-3}/4| \leq 1.02 \times 10^{-43}$ (the skeptic's value is about 1.0195×10^{-43}).

Reference unresolved. The free value $e^{-3}/4$ lies inside the interval (64). The node is an upper certificate: no interaction shift, sign or coefficient of $C(s)$ follows, and `correlation_shift_resolved` is false. At $\tau = -10^{-8}$ the same datum and radius hold for the limit at that coupling, which BB2 identifies with the AQ1 subsequential state there. Every term of R depends on $|\tau|$ only, so this is a replay of the same $|\tau|$ formula, never a second confirmation. Only $s = 1$ is restated. The AV2 crossover $s^* \approx 6.2368634460$ is a property of the radius formula, not a node certificate, and no grid or other node is restated (control `post_hoc_node_rejected`).

6.4 The static Wilson mean restated

The AW2 enclosure is read with the AW1 bound

$$K_2^+ = \frac{81108864767825329926713064490531229475390625}{24176936535511801466930759024724079017984} \approx 3354.80322946. \quad (65)$$

With $D_W = 618929575309102117553427431032936422860390400000000$ (the gate's D), the second-order remainder at the cap is $K_2^+ \tau^2 = 207638693805632844612385445095759947457/D_W \approx$

$3.35480322946 \times 10^{-13}$, and at $\tau = +10^{-8}$ (HNM-BC1-F03)

$$\begin{aligned}\omega_\infty(W) &\in [L, U] = \left[\frac{\tau}{144} - K_2^+ \tau^2, \frac{\tau}{144} + K_2^+ \tau^2 \right], \\ L &= \frac{42773581813770903096597852821080380528959}{D_W}, \\ U &= \frac{43188859201382168785822623711271900423873}{D_W},\end{aligned}\tag{66}$$

contained in $[6.91089641214 \times 10^{-11}, 6.97799247674 \times 10^{-11}]$. At $\tau = -10^{-8}$ the enclosure is the mirror $[-U, -L]$, by the AW1 remainder bound at both signs and the AW1 flip lemma; it is a replay, not a second confirmation. The proof is again the identification: AW2 admits the enclosure for every state in the whole set of AQ1 subsequential limits, and that set is $\{\omega_\infty\}$ at each sign. The checker recomputes $\tau/144 \mp K_2^+ \tau^2$ from the parsed K_2^+ , requires equality with the gate endpoints, with their mirror and with $(L + U)/2 = \tau/144$, and keeps the full remainder at both endpoints (control `second_order_remainder_kept`).

The gate's margins are restated unchanged: the exclusion margin and the sign margin are

$$\begin{aligned}\frac{|\tau|/144 - K_2^+ \tau^2}{K_2^+ \tau^2} &= \frac{42773581813770903096597852821080380528959}{207638693805632844612385445095759947457} \approx 206.00005, \\ \frac{1}{144K_2^+ |\tau|} &= \frac{42981220507576535941210238266176140476416}{207638693805632844612385445095759947457} \approx 207.00005.\end{aligned}\tag{67}$$

Hence $\text{sign } \omega_\infty(W) = \text{sign } \tau$ at both signs, and the Haar reference 0 is strictly excluded. The enclosure concerns a static equal-time mean (`static_not_dynamic`); AW2's resolved interaction shift refers to $\omega(W)$ only, and nothing is said about $C(s)$.

6.5 The finite-box sign corollary for the padded family

This is the one new constant family of BC1 (HNM-BC1-F04). For every $N \geq 2$, both signs and the untruncated F2 ground state $\omega^{F2,N}$ on Λ_N at fixed N ,

$$|\omega^{F2,N}(W) - \omega_\infty(W)| \leq \|W\| \left\| \rho_R^{F2,N} - \rho_R^\infty \right\|_1 \leq C' q^{N-1},\tag{68}$$

where ρ_R^∞ is the R -density of ω_∞ . The proof has five steps.

- (i) BB2 item 1 with $F = F2$ gives (60) for every $M > N$, for the untruncated ground vectors at fixed N and M .
- (ii) By BB2 items 1–2 the whole sequence $\rho_R^{F2,M}$ converges in trace norm to ρ_R^∞ . The trace norm is continuous, and the closed trace-norm ball of radius $C' q^{N-1}$ about $\rho_R^{F2,N}$ contains the limit, so $\left\| \rho_R^\infty - \rho_R^{F2,N} \right\|_1 \leq C' q^{N-1}$.
- (iii) The original xz Wilson loop W lies in $B(\mathcal{H}_R)$; it is real and at most 1 in absolute value pointwise, so $\|W\| \leq 1$. Trace duality gives the first inequality of (68). The factor $\frac{1}{2}$ available for an effect does not apply, because W is not an effect.
- (iv) Adding (68) to (66) on both sides gives the widened enclosure

$$\omega^{F2,N}(W) \in \begin{cases} [L - C' q^{N-1}, U + C' q^{N-1}], & \tau = +10^{-8}, \\ [-U - C' q^{N-1}, -L + C' q^{N-1}], & \tau = -10^{-8}. \end{cases}\tag{69}$$

- (v) The widening $C' q^{N-1}$ is strictly decreasing in N . Define N_{sign} as the least $N \geq 2$ with $C' q^{N-1} < L$, strictly. For every $N \geq N_{\text{sign}}$ the lower end at $+\tau$ is positive and the upper end at $-\tau$ is negative.

The constant is the bound value $C' = 4/984375$ of the BB2 gate decision (assembly `nested_telemscoping`, hypothesis source `bb1_frozen_targets`, BB1 route `polymer_kp`), evaluated at the hypothesis values. The labelled union value $1/250000$ and the labelled re-evaluated value, about 9.0465×10^{-7} , are not used.

N_{sign} , **exactly.** Table 17 lists the widening at $\tau = 10^{-8}$, where $L \approx 6.91089641214 \times 10^{-11}$. The widened interval contains 0 at $N = 2$ and $N = 3$ and excludes it from $N = 4$ on, so

$$N_{\text{sign}} = 4, \quad C'q^3 = \frac{1}{64512000000} \approx 1.55009920634 \times 10^{-11}. \quad (70)$$

At $N = N_{\text{sign}}$ the widened enclosure at $\tau = +10^{-8}$ is

$$\begin{aligned} \omega^{F2,4}(W) &\in [L - C'q^3, U + C'q^3], \\ L - C'q^3 &= \frac{232256915653307984268043972456662444208313}{E}, \\ U + C'q^3 &= \frac{369480171452763518908899363269803522461511}{E}, \\ E &= 4332507027163714822873992017230554960022732800000000, \end{aligned} \quad (71)$$

contained in $[5.36079720580 \times 10^{-11}, 8.52809168309 \times 10^{-11}]$, with its mirror at $\tau = -10^{-8}$. The sign of $\omega^{F2,N}(W)$ equals the sign of τ for every $N \geq 4$ at both signs. The rate of the widening is the BB2 density rate q^{N-1} per coarse step at fixed spacing, for every $N \geq 2$. The new family carries the tier **exact_first_order**. In place of a route it records its inputs: the AW2 gate endpoints, with K_2^+ at the AW1 exact tier, and C' with the labels above. The restated constants keep their Round32 labels.

Table 17: The widening $C'q^{N-1}$ of the static Wilson mean of the untruncated F2 ground state and the widened lower end at $\tau = +10^{-8}$ (decimals are truncated previews of exact rationals). The widened upper end at $N = 3$ is about 1.06184×10^{-9} , so the interval contains 0 there; the rows at $-\tau$ are the mirror.

N	$C'q^{N-1}$ (exact)	preview	widened lower end at $+\tau$	sign certified
2	1/15750000	$6.34920634920 \times 10^{-8}$	$-6.34229545279 \times 10^{-8}$	no
3	1/1008000000	$9.92063492063 \times 10^{-10}$	$-9.22954527941 \times 10^{-10}$	no
4	1/64512000000	$1.55009920634 \times 10^{-11}$	$5.36079720580 \times 10^{-11}$	yes
5	1/4128768000000	$2.42203000992 \times 10^{-13}$	$6.88667611205 \times 10^{-11}$	yes
6	1/264241152000000	$3.78442189050 \times 10^{-15}$	$6.91051796996 \times 10^{-11}$	yes

Labelled information. The following values are recorded and not claimed. The ratio $L/(C'q^3)$ is about 4.4584, and the widened exclusion margin at $N = 4$, $(L - C'q^3)/(K_2^+\tau^2 + C'q^3)$, is about 3.3851. The skeptic notes that $N_{\text{sign}} = 4$ for every C' in $[64^2L, 64^3L) \approx [2.8307 \times 10^{-7}, 1.8117 \times 10^{-5})$, so the two labelled BB2 values would give the same integer; the constant must still be read as the bound value, and the checker rejects both. Below the cap nothing is claimed. At $\tau/100$ the half-width $K_2^+(\tau/100)^2 + C'q^3$ would be about 1.5501×10^{-11} , larger than the first-order value 6.944×10^{-13} . C' at the hypothesis values does not depend on τ , and AW2 admits the enclosure at the cap only.

F1 boxes. The AW2 gate states the enclosure “in every centered whole-star box $N \geq 2$ at every on-site cutoff”, and the AV1 gate removes the on-site cutoff for the ground vector itself in each fixed box. The untruncated F1 value therefore lies in the same closed interval, and the sign of $\omega^{F1,N}(W)$ is certified by AW2 at every $N \geq 2$. BC1 records this as agreement and does not re-derive it.

What is not claimed. Nothing is claimed for F2 boxes with $N = 2$ or $N = 3$, where the widened interval contains 0; this is the open row N4 of Table 18, not a failure. Cutoff- L F2 states are not claimed, because BB2 identifies only the untruncated limit. The row records a candidate route, which is labelled and not claimed. AY1’s forward statements F13–F14, as quoted in the AY2 forward

report, bound the R -local remainder by $K'_2\tau^2$ in every box of either family at every cutoff, and the first-order term gives $\text{Tr}(\rho_R^{(1)}W) = +\tau/144$. A gated box-level form of that bound for the untruncated F2 vector would place $\omega^{F2,N}(W)$ within $K'_2\tau^2$ of $\tau/144$, with sign margin $1/(144K'_2\tau)$ about 51.76. The AY1 and AY2 gates admit K'_2 for subsequential limits only, so this number rests on an ungated box-level reading of AY1 F13–F14 and on the untruncated F2 passage. It is not admitted and certifies nothing for $N = 2, 3$.

Traps. Each is an exact fixture in the forward checker. The BB1 per-comparison constant, about 8.9051×10^{-7} , bounds one comparison and says nothing about ω_∞ without the whole-sequence passage. The harmonic sums $a_N = \sum_{k=2}^N 1/k$ have steps $1/N \rightarrow 0$ yet first exceed 4 at $N = 83$, and the alternating densities $\text{diag}(\frac{1}{2} \pm \frac{1}{100}, \frac{1}{2} \mp \frac{1}{100})$ stay in a fixed ball with two subsequential limits. Taking $N_{\text{sign}} = 3$ is off by one: the widened lower end there is about $-9.22954527941 \times 10^{-10}$. A one-sided widening or a dropped remainder changes an endpoint. In a 2×2 fixture with $\Delta = \text{diag}(\frac{1}{2}, -\frac{1}{2})$, $X = \text{diag}(1, -1)$ attains $|\text{Tr}(\Delta X)| = \|\Delta\|_1$ and $2X$ doubles the difference, so the factor $\|W\| \leq 1$ is needed.

6.6 No finite-box node

The contract settled this item from the admitted record before the freeze, and the forward checker and the skeptic verified each cited fact in the snapshots.

- The AV2 gate quantifies over “every AQ1 subsequential state of the centered whole-star construction (each chosen state separately)”.
- The window lemma and the real-time comparison are stated for the GNS vector of that state, with AQ1 §5 nonnegativity: AV2 forward HNM-AV2-F01, F02, F07 and F15, and AV2 reverse HNM-AV2-R01, R02, R08 and R13.
- The only box-by-box step is the seven-star slope $k = 49|\tau|/4$ (AV2 forward F13–F14, reverse R12), which the record passes to the AQ evolution.
- The AV2 reverse executes a finite-box provenance as a damaging mutation of `changed_model_relabelled`.
- The AW1 gate item (1) gives the finite-box $C_N(s)$ only with an explicitly non-uniform τ^2 constant.

The admitted record therefore proves no finite-box node, and BC1 restates none, for F1 or for F2 boxes; `finite_box_node_claimed` is false. A finite-box node presented as a restatement, obtained by applying the window lemma box by box, or obtained by re-deriving or retuning an AV2 constant is rejected (control `finite_box_node_only_from_record`). Two open rows record what is missing. For finite F1 boxes (row N2) the missing premise is the window lemma for the untruncated finite-box ground vector with generator $G_N - E_N$ at the unchanged $M_0 = 2$ and $M_1 = 4s/\pi$, with AV1’s per-box state bound and AV2’s per-box slope. The route is short, since the spectral measure of χ_N for $G_N - E_N$ lies on $[0, \infty)$ because E_N is the least eigenvalue, but the skeptic notes that it is a new derivation. For F2 boxes (row N3) the F2 incidence of the groups meeting R and the F2 slope are missing as well; they are not in the AV2 record.

6.7 The common GNS item

At each sign of τ the forward states, and the gate admits within its scope (HNM-BC1-F05), the following.

- The F1 limit and the F2 limit are one state, ω_∞ (BB2 items 2–3), with one GNS triple $(\pi_\infty, \mathcal{H}_\infty, \Omega_\infty)$ determined up to unitary equivalence; the triple of every AQ1 and every F2 subsequential state is this one, up to the canonical unitary.

- Its GNS dynamics is the AQ1 limit dynamics T_θ implemented in that representation,

$$U_\theta \pi_\infty(A) \Omega_\infty = \pi_\infty(T_\theta(A)) \Omega_\infty, \quad (72)$$

strongly continuous with a nonnegative self-adjoint generator. T_θ is the limit of the F1 evolutions (AQ1) and of the F2 evolutions (BA2 item (4)); stationarity and the generator come from AQ1 §§4–5 and from BA2’s rerun for F2 limits, inherited through BB2 item 3.

- For $A \in B(\mathcal{H}_R)$ and $|\theta| \leq 8$ the correlation functions $c_A^\infty(\theta)$ of (56) are the limits along the whole sequence of the finite-box correlation functions of either family (BB2 item 5). The inequality (58) gives the rate K_5/N , $K_5 \approx 6.10 \times 10^{-9}$, only on $5 \leq N \leq 14000$; it is vacuous from $N = 14419$, and beyond the certified range only convergence without a rate is admitted.

This is one state with one representation and one dynamics. It is not equality of GNS dynamics of different states, and `gns_dynamics_equality_claimed` is false. The fixture of control `same_state_not_different_states` shows why the distinction matters. On M_2 with the Heisenberg dynamics of $H = \text{diag}(0, 1)$ at the rational phase $v = (3 + 4i)/5$, both basis vector states are invariant, yet the correlation of σ_x is v in one and $\bar{v}$ in the other: one algebraic dynamics, two states, two GNS dynamics. Nothing here is uniform in time.

6.8 The updated obligations table

The forward table has 18 rows: the six AY2 rows O1–O6, the eleven rows N1–N11 that the contract lists, with its item-4 entry split into two, and a row N12 carried from Round32. The skeptic’s pre-comparison table differed in O1, which it kept open, and in three rows. The reviewed table in [the BC1 review](#) has 21 rows. It splits AY2 row O1 into O1a and O1b, keeps O2–O6 and the forward’s N1–N12, and adds by review N13 (analyticity of the reduced density) and N14 (untruncated creation coefficients). The contract’s minimum list is met by the forward table; the two additions are non-blocking. Table 18 reproduces the reviewed table with light condensation; the missing-premise and route texts are those of the review.

Table 18: The reviewed obligations table of the BC1 gate (21 rows). For a closed row the last column gives the closing gate and scope; for an open row it gives the missing premise and a candidate route. No row is closed beyond its named scope.

Row	Obligation	Status	Closing gate and scope, or missing premise and candidate route
O1a	All subsequential limits of F1 and F2 coincide on every finite region (one limit of the named constructions, ω_∞); from AY2 O1, uniqueness of the limit	closed within scope	BB2 items 2–3: every AQ1 and every F2 subsequential limit equals ω_∞ on every finite region. Scope: the named constructions F1 and F2 only, the zero-selected family, fixed spacing, $ \tau \leq 10^{-8}$, both signs. Executed in BB2 (the item-1 Cauchy bound and the BB1 comparison c3).
O1b	Uniqueness of any ground state (every infinite-volume ground state), states outside the named constructions and other boundary conditions; the general part of AY2 O1	open	Missing: a statement covering every ground state, that is, a two-state estimate or a classification of every state in the class (AY2 O1), or a comparison of an arbitrary ground state or boundary prescription with ω_∞ (row N1). Route: HTW-type local stability or a Dobrushin-type condition with evaluated constants at the cap, or a BB1-type comparison for a pre-frozen class of boundary prescriptions (row N1); not executed.

Row	Obligation	Status	Closing gate and scope, or missing premise and candidate route
O2	Whole-sequence convergence	closed within scope	BB2 item 1: the reduced densities of F1 and F2 converge as whole sequences on every finite region (supremum over all $M > N$, $C' = 4/984375$, $q = 1/64$); BA2 items (3)–(4): the finite-box Heisenberg evolutions on $ \theta \leq 8$; BA1: the AM2 creation coefficients in each on-site cutoff space. Scope: the named constructions. Cutoff- L state limits are not used; non-centred F2 volumes, row N8. Executed through nested BB1 comparisons with the full geometric tail (BB2) and Lieb–Robinson Cauchy estimates for all $M > N$ (BA2).
O3	Translation invariance under coarse translations	closed within scope	BB2 item 4: the limit of the named constructions is invariant under every coarse translation on every finite region, from the direct BB1 comparison c5 with $C_h = 1/250000$ and exponent $N - v _\infty - 1$; non-coarse translations are rejected. Scope: coarse translations and the limit of the named constructions. Non-centred F2 volumes in c5 rest on the AY1 local reading (row N8); states outside the named constructions, row N1.
O4	A rate in N	closed within scope	BB2 item 1: the density rate q^{N-1} with $C' = 4/984375$ for every $N \geq 2$; BA2: the dynamics comparison $K_{\text{cmp}}(5N + 1)N^{-3}$ and the Cauchy rate $K_F/(N - 1)$ for every $N \geq 2$ on $ \theta \leq 8$; BB2 item 5: the correlation rate K_5/N only on $5 \leq N \leq 14000$. Scope: the named constructions at fixed spacing, per coarse step. The correlation rate beyond $N = 14000$, row N5; nothing in the lattice spacing, row N10.
O5	Boundary independence of dynamics on compact time windows	closed within scope	BA2 items (1)–(4): the F1 and F2 finite-box evolutions of $A \in B(\mathcal{H}_R)$ differ by at most $K_{\text{cmp}}(5N + 1)N^{-3}$ in norm for $ \theta \leq 8$ and every $N \geq 2$, and the F2 limit dynamics equals T_θ ; BB2 item 5: the correlation functions of the one limit state. Scope: algebraic dynamics of the named constructions on the compact window. Other boundary conditions (N1), uniform in time (N7) and GNS dynamics of different states (N6) stay open. Executed in BA2 with the Nachtergaele–Sims Lieb–Robinson bound and the AQ1 constants $F(r) = (1 + r)^{-4}$, $C \leq 224$.
O6	Dynamics of the padded family itself	closed within scope	BA2 items (4)–(5), decision “AY2 row O6 closed within scope”: the F2 evolutions converge as a whole sequence to T_θ on $ \theta \leq 8$, and AQ1 §§4–5 are rerun for every F2 subsequential limit state; BB2 item 3 identifies those states with ω_∞ . Scope: the named constructions. Executed in BA2 (Nachtergaele–Sims Theorem 4.1 for the owner-set interaction, and the AQ1 rerun).
N1	States outside the named constructions and other boundary conditions	open	Missing: a comparison of an arbitrary infinite-volume ground state, or of another boundary prescription (literal vertex boxes, periodic or orthant boxes), with ω_∞ on R ; every admitted comparison is between F1 and F2 boxes or one-prescription volumes containing Λ_N (BB1 c1–c5). Route: HTW-type local stability or a Dobrushin-type condition with evaluated constants at the cap (the Yarotsky constants of I1 are not evaluated), or a BB1-type comparison for a pre-frozen AM2-admissible class of boundary prescriptions; not executed.

Row	Obligation	Status	Closing gate and scope, or missing premise and candidate route
N2	The finite-box Euclidean node for finite F1 boxes	open	Missing: the window lemma for the untruncated finite-box ground vector with generator $G_N - E_N$ at the unchanged $M_0 = 2$, $M_1 = 4s/\pi$, with the AV1 per-box state bound D and the AV2 per-box seven-star slope; the AV2 record states the lemma and the comparison only for the GNS vector of an AQ1 subsequential state, and the AV2 reverse rejects a finite-box provenance as a changed model. Route: a new contract stating the window lemma for the finite-box vector state (the spectral measure of χ_N for $G_N - E_N$ lies on $[0, \infty)$ because E_N is the least eigenvalue, AM2), reusing F13–F14 box by box and the AV1 per-box D , reviewed as a new statement; not executed, and no constant is re-derived.
N3	The finite-box Euclidean node for F2 boxes	open	Missing: everything in N2, and the F2 incidence of the groups meeting R and the F2 slope (the norm of the F2 interaction terms meeting R in $G = H/\alpha$ units), which are not in the AV2 record; AY1 records the same 82 faces meeting R for $N \geq 2$, but no F2 relative-unitary slope is admitted. Route: the N2 route with the F2 grouping (enumerate the F2 groups meeting R from the I1 table as in the AY1 itemization, bound their norm, apply the finite-box window lemma); not executed, and no constant is evaluated.
N4	The F2 finite-box sign for $2 \leq N < N_{\text{sign}}$	open	Missing: $N_{\text{sign}} = 4$, so $N = 2$ and $N = 3$ are open; there $C'q^{N-1}$ is $1/15750000$ and $1/1008000000$, both above $\tau/144 - K_2^+\tau^2$, and the widened interval contains 0. A box-level bound on the Wilson mean of the untruncated F2 ground vector is missing. Route: gate a box-level form of the AY1 remainder bound (AY1 forward F13–F14 as quoted in the AY2 forward report §4.2) for the untruncated F2 vector through AV1 F22 via AY1; by duality it would place $\omega^{F2,N}(W)$ within $K_2'\tau^2$ of $\tau/144$, with sign margin $1/(144K_2'\tau)$ about 51.76 (labelled preview, not claimed and not admitted: it rests on an ungated box-level reading of AY1 F13–F14 and on the untruncated F2 passage); or an F2 itemization of the AW1 bound K_2^+ .
N5	The correlation-function rate in N beyond $N = 14000$ with the frozen r_N	open	Missing: with $r_N = \lfloor (N-1)/2 \rfloor$ and the frozen region form, the BB2 item-5 bracket gives K_5/N only on $5 \leq N \leq 14000$ and is vacuous from $N = 14419$, because the factor $e^{ \Lambda_{r_N} /10^8}$ eventually outgrows q^{N-r_N} ; beyond $N = 14000$ only convergence without a rate is admitted. Route: a region form with a polynomial volume factor (the BB1 iterated_split form $c_{\text{site}} \sum_{y \in Y} q^{N-\ y\ _\infty}$, as the BB2 skeptic advises) admitted at a gate, then the item-5 argument unchanged; or a re-frozen r_N in a new contract; not executed.
N6	GNS dynamics of different states	open	Missing: nothing compares the GNS dynamics of ω_∞ with that of any other state; for a state outside the named constructions neither stationarity under T_θ nor a relation between its GNS representation and that of ω_∞ is admitted. Route: identify the other state with ω_∞ (row N1), or prove its stationarity under T_θ and its quasi-equivalence with ω_∞ ; not formulated.
N7	Uniform in time	open	Missing: every admitted dynamics constant holds on $ \theta \leq 8$ ($U = 1$ in $u = \theta/8$) and grows like $U^2/(1 - vU/3)$ (BA2); no estimate holds for all θ . Route: a bound for all θ , for example from spectral information (the AM2 finite-box gap $1/2$ and the AQ2 gap $\alpha/16$ on the invariant-local cyclic completion) with a real-time transfer; not formulated.

Row	Obligation	Status	Closing gate and scope, or missing premise and candidate route
N8	Non-centred F2 volumes in the fifth BB1 comparison, which rest on the AY1 itemization as a reviewed local reading	open	Missing: a gated statement of the AY1 itemization (on-site operator, per-anchor grouping inside the star, supports of at most 4 sites, per-site sum through $b \in u - S$, cutoff compression) for every F2 volume containing Λ_N , not only the centred cubes; carried by BB2 item 4. Route: a statement loop reviewing the AY1 items H1–H5 for general F2 volumes and recording them in a gate; not executed.
N9	The route-B uniform model (BC2)	open	Missing: no BC2 gate was recorded when BC1 was produced, and BC1 restates nothing for the uniform model (route B, AX1/AX2). Route: BC2, produced in parallel; the row cites the BC2 gate only once it is recorded.
N10	Anything uniform in the lattice spacing a	open	Missing: every constant is at fixed spacing and strong bare coupling (a coarse step is $(4a, 2a, a)$); by the AL1 dictionary of a different model, as quoted in AY1 and AY2, the cap corresponds to $g^4 \geq 9.6 \times 10^9$; no estimate in a . Route: none within this family at fixed coupling; a statement in a needs a coupling that moves with a and a scaling analysis; not formulated.
N11	The continuum problem	open	The four-dimensional Yang–Mills existence and mass-gap problem remains open, and nothing here constructs a continuum theory. No route is executed or claimed here.
N12	An interaction shift, sign or coefficient of the Euclidean correlation $C(s)$ (carried from Round32)	open	Missing: the AV2 interval contains $e^{-3}/4$ (reference_unresolved); by AW1 the first-order term of $C(s)$ vanishes and its τ^2 constant is not uniform in N , so no second-order enclosure of $C(1)$ for ω_∞ exists. Route: a second-order refinement with the C^2 window’s finite $M_2 = 2s^2$ (AV2 skeptic advice 7) and a τ^2 constant for $C(s)$ uniform in N (AW1: the 40-star relative-unitary step), certified only if it excludes $e^{-3}/4$; not executed.
N13	Analyticity of the reduced density in the coupling (added by review; BA1 and BB1 limitations)	open	Missing: a zero-free region of the complexified normalization; the BA1 disc controls coefficients only. Route: a lower bound on the complexified norm on the BA1 disc; not formulated.
N14	Untruncated creation coefficients (added by review; BA1 limitation)	open	Missing: coefficient bounds after the on-site cutoff is removed (BA1 is per cutoff space). Route: AM2 §6 applied to the coefficient collection; not needed by BB1, BB2 or BC1.

The AY2 falsifying scenario. AY2 exhibited two admissible density matrices on $\mathcal{H}_R$ that differ by at least $(1 - 2 \times 10^{-6}) 2K'_2 \tau^2$ and asked whether two subsequential limits could differ on R . For the named constructions this is now excluded, since every AQ1 and every F2 subsequential limit equals ω_∞ (BB2 items 2–3). It is not excluded for states outside the named constructions or for other boundary conditions (rows O1b and N1). The AY2 witness remains a valid pair of density matrices; it is not a pair of limits of F1 or F2.

Round32 limitations. The forward records, with each limitation anchored as a verbatim substring of its Round32 gate, which limitations the Round33 gates lift and which remain. Lifted in part: AV2’s “Each AQ1 subsequential state separately; no uniqueness, whole-sequence convergence or rate in N ” and AV1’s uniform local closeness, by BB2 items 1–3 within the named constructions; AY1’s limitation that two limits may differ by up to $2K'_2 \tau^2$ on R , and AY2’s scenario, for limits of F1 and F2 only; AW2’s pairing of individual states at $-\tau$ along a common subsequence, since each sign now has one limit (recorded and used for no value; the minus-sign enclosure stays a replay). AY2’s “all six obligations unproved” is updated by Table 18. Remaining: AV2’s upper certificate with the free value inside, its sign-blind replay at $-\tau$ and its crossover s^* ; AW2’s resolved interaction shift for $\omega(W)$ alone; AW1’s finite-volume Taylor statement for $C(s)$ and $c(\theta)$; AY1’s exclusion of literal vertex, periodic and orthant boxes; and AY2’s constants, uniform in N at fixed spacing and never in the lattice spacing. In each lifted case the exclusion of uniqueness of any ground state and of

states outside the named constructions stays in force.

6.9 Evidence and review

The forward checker runs 50 exact checks and implements all 25 contract controls as damaging mutations, with 104 rejections inside the control checks and 113 in the whole checker. Its outputs are byte-identical under `-B` and `-B -O`. The skeptic’s pre-comparison package (55 checks, 25 controls, 86 rejected mutations) replays the restatement and the corollary from the contract and the gates, and predicted $N_{\text{sign}} = 4$, the widening table and the widened enclosures exactly. The post-comparison checker runs 17 checks and 41 source-edit runs: 25 validator weakenings, one per control, 12 must-abort edits and 4 silent edits. Of the silent edits one is harmless (the producer’s least- N rule tightened to $L/4$, which returns the same $N_{\text{sign}} = 4$). The other three alter the O1 scope qualifier, the N4 preview and the ratio $L/(C'q^3)$, which the forward checker does not pin; the skeptic’s validator rejects them. The producer replays reproduce the frozen outputs byte for byte, the closure of 44 files verifies, and the forward holds exactly its 40 declared inputs. The review’s supported statement and limitations pass the round’s negation-aware phrase scanner with the BC1 contract with no hit, negated or not, and the mandatory sentence occurs once as one unbroken span.

Order and isolation. The contracts froze at 04:21:44 UTC. The skeptic’s values were final at 04:53:13 and the forward checker was last written at 04:54:30. The forward commit came first, at 04:57:15, and the skeptic’s pre-comparison package followed 3 min 32 s later. The skeptic saw the subject line of the forward commit, which states $N_{\text{sign}} = 4$, in a `git log` after its values and first freeze were final; this name-only exposure is disclosed. It opened the forward directory only after the coordinator’s post-comparison message. Isolation rests on disclosure and content, and all agents are correlated model agents. The gate was recorded at 05:24 UTC on 25 September. The forward disclosed that the file-writing tool refused its report, which was therefore written through a shell heredoc and is bound by its freeze record. It also disclosed that its first report check let two damaging edits run silently (one of the two N_{sign} claim sites changed to 3, and one repeated preview changed); it fixed the check before freezing, and the skeptic re-ran both edits on the frozen checker, where both abort. The review also notes (P1) that the BB2 forward report describes its harmonic fixture as exceeding 4 at $N = 119$; that is literally true, but the first exceedance is $N = 83$, and no BB2 value depends on it.

The rejected wording on O1. The review found no blocking issue. It did not adopt one piece of producer wording for the gate text. The forward table records AY2 row O1, “uniqueness of the limit”, as `closed_within_scope`, with closing cell “BB2 gate items 2-3: every AQ1 subsequential limit and every F2 subsequential limit equals `omega_inf` on every finite region; scope: the named constructions F1 and F2 only”, and its verdict summarizes “AY2 rows O1–O6 are closed within named scopes by the BB2 and BA2 gates”. The review’s reason is that AY2 row O1 mixes two contents. The narrow one is that two subsequential limits of the named families agree. The broad one appears in the row’s missing premise, which asks for a classification of every state in the class, and in its candidate routes, which are routes to uniqueness of states. BB2 items 2–3 settle only the narrow content, and a closed status under the AY2 row name reads, out of context, as a closure of the broad one. The cell itself carries its scope and points to row N1, every other uniqueness clause of the report is negated or excluded, and `uniqueness_of_ground_state_claimed` is false, so the finding is non-blocking. The reviewed split replaces the wording in the gate: O1a is closed within the BB2 scope, and O1b (uniqueness of any ground state, states outside the named constructions and other boundary conditions) is open. The forward checker hard-codes O1 as closed, so the reviewed status cannot enter the frozen packet; it enters the gate, and the frozen producer files are retained as evidence.

The scanner does not guard that phrase. The skeptic classified every clause of the forward report that mentions uniqueness. Each one is negated or excluded, a field or control identifier, a quotation of the verbatim claim-exclusion lists, a description of a rejected mutation, the pointer to N1, or the single O1 row. The negation-aware scanner is silent on the O1 row by vocabulary, not by negation. Neither the round list nor the BC1 contract’s forbidden phrasings contain the AY2 row name or the every-ground-state wording of the claim exclusions. An affirmative sentence built on the every-ground-state wording therefore produces no hit, while the same sentence with the definite article does. The reviewed wording, not the scanner, guards O1, and the gate records this gap as a limitation. The skeptic’s planning advice, recorded for later contracts and not a result, is to add the two phrasings to their forbidden lists and to freeze the reviewed name of any carried row whose content an admitted result settles only in part.

Contract readings. The review records seven non-blocking contract defects, W1–W5 from the forward and D1 and D3 from the skeptic. The selection reason omits that AW2 was also stated in every centered whole-star box (W1). The two claim-exclusion lists differ in one entry, and the wider entry was applied (W2). The inherited parameter control asks for fields that a statement loop without a decay estimate does not have (W3, equal to D2). Item 4 needs two open rows where the parameter list has one entry (W4). The template leaves the finite-box state implicit, and the scope is the untruncated F2 ground vector at fixed N (W5, equal to R3). There is no frame for limited outcomes (D1), and the inherited clock control is loosely fitted to a statement loop but executed in full (D3). The frozen contract is not amended.

Meaning. BC1 turns two Round32 certificates, stated for subsequential states, into statements about one state: the limit of the named constructions satisfies the AV2 node certificate at $s = 1$ and the AW2 enclosure of the static Wilson mean, with every constant unchanged. The whole-sequence rate of BB2 adds a sign certificate for the untruncated padded boxes from $N_{\text{sign}} = 4$ on, at both signs of τ . These are restatements and a finite-box corollary. They are not uniqueness of any ground state, not an interaction shift of the Euclidean correlation, not a finite-box node and not uniform in the lattice spacing. The node keeps the free reference inside its interval, and the enclosure remains a static equal-time mean.

6.10 Limitations recorded by the gate

The BC1 gate records the following limitations verbatim.

- Scope: only the zero-selected patterned family (selected triple $(0,0,0)$, Haar reference), fixed spacing, $\tau=10^{-8}$ with $\tau=-10^{-8}$ a mirror replay of the same $|\tau|$ formula (never a second confirmation), the cover $R=\{0,e_z\}$ and the original xz Wilson loop W , the single node $s=1$, the limit of the named constructions ω_{inf} and the untruncated F2 finite-box ground states at fixed N ; nothing transfers to other selected triples, the route-B uniform model (BC2), other boundary conditions, states outside the named constructions, literal vertex boxes, weak coupling or the continuum, and no estimate is uniform in the lattice spacing a .
- Restatement only: the node certificate is the AV2 certificate with d , R and the interval unchanged and the free reference $e^{-3}/4$ inside the interval (reference_unresolved), so no interaction shift, sign or coefficient of $C(s)$ is claimed; the static Wilson mean is the AW2 enclosure with its endpoints and the full second-order remainder $K_2^+ \tau^2$ unchanged (static_not_dynamic); no constant of either certificate is re-derived or retuned, and no Round32 gate, report or snapshot is edited.
- Identification: the AV2 and AW2 certificates are stated for every state in the whole set of AQ1 subsequential limits (AW2 also in every centered whole-star box at every on-site cutoff); they pass to ω_{inf} only because BB2 items 2–3 admit that this set is the one limit of the named constructions. BB2 carries its own limitations: the BB1 discharge at the frozen targets, the Kotecky–Preiss citation checked post

- hoc against the committed excerpt, and the AY1 itemization as a reviewed local reading for non-centred F2 volumes in the fifth BB1 comparison.
- Finite-box sign: certified only for the untruncated F2 finite-box ground states at fixed N with N at least $N_{\text{sign}}=4$, at both signs, through the BB2 whole-sequence bound at the BB2 gate bound value $C'=4/984375$ with $q=1/64$ and $||W||$ at most 1 (the labelled union_comparison value and the labelled re-evaluated values are not used); $N=2$ and $N=3$ are open (row N4) because the widened interval contains 0 there; cutoff-L F2 states are not claimed; the F1 boxes are as certified by AW2, recorded as agreement and not re-derived. C' at the hypothesis values does not depend on τ and AW2 admits the enclosure at the cap only, so nothing below the cap is claimed.
 - No finite-box node: the admitted AV2 record states the window lemma and the comparison only for the GNS vector of an AQ1 subsequential state, so no node certificate for F1 or F2 boxes is restated (rows N2 and N3, open).
 - O1 in the narrow form only: AY2 row O1 is closed only as the statement that all subsequential limits of F1 and F2 coincide on every finite region (one limit of the named constructions, row O1a); uniqueness of any ground state, states outside the named constructions and other boundary conditions are not proved and stay open (rows O1b and N1). The producer status closed_within_scope for the whole AY2 row O1 and its summary sentence that AY2 rows O1–O6 are closed within named scopes are not adopted for the gate text; the frozen producer files are retained as evidence.
 - Common GNS item and rates: one state with one GNS triple and the AQ1 limit dynamics T_{theta} ; correlation functions only on $|\text{theta}|$ at most 8, with a rate in N only on $5 \leq N \leq 14000$ (BB2 item 5; the bound is vacuous from $N=14419$ with the frozen r_N) and convergence without a rate beyond; no equality of GNS dynamics of different states and no uniform-in-time statement. The density rate $q^{(N-1)}$ holds for every N at least 2, per coarse step at fixed spacing.
 - Labelled only, not claimed: the ratio $L/(C' q^3)$ about 4.4584, the widened exclusion margin about 3.3851, the half-width $K_2^+ (\tau/100)^2 + C' q^3$ about $1.5501e-11$ against the first-order value about $6.944e-13$ at $\tau/100$, and the row-N4 preview sign margin $1/(144 K_2' \tau)$ about 51.76, which rests on an ungated box-level reading of AY1 F13–F14 and on the untruncated F2 passage and is not admitted.
 - Contract wording read as follows (non-blocking): W1 the selection reason omits that AW2 was also stated in every centered whole-star box; W2 the two claim-exclusion lists differ in one entry (the wider contract entry applied); W3 (equal to skeptic D2) the inherited parameter control asks for d_X , N_0 and weights, and BC1 derives no decay estimate (N_{min} plays the role of N_0); W4 item 4 needs two open rows where the parameter list has one entry; W5 (equal to skeptic R3) the template leaves the finite-box state implicit (the untruncated F2 ground vector at fixed N); D1 no frame for limited outcomes (moot at this verdict); D3 the inherited clock control is loosely fitted to a statement loop (executed in full).
 - Phrase-list gap: neither the round list nor this contract's forbidden phrasings contain the AY2 row name or the every-ground-state wording of the claim exclusions, so the negation-aware scanner does not guard the O1 row; the reviewed wording does.
 - Independence: one forward producer plus the skeptic's pre-comparison replay, all correlated model agents; the forward commit 5816e0b came first and the skeptic pre-comparison package 5ec599f followed 3 min 32 s later, with the skeptic values final before the forward check.py was last written; isolation rests on disclosure and content, including a disclosed name-only exposure of the skeptic to the forward commit subject; the forward report was written by a shell heredoc and is bound by its freeze record.
 - Scientific priority is unverified.

Full derivations and reviewed implementations: [BC1 forward](#), [independent model-agent review](#), [skeptic pre-comparison derivation](#) and the [frozen contract](#); the restated certificates are those of the AV2 and AW2 gates, which are not edited.

7 BC2: convergence for the route-B uniform model

Contribution HNM-C-BC2. Round32 also handled a second model: the uniform Kogut–Susskind $SU(2)$ Hamiltonian at fixed spacing and strong bare coupling, in which every elementary face, the three selected faces of each factor included, carries the coupling. AX1 regrouped its interaction

(route B), and AX2 certified its Euclidean Wilson node at $s = 1$ for each AQ1-type subsequential state of its one named construction separately ([Round32 addendum](#)). The BC2 contract, recorded under the title “Route-B uniform model: coefficient decay, marginal locality and whole-sequence convergence of its named construction, and the node certificate restated for the limit”, re-derives the chain of BA1, BB1 and BB2 for that model with its own constants, identifies the limit, and restates the AX2 node for it. The plan ran BC2 only if BB1 and BB2 were accepted within scope; both were, and BC2 was selected as investigation 6 of 8 ([selection note](#)). It was frozen together with BC1 on 25 September 2026 at 04:21:44 UTC, after the skeptic’s pre-freeze review of the BC drafts. Its gate is recorded under the title “Hruday route-B convergence of the uniform model’s named construction and its node”.

BC2 is a single-direction investigation (direction S of Table 4). One forward producer proves every item; the skeptic’s pre-comparison replay from the contract is the second admission input. The forward producer received `AGENTS.md`, the contract and 40 shared premises, among them the AX1 and AX2 gates, the admitted AX2 calculator, the AX1 forward report, the BA1 forward and reverse reports, the BB1 reverse report, the BB2 reports and the I1 class table. The contract names three shared methods: the analytic disc of the BA1 reverse (Section 2.6), the iterated split of the BB1 reverse (Section 4.4) and the whole-sequence and identification scheme of BB2 (Section 5). It forbids citing any BA1, BB1 or BB2 value or form as admitted for route B, and it forbids the polymer route, since no Kotecký–Preiss excerpt is a premise of this contract. Every constant is evaluated at the constants proved in the packet, with no hypothesis layer.

The contract names seven items: route-B incidence and admissibility, the coefficient input on a complex disc, marginal locality by the iterated split, whole-sequence convergence with four further items, the node restated for the limit, the obligations, and the fixtures with the mandatory sentence, the gate fields and an exact checker. Its targets are $K_B \leq 1/1000000$, $C_B \leq 1/250000$ and $c_{\text{site},B} \leq 1/500000$, frozen as blind discriminating thresholds with a margin of at least 2 against the advisor’s recorded previews. The whole-sequence targets, $C'_B \leq 1/250000$ and $c'_{\text{site},B} \leq 1/400000$, follow from the first two up to the assembly factor, and the contract calls them consequences, not independent thresholds. The node values are a format check by equality. The $\tau \rightarrow \tau/100$ bracket is $[95, 105]$ for every constant, exactly 1 for q , for $\rho/|\tau|$ and for the node datum, and $[95, 105]$, recorded and not a target, for the node radius replay. The contract itemizes ten error terms and lists 52 controls.

7.1 Reviewed scope

The [BC2 gate](#) records the verdict `accepted_within_scope`, with sub-label `convergence_of_named_constructions` and secondary sub-labels `certificate_restated_for_limit`, `reference_unresolved` and `static_not_dynamic`. The review reads the last as covering state statements and the inherited Euclidean node, with no route-B real-time dynamics. The gate fields `whole_sequence_claimed`, `state_convergence_claimed`, `state_decay_claimed`, `rate_in_N_claimed`, `translation_invariance_claimed` and `node_certificate_restated_for_limit` are true, each within its scope string: the whole sequence “reduced densities of the centered whole-star-plus-single-group construction of the uniform route-B model on every finite region”, the translations “coarse translations; the limit of the named construction (route B)”, and the state decay “reduced densities of route-B finite-box ground vectors on $\mathbb{R}$ and on regions Y , per comparison (c1B, c4B, c5B)”. Nine fields are false: `common_limit_claimed` and `uniqueness_of_ground_state_claimed` for the limit; `gns_dynamics_equality_claimed` and `uniform_in_time_claimed` for the dynamics; `resolved_interaction_shift` for the node; `rate_in_a_claimed`; and `continuum_claim`, `weak_coupling_claim` and `scientific_priority_verified`. No dynamics level is exported. The accepted statement, quoted verbatim below in the gate’s ASCII notation, is the scope of every claim in this section.

In the uniform Kogut–Susskind $SU(2)$ model at fixed spacing and strong bare coupling

handled by route B (model AQ_uniform_routeB, labelled uniform Kogut-Susskind SU(2) at fixed spacing and strong bare coupling ($g^4=9.6 \times 10^{-9}$) by the AL1 dictionary $\tau=96/g^4$ of this model; every elementary face $\nu=\alpha \tau/24$, selected triple $(\tau/24, \tau/24, \tau/24)$; Haar reference $h_b=8 \sum_e C_e$; whole stars ϕ_b (21 omitted faces, norm $7|\tau|$) and single-factor groups ψ_b (3 selected faces, support $\{b\}$, norm at most $|\tau|$), per-site $\sum J'=29|\tau|$ (AX1 gate sha256 1b8fb152696659439bb01fc7df429fc76fa3132d4936df450d8131256015d177); both signs $|\tau| \leq 10^{-8}$, with $\tau < 0$ the U_E mirror and no real g ; coarse l -infinity metric on factor sites, a whole star of diameter 1 and a single-factor group of diameter 0; trace norm on $B(H_Y)$; cover $R=\{0, e_z\}$; Euclidean clock $s=\alpha t_E/\hbar$ with $G=H/\alpha$ at the single node $s=1$; real-time clock $\theta=\alpha t/\hbar$ for the inherited limit dynamics only), with the named construction FB = centered whole-star-plus-single-group boxes $\Lambda_N=[-N, N]^3$, N at least 2, and the finite complete-factor route-B volumes containing Λ_N (stars inside the volume, every single group kept), every bound holding in each on-site cutoff space uniformly in the cutoff and then, at fixed N , for the untruncated ground vectors (AV1 F20-F23 with the route-B gap $1/2$ of the AX1 gate; the N and cutoff limits are never exchanged): For the uniform Kogut-Susskind SU(2) model at fixed spacing and strong bare coupling handled by route B, at the same coupling $|\tau| \leq 10^{-8}$, the reduced densities of the named construction (centered whole-star-plus-single-group boxes on centered coarse cubes) converge as a whole sequence on every finite region, at a rate in N that holds for every N at least 2 with the region inside the box, and to the same limit along every exhausting sequence of finite complete-factor volumes of the same prescription; this limit coincides with every AQ1 subsequential state of that construction, is invariant under coarse translations, and at $\tau=+10^{-8}$ has centered Euclidean Wilson correlation $C(1)$ within the AX2 radius of the AX2 datum at the node $s=1$; this is convergence of the named constructions (one family for route B), not uniqueness of any ground state, not a statement about other boundary conditions or about states outside the named construction, not weak coupling or continuum, and not a statement uniform in the lattice spacing a . Constants (exact; $q=1/64$; both signs; the comparisons $c1B$, $c4B$ and $c5B$ compared directly; every constant recomputed from the route-B inputs, with no hypothesis layer). T0 (tier exact_first_order, route analytic_disc on the disc $|z| \leq 64|\tau|$, proved in the packet from 52 first-order faces per factor and the per-site $\sum 29|z|$, with the route-B circle bound $T_B(\rho)=(52 \rho/144)/(1-29 \cdot 352 \rho)$ and $T_B(|\tau|)=13/3599632512$ equal to the AX1 gate T'): at every site u of the union volume and for every N at least 2, $\sum_I$ containing u of $||c^A_I - c^B_I|| \leq K_B q^{((N-|u|_\infty)_+)} with $K_B=2T_B(64|\tau|)=13/27941256$ (about $4.6526e-7$; margin about 2.149 against $1/1000000$), in each on-site cutoff space uniformly in the cutoff (no untruncated creation coefficients are asserted). T1 and T2 (tier exact_first_order, route iterated_split with $W=1024$ and $\lambda=2/W=1/512$; $t_0=13/1799816256$, $t_W=26/3148137$, $\beta=132572461259098000347351859/4012775859323049054050909456204$, $S_\lambda=2169/343$): for every N at least 2, $||\rho^{\{box1\}}_R - \rho^{\{box2\}}_R||_1 \leq C_B q^{(N-1)}$ with $C_B=2326328761843649826272217312701707175/2461302090348550272620124432958434944821248$ (about $9.4516e-7$; margin about 4.232 against $1/250000$), and for every N at least 2 and every finite complete-factor region Y inside Λ_N , $||\rho^{\{box1\}}_Y - \rho^{\{box2\}}_Y||_1 \leq c_{site,B} |Y| e^{\{|Y|/10^8\}} q^{\{d_Y\}}$, $d_Y=N-\max\{y \text{ in } Y\}|y|_\infty$, with $c_{site,B}=35789673259133074250341804810795495/38457845161696098009689444264975546012832$ (about $9.3062e-7$; margin about 2.149 against $1/500000$) and $C_B=c_{site,B}(1+q)$. T3 and T4 (assembly union_comparison: each M greater than N compared with N by one direct $c4B$ comparison of Λ_N and Λ_M , factor 1): for every N at least 2, $\sup$ over all M greater than N of $||\rho^{\{FB,M\}}_R - \rho^{\{FB,N\}}_R||_1 \leq C'_B q^{(N-1)}$ with $C'_B=C_B$ (margin about 4.232 against $1/250000$), and for every N at least 2 with Y inside Λ_N , $\sup$ over all M greater than N of $||\rho^{\{FB,M\}}_Y - \rho^{\{FB,N\}}_Y||_1 \leq c'_{site,B} |Y| e^{\{|Y|/10^8\}} q^{\{d_Y\}}$ with $c'_{site,B}=c_{site,B}$ (margin about 2.686 against $1/400000$); the nested-telescoping values $C_B/(1-q)$ and $c_{site,B}/(1-q)$ are labelled only. The limit ω_∞ exists on every finite region by this Cauchy bound and trace-class completeness, without compactness, and $||\rho^{\{FB,N\}}_Y - \rho^\infty_Y||_1 \leq c'_{site,B} |Y| e^{\{|Y|/10^8\}} q^{\{d_Y\}}$ for every N at least 2 with Y inside Λ_N . Item 2a (exhaustion within the prescription): for finite complete-factor route-B volumes V_k with N_k the largest N with Λ_N inside V_k , for every N_k at least 2 and every finite complete-factor region Y inside Λ_{N_k} , $||\rho^{\{V_k\}}_Y - \rho^\infty_Y||_1 \leq (c_{site,B}+c'_{site,B}) |Y| e^{\{|Y|/10^8\}} q^{(N_k-\max\{y \text{ in } Y\}|y|_\infty)}$, with $c_{site,B}+c'_{site,B}=2c_{site,B}$ (about $1.8612e-6$), from one direct $c5B$ comparison of V_k with Λ_{N_k} and the whole-sequence bound; this is exhaustion within one prescription, not a comparison of boundary conditions. Item 2b:$

$\omega_{-\tau} = \omega_{\tau} \circ \alpha_E$ on every finite region, from the AX1 identity $U_E H_N(\tau) U_E^* = H_N(-\tau)$ in every centered box and cutoff (U_E a product of one-link unitaries) and whole-sequence convergence at both signs; the $-\tau$ side is a mirror replay, not a second coupling. Item 3: ω_{τ} equals every AQ1-type subsequential state of the AX1 construction on every finite region, and only then inherits exactly the AX1 gate's route-B list. Item 4: for every coarse translation v and N at least $|v|_{\infty} + 2$, $\|\rho_{\Lambda_N + v} - \rho_{\Lambda_N}\|_1 \leq C_B q^{(N - |v|_{\infty} - 1)}$ by one direct c5B comparison, and ω_{τ} is invariant under coarse translations on every finite region by the region form of c5B; nothing is claimed for non-coarse fine translations. Node (AX2 gate sha256 1db36627b9eab00915b0cb38a0e9fd9cf9dd4dbf3ceb7eab969336f3f9f44be4, sub-label reference_unresolved): at $\tau = 10^{-8}$ and $s=1$, $C_{\infty}(1) = (\chi, e^{-G} \chi)$ with $\chi = (\pi(W) - \omega_{\tau}(W)) \Omega_{\omega_{\tau}}$ in the GNS representation of ω_{τ} satisfies $|C_{\infty}(1) - d| \leq R'$ with $d = 497870683678639429793424156500617766317/400000000000000000$ (about $1.24467670920e-2$) and $R' = 1912298807996871790146581299723633/100$ (about $1.91229880800e-7$), the AX2 gate rationals, returned exactly by the admitted AX2 calculator at its fixed design (the AX1 bound value D' , $k' = 51|\tau|/4$, $M_0 = 2$, $M_1 = 4s/\pi$); the free value $e^{-3}/4$ lies inside $[d - R', d + R']$, so no interaction shift, sign or coefficient of $C(s)$ is claimed. The τ to $\tau/100$ ratios are $39059948/388073$ (about 100.651) for K_B and about 100.661 for C_B , $c_{\text{site},B}$, C'_B and $c'_{\text{site},B}$. Rates are per coarse step in N at fixed spacing.

The rest of this section derives that statement, and the limitations recorded by the gate close it (Section 7.13).

7.2 The route-B model and its single named construction

The model. The model identifier is `AQ_uniform_routeB`, with the label “uniform Kogut–Susskind $SU(2)$ at fixed spacing and strong bare coupling ($g^4 = 9.6 \times 10^9$)”. Every elementary face has $\nu = \alpha\tau/24$, so the selected triple is $(\tau/24, \tau/24, \tau/24)$. By the AL1 dictionary of this model, $\tau = 96/g^4$, the cap $|\tau| \leq 10^{-8}$ corresponds to $g^4 \geq 9.6 \times 10^9$; the model is never at weak coupling or in the continuum. Both signs are treated, and $\tau < 0$, which has no real g , is the U_E mirror. Route B keeps the Haar reference $h_b = 8 \sum_e C_e \geq 6Q_b$, independent of τ , and groups the interaction into

$$\begin{aligned} \phi_b & \text{ (whole star: 21 omitted faces, } \|\phi_b\| \leq 7|\tau|), \\ \psi_b & \text{ (single-factor group: 3 selected faces, support } \{b\}, \|\psi_b\| \leq |\tau|), \\ J' = \sum_{X \ni u} \|V_X\| &= 4 \cdot 7|\tau| + |\tau| = 29|\tau|, \quad J'_0 = \frac{29}{10^8}, \end{aligned} \tag{73}$$

where the four stars through a factor include the incoming ones (AX1 gate). AX1 admits that every finite complete-factor route-B volume has a simple gauge-invariant ground and a full-space gap of at least $1/2$ in normalized units. The metric is the coarse ℓ^∞ metric on factor sites, in which a whole star has diameter 1 and a single-factor group diameter 0.

One named construction. The named construction FB consists of the centered whole-star-plus-single-group boxes $\Lambda_N = [-N, N]^3$, $N \geq 2$, and the finite complete-factor route-B volumes containing Λ_N , with every star inside the volume and every single group of the volume kept. Three comparisons are made directly: c1B (Λ_N against Λ_{N+1}), c4B (any two centered boxes $\Lambda_M, \Lambda_{M'}$ with $M, M' \geq N$) and c5B (any two finite complete-factor route-B volumes containing Λ_N). The admitted record has no second route-B family. The contract therefore replaces BB2's common-limit item by exhaustion within the prescription (item 2a) and the pointwise sign mirror (item 2b), and the control `one_family_no_common_limit` rejects a common limit, a second route-B family and an identification with an all-contained boundary. The gate field `common_limit_claimed` is false. Nothing is identified with the zero-selected limit of BB2, and no BA1, BB1 or BB2 constant is applied to route B (control `model_crossing_rejected`).

7.3 What changes from route A

The forward checker derives every route-B input from the fine geometry, with factor b owning the 24 positive links with tails $(4b_x + r, 2b_y + s, b_z)$, $r = 0, \dots, 3$, $s = 0, 1$. Over the 3000 plaquettes with base in the fine region of Λ_2 , every plaquette lies in exactly one route-B group: the single-factor group of its anchor if it is selected, and otherwise its anchor's star. Table 19 compares the inputs with those of the zero-selected family.

Table 19: Inputs of the zero-selected family (route A, BA1 and BB1) and of the route-B uniform model (BC2). Every route-B entry is enumerated or recomputed in the BC2 packet; substituting a route-A entry aborts the forward checker.

Quantity	Route A	Route B
per-site interaction sum	$28 \tau $	$J' = 29 \tau $
first-order faces per factor	49	52 (49 omitted, 3 selected)
interaction groups per site	4 stars	5 (4 stars, 1 single-factor group)
groups of support one	none	one per factor, diameter 0
groups meeting R	7 stars	7 stars, 2 single-factor groups
faces meeting R	82	88 (82 omitted, 6 selected)
faces inside R (first-order R -marginal)	10	16 (10 omitted, 6 selected)
largest first-order site energy	18	24 (selected faces)
circle bound at $ \tau $	$49/14398580736$	$13/3599632512$ (AX1 T')
coefficient constant at $q = 1/64$	$K = 49/111790368$	$K_B = 13/27941256$
node slope	$49 \tau /4$	$k' = 51 \tau /4$

Per-site inputs. For a factor u , the faces with u in their owner set number $21 + 4 + 8 + 16 = 49$ omitted and 3 selected, 52 in all. The owner sets containing u number 16, and the interaction groups containing u number 5: the stars at $u - S$ and the single group at u . The per-site sum is $(4 \cdot 21 + 3)|\tau|/3 = 29|\tau|$. Boundary counts never exceed the bulk value (checked on Λ_1 to Λ_3).

Incidence on R . For every $N \geq 2$ (enumerated for $N = 2, 3, 4$) exactly 7 whole stars, with anchors $R - S$, and 2 single-factor groups, at 0 and e_z , meet $R = \{0, e_z\}$. They charge 153 faces, of which 88 meet R (82 omitted and 6 selected), 16 lie inside R (the 10 omitted faces with owner set R and the 6 selected faces of the two single sites) and 72 straddle; 6 of the 72 have an owner set strictly containing R .

Support-one groups. A single-factor group has support $\{b\}$ and diameter 0. It is never clipped by a route-B volume containing b , and it never straddles a region, since a one-site support is either inside or outside. Its first-order creation is nonzero, with $\|c_{\{b\}}^{(1)}\| = \sqrt{3}|\tau|/144$. It enters the order-versus-distance count at a cost of one order and zero distance.

The first-order R -marginal. The first-order ground vector is $(\tau/72) \sum_f W_f \Omega_0$ over the retained faces. After the partial trace over R^c a face term survives only if its owner set lies in R , because the Haar mean of a spin- $\frac{1}{2}$ link outside R vanishes. Exactly the 16 faces inside R survive. Their vectors are orthogonal with norm $\frac{1}{2}$, so the first-order marginal has trace norm exactly $|\tau|/18$, while all 72 straddling faces, including the 6 whose owner set strictly contains R , have zero first-order R -marginal. The route-A ten-face value $\sqrt{10}|\tau|/72$ is rejected.

Selected-face site energy. A selected face puts spin $\frac{1}{2}$ on four links of one factor, so its vector has site energy $8 \cdot 4 \cdot \frac{3}{4} = 24$ there, while an omitted face has at most three links on one factor and

site energy at most 18. The selected-face vectors are therefore kept in the cutoff space Q_L exactly for $L \geq 24$; the route-A statement that first-order site energies are at most 18 is false for route B.

7.4 Admissibility of the disc and the weight

The AM2 majorant has $G(R) < 148/7$ and $G'(R) < 352$ at the ball radius $R = 1/64$, from $e^{1/8} < 8/7$. Under $J' = 29|\tau|$, the disc radius $\rho = 64|\tau|$ and the split weight $W = 1024$ are admissible, checked as exact rationals before any constant is evaluated (HNM-BC2-F01):

$$\begin{aligned} 29\rho G(R) &\leq \frac{1073}{2734375} < \frac{1}{64}, & 29\rho G'(R) &\leq \frac{2552}{390625} < 1 \quad (\rho = 64|\tau|), \\ 29W|\tau| G(R) &\leq \frac{17168}{2734375} < \frac{1}{64}, & 29W|\tau| G'(R) &\leq \frac{40832}{390625} < 1 \quad (W = 1024), \end{aligned} \quad (74)$$

and $\lambda = 2/W = 1/512$ lies below $q = 1/64$. The route-B maximal disc radius is $R/(29 \cdot 148/7) = 7/274688$, so the admissible weights are $1 \leq W \leq 7/(274688|\tau|)$, about 2548.35 at the cap. The rate $q = 1/64$ is frozen and never optimized in N .

The route-A extremes are rejected, by a factor of only 29/28. The BA1 disc radius $\tau_* = 1/37888$ and the BB1 secondary weight $1/(37888|\tau|)$ sit exactly on route A's own boundary, $28 \cdot (148/7)/37888 = 1/64$. Under route B they give

$$29 \cdot \frac{148}{7} \cdot \frac{1}{37888} = \frac{29}{1792} > \frac{28}{1792} = \frac{1}{64}, \quad (75)$$

so they fail the route-B self-map (HNM-BC2-F02). The contraction alone would accept them ($319/1184 < 1$), so the self-map is the test that rejects them. The rejection is exact but thin: the factor is 29/28 (producer reading W4, equal to the skeptic's R4). A weight admissible only under $28|\tau|$, such as $W = 2600$, is rejected as well, and $W = 64$ fails because λ is then not below q .

7.5 The coefficient input

The route-B coefficient input is proved in full in the packet, on the pattern of BA1's reverse route; no BA1 statement is cited as admitted. For complex z the complexified route-B map satisfies

$$F_z(c) = \sum_{k=0}^8 \frac{L_k^{(z)}(c, \dots, c)}{k!}, \quad \|L_k^{(z)}(c_1, \dots, c_k)\|_a \leq 29|z| 16 \cdot 8^k (1 + \frac{5k}{4}) \prod_j \|c_j\|_a, \quad (76)$$

since z enters only through $\|V_X(z)\| = |z| \|V_{X,1}\|$, with the per-site sum $29|z|$ and termination above order 8. The constants of the AM2 steps increase with the support size, so the one-site groups are covered by the bound for support 4 (checked for supports 1 to 4 and orders 0 to 8). For $\rho \leq 7/274688$ the map sends the anchored ball into itself and contracts it. Its fixed point $c(z)$ is continuous on the closed disc, holomorphic inside, and equal to the route-B AM2 fixed point at real τ . Only the creation coefficients are continued in z ; no zero-free region of the normalization is proved, and no analyticity of the reduced density is claimed.

The circle bound. The first-order coefficient counts 52 faces per factor, selected ones included, each with face vector norm $\frac{1}{2}$ and energy 24, so on the circle $|w| = \rho$

$$\begin{aligned} \|c(w)\|_a &\leq T_B(\rho) = \frac{52\rho/144}{1 - 29 \cdot 352\rho} = \frac{13\rho/36}{1 - 10208\rho}, \\ T_B(|\tau|) &= \frac{13}{3599632512}, \quad T_B(64|\tau|) = \frac{13}{55882512}. \end{aligned} \quad (77)$$

The value at $|\tau|$ is exactly the AX1 gate value T' , and the checker pins it: a route-A substitution of (28, 49) gives 49/14398580736 and aborts at the pin.

Order against distance, and the common core. In the piece graph of route B, omitted faces have owner sets of diameter 1 and selected faces owner sets of diameter 0. A nonzero term of order n on a support through u that contains a piece at distance $d \geq 1$ from u has $n \geq 1 + d$, since a single-site piece costs one order and no distance. A breadth-first search on the 1639 pieces of Λ_3 , 343 of them single sites, finds the bound attained, with slack 0, from five sites. Every face, omitted or selected, with a site at distance at most $N - |u|_\infty - 1$ from u is retained by every route-B volume containing Λ_N , so every source face of a comparison lies at distance at least $N - |u|_\infty$ from u . The enumeration covers c1B, c4B and c5B; the source faces of c1B include 654, 1158 and 1806 selected faces at $N = 2, 3, 4$.

The every-site bound. Schwarz’s lemma on the disc then gives, for c1B, c4B and c5B, both signs, every on-site cutoff and every site u of the union volume (HNM-BC2-F09),

$$\sum_{I \ni u} \|c_I^A(\tau) - c_I^B(\tau)\| \leq K_B q^{(N-|u|_\infty)_+}, \quad K_B = 2T_B(64|\tau|) = \frac{13}{27941256} \approx 4.65262 \times 10^{-7}, \quad (78)$$

with margin about 2.1493 against $1/1000000$ (tier `exact_first_order`, route `analytic_disc`). The bound holds in each cutoff space uniformly in the cutoff, and no untruncated creation coefficients are asserted (producer reading W3). The proved vanishing order is at least $N - |u|_\infty + 1$; the contract form $(N - |u|_\infty)_+$ is used, and the sharper exponent is not substituted (W2, equal to the skeptic’s R5). The crude tier $2 \cdot 29\rho G(R) = 2146/2734375 \approx 7.848 \times 10^{-4}$ fails and is retained. The global Lipschitz constant of the route-B map, at most $319/3125000$, and the no-decay bound $1073/174982136$ carry no distance and appear in no constant.

7.6 Marginal locality by the iterated split

The split route is BB1’s reverse route (eqs. (45)–(47)), re-derived with the route-B sums. Nothing in it uses the shape of a support beyond its diameter and size, so single-site supports enter like any other and never straddle. With the per-site sums of a comparison,

$$\begin{aligned} t_0 &= 2T_B(|\tau|) = \frac{13}{1799816256}, & t_W &= 2T_B(1024|\tau|) = \frac{26}{3148137}, \\ 8t_W &= \frac{208}{3148137} \approx 6.607 \times 10^{-5}, \end{aligned} \quad (79)$$

the weighted sum carrying the loss W^{d_X} once per interaction term, with $d_X = 1$ for stars and 0 for single groups. The closure (47) gives

$$\beta^* = \frac{132572461259098000347351859}{4012775859323049054050909456204} \approx 3.30375944 \times 10^{-5}, \quad (80)$$

with $c_2 \approx 1.909 \times 10^{-12}$. The lattice sum at $x = \lambda/q = 1/8$ is $S_\lambda = 1 + 24x(1+x)/(1-x)^3 + 2x/(1-x) = 2169/343$. For c1B, c4B and c5B, compared directly, both signs, every cutoff and every finite complete-factor region $Y \subset \Lambda_N$ (HNM-BC2-F17),

$$\|\rho_Y^A - \rho_Y^B\|_1 \leq c_{\text{site},B} \sum_{y \in Y} q^{N-|y|_\infty} \leq c_{\text{site},B} |Y| e^{|Y|/10^8} q^{d_Y}, \quad \|\rho_R^A - \rho_R^B\|_1 \leq C_B q^{N-1}, \quad (81)$$

with $d_Y = N - \max_{y \in Y} |y|_\infty$ and

$$\begin{aligned} c_{\text{site},B} &= K_B(2 + \beta^* S_\lambda) = \frac{35789673259133074250341804810795495}{38457845161696098009689444264975546012832} \approx 9.30621 \times 10^{-7}, \\ C_B &= c_{\text{site},B}(1 + q) = \frac{2326328761843649826272217312701707175}{2461302090348550272620124432958434944821248} \approx 9.45162 \times 10^{-7}. \end{aligned} \quad (82)$$

The margins are about 2.1491 against $1/500000$ and 4.2321 against $1/250000$ (tier `exact_first_order`, route `iterated_split`). The same bounds hold for the untruncated ground vectors at fixed

N . The order is fixed: the bound in each cutoff space uniformly in L , then $L \rightarrow \infty$ at fixed N by AV1 F20–F23 with the route-B gap $1/2$, then the bound read as a function of N ; the two limits are never exchanged.

Where the constant comes from. The ledger adds exactly to C_B . The near term $2K_B(1+q) = 845/894120192 \approx 9.4506 \times 10^{-7}$ is almost all of it. The straddling channel contributes about 9.872×10^{-11} , the normalization channel about 7.13×10^{-19} , and the split remainder less still, so the far terms are about 1.04×10^{-4} of the near term. This is why the two thin margins agree: $c_{\text{site},B}$ is essentially $2K_B$, and its margin is inherited from the coefficient input. The split bound needs no factor $e^{|Y|/10^8}$; the contract still asks for $(1+t)^2 \leq 1+10^{-8}$ as a fixture with the route-B anchored bound, and with $t = T'$ the value is $93590445481/12957354221447430144 \approx 7.223 \times 10^{-9}$. The union-volume form $2C_B$, the weighted first order $(49W+3)|\tau|/144$ and the nested values are labelled only, and the crude tier (about 1.876×10^{-3} for C_B and 1.847×10^{-3} for $c_{\text{site},B}$) fails and is retained.

7.7 Whole-sequence convergence

Each $M > N$ is compared with N by one direct c4B comparison of Λ_N and Λ_M , both of which contain Λ_N (assembly `union_comparison`, factor 1). Hence, for every $N \geq 2$, both signs, each cutoff space and then the untruncated vectors (HNM-BC2-F18),

$$\sup_{M>N} \left\| \rho_R^{FB,M} - \rho_R^{FB,N} \right\|_1 \leq C'_B q^{N-1}, \quad \sup_{M>N} \left\| \rho_Y^{FB,M} - \rho_Y^{FB,N} \right\|_1 \leq c'_{\text{site},B} |Y| e^{|Y|/10^8} q^{d_Y}, \quad (83)$$

with $C'_B = C_B$ and $c'_{\text{site},B} = c_{\text{site},B}$, the second for every N with $Y \subset \Lambda_N$. The margins are about 4.2321 against $1/250000$ and 2.6864 against $1/400000$. With this assembly the whole-sequence bound on R carries nothing beyond the comparison bound (W5, equal to the skeptic's R6). Nested telescoping would give $C_B/(1-q) \approx 9.6016 \times 10^{-7}$ and $c_{\text{site},B}/(1-q) \approx 9.4539 \times 10^{-7}$; these are labelled only.

For a fixed region the whole sequence $\rho_Y^{FB,N}$ is Cauchy in the trace class, so it converges in trace norm without compactness, and the limits on nested regions are compatible. They define a locally normal state ω_∞ that extends to the quasi-local algebra, and letting $M \rightarrow \infty$ in (83) gives $\left\| \rho_R^{FB,N} - \rho_R^\infty \right\|_1 \leq C'_B q^{N-1}$ and the region analogue. Example values of $C_B q^{N-1}$ are about 1.4768×10^{-8} ($N = 2$), 2.3075×10^{-10} ($N = 3$) and 3.6055×10^{-12} ($N = 4$). Every rate is per coarse step in N at fixed spacing, and it holds for every $N \geq 2$ (on a region Y , for every N with $Y \subset \Lambda_N$; for the exhaustion below, for every $N_k \geq 2$). No correlation-function rate in N is claimed, and no BB2 rate is transferred.

7.8 Items 2a, 2b, 3 and 4

Item 2a: exhaustion within the prescription. Let V_k be finite complete-factor route-B volumes, with every star inside the volume and every single group kept, and let N_k be the largest N with $\Lambda_N \subset V_k$, tending to infinity. For every $N_k \geq 2$ and every finite complete-factor region $Y \subset \Lambda_{N_k}$, one direct c5B comparison of V_k with Λ_{N_k} and the whole-sequence bound give (HNM-BC2-F19)

$$\left\| \rho_{Y_k}^{V_k} - \rho_Y^\infty \right\|_1 \leq (c_{\text{site},B} + c'_{\text{site},B}) |Y| e^{|Y|/10^8} q^{N_k - \max_{y \in Y} |y|_\infty}, \quad (84)$$

$$c_{\text{site},B} + c'_{\text{site},B} = 2c_{\text{site},B} \approx 1.8612 \times 10^{-6}.$$

Every such exhausting sequence therefore has the same limit on every finite region. This is exhaustion within one prescription, not a comparison of boundary conditions. The contract's clause cites "item 1" for the second ingredient; the producer reads it as the whole-sequence bound, not BC2's item 1 (W1).

Item 2b: the pointwise sign mirror. In every centered box and every cutoff, $U_E H_N(\tau) U_E^* = H_N(-\tau)$ (AX1 forward F19–F20): every face coefficient, selected faces included, is $-\tau/3$, the on-site operator and its cutoffs commute with U_E , and the flip set meets every plaquette in 1 or 3 links (864 plaquettes of a period window re-enumerated, 108 of them selected). The grounds are simple, and U_E is a product of one-link unitaries, hence of factor unitaries $U_{E,x}$, so $\rho_Y^N(-\tau) = U_{E,Y} \rho_Y^N(\tau) U_{E,Y}^*$ for every finite Y , every N and every cutoff. Whole-sequence convergence at both signs then gives (HNM-BC2-F20)

$$\omega_\infty^{-\tau} = \omega_\infty^\tau \circ \alpha_E \quad \text{on every finite region,} \quad \alpha_E(A) = U_E A U_E^*. \quad (85)$$

AX1’s whole-set statement $S(-\tau) = S(\tau) \circ \alpha_E$ becomes pointwise. The $-\tau$ side is the mirror replay with no real g , not a second coupling and not a second observation.

Item 3: identification before inheritance. An AQ1-type subsequential state of the AX1 construction is a limit, along some subsequence, of the densities of the centered whole-star-plus-single-group boxes on every finite region (AQ1’s diagonal extraction, re-applied in route B by AX1). The whole sequence converges, so every subsequence has the same limit, and the set of these states, nonempty by compactness, is the single state ω_∞ (HNM-BC2-F21). Only then does ω_∞ inherit exactly the AX1 gate’s route-B list: a compatible, locally normal, gauge-invariant state, stationary under the route-B limit dynamics T_θ , with a strongly continuous GNS evolution and a nonnegative physical generator; the AQ2-type inequality $H_{\text{phys}} \geq (\alpha/16)(I - P_\Omega)$ on the invariant-local cyclic completion, with the full-GNS strengthening only as qualified there; the reset $\omega(h_R) \leq 102|\tau|$ and $\epsilon_R \leq 17|\tau|$; and the AX1 state lemma with $D' = 2425369125199104794263242601250/167893028420061547330293754713793182097 \approx 1.44459 \times 10^{-8}$. Not inherited: the zero-selected constants, the BB2 list, a uniform $\omega(W^2)$ constant or Wilson-mean sign, any route-B dynamics comparison or correlation-function statement, and uniqueness of any ground state. The inherited stationarity is a property of the identified state, not a route-B dynamics claim (W6).

Item 4: coarse translations. A coarse translation v is the fine translation $(4v_x, 2v_y, v_z)$. It preserves the residues, the face classes, the selected set and link ownership, so it maps route-B volumes to route-B volumes. For every coarse v and every $N \geq |v|_\infty + 2$, both $\Lambda_N + v$ and Λ_N contain $\Lambda_{N-|v|_\infty}$, and one direct c5B comparison gives (HNM-BC2-F22)

$$\left\| \rho_R^{\Lambda_N + v} - \rho_R^{\Lambda_N} \right\|_1 \leq C_B q^{N-|v|_\infty-1} \quad (N \geq |v|_\infty + 2). \quad (86)$$

The region form of c5B makes ω_∞ invariant under every coarse translation on every finite region. Nothing is claimed for non-coarse fine translations, neither invariance nor its failure. Among the 64 fine translations of a period window exactly the 8 coarse ones preserve the factor partition and the grouping. A non-coarse translation keeps the uniform coefficient $-\tau/3$ of every face but not the grouping: $(1, 0, 0)$ maps the selected face at $(2, 0, 0)$, a single group with one owner, to the face at $(3, 0, 0)$ with two owners. The route-A reason for rejecting such translations, a selected face mapped to an omitted face of another coefficient, does not apply here.

7.9 The AX2 node restated for the limit

AX2 certifies, for every AQ1-type subsequential state of the construction, each separately, at $\tau = +10^{-8}$ and $s = 1$, that $|C(1) - d| \leq r \leq R'$. By item 3 the limit is that state, with the same

GNS representation and generator. With $\chi = (\pi(W) - \omega_\infty(W))\Omega$ in that representation,

$$\begin{aligned} |C_\infty(1) - d| &\leq R', \quad C_\infty(1) = (\chi, e^{-G}\chi), \\ d &= \frac{497870683678639429793424156500617766317}{4 \cdot 10^{40}} \approx 1.24467670920 \times 10^{-2}, \\ R' &= \frac{1912298807996871790146581299723633}{10^{40}} \approx 1.91229880800 \times 10^{-7}, \end{aligned} \quad (87)$$

so $C_\infty(1)$ lies in $[d - R', d + R']$ with

$$\begin{aligned} d - R' &= \frac{99572606896681488461252714035083774357}{8 \cdot 10^{39}} \approx 1.24465758620 \times 10^{-2}, \\ d + R' &= \frac{497878332873871417280584742825816660849}{4 \cdot 10^{40}} \approx 1.24469583218 \times 10^{-2}. \end{aligned} \quad (88)$$

The datum is the AV2 datum, unchanged because the Haar reference and the free z link are unchanged. Both values are read by hash from the AX2 gate. The admitted AX2 calculator, a declared premise loaded from the producer’s input snapshot, is replayed at its fixed design ($\tau = +10^{-8}$, $s = 1$, the gate-bound D' , $k' = 51|\tau|/4$, $M_0 = 2$, $M_1 = 4s/\pi$). It returns d and its outward 10^{-40} radius equal to the gate rationals exactly, the label `reference_unresolved`, and at $-\tau$ the same values as the mirror replay. The skeptic ran the calculator from its source bytes and obtained the same values. The producer’s own re-evaluation, with its own Machin enclosure of π and bracket for e^{-3} , lies about 2.7×10^{-41} below R' , and so does the skeptic’s; both are labelled cross-checks, not the headline.

The free value $e^{-3}/4$ lies inside the interval, so no interaction shift, sign or coefficient of $C(s)$ is claimed, and `resolved_interaction_shift` is false. No other node, no convergence of the finite-box nodes $C_N(1)$ to $C_\infty(1)$ and no smaller state term for the limit is claimed; the first would need a route-B dynamics comparison. The region form of (83) is load-bearing here. $C(1)$ is a functional of the state on the quasi-local algebra, so the restatement needs the identification on every finite region, for the untruncated vectors; with the R form only, or inside each cutoff space only, it would fail (control `region_form_load_bearing_for_node`). The datum ratio under $\tau \rightarrow \tau/100$ is exactly 1, and the radius replay ratio, about 100.0015, is recorded and not a target.

7.10 Constants and scaling

Table 20 collects the bound values. The binding margins, both about 2.149, are those of K_B and $c_{\text{site},B}$.

Table 20: The BC2 constants at $q = 1/64$, both signs, every $N \geq 2$ (on a region, every N with $Y \subset \Lambda_N$). Exact values are in (78), (82) and (87); every constant is tier `exact_first_order`. The ratio is the constant at τ over its value at $\tau/100$, from the same exact formula.

Constant	Preview	Target	Margin	Route or assembly	Ratio
K_B (T0)	4.6526×10^{-7}	10^{-6}	2.1493	<code>analytic_disc</code>	100.651
C_B (T1)	9.4516×10^{-7}	$1/250000$	4.2321	<code>iterated_split</code>	100.661
$c_{\text{site},B}$ (T2)	9.3062×10^{-7}	$1/500000$	2.1491	<code>iterated_split</code>	100.661
$C'_B = C_B$ (T3)	9.4516×10^{-7}	$1/250000$	4.2321	<code>union_comparison</code>	100.661
$c'_{\text{site},B} = c_{\text{site},B}$ (T4)	9.3062×10^{-7}	$1/400000$	2.6864	<code>union_comparison</code>	100.661
exhaustion $2c_{\text{site},B}$	1.8612×10^{-6}	—	—	<code>c5B and T4</code>	—
node d, R'	AX2 gate values	equality	equal	<code>calculator replay</code>	1 (d)

The ratio for K_B is exactly 39059948/388073. Every ratio lies in the frozen bracket [95, 105]: each constant is linear in τ with a small positive correction from $1/(1 - 10208 \cdot 64|\tau|)$ and the split terms. A quadratic, constant or square-root headline and a bracket chosen after evaluation are rejected controls. The skeptic’s pre-comparison table printed the density ratio as about 100.662; the exact value is 100.6614518, and the review records the slip against itself.

7.11 Evidence and review

The forward checker runs 111 exact checks and implements all 52 contract controls as damaging mutations, with 219 rejections. Its outputs are byte-identical under `-B` and `-B -0`. The skeptic’s pre-comparison package (110 checks, 52 controls as 138 rejected mutations with 5 positives) was final by 04:58:27 UTC and committed first, at 05:00:47. The forward checker was last written at 05:02:17, and the forward commit followed at 05:04:31, 3 min 44 s after the skeptic’s. The skeptic’s exposure was to names only: untracked file names in a `git status`, and the subject line of the forward commit, which states K_B , seen after its BC2 package was committed. It opened the forward directory only after the coordinator’s post-comparison message. The post-comparison checker runs 21 checks and 94 source-edit runs. The forward replays reproduce the frozen outputs byte for byte, the closure of 46 files verifies, and the forward holds exactly its 42 declared inputs. No interpreter cache is written under the AX2 calculator’s directory or in the closure, even by a run without `-B`. The gate was recorded at 05:57 UTC on 25 September.

What the exact agreement means. Every route-B constant, including t_0 , t_W , c_1 , c_2 , β^* and S_λ , equals the skeptic’s frozen pre-comparison rational exactly, not only to the printed digits. Both codes follow the same frozen formula chain: the contract’s route-B inputs, the BB1 reverse covering-chain closure and T_B . The agreement therefore checks arithmetic and code. It is not a second method. The skeptic’s code reproduces both BB1 gate `iterated_split` rationals exactly from route-A inputs, which anchors its implementation.

Twenty-two route-A substitutions abort. The review substituted route-A inputs and gate constants into the producer’s checker 22 times, and each run aborted without output. The substitutions were:

- the per-site sum $28|\tau|$, 49 faces or 4 groups;
- every constant evaluated with (28, 49), or K_B replaced by the BA1 value;
- the contraction coefficient $9856 = 28 \cdot 352$ in T_B , or $G'(R)$ rescaled so that $29G'(R) = 9856$;
- BB1 or BB2 gate constants in place of C_B , $c_{\text{site},B}$, C'_B or $c'_{\text{site},B}$;
- the route-A disc radius or split weight, or $W = 2600$;
- the route-A ten faces on R ;
- D' recomputed with $28|\tau|$ or 49 faces, or the node slope $49|\tau|/4$;
- a changed selected predicate, clipped single groups, or an emptied route-A rejection list;
- both guards removed, in which case the derived per-site face counts still expose 49.

Readings, silent edits and disclosures. The review adopts reading R1: the $(1+t)^2$ fixture uses $t = T'$, the per-box anchored bound, and not $2T'$, with which the value, about 1.4446×10^{-8} , would fail. Of six silent source edits, one is harmless (a 30-term enclosure of $e^{1/8}$). The other five run silently in the producer’s checker and are rejected only by the skeptic’s validator. They were the example exponent N in place of $N - 1$, a translation row with exponent 3, the labelled own radius computed with slope $49|\tau|/4$, a changed K_B margin in the verdict table, and a changed digit of C_B at one of its two report sites. The review’s advice for later single-direction loops is to freeze an exact-value pin list for every exported number, examples and labelled cross-checks included. The forward disclosed that the file tool refused its report, which was written through a shell heredoc after a briefly written stub; the stub was disclosed to the coordinator and not in the frozen packet, so the review records it. It also disclosed a listing of tool file names and a reading of parts of the gated BB1 reverse checker, outside its inputs, for coding conventions; no BC2 constant depends on either. The review records the contract readings W1–W7 and its own D1–D3 as non-blocking, and found no blocking issue.

7.12 Obligations

The review records five open obligations for route B, as contract item 6 requires (Table 21). Row N9 of Table 18, which was to cite the BC2 gate only once recorded, stays as the BC1 review froze it; what the BC2 gate admits for the route-B model is stated in this section, and these five rows are what it leaves open.

Table 21: Open obligations recorded by the BC2 review for the route-B model.

Row	Obligation	Missing premise and candidate route
B1	A second route-B boundary prescription (an all-contained analogue with its own itemization)	An all-contained route-B prescription with its own grouping, per-site sum and retention, compared with FB. Route: re-run the coefficient input and the split with the new source set; not executed.
B2	Route-B dynamics and correlation functions (a route-B BA2)	A route-B Lieb–Robinson/Duhamel comparison; the BA2 constants use $2268 \tau $, while route B has $\ \Phi'\ _F \leq 2349 \tau $. Route: the BA2 route with the route-B constants; not executed.
B3	Uniqueness of any ground state	A mechanism covering states outside the named construction. Route: local stability with evaluated constants; not formulated.
B4	Anything uniform in the lattice spacing a	Every constant is at fixed spacing and strong bare coupling. No route within this model.
B5	The continuum problem	Outside this investigation; no route claimed.

Meaning. BC2 carries the whole locality and convergence chain of BA1, BB1 and BB2 over to the uniform model handled by route B, with every constant recomputed from route-B inputs and each at most its frozen target. The reduced densities of the one named construction converge as a whole sequence at the rate q^{N-1} per coarse step at fixed spacing, to one limit that is also reached along every exhausting sequence of the same prescription. That limit is the AQ1-type subsequential state of the AX1 construction, is mirrored pointwise between the two signs, is invariant under coarse translations, and satisfies the AX2 node certificate at $s = 1$ with unchanged constants. It is convergence of one named construction. It is not uniqueness of any ground state, not a statement about other boundary prescriptions or about route-B dynamics, not a finite-box node, not weak coupling or continuum, and not uniform in the lattice spacing.

7.13 Limitations recorded by the gate

The BC2 gate records the following limitations verbatim.

- Scope: only the uniform route-B model at fixed spacing and strong bare coupling ($|\tau|$ at most 10^{-8} , g^4 at least 9.6×10^9 by the AL1 dictionary of this model), the named construction FB and the finite complete-factor route-B volumes containing Λ_N with N at least 2, finite complete-factor regions and the cover R , coarse translations and the single node $s=1$ at $\tau=+10^{-8}$; route B has one named family, so no common limit of two route-B families is claimed; nothing transfers to the zero-selected family (no identification with the BB2 limit and no BB1 or BB2 constant applied), to other boundary prescriptions, to weak coupling or the continuum; the constants are uniform in N at fixed spacing and in the on-site cutoff, and no estimate is uniform in the lattice spacing a .
- Convergence of the named construction only: no uniqueness of any ground state and no statement about states outside the named construction; the limit is identified only with the AQ1-type subsequential states of the AX1 construction and inherits only the AX1 gate's route-B list; the AX1 finite-volume statement (a simple ground and gap $1/2$ in every finite complete-factor route-B volume) is a finite-volume premise, not an infinite-volume statement.
- Dynamics: no route-B dynamics comparison and no correlation-function statement (dynamics_level is not exported); stationarity under the route-B limit dynamics

- T_theta and the GNS generator are inherited properties of the identified state from the AX1 gate (producer reading W6), not a route-B BA2; no equality of GNS dynamics of different states and no uniform-in-time statement; the secondary label static_not_dynamic is read in this sense (state statements and the inherited Euclidean node, no route-B real-time dynamics).
- Node restatement only: d and R' are the AX2 gate rationals; the free value $e^{-3}/4$ lies inside the interval (reference_unresolved); no interaction shift, sign or coefficient of C(s), no other node and no convergence of finite-box nodes C_N(1) to C_inf(1) is claimed, and the -tau value is the U_E mirror replay; the labelled own re-evaluations of the radius (the producer's and the skeptic's, each about $2.7e-41$ below R') are cross-checks, not the headline.
 - Coefficient input: T0 is a statement in each on-site cutoff space (untruncated creation coefficients are not asserted; producer reading W3); only the creation coefficients are continued in the coupling, and no analyticity of the reduced density or of the normalization is claimed (no zero-free region); the analytic steps of the disc lemma (Weierstrass, the maximum principle) and the AM2 majorant steps are proved in the report and are not machine-checked.
 - Constants are upper bounds only, evaluated at the constants proved in the packet (no hypothesis layer); the thin margins are T0 (about 2.1493) and T2 (about 2.1491), because $c_{\text{site},B}=K_B(2+\beta \cdot S_{\text{lambda}})$ and the far term is about $1.04e-4$ of the near term; the route-A extremes fail the route-B self-map by the factor 29/28 only (producer reading W4); the crude tier fails every target and is retained; the nested-telescoping and union-volume variants, the telescoped coefficient input and the weighted first order $(49W+3)|\tau|/144$ are labelled only.
 - Rates: every rate is per coarse step in N at fixed spacing (a coarse step is $(4a, 2a, a)$) and holds for every N at least 2 (on a region Y, for every N with Y inside Λ_N ; for the exhaustion, for every N_k at least 2); no correlation-function rate in N and no rate transferred from BB2 is claimed.
 - Contract wording read as follows (non-blocking): W1 clause 2a's item 1 is the whole-sequence bound T4; W2 (equal to skeptic R5) the sharper vanishing order $N-|u|_{\text{inf}}+1$ is not substituted; W3 T0 holds only in each Q_L; W4 (equal to skeptic R4) the route-A extremes fail only the self-map, by 29/28; W5 (equal to skeptic R6) with union_comparison T3 carries nothing beyond T1; W6 the inherited AX1 stationarity is not a route-B dynamics claim; W7 the factor 2 of $8 \cdot 2^{\{\text{diam } I\}}$ is absorbed into $\text{lambda}=2/W$; skeptic D1 (T1 has no form field), D2 (c1B is a special case of c4B) and D3 (the inherited parameter control asks for d_X and N_0) stand.
 - Premises: inherited without re-proof are the AM2 majorant steps and fixed point, the AX1 route-B grouping, finite-volume theorem, state lemma, flip identity and route-B AQ1/AQ2 re-instantiation list, AV1 F20-F23, the AX2 certificate and calculator, and the I1 dictionary; the analytic disc (BA1 reverse Theorem 4.1), the iterated split (BB1 reverse) and the whole-sequence scheme (BB2) are shared premises named by the contract. The producer read parts of the gated BB1 reverse check.py outside its inputs for conventions (disclosed); no BC2 constant depends on it.
 - Independence: one forward producer plus the skeptic's pre-comparison replay, all correlated model agents; the skeptic pre-comparison package (5ec599f) was committed 3 min 44 s before the forward commit (ee45eed); the pre-comparison rationals equal the producer's exactly because both follow the frozen formula chain, so the agreement checks code and arithmetic, not a second method; isolation rests on disclosure and content; the report was written by a shell heredoc after a briefly written stub (disclosed to the coordinator, not in the frozen packet) and is bound by its freeze record.
 - Scientific priority is unverified.

Full derivations and reviewed implementations: [BC2 forward](#), [independent model-agent review](#), [skeptical pre-comparison derivation](#) and the [frozen contract](#); the restated node is that of the [AX2 gate](#), and the route-B grouping that of the [AX1 gate](#), neither of which is edited. With BC1 and BC2 gated, sub-round 3 is complete.

8 Applications: transfers and recorded obstructions

The fourth stage of the plan applies admitted Round32 and Round33 equations to related problems and records, for each equation it applies, either a labelled transfer to a named model with its own

exact checks or the obstruction that prevents the transfer (Section 1.4). The BD1 and BD2 contracts were drafted during the BC production, since no BD item depends on a BC outcome, and were frozen together on 25 September 2026 at 05:09:10 UTC, after the skeptic’s pre-freeze review of the drafts (34 blocking and 21 non-blocking edits applied). The review found the drafted BD2 width target of $1/10^{10}$ non-discriminating, and the advisor tightened it to $1/10^{16}$ at the cutoff $D = 8$ before the freeze. BD1 applies the Round32 centre-symmetry equations (the AW1 link-flip lemma, parity theorem and first-order Wilson coefficient) to other gauge groups. BD2 applies admitted SU(2) equations to new observables, to a finite graph with independent couplings and to a new dimension. Both gates were recorded at 07:52 and 07:53 UTC on 25 September; with them all eight Round33 investigations are gated. Table 29 at the end of the section summarizes what transferred and what did not.

8.1 BD1: centre symmetry across gauge groups

Contribution HNM-C-BD1. The BD1 contract, recorded under the title “Applications I: centre-symmetry transfer of the parity theorem, the link-flip lemma and the first-order Wilson mean across gauge groups, with the SU(3) and SO(3) obstructions and the SU(2) area-parity corollary”, asks which of three AW1 statements survive when SU(2) is replaced by SU(3), SU(4), SU(5), U(1), Z2 or SO(3), and asks for an exact counterexample wherever one does not ([selection note](#)). It is investigation 7 of 8 and paired. The forward producer computes by characters: Peter–Weyl, the Pieri rule, Clebsch–Gordan series and charge counting. The reverse producer, under premise isolation, computes by Weyl integration: constant terms on the maximal torus. Both received AGENTS.md, the contract and 21 shared premises. The target is exact agreement of the two routes on every moment and coefficient, and nonzero SU(3) and SO(3) obstruction coefficients; the contract notes that the obstruction cells are discriminating, since a zero value would falsify them. It lists 22 controls. Its gate is recorded under the title “Hruday centre-symmetry transfer across gauge groups with recorded obstructions”.

8.1.1 Reviewed scope

The [BD1 gate](#) records the verdict `accepted_within_scope`, with sub-label `transfer_to_named_model` and secondary sub-label `obstruction_recorded`. Its gate fields set `flip_transfer_claimed`, `parity_transfer_claimed`, `obstructions_recorded`, `area_parity_limit_claimed` and `model_is_finite_graph` to true, and `transfers_to_aq`, `uniqueness_of_ground_state_claimed`, `rate_in_a_claimed`, `continuum_claim`, `weak_coupling_claim` and `scientific_priority_verified` to false. The flip scope names SU(2), SU(4), U(1) and Z2 on their one-plaquette and box models, with no AM2, AV1 or AQ statement for any group other than SU(2). The accepted statement, quoted verbatim below in the gate’s ASCII notation, is the scope of every claim in this subsection.

```
Under the frozen BD1 convention (per-link electric term  $8 C_2$  with  $C_F=(N^2-1)/(2N)$  for
SU(N),  $C_2(n)=n^2$  for U(1),  $C_2(1)=1(1+1)$  for SO(3) and  $C_2=1$  on the odd Z2 state; face
energy  $32 C_F$ ; magnetic term  $-(\tau/3) W$  with  $W = \text{Re } \chi_{\text{fund}}/\text{dim fund}$ ; creation sign
 $\psi=e^{-\{C\} \Omega_0}$  and  $\omega(W) = -2 \text{Re } (W \Omega_0, c^{(1)}) + 0(\tau^2)$ ), on the
one-plaquette finite graphs  $H_{FG}(G)$  ( $|\tau|$  below  $48 C_F$ ) and on the group-G whole-star
box models  $H^G_N$  on the open centered boxes  $\Lambda_N$ ,  $N$  at least 2, each box separately
for  $|\tau|$  below its Kato radius  $12 C_F/(21(2N)^3)$ , for  $G$  in {SU(2), SU(3), SU(4), SU(5),
U(1), Z2, SO(3)}: the Haar moments  $E[W^k]$ ,  $k=1..4$ , and the first-order coefficients of
 $\omega(W)/\tau$  (1/144, 1/1152, 1/2880, 1/5760, 1/96, 1/48 and 1/864 in that order) agree
exactly between the characters route (forward) and Weyl integration on the SU(N) torus
(reverse); the link-flip lemma holds on these models for SU(2), SU(4), U(1) and Z2,
whose central elements  $-I$ ,  $-I$ ,  $e^{i\pi}$  and the nontrivial element act as  $-1$  on the
Wilson representation, with  $E_3$  verified on  $\Lambda_2$ ,  $\Lambda_3$ ,  $\Lambda_4$  and the 24-link
coarse factors and  $E_2$  on  $[-N,N]^2$  by exact enumeration; the first-order parity theorem
holds for SU(2), SU(4), SU(5), U(1) and Z2 ( $E[W^3]=0$ ); SU(3) is an obstruction with
 $E[W^3]=1/108$ ,  $d \omega(W^2)/d\tau=1/6912$  and second-order coefficient 1/589824, SO(3)
```

with $1/27$, $1/2592$ and $1/331776$, and $SU(5)$ a flip obstruction with fourth-order coefficient $1/63403380965376=1/(2^{30} 3^{10})$ on $H_{FG}(SU(5))$; for $SU(2)$ in the AM2/AQ1 zero-selected patterned family at $\kappa=0$, $|\tau|$ at most 10^{-8} and both signs, $\omega(W_C)(-\tau)=(-1)^{A(C)} \omega(W_C)(\tau)$ in every open centered whole-star box and every on-site cutoff, and pointwise for the limit of the named constructions through the BB2 whole-sequence bound at each sign ($c_site=2/984375$, $q=1/64$). Under the frozen convention, the link-flip lemma transfers exactly to the listed gauge groups that have a central element acting as -1 on the Wilson representation and the first-order parity theorem exactly to those with vanishing third Haar moment of W , each with its own first-order Wilson coefficient, on one-plaquette models and on group box models in each box separately; $SU(3)$ and $SO(3)$ are recorded obstructions with exact nonzero coefficients, and $SU(5)$ is a recorded flip obstruction with an exact nonzero fourth-order coefficient; for $SU(2)$ in the zero-selected family at $\kappa=0$ the Wilson-loop mean is odd or even in τ according to the parity of the number of plaquettes of any spanning surface of the loop, in every open centered whole-star box and for the limit of the named constructions; these are statements about named models and one-plaquette finite graphs, not AM2 or continuum statements for other groups, not statements for periodic boxes with an odd side, not uniqueness of any ground state, and not a statement uniform in the lattice spacing a .

8.1.2 The frozen convention and the models

Energies are in units $\delta = \alpha/8$. The per-link electric term is $8C_2$, with $C_F = (N^2 - 1)/(2N)$ for $SU(N)$, $C_2(n) = n^2$ for $U(1)$, $C_2(l) = l(l+1)$ for $SO(3)$ and $C_2 = 1$ on the odd Z2 state. A face whose four links carry the Wilson representation F therefore has energy $32C_F$. The magnetic term is $-(\tau/3)W$ with $W = \text{Re } \chi_F / \dim F$. With the AM2/AV1 creation sign $\psi = e^{-C}\Omega_0$,

$$c^{(1)} = -\frac{\tau/3}{32C_F} \sum_f W_f \Omega_0, \quad a = \frac{1/3}{32C_F}, \quad (89)$$

$$\omega(W) = -2 \text{Re}(W\Omega_0, c^{(1)}) + O(\tau^2) = 2a \mathbb{E}[W^2] \tau + O(\tau^2).$$

The inherited AW1 display $2(W\Omega_0, c^{(1)})$ has the opposite sign; the contract records it, and both producers reject it (contract defect C7). The Z2 normalization is the frozen one, $8C_2 = 8$ on the odd link state; the readings with $C_2(\text{odd}) = 1/8$ (coefficient $1/6$) and with $1 - \sigma^x$ (coefficient $1/12$) are rejected controls.

The models are named. The one-plaquette finite graph is $H_{FG}(G) = 32C_2 - (\tau/3)W$ on class functions of one plaquette holonomy (`model_is_finite_graph` true, `transfers_to_aq` false). The group box model $H_N^G = \sum_e 8C_2(e) - (\tau/3) \sum_f W_f$ lives on the open centered whole-star box Λ_N , $N \geq 2$, with the AQ1 geometry: coarse 24-link factors, the 21 omitted faces of an anchor retained when its whole star lies in the box, and the selected faces at coefficient 0. For $SU(2)$ this is exactly the AW1 box Hamiltonian of the zero-selected family. For every other group it carries no AM2, AV1 or AQ statement.

The Kato radius. For every listed group C_F is the smallest nonzero Casimir, attained exactly by F and $\bar{F}$. The free gap is therefore $8C_F$ in a box (one excited link) and $32C_F$ on the plaquette, and $\|W_f\| \leq 1$. A simple isolated ground eigenvalue persists, analytically in complex τ , for

$$|\tau| < 48C_F \quad \text{on } H_{FG}(G), \quad |\tau| < \frac{12C_F}{21(2N)^3} \quad \text{on } H_N^G, \quad (90)$$

with $21(2N)^3$ retained faces: $3/448$ for $SU(2)$ at $N = 2$. This radius is sufficient, not sharp, and it shrinks like N^{-3} , so nothing is claimed for all N at once for any group other than $SU(2)$, whose AM2 regime $|\tau| \leq 10^{-8}$ is admitted separately. The contract's word "unique" in items 1 and 6 means this simplicity of a finite-box ground eigenvalue, used as a hypothesis (contract defect C3); it is not uniqueness of any ground state.

8.1.3 Criterion A: a central element acting as -1

Let z be central in G with $\rho_F(z) = -I$, and let E be a set of links meeting every plaquette an odd number of times. Multiplying U_e by z on every link of E defines a unitary U_E . Haar measure is invariant, so U_E is unitary and fixes Ω_0 . Because z is central, U_E commutes with every translation of each link variable, hence with every endpoint gauge action; by Schur’s lemma it acts on each Peter–Weyl block as a scalar, hence commutes with every link Casimir and every on-site cutoff projection. A flipped link multiplies the holonomy of each plaquette through it by z or z^{-1} , which reverses χ_F , and an odd number of flipped links per plaquette gives $U_E W_f U_E^* = -W_f$. Hence (HNM-BD1-F05)

$$U_E H_N^G(\tau) U_E^* = H_N^G(-\tau) \quad \text{in every box and cutoff,} \quad U H_{FG}(G)(\tau) U^* = H_{FG}(G)(-\tau), \quad (91)$$

and $\omega_{-\tau}(W) = -\omega_\tau(W)$ wherever the ground eigenvalue is simple, that is, inside (90). Such an element exists for exactly four of the seven groups (Table 22).

Table 22: Central elements acting as -1 on the Wilson representation.

Group	Centre	Element with $\rho_F(z) = -I$	Reason
SU(2)	$\{I, -I\}$	$-I$	$\det(-I) = (-1)^2 = 1$
SU(3)	$\mathbb{Z}_3$	none	cube roots of unity are never -1
SU(4)	$\mathbb{Z}_4$	$-I = e^{2\pi i \cdot 2/4} I$	$\det(-I) = (-1)^4 = 1$
SU(5)	$\mathbb{Z}_5$	none	fifth roots of unity are never -1
U(1)	U(1)	$e^{i\pi}$	charge-1 character value -1
Z2	Z2	the nontrivial element	sign character value -1
SO(3)	trivial	none	$-I \notin \text{SO}(3)$, since $\det(-I) = -1$

The absence of a central -1 is recorded as the reason, not as the proof, of non-transfer. The proof is an exact nonzero even-order coefficient (Section 8.1.6): any unitary that commutes with $32C_2$ and reverses W on $H_{FG}(G)$, central or not, would make $\omega(W)$ odd in τ and kill every even coefficient.

8.1.4 Criterion B: the third Haar moment

Single-occurrence orthogonality. In every listed group a product of face variables has Haar mean 0 if some link lies in exactly one of the faces, because as a function of that link the product is a combination of matrix coefficients of F and $\bar{F}$. Two distinct plaquettes share at most one link.

The degenerate level. The gauge-invariant eigenspace of $\sum 8C_2$ at energy $32C_F$ is spanned by $\chi_F(U_g)\Omega_0$ and $\overline{\chi_F(U_g)}\Omega_0$ over the plaquettes g of the box. Each excited link costs at least C_F , so at most four links are excited; gauge invariance forbids a vertex of degree one; $\mathbb{Z}^3$ is bipartite, so the excited links form a 4-cycle, which is a plaquette. For U(1) and SU(N), $N \geq 3$, the level therefore contains the $\text{Im } \chi_F$ directions as well as $W_g\Omega_0$.

The criterion. On $H_{FG}(G)$ and on every H_N^G inside its Kato radius, if $\mathbb{E}[W^3] = 0$ then the first-order τ -derivatives at 0 of $\omega(W^2)$, of the centred Euclidean correlation $C(s)$ for every $s \geq 0$ and of the centred real-time correlation $c(\theta)$ for every real θ vanish, and the first-order splitting of the whole level vanishes. Every first-order term (state, Duhamel, energy and vector centring) is a Haar mean of three face variables, which reduces to $\mathbb{E}[W^3]$ by single-occurrence orthogonality. Since $\mathbb{E}[W^3] = (2 \dim F)^{-3} \sum_a \binom{3}{a} \mathbb{E}[\chi^a \bar{\chi}^{3-a}]$ is a positive combination of nonnegative integers, $\mathbb{E}[W^3] = 0$ exactly when every cubic moment vanishes, which also kills the splitting on the complex directions. Conversely, on $H_{FG}(G)$,

$$\left. \frac{d\omega(W^2)}{d\tau} \right|_0 = 2a \mathbb{E}[W^3], \quad C(s)^{(1)} = e^{-32C_F s} \mathbb{E}[W^3] (2a + \frac{s}{3}), \quad (92)$$

which are nonzero for every $s \geq 0$ when $\mathbb{E}[W^3] \neq 0$. The vanishing third moment holds for $\text{SU}(2)$, $\text{SU}(4)$, $\text{SU}(5)$, $\text{U}(1)$ and Z_2 . The flip and the parity are separate columns: $\text{SU}(5)$ has the parity without the flip.

8.1.5 Moments and first-order coefficients

Table 23 gives the Haar moments and the first-order coefficients of $\omega(W)/\tau$ from (89). For $\text{SU}(N)$, $N \geq 3$, only $\mathbb{E}[|\chi_F|^2] = 1$ survives in $\mathbb{E}[W^2]$, so $\mathbb{E}[W^2] = 1/(2N^2)$ and the coefficient is $1/(48N(N^2 - 1))$. The $\text{SU}(2)$ value $1/144$ reproduces the admitted AW1 coefficient; it is a check of the convention, not a transferred value, and no $\text{SU}(2)$ value enters any other cell. The first-order coefficients carry the tier `exact_first_order`; the moments and the higher-order coefficients carry a route and no tier. The same coefficients hold on the box models, because every retained face other than W shares at most one link with it.

Table 23: Haar moments $\mathbb{E}[W^k]$, $k = 1, \dots, 4$, and first-order Wilson coefficients under the frozen convention, identical on both routes.

Group	$\mathbb{E}[W]$	$\mathbb{E}[W^2]$	$\mathbb{E}[W^3]$	$\mathbb{E}[W^4]$	C_F	$\omega(W)/\tau$ at first order
$\text{SU}(2)$	0	1/4	0	1/8	3/4	1/144
$\text{SU}(3)$	0	1/18	1/108	1/108	4/3	1/1152
$\text{SU}(4)$	0	1/32	0	7/2048	15/8	1/2880
$\text{SU}(5)$	0	1/50	0	3/2500	12/5	1/5760
$\text{U}(1)$	0	1/2	0	3/8	1	1/96
Z_2	0	1	0	1	1	1/48
$\text{SO}(3)$	0	1/9	1/27	1/27	2	1/864

The cubic moments $\mathbb{E}[\chi^a \bar{\chi}^b]$, $a + b = 3$, all vanish except for $\text{SU}(3)$, where the invariant ε tensor in $F^{\otimes 3}$ gives $\mathbb{E}[\chi^3] = \mathbb{E}[\bar{\chi}^3] = 1$, and $\text{SO}(3)$, where the real vector representation carries ε_{ijk} and every cubic moment is 1. The forward also exports $\mathbb{E}[W^5]$ (5/1296 for $\text{SU}(3)$, 1/50000 for $\text{SU}(5)$, 2/81 for $\text{SO}(3)$, zero otherwise).

8.1.6 The obstruction cells

On $H_{FG}(G)$ the second-order coefficient of $\omega(W)$ is $(\psi_1, W\psi_1) + 2(W\Omega_0, \psi_2) = 3a^2 \mathbb{E}[W^3]$, so a nonzero third moment gives a nonzero even coefficient.

- **SU(3)** ($a = 1/128$): $\mathbb{E}[W^3] = 1/108$, $d\omega(W^2)/d\tau = 1/6912$, and second-order coefficient $1/589824$. The level $\{\chi_F, \chi_{\bar{F}}\}$ splits at first order by $\pm 1/18$.
- **SO(3)** ($a = 1/192$): $\mathbb{E}[W^3] = 1/27$, $d\omega(W^2)/d\tau = 1/2592$, and second-order coefficient $1/331776$.
- **SU(5)** ($a = 5/1152$): $\mathbb{E}[W^3] = 0$, so the parity column is a transfer. Each application of W changes the 5-ality by ± 1 , so the order- m coefficient involves Haar means of $m + 1$ factors χ_F or $\bar{\chi}_F$ with $a - b$ divisible by 5; orders 1 to 3 give no even contribution, and the first even order that can be nonzero is 4. There

$$c_4 = \frac{1}{63403380965376} = \frac{1}{2^{30}3^{10}} \approx 1.57720295790 \times 10^{-14}, \quad (93)$$

by three evaluations: the direct Rayleigh–Schrödinger series, Hellmann–Feynman $c_4 = -15E_5$ with $E_5 = -1/951050714480640$, and a closed form. The forward’s closed form uses the single channel $\Lambda^2 \leftrightarrow \Lambda^3$ inside ψ_2 ; the reverse’s uses the column chain $\Lambda^1 \rightarrow \Lambda^2 \rightarrow \Lambda^3 \rightarrow \Lambda^4 \rightarrow \text{trivial}$, which gives $c_4 = (10/81) 10^{-5} / \prod_k E(\Lambda^k)$ with $E(\Lambda^k) = 384/5, 576/5, 576/5, 384/5$. $\text{SU}(5)$ is therefore a flip obstruction with the parity column a transfer.

The one-plaquette series through order 5 are exported for all seven groups; the $\text{SU}(2)$ row reproduces the admitted AW1 fixture $\tau/144 - (5/11943936)\tau^3$. As a supplementary argument, not used in

place of the coefficients, a unitary reversing W on $H_{FG}(G)$ needs the range of W to be symmetric; it is $[-1/2, 1]$ for $SU(3)$ and $[-1/3, 1]$ for $SO(3)$, and -1 is not attained for $SU(5)$.

8.1.7 The transfer ledger

Table 24 is the ledger both producers export. Every transfer is to $H_{FG}(G)$ and H_N^G : the operator identity in every box and cutoff, and the oddness and first-order statements in each box inside its Kato radius.

Table 24: The BD1 transfer ledger. T: transfer to the named models with its own exact checks. O: obstruction, with its exact counterexample on $H_{FG}(G)$.

Group	Flip lemma	Parity theorem	First order	AM2/AQ chain, dictionary
SU(2)	T ($-I$; the admitted AW1 lemma)	T	1/144 (check)	admitted reference
SU(3)	O: $c_2 = 1/589824$ (reason $\mathbb{Z}_3$)	O: $d\omega(W^2)/d\tau = 1/6912$	1/1152	obligation
SU(4)	T ($-I$)	T	1/2880	obligation
SU(5)	O: $c_4 = 1/(2^{30}3^{10})$ (reason $\mathbb{Z}_5$)	T	1/5760	obligation
U(1)	T ($e^{i\pi}$)	T	1/96	obligation
Z2	T (nontrivial element)	T	1/48	obligation
SO(3)	O: $c_2 = 1/331776$ (trivial centre)	O: 1/2592	1/864	obligation

For every group other than $SU(2)$ the AM2/AQ chain is an obligation: the AM2 re-instantiation with its finite-volume ground state and gap, the AV1 product split, the AQ1 construction and dynamics with the AQ2 gap, the BB1 and BB2 comparisons, and a dictionary to a bare coupling. The review records these as rows O1–O7 and O9 of its obligations table; row O8 covers F2 and literal vertex boxes, O10 $SU(2)$ at a nonzero selected triple and O11 periodic boxes with an odd side.

8.1.8 Flip sets

The flip set is $E_3 = \{(p, x) : p_y \text{ even}\} \cup \{(p, y) : p_z \text{ even}\} \cup \{(p, z) : p_x \text{ even}\}$, the admitted AW1 set, and in two dimensions $E_2 = \{(p, x) : p_y \text{ even}\}$. The reverse derives E_3 before reading the contract: of the 27 rules “link (p, d) in E if and only if $p_{\sigma(d)}$ is even”, exactly two are odd on all 24 plaquette classes, the cyclic rule, which is E_3 , and the anticyclic rule, and they differ by a centre gauge transformation. Table 25 gives the enumeration on the boxes.

Table 25: E_3 on the open centered whole-star boxes: every owned plaquette and every retained face meets E_3 an odd number of times.

N	Owned links	Plaquettes	Met once	Met three times	Retained faces (once / three times)
2	3000	2335	1082	1253	1344 (672 / 672)
3	8232	6909	3630	3279	4536 (2268 / 2268)
4	17496	15291	7348	7943	10752 (5376 / 5376)

The $N = 2$ counts reproduce the AW1 forward report. On seven probed coarse factors every one of the 52 faces meeting the factor (49 omitted, 3 selected) meets E_3 oddly; E_3 holds 16 links of a factor with even z and 8 with odd z , so it is not invariant under odd coarse translations in z , which the flip lemma does not need. E_2 meets every plaquette of $[-N, N]^2$ exactly once (16, 36 and 64 plaquettes for $N = 2, 3, 4$).

Periodic tori: the GF(2) reading. On periodic tori with an odd side, E_3 and E_2 meet seam plaquettes evenly: 16 even plaquettes on $3 \times 4 \times 4$, $4 \times 3 \times 4$ and $4 \times 4 \times 3$, 24 on $3 \times 3 \times 4$, 27 on $3 \times 3 \times 3$, and 4 on the 4×3 torus. E_2 , keyed to p_y only, still meets every plaquette of the 3×4 torus once. Gaussian elimination over GF(2) shows more: a link set odd on every plaquette of a torus exists exactly when every coordinate plane has an even plaquette count, that is, when at most one side is odd in three dimensions, or when $L_x L_y$ is even in two. The contract’s periodic “obstruction” therefore concerns E_3 and E_2 at the seam, not every flip set (contract defect C4). These are recorded findings, not claims: no flip or parity statement is made on any periodic box with an odd side.

8.1.9 SU(2) area parity

For SU(2) in the zero-selected family at $\kappa = 0$, $|\tau| \leq 10^{-8}$ and both signs, let W_C be the half-trace of the holonomy of a closed path C of plaquette edges and $A(C)$ the number of plaquettes of a spanning surface. Every plaquette meets E_3 oddly and every interior link of a surface is counted twice, so for every spanning surface S of C , $|S| \equiv |C \cap E_3| \pmod{2}$, independently of S ; in particular every closed surface of plaquettes has an even number of them. In every open centered whole-star box, every on-site cutoff and for the untruncated ground vectors, the AW1 identity $U_E H_N(\tau) U_E^* = H_N(-\tau)$ and AM2 simplicity give

$$\omega_{N,-\tau}(W_C) = (-1)^{A(C)} \omega_{N,\tau}(W_C), \quad (94)$$

because -1 is central and each traversal of a flipped link contributes one factor -1 . For the limit of the named constructions the identity holds pointwise at each sign: U_E is a product over links, so $\rho_Y^{F1,N}(-\tau) = U_{E \cap Y} \rho_Y^{F1,N}(\tau) U_{E \cap Y}^*$ for the region Y owning the links of C , and the BB2 whole-sequence bound (52) with $c'_{\text{site}} = 2/984375$ and $q = 1/64$ carries it to the limit at each sign separately. More generally $\omega_\infty^- = \omega_\infty^+ \circ \alpha_E$. The AW1 whole-set statement alone would give this only as sets of subsequential limits, and using it for a pointwise claim is a rejected mutation. Observables fixed by U_E , such as W_C^2 , are even in τ .

The surface identity is audited on nine loops with two disjoint spanning surfaces each and five closed surfaces (6, 10, 14, 24 and 32 plaquettes, including a ring of eight cubes of Euler characteristic 0) in the forward, on 6480 rectangles and 400 random surfaces in the reverse, and on 216 planar loops in the skeptic’s pre-comparison check. The audits check the identity; they do not replace the proof. The statement is made only for SU(2), the zero-selected family, $\kappa = 0$, open centered whole-star boxes, their cutoffs and the limit of the named constructions; not for periodic boxes with an odd side, F2 or literal vertex boxes, nonzero selected triples or other groups.

8.1.10 Two routes and the review

The forward route computes each moment three ways inside the character route (iterated action on the character basis, iterated tensor multiplicities, and closed-form counts by Schur–Weyl duality with the hook-length formula, charge balance, parity and Catalan numbers) and adds a labelled internal Weyl cross-check that is never an admission value. The reverse route never uses a character table: it integrates on the SU(N) torus by constant terms against the Weyl density, with exponents taken modulo the diagonal, on the SO(3) torus with density $(2 - z - z^{-1})/2$, by the binomial constant term for U(1) and by summation for Z2, and it applies the electric operator to Weyl numerators. The forward checker runs 37 exact checks and the reverse 64, each with all 22 controls as damaging mutations (111 and 110 rejections); both replay byte for byte under $-B$ and $-B -0$.

The skeptic computed every value by three routes of its own (characters, Weyl constant terms, and Frobenius and hook-length counts). The moments for $k = 1, \dots, 4$, the first-order coefficients and derivatives, the one-plaquette series and the SU(5) value $\mathbb{E}[W^5]$ agree exactly across forward, reverse and skeptic, 141 cells in all, with no single-route cell. The review records a trap it tested. On the U(N) torus instead of the SU(N) torus the SU(3) third moment would be 0 and the SU(4) fourth

moment 3/1024; both producers reduce modulo the diagonal, return 1/108 and 7/2048, and abort under a $U(N)$ substitution. The reverse committed at 05:53:54 UTC and the forward at 05:54:17. The skeptic's values were final at 05:45:05, before either producer commit, and its package was committed last, at 06:10:08; its exposure before that was to file names, hashes and possibly commit subject lines. The post-comparison checker runs 29 checks and 71 source-edit runs: 44 validator weakenings, 17 must-abort substitutions, and 6 silent edits, of which 2 are harmless and 4 are caught only by the skeptic's validator. It also rejects 27 damaged packets. Both closures of 27 files verify, each with its 23 declared inputs. The review found no blocking issue. It records a wrong preview digit string for c_4 in the reverse report, whose results carry the exact value; the forward's literal flip-scope field beside its named groups; two reverse fields outside the contract list; and both reports written through the shell after a tool refusal.

Contract defects. Seven are recorded, none blocking: the preregistered tier rule conflicts with the tier control for second- and fourth-order coefficients, and the control governs (C1); the frozen flip scope field is a description, so the gate names the groups (C2); “unique” means finite-box simplicity (C3); the periodic obstruction concerns E_3 and E_2 at the seam (C4); the two-route table comparison is a pair-level comparison in a paired loop (C5); the contract does not state that the box coefficient equals the plaquette value, which both producers prove (C6); and the inherited AW1 display sign (C7).

8.1.11 Limitations recorded by the gate

The BD1 gate records the following limitations verbatim.

- Scope: statements about named finite models only, $H_FG(G)$ (finite graphs; `model_is_finite_graph true, transfers_to_aq false`) and H^G_N box by box inside a Kato radius that shrinks like N^{-3} ; nothing is claimed for all N at once for any group other than $SU(2)$, no limit state is constructed for any other group, and no AM2, AV1, AQ, BB1 or BB2 statement and no dictionary to a bare coupling is transferred to $SU(3)$, $SU(4)$, $SU(5)$, $U(1)$, Z_2 or $S_0(3)$; each is recorded as an obligation.
- The group-cell oddness statements use simplicity of the finite-box ground eigenvalue from Kato's theorem with a sufficient, not sharp, radius (12 C_F/F_N with $F_N=21(2N)^3$ retained faces; 48 C_F on one plaquette); this is simplicity in a finite box, not uniqueness of any ground state, and the word “unique” in contract items 1 and 6 is read that way.
- No flip or parity statement is made on any periodic box with an odd side. The recorded obstruction is that E_3 and E_2 meet seam plaquettes evenly when the keyed coordinate has an odd side (16 on $3 \times 4 \times 4$, $4 \times 3 \times 4$ and $4 \times 4 \times 3$; 24 on $3 \times 3 \times 4$; 4 on the 4×3 torus); E_2 still meets every plaquette of the 3×4 torus once, and by $GF(2)$ elimination some flip set exists whenever every coordinate plane has an even plaquette count (at most one odd side); these are recorded findings, not claims.
- The $SU(2)$ corollary inherits the AW1 flip lemma, AM2 simplicity and cutoff removal and the BB2 whole-sequence convergence with their own limitations (the named constructions F1 and F2 only; no statement about other states or boundary conditions); it is made at $\kappa=0$ only, not at a nonzero selected triple and not for F2 or literal vertex boxes; the loop and surface enumerations audit the identity $|C \cap E_3| = A(C) \bmod 2$ and do not replace the proof.
- Exact rational arithmetic decides every value; decimals are truncated previews (the reverse report prints 1.57720727258e-14 for ω_4 of $SU(5)$, a text slip; the exact value 1/63403380965376 and the results preview 1.57720295790e-14 stand). Kato's theorem, Peter-Weyl, the Weyl integration and character formulas, Schur-Weyl duality and Schur's lemma are cited, not machine-checked; the finite enumerations and fixtures audit their finite consequences.
- Contract defects recorded, none blocking: the preregistered `tier_label_rule` conflicts with the `tier_mixing_rejected` control for second- and fourth-order coefficients (both producers follow the control: route named, no tier); `gate_fields_required flip_transfer_scope` is a description, so the gate names $SU(2)$,

- SU(4), U(1) and Z₂; items 1 and 6 say "unique" for finite-box simplicity; the periodic-torus obstruction concerns E₃ and E₂ at the seam, not every flip set.
- Review standing: model-agent skeptic with correlated ancestry (same model family as both producers and the advisor); its pre-comparison package was committed after both producer packets, with its values final before either producer commit; not human peer review; scientific priority is not verified; weak coupling and the continuum are not addressed, and no estimate is made uniformly in the lattice spacing a .

Full derivations and reviewed implementations: [BD1 forward](#), [BD1 reverse](#), [independent model-agent review](#) and the [frozen contract](#); the transferred statements are those of the [AW1 gate](#), which is not edited.

8.2 BD2: new observables and a new dimension in SU(2)

Contribution HNM-C-BD2. The BD2 contract, recorded under the title “Applications II: new observables and a new dimension in SU(2) - the 1x2 Wilson loop, the electric energy band and the two-plaquette graph with independent couplings, and the 2+1D AM2 constants”, applies admitted SU(2) equations to four new cases ([selection note](#)). It is investigation 8 of 8 and single-direction: one forward producer, with 33 shared premises, and the skeptic’s replay from the contract. Its blind discriminating target is the relative width of the certified enclosure of the 1x2 loop on the graph at $D = 8$, at most $1/10^{16}$; the other items are exact computations and format checks. It lists 21 controls. Its gate is recorded under the title “Hruday new observables and dimensions in SU(2): the two-plaquette graph, the 1x2 loop, the electric band and 2+1D”.

8.2.1 Reviewed scope

The [BD2 gate](#) records the verdict `accepted_within_scope`, with sub-label `sign_certified_finite_graph` and secondary sub-labels `transfer_to_named_model`, `obstruction_recorded` and `static_not_dynamic`. Its gate fields set `graph_sign_certified`, `z3_1x2_formal_only`, `electric_band_claimed` and `model_is_finite_graph` to true, with the band scope “ $\omega(h_R)$ for every finite box of F1 and F2 (untruncated at fixed N) and the limit of the named constructions, both signs”. Seven fields are false: `area_law_claimed`, `transfers_to_aq`, `uniqueness_of_ground_state_claimed`, `continuum_claim`, `rate_in_a_claimed`, `weak_coupling_claim` and `scientific_priority_verified`. The accepted statement, quoted verbatim below in the gate’s ASCII notation, is the scope of every claim in this subsection.

On the Round11 two-plaquette graph $H_{FG}(l1,l2) = K - l1 W_1 - l2 W_2$ (open two-square patch, 6 vertices, 7 links, gauge-invariant sector, K the sum of the seven link Casimirs at $\rho=1$, alpha units, trace-monomial cutoffs $D=6$ and $D=8$), the coefficients through total order 4 are identical at both cutoffs: $\langle W_1 \rangle = 11/6 - 5 l1^3/864 + l1 l2^2/4212$, $\langle z \rangle = 7 l1 l2/216 - 349 (l1^3 l2 + l1 l2^3)/303264$ for the 1x2 loop z , and $\langle C_{shared} \rangle = (l1^2 + l2^2)/48 - 5 (l1^4 + l2^4)/4608 - 49 l1^2 l2^2/438048$; $\langle W_1 \rangle$ is odd in $l1$ and even in $l2$ by the one-link flips of $h1$ and $h2$, $\langle z \rangle$ is odd in each coupling and $\langle C_{shared} \rangle$ even in each; at $l1=l2=1$ with $|l|$ in $\{1/10, 1/100, 1/1000\}$ and both signs the certified enclosures of $\langle z \rangle$ are positive at $D=6$ and $D=8$, with $D=8$ relative width at most $1.79645e-17$ against the frozen $1/10^{16}$. In the zero-selected SU(2) family (AM2/AQ1, $|\tau|$ at most 10^{-8} , cover $R=\{0, e_z\}$) the 1x2 Wilson loop $W_{\{1x2\}}$ of the omitted xz faces at 0 and e_x has first-order coefficient 0 in every box, satisfies $|\omega(W_{\{1x2\}})| \leq K_2' \tau^2$ (K_2' the AY1 gate constant; about $1.34175e-12$ at $|\tau|=10^{-8}$) for every F1 and F2 box and for the limit of the named constructions at both signs, is even in τ in every F1 box and for the limit, and has the formal second-order coefficient $7/124416$ (labelled `formal_second_order_coefficient`: no sign and no certified remainder); the electric energy $\omega(h_R)$ on R lies in $[(3/2)L^2, 98|\tau|]$, about $[2.87586637818e-19, 9.8e-7]$ at $|\tau|=10^{-8}$, with L the AY2 lower distance, for every F1 and F2 box (untruncated at fixed N) and for the limit, at both signs. In the declared 2+1-dimensional model the AM2 counting gives $p=3$, $J=|\tau|$, termination order 6, $G_3(R)=9 e^{\{3/32\}}$ and $G_3'(R)=118 e^{\{3/32\}}$ at $R=1/64$, and the cap

$|\tau| \leq 1/(576 e^{\{3/32\}})$ (directed lower bound $1580747155173/10^{15}$), with no finite-volume theorem. On the Round11 two-plaquette graph with independent couplings, the exact low-order coefficients of the face mean, the 1x2 loop and the shared-link Casimir are computed with zero truncation error, the face mean is odd in its own coupling and even in the other, and the 1x2 loop has a certified sign at every grid point; in the zero-selected SU(2) family the 1x2 Wilson mean vanishes at first order and is even in τ , with only a formal second-order coefficient, the electric energy on the cover lies in a certified two-sided band for every box and for the limit of the named constructions, and the 2+1-dimensional AM2 constants are recounted; these are statements about named models and a finite graph, not an area law, not a certified sign of the 1x2 mean, not uniqueness of any ground state, and not a statement uniform in the lattice spacing a .

8.2.2 The two-plaquette graph with independent couplings

The graph is the Round11 open two-square patch [5]: vertices TL, TM, TR, BL, BM, BR and seven links h_1 (TL→TM), h_2 (TM→TR), h_3 (BL→BM), h_4 (BM→BR), v_L , v_M and v_R , with holonomies $U = v_M h_3^{-1} v_L^{-1} h_1$ and $V = h_2 v_R h_4^{-1} v_M^{-1}$, face variables $x = W_1 = \frac{1}{2} \text{Tr } U$ and $y = W_2 = \frac{1}{2} \text{Tr } V$, and the 1x2 loop $z = \frac{1}{2} \text{Tr}(UV)$, in which the shared link v_M cancels. In alpha units, on the gauge-invariant sector,

$$H_{FG}(l_1, l_2) = K - l_1 W_1 - l_2 W_2, \quad K = 3C_U + 3C_V + C_{\text{shared}}, \quad (95)$$

with K the sum of the seven link Casimirs at $\rho = 1$ and C_{shared} the Casimir of v_M . AZ2 treated the equal couplings $l_1 = l_2$ (Round32 addendum); here the two faces are coupled independently.

Exact coefficients. K is triangular in the degree, so every Rayleigh–Schrödinger vector ψ_{mn} lies in the polynomials of degree $m+n$ and the equations hold as polynomial identities for $m+n \leq 5$. The truncation error is therefore exactly 0, and the tables computed at the trace-monomial cutoffs $D = 6$ and $D = 8$ are identical. The first vectors are $\psi_{10} = x/3$, $\psi_{01} = y/3$ and $\psi_{11} = 4xy/39 + 4z/351$. Through total order 4,

$$\begin{aligned} \langle W_1 \rangle &= \frac{l_1}{6} - \frac{5l_1^3}{864} + \frac{l_1 l_2^2}{4212} + \dots, \\ \langle z \rangle &= \frac{7l_1 l_2}{216} - \frac{349(l_1^3 l_2 + l_1 l_2^3)}{303264} + \dots, \\ \langle C_{\text{shared}} \rangle &= \frac{l_1^2 + l_2^2}{48} - \frac{5(l_1^4 + l_2^4)}{4608} - \frac{49l_1^2 l_2^2}{438048} + \dots, \\ E_0 &= -\frac{l_1^2 + l_2^2}{12} + \frac{5(l_1^4 + l_2^4)}{3456} - \frac{l_1^2 l_2^2}{8424} + \dots, \end{aligned} \quad (96)$$

with every other coefficient of total order at most 4 exactly zero. The checks are exact: Hellmann–Feynman $\langle W_1 \rangle_{(m,n)} = -(m+1)E_{(m+1,n)}$ through total order 5; a second route for $\langle C_{\text{shared}} \rangle$ at second order, from $E_{20}(\rho) = -1/(3(3+\rho))$ and its ρ -derivative $1/48$; the section $l_2 = 0$, which gives the one-face value $-5/864$; and the diagonal $l_1 = l_2$, which reproduces the admitted AZ2 series ($1/6$ and $-187/33696$ for $\langle W_1 \rangle$, $7/216$ and $-349/151632$ for $\langle z \rangle$). At $l_1 = l_2$ the second-order coefficients are $7/216$ for $\langle z \rangle$ and $1/24$ for $\langle C_{\text{shared}} \rangle$.

Flip parities. For a link e the flip $U_e \mapsto -U_e$ commutes with every Casimir, with the six vertex gauge actions and with Haar measure, and reverses every loop through e . Table 26 lists the actions. The ground state of H_{FG} is simple for every real (l_1, l_2) (Round11 Theorem F), so the flip of h_1 , a non-shared link of face 1, makes $\langle W_1 \rangle$ and $\langle z \rangle$ odd in l_1 and $\langle C_{\text{shared}} \rangle$ even, and the flip of h_2 , a non-shared link of face 2, makes $\langle W_1 \rangle$ even in l_2 , $\langle z \rangle$ odd and $\langle C_{\text{shared}} \rangle$ even. Coefficient by coefficient, $F_{h_1} \psi_{mn} = (-1)^m \psi_{mn}$ and $F_{h_2} \psi_{mn} = (-1)^n \psi_{mn}$. The shared-link flip alone gives only the joint statement, and reading a two-variable parity from it, or checking only at $l_1 = l_2$, is a rejected control.

Table 26: One-link centre flips on the two-plaquette graph.

Flipped link	Action on (x, y, z)	Effect on $H_{FG}(l_1, l_2)$
h_1, v_L or h_3 (face 1, non-shared)	$(-x, y, -z)$	$H(-l_1, l_2)$
h_2, v_R or h_4 (face 2, non-shared)	$(x, -y, -z)$	$H(l_1, -l_2)$
v_M (shared)	$(-x, -y, z)$	$H(-l_1, -l_2)$ only

Certified enclosures of $\langle z \rangle$. At $l_1 = l_2 = l$ the Hamiltonian is the AZ2 model, and the ground-state certificate is the admitted AZ2 certificate code, copied verbatim (63 blocks compared textually with the snapshot); only the observable changes. A rational Ritz vector with complete residual, the Weyl gap $E_1 - E_0 \geq 3 - 2|l|$, Temple’s bound, the Round11 Schur-complement tail comparison, and the Eckart and Davis–Kahan angles give $\sin \theta \leq s$, and then $|\langle v, zv \rangle / \langle v, v \rangle - \langle z \rangle| \leq 2s\sigma_z + 2s^2 \|z\|$ with $\sigma_z \leq \|z\| = 1$. Ends are rounded outward to denominators 10^{45} . Table 27 gives the results.

Table 27: Certified enclosures of $\langle z \rangle$ at $l_1 = l_2$ (both signs give the same enclosure, since $\langle z \rangle$ is even under the shared-link flip). Relative width is $(\text{upper} - \text{lower}) / \min(|\text{lower}|, |\text{upper}|)$.

D	$ l $	Lower end (preview)	Relative width	Sign
6	1/1000	$3.240740510578259506053328 \times 10^{-8}$	6.79043×10^{-22}	+
6	1/100	$3.240717724665334529490919 \times 10^{-6}$	6.81088×10^{-17}	+
6	1/10	$3.238440859055494033169085 \times 10^{-4}$	7.02377×10^{-12}	+
8	1/1000	$3.240740510578259506054428 \times 10^{-8}$	1.73689×10^{-31}	+
8	1/100	$3.240717724665334639851747 \times 10^{-6}$	1.74212×10^{-24}	+
8	1/10	$3.238440859066867050597032 \times 10^{-4}$	1.79645×10^{-17}	+

Every enclosure excludes 0 with a positive sign, and every $D = 8$ enclosure lies inside its $D = 6$ one, which is observed and not a target. The worst $D = 8$ relative width is 1.79645×10^{-17} , at $|l| = 1/10$, against the frozen $1/10^{16}$: a margin of about 5.57. At $l = +1/10$ and $D = 8$ the exact ends are

$$\begin{aligned} \text{lower} &= \frac{40480510738335838132462907353125470123319}{125 \cdot 10^{42}}, \\ \text{upper} &= \frac{64768817181337342175480593943907904225707}{2 \cdot 10^{44}}. \end{aligned} \tag{97}$$

Each point carries the five-part AZ2-type residual ledger: seven per-link rows (thresholds $123/2$ and 96 on the non-shared links, 45 and 69 on v_M), the joint product channel with its corner overlaps and the positive interference, the gauge projection, which is exactly 0 with its reason, the Ritz residual with both angles and the Weyl gap, and the directed arithmetic. The skeptic re-evaluated the lemma in its own spin-network basis with its own Ritz vectors and reproduced every angle and width to 8 digits. Its own sharper tail certificate gives a worst width of 3.56×10^{-18} (margin 28.09), with enclosures inside the producer’s, and Davis–Kahan alone would give 8.72×10^{-17} ; any of the three meets the target.

8.2.3 The $\mathbb{Z}^3$ 1x2 rectangle

In the zero-selected family the frozen rectangle is the union of the omitted xz faces at 0 and e_x , with links $(0, x)$, (e_x, x) , $(2e_x, z)$, $(e_x + e_z, x)$, (e_z, x) and $(0, z)$; the shared middle link cancels. All six links are owned by $R = \{0, e_z\}$, so $W_{1 \times 2} \in B(\mathcal{H}_R)$ with $\|W_{1 \times 2}\| \leq 1$, and on the seven links of the two faces $(W_{F1}, W_{F2}, W_{1 \times 2})$ is the graph’s (x, y, z) .

The first-order zero. Every link of $W_{1 \times 2}$ occurs once and a single face covers at most three of its six links, so $\mathbb{E}[W_{1 \times 2} W_f] = 0$ for every face and the first-order coefficient is exactly 0 in every box of F1 and F2.

The bound. Writing $\omega(W_{1 \times 2}) = \text{Tr}((\rho_R - P_R - \rho_R^{(1)})W_{1 \times 2})$, since the product and the first-order density contribute 0,

$$|\omega(W_{1 \times 2})| \leq K'_2 \tau^2 \approx 1.34175 \times 10^{-12} \quad (|\tau| = 10^{-8}), \quad K'_2 \approx 13417.5275439, \quad (98)$$

at both signs, with K'_2 the AY1 gate constant. It holds for every F1 and F2 box through the AY1 per-box ball (AY1 forward F11–F14, a reviewed step of the admitted proof, passed to the untruncated vector by AV1 F22) and for the limit of the named constructions through the AY1 and AY2 gate ball and BB2 items 2–3. The bound carries no sign.

Evenness, for F1 and the limit only. The flip set E_3 meets each of the two faces in 3 links and the rectangle in all 6, so $U_E W_{1 \times 2} U_E^* = W_{1 \times 2}$ (area 2). With the AW1 identity at $\kappa = 0$, $\omega_{N,-\tau}(W_{1 \times 2}) = \omega_{N,\tau}(W_{1 \times 2})$ in every open centered whole-star box (F1) and every cutoff, and the limit is even by BB2 at each sign. Evenness is not claimed for F2 boxes, and it gives no remainder bound.

The formal second-order coefficient. Only the ordered pairs of the two faces cover the rectangle evenly, with $\mathbb{E}[W_{F1}W_{F2}W_{1 \times 2}] = 1/16$, and $W_{1 \times 2}\Omega_0$ has energy 36, so formally

$$\omega(W_{1 \times 2}) = \left(\frac{1}{31104} + \frac{1}{41472} \right) \tau^2 + \dots = \frac{7}{124416} \tau^2 + \dots. \quad (99)$$

This is labelled `formal_second_order_coefficient`: it has no sign, no certified value and no remainder. The identity $7/124416 = (7/216)/24^2$ relates it to the graph coefficient of $\langle z \rangle$ under $\tau_{FG} = \tau/24$; it is a labelled identity, not a transfer. The certified bound exceeds the formal term by a factor of about 2.38479×10^8 , so a certified sign of the $\mathbb{Z}^3$ 1x2 mean is a recorded obstruction.

8.2.4 The electric band

The observable $h_R = 8 \sum C_e$ over the 48 links of R is nonnegative and unbounded; $\omega(h_R)$ is the monotone limit over its spectral cutoffs. For every F1 and F2 box (untruncated at fixed N) and for the limit, at both signs,

$$\frac{3}{2}L^2 \leq \omega(h_R) \leq 98|\tau|, \quad \frac{3}{2}L^2 \approx 2.87586637818 \times 10^{-19}, \quad 98|\tau| = \frac{49}{50000000} \quad (|\tau| = 10^{-8}). \quad (100)$$

Lower end. The spectrum of h_R is $\{0\} \cup [6, \infty)$ with kernel P_R , since the smallest nonzero value of $8 \sum j_e(j_e + 1)$ is $8 \cdot \frac{3}{4} = 6$; hence $\omega(h_R) \geq 6 \text{Tr}(Q_R \rho_R)$ with $Q_R = I - P_R$. For a pure reference the Fuchs–van de Graaf inequality, proved inline, gives $\frac{1}{2} \|\rho_R - P_R\|_1 \leq \sqrt{\text{Tr}(Q_R \rho_R)}$, so $\omega(h_R) \geq 6(L/2)^2 = \frac{3}{2}L^2$ whenever $\|\rho_R - P_R\|_1 \geq L$. The distance $L \approx 4.37863 \times 10^{-10}$ is the AY2 gate lower end for the limit, and the same number from the AY1 per-box ball for every box (tier `first_order_distance_from_product`).

Upper end. The AQ1 reset on $F = R$ charges the seven incident anchors, each whole star of norm $7|\tau|$, giving $\omega_N(h_R) \leq 2 \cdot 7 \cdot 7|\tau| = 98|\tau|$ for every F1 box; for F2 boxes the same budget is AY1 gate item (1) (tier `crude_majorant`). The bound passes to the limit only by lower semicontinuity through the cutoff, and an exact fixture shows why the other direction fails: $\rho_k = (1 - 1/k)P_0 + (1/k)P_k$ with $h = \text{diag}(n)$ has energy 1 for every k and a limit of energy 0.

The tight-enclosure obstruction. Formally, each link e of R contributes $n_e/3456$ at second order, with n_e the number of omitted faces containing it (28 links with $n_e = 4$, 16 with 3 and 4 with 2), which totals $168/3456 = 7/144$: formally $\omega(h_R) = (7/144)\tau^2 + \dots$, about 4.861×10^{-18} at the

cap. The upper end exceeds this by about 2.016×10^{11} and the lower end lies below it by a factor of about 16.9; the band is about 3.4×10^{12} wide in ratio. A tight enclosure needs a second-order upper bound on the unbounded h_R , an energy-weighted state estimate that is not admitted, and it is recorded as an obstruction.

8.2.5 The 2+1-dimensional recount

In the declared model on $\mathbb{Z}^2$ each site owns the links (p, x) and (p, y) , the face at base p has owner set $\{p, p+e_x, p+e_y\}$, and each face is its own interaction term, $V_X = -(\tau/3)W_f$. The AM2 counting is redone for this geometry (Table 28): the maximal support is $p = 3$, three faces contain each site, the per-site sum is $J = |\tau|$ and the creation expansion terminates at order 6, with

$$\begin{aligned} G_3(t) &= 8e^{6t}(1+8t), & G'_3(t) &= 8e^{6t}(14+48t), \\ G_3(\tfrac{1}{64}) &= 9e^{3/32} \approx 9.88457, & G'_3(\tfrac{1}{64}) &= 118e^{3/32} \approx 129.598. \end{aligned} \quad (101)$$

The numerators $2^p(2p)^k(1+k(p+1)/p)$ for $k = 0, \dots, 6$ are 8, 112, 1056, 8640, 65664, 476928 and 3359232, and a three-qubit fixture checks termination ($\text{ad}_C^6(V)\Omega = -720|111\rangle$, $\text{ad}_C^7(V) = 0$). With a directed series enclosure of $e^{3/32}$, the self-map condition $JG_3(R) \leq R$ at $R = 1/64$ binds and gives the cap

$$|\tau| \leq \frac{1}{576e^{3/32}}, \quad \text{directed lower bound } \frac{1580747155173}{10^{15}} \approx 1.5807 \times 10^{-3}, \quad (102)$$

while the exclusion condition $2JG'_3(R) < 1$ alone would allow $1/(236e^{3/32}) \approx 3.858 \times 10^{-3}$. These constants are recounted, not a theorem: no finite-volume ground-state or gap statement, no AQ chain, no node and no dictionary is claimed in 2+1 dimensions, where g^2 has mass dimension and the 3+1D dictionary does not apply.

Table 28: The AM2 constants recounted in 2+1 dimensions (3+1D values not reused).

Quantity	2+1D (recounted)	3+1D (AM2)
star and faces per anchor	$\{0, e_x, e_y\}$, one face	$\{0, e_x, e_y, e_z\}$, 21 faces
terms containing a site	3 faces	4 stars, 49 faces
per-site sum J	$ \tau $	$28 \tau $
maximal support, termination order	3, 6	4, 8
majorant $G_p(t)$	$8e^{6t}(1+8t)$	$16e^{8t}(1+10t)$

8.2.6 Obligations, no-transfer rows and code provenance

The review records six rows, each open or a no-transfer row with its missing premise:

- *area law*: analyticity of the reduced density in a complex τ -disc uniformly in N (a zero-free region of the complexified normalization) and a family of growing loops; no zero-free region is admitted;
- *a certified sign of the $\mathbb{Z}^3$ 1x2 mean*: a uniform remainder $|\omega(W_{1 \times 2}) - (7/124416)\tau^2| \leq K_4\tau^4$ with $K_4\tau^2$ below $7/124416$ for F1 boxes and the limit, and a third-order remainder for F2 boxes;
- *a tight electric enclosure*: a second-order upper bound on the unbounded $\omega(h_R)$;
- *the site-blocked uniform 3+1D regime*: its own contract;
- *the finite-volume theorem in 2+1 dimensions*: a two-dimensional finite-volume prescription, the on-site domains and AM2 sections 4–6 re-verified, then the AQ chain, node and dictionary;
- *the Einstein–QED evidence rounds*: a no-transfer row, since they share no equation with this $\text{SU}(2)$ chain.

Code provenance. The forward checker imports only the standard library. The AZ2 certificate algebra is copied verbatim with attribution and compared block by block with the pinned snapshot, and the copied code reproduces the AZ2 gate enclosure of $\langle W_1 \rangle$ exactly. The Round11 solver is read as bytes from its snapshot to bind 13 exact source strings, and it is never imported or executed; the producer agent did not open it. No interpreter cache is written under the Round11 or Round32 directories or in the closure.

Evidence and review. The forward checker runs 41 exact checks and implements all 21 controls as damaging mutations, with 107 rejections; its outputs are byte-identical under `-B` and `-B -O`. The skeptic’s pre-comparison package (64 checks, 21 controls, 85 rejected mutations with 4 positives) had its values final before the forward commit at 06:00:06 UTC and was committed 10 min 2 s after it, at 06:10:08; its exposure was to input file names, hashes and possibly the commit subject. Every exported value equals, or certifiably contains, the skeptic’s own. The post-comparison checker runs 22 checks and 38 source-edit runs: 21 validator weakenings, 12 must-abort edits, and 3 silent edits, of which 1 is harmless and 2 are export-only edits caught only by the skeptic’s validator. It also rejects 19 damaged packets. The closure of 39 files verifies, with its 35 declared inputs. The review found no blocking issue. It records the producer’s contract readings W1–W7: the single model id names only the graph; evenness is admitted for F1 and the limit only, so a certified sign for F2 would need a third-order remainder; “three faces per site” counts the faces whose owner set contains the site; the per-site sum is linear in $|\tau|$; the tier name of K'_2 ; and the sign restriction of the Round11 solver interface. It also keeps separate the two scopes of the 1x2 loop, the bound for F1, F2 and the limit and the evenness for F1 and the limit only.

8.2.7 Limitations recorded by the gate

The BD2 gate records the following limitations verbatim.

- Graph scope: the Round11 two-square patch only, at $D=6$ and $D=8$ and the declared grid; static means with no dynamical or mass-gap reading; `transfers_to_aq` false; the graph coefficients are finite-graph values, and $7/124416 = (7/216)/24^2$ is a labelled identity between the graph and the formal Z^3 coefficient, not a transfer.
- Z^3 scope: the zero-selected family at $|\tau|$ at most 10^{-8} , the fixed cover R and fixed spacing; no certified value or sign of the Z^3 1x2 mean (the admitted bound $K'_2 \tau^2$ exceeds the formal term by about $2.38479e8$; a certified sign needs a uniform fourth-order remainder, and for F2 boxes a third-order one); evenness in τ is not claimed for F2 boxes.
- The band is two-sided but loose (ratio about $3.4e12$) and is not a tight enclosure, which would need a second-order upper bound for the unbounded h_R ; its lower end rests on the AY1 forward F11–F14 per-box ball (a reviewed step, not a gate sentence) for boxes and on AY2 item (2) for the limit; its upper end passes to the limit by lower semicontinuity only.
- No area law and no string-tension statement is made: no zero-free region of the reduced density is admitted.
- 2+1 dimensions: recounted AM2 constants only; no finite-volume ground-state or gap theorem and no AQ chain, node or dictionary in 2+1 dimensions (g^2 has mass dimension); each needs its own contract.
- The enclosures use the admitted AZ2 certificate code copied verbatim (Weyl gap 3-2|||, Temple, the Round11 Schur-complement tail comparison, Eckart and Davis-Kahan angles); the skeptic re-evaluated the lemma in its own basis and has its own sharper tail certificate (worst $D=8$ relative width $3.56e-18$); Kato, Weyl, Temple, Eckart, Davis-Kahan, Peter-Weyl and Fuchs-van de Graaf are cited, not machine-checked.
- Review standing: model-agent skeptic with correlated ancestry (same model family as the producer and the advisor); its pre-comparison package was committed after the producer packet, with its values final before the producer commit; not human peer review; scientific priority is not verified; weak coupling and the continuum are not addressed, and no estimate is made uniformly in the lattice spacing a .

Full derivations and reviewed implementations: [BD2 forward](#), [independent model-agent review](#) and

the [frozen contract](#); the graph certificate is that of the [AZ2 gate](#), which is not edited.

What transferred and what did not

Table 29 collects the outcome of the applications stage. Its rows are the admitted equations that BD1 and BD2 applied, and its columns the models they were applied to. “Source” marks the model in which an equation was admitted; every other cell reads transfer, obstruction or not attempted, with its reason. A transfer is a labelled statement about the named model with its own exact checks, never a statement about another model, and “not attempted” records no failure. The Einstein–QED evidence rounds share no equation with this chain and are a no-transfer row of BD2; they are not a column.

9 Process history of the Round33 cycle

This section records how the cycle was run, using the repository files named in each paragraph. It adds no mathematical claim, and none of the planning numbers below is an admitted value. All times are UTC. The contract freezer wrote the `frozen_at` stamps and the gate recorder the `completed_at` stamps, both from the machine clock; like commit dates, they are self-reported and are not trusted timestamps. The roles are those of the status section: an advisor who owns contracts, gates and publication; a skeptic; three methodological lenses; one research assistant per lens after each sub-round; and forward and reverse producers. All of them are model agents with correlated ancestry, working under the project’s workflow contract [1].

9.1 Authorization and brief

After the merged Round32 cycle [4], the user asked on 24 September 2026 to “continue with the next 3 rounds - and then apply all these solutions and new equations to solve their related problems - if any”. As in Round32, a round was read as a sub-round of two investigations with the advisor, the skeptic and the expert panel, one research assistant per lens after each sub-round, and goal changes allowed after each. The advisor’s brief [6] therefore fixed a panel deliberation of two loops, three research sub-rounds (BA, BB, BC), an applications stage (BD) that applies the admitted equations where they apply and records an explicit obstruction where they do not, and then integration and release. The brief restated the admitted Round32 equations and constants, the ranked Round32 roadmap and the ten closing-panel rules of Round32. The contract freezer, committed with the Round33 skeleton at 20:54 on 24 September, enforces rules R1–R5 and R8; the gate recorder enforces R6 and R7. The brief also listed the advisor’s candidate directions, boundary-influence decay, a Lieb–Robinson comparison of the dynamics, restated certificates and a list of applications, as triage input, not decisions.

9.2 Two deliberation loops and plan v2

Loop 1: memos, triage and plan v1. Each lens wrote a memo, a source ledger and a structured recommendation [8], the Jung lens also kept the occult and government-record reading ledger [9], and the skeptic wrote a prospective triage with a control catalogue [7]. The advisor’s synthesis records six points of convergence. The first corrected the brief: the smallness $JG(R) < 1/64$ is absolute, and the per-step rate is the ball ratio $64JG(R) = 37888|\tau|$, about 3.79×10^{-4} at the cap, while the global Lipschitz constant is never a decay factor. The others were that a weighted norm and complex-coupling analyticity give the same coefficient decay; that coefficient decay is not decay of the reduced density, so marginal locality needs a new lemma and is the round’s main risk; that the dynamics comparison is mechanical once the extra faces and their distances are counted;

Table 29: What transferred and what did not. Columns: the SU(2) zero-selected family (boxes and the limit of the named constructions); the one-plaquette and box models $H_{FG}(G)$, H_N^G of the other groups; the Round11 two-plaquette graph with independent couplings; the declared 2+1D SU(2) model.

Equation	SU(2) family	SU(4), U(1), $\mathbb{Z}_2$	SU(5)	SU(3), SO(3)	Two-plaquette graph	2+1D model
Link-flip lemma (AW1)	Source; used for the area parity	Transfer: central element acting as -1 ; box by box	Obstruction: $c_4 = 1/(2^{30}3^{10})$; no central -1	Obstruction: $c_2 = 1/589824$, $1/331776$; no central -1	Transfer: one-link flips give the parity in each coupling	Not attempted: constants recounted only
First-order parity theorem (AW1)	Source (finite boxes)	Transfer: $\mathbb{E}[W^3] = 0$	Transfer: $\mathbb{E}[W^3] = 0$ by the $\mathbb{Z}_5$ grading	Obstruction: $d\omega(W^2)/d\tau = 1/6912, 1/2592$	Not attempted: parities come from the flips	Not attempted: no finite-volume theorem
First-order Wilson coefficient $\tau/144$ (AW1)	Source; reproduced as a convention check	Transfer of the method, own values $1/2880, 1/96, 1/48$; $1/144$ not transferred	Transfer of the method, own value $1/5760$	Transfer of the method, own values $1/1152, 1/864$	Transfer: $\langle W_1 \rangle = l_1/6 + \dots$ with independent couplings	Not attempted
Area parity (AW1 flip with BB2)	Transfer at $\kappa = 0$: every open centered whole-star (F1) box and pointwise for the limit; not claimed for F2 boxes (so the 1×2 loop is even for F1 and the limit, not F2)	Not attempted: no limit and no regime for all N	Not attempted: no flip	Not attempted: no flip	Not attempted	Not attempted
AM2, AV1, AQ1–AQ2, BB1–BB2 chain	Source	Not attempted: group-specific AM2 re-instantiation is an obligation	Not attempted: obligation	Not attempted: obligation	Not attempted: finite graph, no AQ transfer	Not attempted: AM2 constants recounted, cap about 1.5807×10^{-3} ; no theorem
Coupling dictionary $\tau = 96/g^4$	Source	Not attempted: a group-specific dictionary is needed	Not attempted	Not attempted	Not attempted	Not attempted: g^2 has mass dimension
R -local bound $K_2'\tau^2$ (AY1)	Transfer: $ \omega(W_{1 \times 2}) \leq K_2'\tau^2$ for F1, F2 and the limit; certified sign an obstruction	Not attempted	Not attempted	Not attempted	Not attempted: $7/124416 = (7/216)/24^2$ is a labelled identity only	Not attempted
Reset budget and product distance (AQ1, AY1, AY2)	Transfer: $\frac{3}{2}L^2 \leq \omega(h_R) \leq 98 \tau $; tight enclosure an obstruction	Not attempted	Not attempted	Not attempted	Not attempted	Not attempted
Two-plaquette certificate (AZ2)	Not attempted: finite-graph certificate, no AQ transfer	Not attempted	Not attempted	Not attempted	Transfer: independent-coupling tables through total order 4; $\langle z \rangle$ certified positive only at $l_1 = l_2 = \pm 10^{-k}$, $k = 1, 2, 3$ ($D = 6, D = 8$), $D = 8$ relative width $\leq 1.79645 \times 10^{-17}$	Not attempted

that consequences restate and never strengthen; and a first list of applications, with the $SU(3)$ obstruction and the first-order coefficients per group recorded as previews for the producers to derive. Four dissents were recorded. The Jung lens asked for a separate Dobrushin/HTW loop, which the advisor did not schedule. The historical lens asked to split the dynamics into a derivation and a certification, adopted in substance. The modern lens reserved the word uniqueness for its strong form and set the Cauchy rate at N^{-1} , both adopted. The skeptic asked for margins of at least 2, frozen rate–constant pairs and per-constant scaling brackets, adopted as contract rules. Plan v1 fixed the four sub-rounds of Table 4.

Loop 2: sign-offs and the vetoed freeze. All three lenses signed off plan v1. The skeptic signed off the architecture with thirteen required changes, P1–P13, and vetoed the BA1 and BA2 freezes until its blocking edits were applied; its loop-2 review lists 16 blocking and 9 non-blocking entries for BA1 and 14 and 6 for BA2 [7]. Among the edits: the BA1 floor target moved from $1/16$, whose margin was 1.99924, to $1/12$; the BA2 reverse route was replaced, since a triangle inequality through the two limits misses the comparison target at every N ; the within-family Cauchy targets were quantified over all $M > N$; and the Nachtergaele–Sims theorem was committed as a source excerpt [11]. The freezer’s placeholder detector had misread inequality notation as a placeholder span and was repaired before the freeze (rule R1). The Jung lens kept a narrowed dissent, answered in the BB selection note. The loop-1 panel previews that justify the frozen margins are recorded only in the loop-2 synthesis, which is not a producer premise, so the producers derive their constants blind.

Plan v2. The changes P1–P13 were recorded in `plan.json` and plan v2 was frozen before BA1 and BA2 (Section 1.4). They fixed the forbidden phrasings of the round (P1), one tier and one route per constant (P2), a closed gate-field vocabulary (P3), the bracket, margin, excerpt and Cauchy rules (P4), the plan hash recorded at each freeze (P5), the directions of BB2, BC1, BC2, BD1 and BD2 (P6 and P8–P11), the isolation of the BB1 reverse from the lens memos and the triage (P7), and the recording of the changes (P12, P13). The plan history also carries a later correction entry for the BB1 secondary pair, recorded beside the earlier entry without editing it.

9.3 Contracts, gates and chronology

Table 30 lists the eight contracts in freeze order. Each pair was frozen together, after a skeptic pre-freeze review of its drafts whose blocking edits were applied field by field. Five investigations are paired (BA1, BA2, BB1, BB2 and BD1), two are single-direction with a skeptic replay (BC2 and BD2) and one is a statement investigation (BC1).

The cycle ran from the BA freeze at 21:56 on 24 September to the BD2 gate at 07:53 on 25 September.

Sub-round 1: locality. BA1 and BA2 were frozen at 21:56:46 on 24 September, after the loop-2 edits. Their gates were completed at 23:08:07 (BA2) and 23:11:31 (BA1), about 71 and 75 minutes after the freeze.

Sub-round 2: the parallel freeze. BB1 and BB2 were frozen together at 23:36:46, after the skeptic’s pre-freeze review of the BB drafts, which vetoed both freezes until 7 blocking edits to BB1 and 15 to BB2 were applied, with 10 and 5 non-blocking ones. BB1 was still to be produced, so BB2 takes BB1’s frozen targets as explicit hypotheses, and its producers marked every dependent statement as conditional on them. The plan history entry for this freeze was written afterwards, as the BB2 skeptic’s defect D8 records. The BB1 gate was completed at 03:01:43 and the BB2 gate at 03:31:54, after the skeptic’s admission record had discharged every BB1 hypothesis item by item

Table 30: Round33 chronology from the contract and gate records (UTC, 2026). Dir.: P = paired, S = single direction plus skeptic replay, T = statement plus skeptic. Every verdict is `accepted_within_scope`.

Loop	Dir.	Contract frozen	Gate completed	Edits (B / NB)	Sub-labels
BA1	P	09-24 21:56:46	09-24 23:11:31	16 / 9	BD; SD
BA2	P	09-24 21:56:46	09-24 23:08:07	14 / 6	DW
BB1	P	09-24 23:36:46	09-25 03:01:43	7 / 10	BD; SD
BB2	P	09-24 23:36:46	09-25 03:31:54	15 / 5	CN; CL
BC1	T	09-25 04:21:44	09-25 05:24:16	30 / 28 (both)	CR; RU, SD
BC2	S	09-25 04:21:44	09-25 05:57:18	(with BC1)	CN; CR, RU, SD
BD1	P	09-25 05:09:10	09-25 07:52:23	34 / 21 (both)	TN; OR
BD2	S	09-25 05:09:10	09-25 07:53:00	(with BD1)	SF; TN, OR, SD

Sub-labels, primary before the semicolon: BD = `boundary_decay_rate_only`; SD = `static_not_dynamic`; DW = `dynamics_on_compact_windows`; CN = `convergence_of_named_constructions`; CL = `common_limit_of_named_constructions`; CR = `certificate_restated_for_limit`; RU = `reference_unresolved`; TN = `transfer_to_named_model`; OR = `obstruction_recorded`; SF = `sign_certified_finite_graph`. Edits: blocking (B) and non-blocking (NB) pre-freeze edits. The BA counts are the loop-2 review’s entries; the BB counts are per contract; the BC and BD counts are the reviews’ totals over both contracts and the plan.

against the BB1 gate: 30 anchored entries, each a verbatim substring of the gate, cross-checked in 40 cells (Section 5.2).

The Kotecký–Preiss repair. The BB1 forward producer cited the Kotecký–Preiss criterion [13] in the form of Ueltschi [12] without a committed excerpt, because the rule of plan v2 that external theorems be quoted from committed excerpts had not been carried into the BB1 contract (the BB1 review’s contract defect D2). The skeptic ruled this a limitation, not a blocking issue. The advisor committed an excerpt of Ueltschi’s paper at 02:51:54 on 25 September, after both BB1 freezes and outside every producer’s inputs, and the skeptic’s repair record compares the cited statement with it post hoc [7]. No frozen file and no bound value changed (Section 4.3).

Sub-round 3: consequences. The BC drafts were written after the BB2 gate. The skeptic’s pre-freeze review found 2 blocking issues for BC1, 5 for BC2 and one shared with the plan, the closure of the gate-field vocabulary, and required 30 blocking edits (6 to BC1, 17 to BC2 and 7 to the plan) and 28 non-blocking ones. BC1 and BC2 were frozen at 04:21:44 on 25 September. BC2 ran only because BB1 and BB2 had been accepted within scope, as the plan required. The gates were completed at 05:24:16 (BC1) and 05:57:18 (BC2).

Sub-round 4: applications. Since no BD item depends on a BC outcome, the BD drafts were written during the BC production, at 04:27. The skeptic’s pre-freeze review found 5 blocking issues for BD1 and 6 for BD2 and required 34 blocking edits (15 and 19) and 21 non-blocking ones. It also found the drafted BD2 width target of $1/10^{10}$ non-discriminating, and the advisor tightened it to $1/10^{16}$ at $D = 8$. BD1 and BD2 were frozen at 05:09:10, and their gates were completed at 07:52:23 and 07:53:00.

The commit-order record. For every loop the skeptic’s pre-comparison package and the producer packets were committed separately, and each gate records the order (Table 31). Where the skeptic’s package was committed after a producer’s, the review records that its values were final before that producer’s commit, and every skeptic discloses its name-only exposures, such as file names in a `git status` or a commit subject line. Isolation rests on this disclosure and on content, not on the order alone.

Table 31: Commit order of the producer packets and the skeptic’s pre-comparison packages (UTC).

Loop	Order
BA1	skeptic 09ae71e (22:36:35) before forward 721ffe5 (22:38:15) and reverse e87511b (22:41:21)
BA2	skeptic 7daf273 (22:29:25) before reverse 9f207fe (22:34:30) and forward 302347b (22:34:46)
BB1	skeptic 09d0bdb (00:28:59) before forward 33e0ae6 (00:49:30) and reverse 13ac349 (02:06:29)
BB2	forward d05efda (00:10:26), then skeptic 8921539 (00:14:56), then reverse 17c0ecf (00:18:36)
BC1	forward 5816e0b (04:57:15), then skeptic 5ec599f (05:00:47); the skeptic’s values were final at 04:53:13
BC2	skeptic 5ec599f (05:00:47) before forward ee45eed (05:04:31)
BD1	reverse a1010a4 (05:53:54) and forward 5a792b0 (05:54:17), then skeptic bf1f531 (06:10:08); the skeptic’s values were final at 05:45:05
BD2	forward 37fa623 (06:00:06), then skeptic bf1f531 (06:10:08); the skeptic’s values were final before the forward commit

Closing statement. The Round33 cycle consists of exactly eight gated investigations, BA1 to BD2, in three research sub-rounds and an applications stage, and all eight are accepted within their stated model scope. Nothing after the BD2 gate, including the findings record, the closing lens updates, the roadmap, this manuscript and the release checks, adds a research loop or changes an admission.

9.4 Sub-round updates, assistants and panel updates

After each sub-round each lens reviewed the gates and wrote an update, one assistant per lens ran the tests that lens had asked for, and the advisor wrote a panel update [6, 8]. The assistant packages count zero research loops, and nothing in them decides an admission. Every panel update kept the goals of the remaining sub-rounds and recorded that no occult, mystical or historical source changed any premise.

Panel update 1 (24 September, after BA1 and BA2). The historical assistant confirmed every count in both gates by an independent enumeration, among them the $28N(5N + 1)$ extra faces, the padding sites and the nested source counts. The Jung assistant found the phrase scan of the BA gate, review and report text clean, and listed four bare “rate” phrasings in the frozen BA contracts, which stay unamended. The modern assistant reproduced every admitted BA1 and BA2 constant inside Arb and mpmath enclosures and reconciled the 0.68% gap between the two loop-1 dynamics previews as two valid directed remainder bounds. The update adopted the modern lens’s BB targets, froze BB2 with BB1 on BB1’s targets, named the fixed- N change of boundary prescription as its own BB1 item, and aligned the scanner with the plan vocabulary.

Panel update 2 (25 September, after BB1 and BB2). The historical assistant enumerated the every-site common-core lemma for all five BB1 comparisons at $N = 2, 3, 4$, found exactly 8 coarse translations among 64 fine ones, and found the first vacuous point of the item-5 bracket at $N = 14419$ independently of the skeptic. The Jung assistant found the four BB packets clean under the scanner and one closed-vocabulary defect, which the BB2 gate records. The modern assistant reproduced every BB1 constant of both routes and found the Kotecký–Preiss side conditions satisfied. The update carried three lessons into the next contracts: every rate claim carries its certified range of N in the same clause; an external theorem that a route will use is a declared premise at freeze with its committed excerpt; and direct comparisons carry the targets, with union-volume and telescoped variants labelled only. It let BC2 run and adopted the modern lens’s BC2 targets.

Panel update 3 (25 September, after BC1 and BC2). The historical assistant enumerated the route-B partition and incidence independently, recomputed $N_{\text{sign}} = 4$ and found the 21 reviewed obligation rows well formed. The Jung assistant found BC1 and BC2 clean under the scanner and ran a vocabulary-gap audit. It found no affirmative unscoped uniqueness phrasing, but identified a mechanism by which a heading followed by a list severs an exclusion cue from the listed items, and it recommended scanner additions. These were recorded for the next round and not applied mid-round, since a scanner change during the round would change the admission test of text already recorded. The modern assistant reproduced every BC2 constant exactly and inside Arb enclosures, and recomputed the AX2 radius exactly. The update kept the applications stage unchanged.

Closing lens updates and the roadmap. After the BD gates each lens wrote a closing update with a ranked proposal of at most five goals [8], and the advisor recorded the findings, the ranked roadmap and the handoff [6]. None of the three closing updates read a new source. The applications-stage assistant packages, one per lens, are recorded under `experts/<lens>/assistant-4/`; like the earlier ones, they count zero investigations and decide no admission.

9.5 Process rules and tools

The shared tools compute no scientific result [10]. The contract freezer enforces rules R1–R5 and R8 at freeze time and requires the pre-registration control list to equal the contract’s. The freezing tool snapshots the declared premises into each producer’s inputs, replays the checker under normal and `-O Python` into fresh external directories, requires byte-identical outputs and writes the freeze record; it also machine-checks the isolation of a reverse producer’s inventory. The negation-aware phrase scanner, added at the start of the round, checks the round’s forbidden phrasings and each contract’s own list, with the mandatory sentence template removed as one literal; the gate recorder applies it and the template-span check at gate time. All admission arithmetic uses exact `fractions.Fraction`.

Four lessons of the round are recorded as rules for the next contracts, in the panel updates and the handoff: every rate claim carries its range of N ; external theorems are committed excerpts declared at freeze; an obligation row that an admitted result settles only in part is split and frozen with its reviewed name, as BC1’s O1a and O1b were; and a single-direction loop names an adversarial substitution of another model’s constants as a required control, as BC2’s 22 route-A substitutions did. The scanner’s vocabulary gap found in BC1 is a methods decision of the roadmap (Section 10.1), not a mid-round change.

9.6 Sources, attribution and disclosure

Two external sources were committed as excerpts with their PDF hashes: the Nachtergaele–Sims paper, before the BA freeze, for the Lieb–Robinson inequality and the limit dynamics used by BA2 and BB2 [11], and Ueltschi’s paper, after the BB1 freezes, as the repair source above [12]. The AQ1 construction and the AM2 creation expansion are inherited with the Round32 attribution sentence [4], which credits their strategy to Gauvin [14]. The lens ledgers record the sources the lenses read in Round33, at depths from search summaries to selected sections: four records in the historical ledger, two in the Jung ledger and 27 in the modern ledger, with the occult and government-record reading ledger kept by the Jung lens [8, 9]. They motivated methodological rules only. *No occult, mystical or historical source changed any premise*, and no historical figure participated in or endorsed the work.

AI assistance. The advisor, the skeptic, the lenses, the assistants and the producers are AI model agents directed by the human author; several producer reports identify their author as a Claude model agent. The text of this addendum was also drafted with AI assistance. The agreement of these roles is correlated model-agent work, not external human peer review.

10 What the Round33 record does and does not establish

All eight Round33 gates record the verdict `accepted_within_scope`, and each acceptance holds only inside the model its gate names. The record uses five models, and they are kept apart:

- the zero-selected patterned family of the SU(2) Kogut–Susskind form at fixed spacing and $|\tau| \leq 10^{-8}$ [15], with the two named construction families F1 and F2, their finite boxes and the limit of the named constructions (BA1, BA2, BB1, BB2, BC1, the SU(2) corollary of BD1, and items 3 and 4 of BD2);
- the uniform Kogut–Susskind SU(2) model at fixed spacing and strong bare coupling, $g^4 \geq 9.6 \times 10^9$ at the cap, handled by route B, with its one named construction (BC2);
- the one-plaquette models and the group box models of seven gauge groups under a frozen convention, each box separately inside its Kato radius (BD1);
- the Round11 two-plaquette graph with independent couplings, a finite model [5] (BD2);
- a declared 2+1-dimensional SU(2) model, for which only the AM2 constants are recounted (BD2).

A result proved in one model is not a claim about another. BC2 re-derived the chain for route B with its own constants; it imported none. BD1 transfers statements to named finite models by criteria, not by analogy, and no one-plaquette or graph value transfers to a lattice limit.

The limit is the limit of the named constructions. BB2 shows that the reduced densities of F1 and F2 converge as whole sequences to one common limit on every finite region, and that this limit equals every AQ1 and every F2 subsequential limit. That is convergence of the named constructions. It is not uniqueness of any ground state, and it says nothing about states outside the named constructions or about other boundary conditions, such as literal vertex, periodic or orthant boxes. BC2 proves the same for the one route-B construction, with exhaustion within its own prescription in place of a common limit. Every certificate restated in BC1 and BC2, and the SU(2) area parity of BD1, holds for these limits only.

The obligations state. The reviewed table of the BC1 gate has 21 rows (Table 18); Table 32 summarizes their standing. Row O1 of Round32’s AY2 table is split: O1a, that all subsequential limits of F1 and F2 coincide on every finite region, is closed within the BB2 scope, and O1b, uniqueness of any ground state together with states outside the named constructions and other boundary conditions, is open. Rows O2–O6 are closed within the scopes of the BB2 and BA2 gates, and rows N1–N14 are open. Row N9, frozen before BC2 was gated, stays as the BC1 review froze it: BC2 admits the convergence of the route-B named construction (Section 7), and its review records five further open rows for route B (Table 21). The BD1 review records eleven open rows, among them an AM2/AQ chain and a dictionary for every group other than SU(2), and the BD2 review six, among them the area law and a certified sign of the $\mathbb{Z}^3$ 1x2 mean.

Table 32: Standing of the 21 reviewed obligation rows after Round33. Closed rows are closed only within the named scope; the full rows, with missing premises and candidate routes, are in Table 18.

Row	Obligation	Standing
O1a	all subsequential limits of F1 and F2 coincide on every finite region	closed within the BB2 scope (items 2–3), named constructions only
O1b	uniqueness of any ground state; states outside the named constructions; other boundary conditions	open
O2	whole-sequence convergence	closed within the BB2 scope (item 1) and for the dynamics within the BA2 scope
O3	translation invariance under coarse translations	closed within the BB2 scope (item 4); no claim for non-coarse translations

Row	Obligation	Standing
O4	a rate in N	closed within the BB2 and BA2 scopes, per coarse step at fixed spacing; the correlation rate only on $5 \leq N \leq 14000$
O5	boundary independence of dynamics on compact time windows	closed within the BA2 scope, $ \theta \leq 8$, algebraic dynamics
O6	dynamics of the padded family	closed within the BA2 scope
N1	states outside the named constructions and other boundary conditions	open
N2	the finite-box Euclidean node for finite F1 boxes	open
N3	the finite-box Euclidean node for F2 boxes	open
N4	the F2 finite-box sign for $N = 2, 3$	open; a candidate route is labelled, not admitted
N5	the correlation rate in N beyond $N = 14000$	open
N6	GNS dynamics of different states	open
N7	uniform in time	open
N8	non-centred F2 volumes in the fifth BB1 comparison	open as a gated statement
N9	the route-B uniform model	open as frozen; see BC2 and its rows B1–B5
N10	anything uniform in the lattice spacing a	open
N11	the continuum problem	open
N12	an interaction shift, sign or coefficient of $C(s)$	open
N13	analyticity of the reduced density in the coupling	open
N14	untruncated creation coefficients	open

What the applications did not transfer. Table 29 records every cell. The link-flip lemma does not transfer to $SU(3)$, $SO(3)$ or $SU(5)$, and the first-order parity theorem not to $SU(3)$ or $SO(3)$; each non-transfer is proved by an exact nonzero coefficient on the one-plaquette model, not by the absence of a central element alone. No AM2 fixed point, AV1 state lemma, AQ construction, BB1 or BB2 comparison or coupling dictionary is transferred to any group other than $SU(2)$, and no statement for any other group holds for all box sizes at once. No flip or parity statement is made for $SU(2)$ at a nonzero selected triple, for F2 or literal vertex boxes, or on periodic boxes with an odd side. The two-plaquette graph transfers nothing to the lattice limit. In the zero-selected family the 1×2 Wilson mean has no certified sign, only a formal second-order coefficient; no area law is claimed, because no zero-free region of the reduced density is admitted; and the electric band is certified but not tight. In $2+1$ dimensions only the AM2 constants are recounted, with no finite-volume theorem, AQ chain, node or dictionary. The Einstein–QED evidence rounds share no equation with this chain.

Nothing is uniform in the lattice spacing. Every lattice result holds at one fixed spacing and at strong bare coupling. Every rate is per coarse step in N at fixed spacing, where a coarse step is $(4a, 2a, a)$, and a rate in N is not a rate in a . The Round32 statement AZ1 already names the one necessary estimate the fixed-spacing family supplies along a trajectory and shows that it supplies none uniform along $a_n \rightarrow 0$ [4]. No Round33 gate changes that.

Standing of the evidence. Every admission uses exact rational arithmetic with directed rounding; the decimals in this addendum are previews, and the Arb and mpmath comparisons never admit a value. The producers, the skeptic, the advisor, the lenses and the assistants are model agents with correlated ancestry. Their agreement is correlated model-agent work, never external human peer review or formal verification; in the single-direction loops, exact agreement with the skeptic’s replay checks arithmetic and code, not a second method. Contract wording defects were recorded as non-blocking by the gates, and no frozen file was edited. The phrase scanner is a vocabulary guard: BC1 showed that it does not see a phrasing it was not given, so the reviewed wording, not the

scanner, guards the O1 split. Inherited premises are cited, not re-proved, and the external theorems named in the text [11, 13, 12, 16, 17] are cited, not machine-checked.

Attribution and priority. No scientific priority is claimed or verified for any Round33 result. The attribution sentence of the Round32 addendum for the AM2 creation expansion and the AQ construction [4, 14] stands unchanged. Centre symmetries of lattice gauge theories, cluster expansions and Lieb–Robinson bounds are known kinds of argument; the contribution here is their application to the named models with exact constants and exact obstruction coefficients.

10.1 The ranked roadmap

The advisor’s [Round33 roadmap](#) ranks five goals after the closing lens updates (Table 33). Every goal is a fixed-lattice question for a named model with a named missing premise, and every one is planned and unexecuted; any of them needs a fresh prospective contract. No Round34 contract, producer or gate exists.

Table 33: The Round33 roadmap: ranked, unexecuted goals.

Rank	Goal	Missing premise
1	Dynamics and correlation functions for the route-B uniform model	no route-B dynamics comparison exists; the BA2 constants use $\ \Phi\ _F \leq 2268 \tau $, route B has $2349 \tau $
2	Uniqueness of the local restriction beyond the named constructions (O1b)	no evaluated two-state estimate, Dobrushin- or HTW-type, covers other boundary conditions
3	Finite-box node certificates and the padded-box sign at $N = 2, 3$	the AV2 window lemma is stated for the limit’s GNS vector; the sign at $N = 2, 3$ needs a gated box-level form of the AY1 bound
4	A correlation-function rate in N beyond $N = 14000$	the BB2 item-5 bracket is vacuous from $N = 14419$ with the frozen r_N
5	The AM2/AQ chain for Z2 with an external cross-check	no AM2/AQ chain exists for any group other than SU(2)

The ranking is the advisor’s decision after the closing updates. Route-B dynamics is first because all three lenses rank it and its route is fully named; uniqueness beyond the named constructions is second as the best-defined discriminating target, with a failed condition recorded in advance as no evidence of non-uniqueness. Two dissents are recorded. The modern lens would rank an area law through a zero-free region of the reduced density in the top five; it is recorded below rank 5, with BD2’s missing zero-free region as the obstruction. The historical lens would rank the finite-box certificates above uniqueness; they stand at rank 3.

The scanner methods decision. Before any Round34 contract is frozen, the phrase scanner’s list is to be extended with the AY2 row name of O1 and its short form, and the every-ground-state wording of the claim exclusions is to be added only together with a fix for the heading-list severance found by the Jung assistant. The Round33 frozen texts are not re-scanned retroactively. This is a methods decision recorded in the roadmap, not an executed change.

Closing. The number of investigations, pages or named contributions is not a measure of progress on the continuum problem. Loop counts are never a fraction of it. The four-dimensional continuum Yang–Mills existence and mass-gap problem remains open [18].

A Contribution and evidence ledger

The names below are internal attribution aliases for this project, taken from the `contribution_id` fields of the advisor’s [findings record](#). Established methods keep their existing authorship: Peter–Weyl theory, Weyl integration and the character formulas, Schur–Weyl duality, Kato perturbation theory, the Duhamel formula, the Weyl, Temple, Eckart and Davis–Kahan inequalities, the Fuchs–van de Graaf inequality, Banach’s fixed-point theorem, Weierstrass’s theorem and Cauchy’s estimates, the Nachtergaele–Sims dynamics [11], the Kotecký–Preiss criterion [13, 12] and the creation-operator strategy inherited from Round32 [4, 14]. No alias establishes scientific priority. The dependency identifiers are those of the findings record: `r33-` marks a Round33 investigation, and `r32-` and `r29-` earlier admitted premises.

Table 34: The eight Round33 contributions. Each summary is scoped by its gate’s accepted statement; decimals are previews of exact rationals.

Alias	Investigation and scoped contribution	Depends on	Gate
HNM-C-BA1	<i>Boundary decay of the AM2 creation coefficients for the named constructions.</i> A change of box boundary moves the coefficients on supports meeting the cover by at most Kq^{N-1} , $K = 49/111790368$ at $q = 1/64$ and $K = 49/2018304$ at $q = 37888 \tau $, in each cutoff space; coefficients only, by a weighted norm and an analytic disc.	<code>r29-am2</code> , <code>r32-av1</code> , <code>r32-aw1</code> , <code>r32-ay1</code>	ba1-gate.json
HNM-C-BA2	<i>Boundary comparison of the Heisenberg dynamics on compact windows.</i> The F1 and F2 evolutions of $A \in B(\mathcal{H}_R)$ differ by at most $K_{\text{cmp}}(5N+1)N^{-3}\ A\ $, $K_{\text{cmp}} \approx 2.549 \times 10^{-11}$, on $ \theta \leq 8$; Cauchy in N at the rate $1/(N-1)$; the F2 limit dynamics is AQ1’s; algebraic dynamics only.	<code>r29-aq1</code> , <code>r32-ay1</code> , <code>r32-ay2</code> , <code>r29-am2</code>	ba2-gate.json
HNM-C-BB1	<i>Marginal locality of the reduced densities of the named constructions.</i> For five comparisons, $C \approx 8.905 \times 10^{-7}$ on R and $c_{\text{site}} \approx 8.768 \times 10^{-7}$ on regions at $q = 1/64$, per comparison, in each cutoff space and untruncated at fixed N ; by a polymer expansion with a mixed-weight lemma and by an iterated split.	<code>r33-ba1</code> , <code>r32-av1</code> , <code>r29-am2</code> , <code>r32-ay1</code>	bb1-gate.json
HNM-C-BB2	<i>Whole-sequence convergence and common limit of the named constructions.</i> $C' = 4/984375$ for every $N \geq 2$; one common limit on every finite region, equal to every AQ1 and F2 subsequential limit and invariant under coarse translations; correlation functions on $ \theta \leq 8$ converge, with a rate only on $5 \leq N \leq 14000$.	<code>r33-bb1</code> , <code>r33-ba2</code> , <code>r33-ba1</code> , <code>r29-aq1</code> , <code>r29-aq2</code> , <code>r32-ay1</code>	bb2-gate.json
HNM-C-BC1	<i>Node and sign certificates for the limit of the named constructions.</i> The AV2 node and the AW2 enclosure restated with unchanged constants; the F2 box sign certified for $N \geq N_{\text{sign}} = 4$ at $\tau = \pm 10^{-8}$; one GNS triple and dynamics; the reviewed 21-row obligations table with O1 split.	<code>r33-bb2</code> , <code>r33-bb1</code> , <code>r33-ba2</code> , <code>r32-av2</code> , <code>r32-aw2</code> , <code>r32-aw1</code> , <code>r32-ay2</code>	bc1-gate.json

Alias	Investigation and scoped contribution	Depends on	Gate
HNM-C-BC2	<i>Route-B convergence of the uniform model's named construction and its node.</i> $K_B = 13/27941256$, $C_B \approx 9.452 \times 10^{-7}$, $c_{\text{site},B} \approx 9.306 \times 10^{-7}$; whole-sequence convergence, exhaustion, sign mirror, identification and coarse translations; the AX2 node at $\tau = +10^{-8}$, $s = 1$ restated for the limit.	r33-bb2, r33-bb1, r33-ba1, r32-ax1, r32-ax2, r29-am2, r32-av1	bc2-gate.json
HNM-C-BD1	<i>Centre-symmetry transfer across gauge groups with recorded obstructions.</i> On one-plaquette and box models under the frozen convention (box by box inside a Kato radius), the flip lemma for SU(2), SU(4), U(1) and Z2, the parity theorem also for SU(5), first-order coefficients for seven groups, exact SU(3), SO(3) and SU(5) obstruction coefficients, and the SU(2) area parity at $\kappa = 0$ in centered whole-star boxes and for the limit.	r32-aw1, r33-bb2, r32-aw2, r32-az2	bd1-gate.json
HNM-C-BD2	<i>New observables and dimensions in SU(2).</i> Exact two-coupling tables through total order 4 and a 1x2 loop certified positive at $l_1 = l_2$ on the grid, on the two-plaquette graph; in the zero-selected family at $ \tau \leq 10^{-8}$ the $\mathbb{Z}^3$ 1x2 mean zero at first order and bounded by $K'_2\tau^2$; the electric band; the 2+1D AM2 constants recounted, with no theorem.	r32-az2, r32-aw1, r32-ay1, r32-ay2, r33-bb2, r29-am2	bd2-gate.json

Roles. The human project author is Hruday N M (BUNZEEY), who directed the cycle. The advisor, the skeptic, the three methodological lenses (Newton/Tesla, Jung/Pauli and Penrose/Feynman), the research assistants and the forward and reverse producers are model-agent roles with shared mathematical ancestry. Their agreement is correlated model-agent work, not external human peer review, independent physical observation or formal verification. Historical figures supply documented methodological lenses only; none participated in or endorsed this work. Scientific priority is unverified.

Process records. The cycle also produced process records. They are methods of working, not scientific results, and they count as zero investigations: the pre-freeze review of every draft contract, applied field by field before freezing; the negation-aware phrase scanner and the gate-time template check; the discharge protocol by which BB2's hypotheses on BB1's frozen targets were checked item by item against the BB1 gate; the split of an inherited obligation row into a closed and an open part, frozen with its reviewed wording; and the route-A substitution test of BC2, in which 22 edits that feed another model's constants into the producer's checker all abort.

Evidence records. The immutable decision records are the eight gates listed above. Each binds its contract, the frozen producer reports, checkers, input snapshots and outputs, and the skeptic's review evidence. Fresh normal and optimized replays reproduce every producer output byte for byte. This addendum's builder refuses to run until all eight gates exist and verifies every gate binding; its build receipt binds the scientific and authoring sources to the PDF, and a separate QA record binds the page review to the PDF's digest. From the repository root:

```
python -B papers/round33-addendum/build.py --render
python -B papers/round33-addendum/build.py --check
```

A hash or a passing program does not replace the stated mathematical hypotheses. Exactly eight investigations are closed. The deliberation loops, the assistant packages, the lens updates, the repairs, the replays, this manuscript and publication add no research loop. Draft03 [2], the Round31 addendum [3], the Round32 addendum [4] and every earlier scientific verdict are preserved.

References

- [1] Hruday N M (BUNZEEY), *Yang–Mills workbench research-workflow contract*, [AGENTS.md](#). It is snapshotted into every Round33 producer closure and bound by every Round33 gate.
- [2] Hruday N M (BUNZEEY), with model-agent assistance, *Hruday gauge-theory workbench: spectral bounds, local corrections, Wilson readouts, and controlled evolution*, Draft03, integrated research checkpoint (Round30), 22 September 2026, [papers/draft-03/main.pdf](#). Unchanged by this addendum.
- [3] Hruday N M (BUNZEEY), with model-agent assistance, *Actual Euclidean certificates in a patterned AQ subfamily*, Round31 scientific addendum (AT4–AT6), 22 September 2026, [papers/round31-addendum/main.pdf](#). Unchanged by this addendum.
- [4] Hruday N M (BUNZEEY), with model-agent assistance, *Linear local state bounds, window certificates and first-order Wilson effects at the original coupling cap*, Round32 scientific addendum (AV1–AZ2), 24 September 2026, [papers/round32-addendum/main.pdf](#). Its gates AV1–AZ2 are shared premises of the Round33 contracts, and its attribution sentence for the AM2 creation expansion and the AQ construction is inherited unchanged. Unchanged by this addendum.
- [5] Hruday N M (BUNZEEY), with model-agent assistance, *Round11 two-plaquette finite graph*, [Round11 README](#); [solver README](#) and [solver/two_plaquette.py](#). These are premises of BD2, bound by text and hash; the solver is never imported or executed by the BD2 checker.
- [6] Round33 advisor, *Brief, deliberations, plan, panel updates, findings, roadmap and handoff*, 2026: [brief](#); [deliberation 1](#) and [2](#); [plan record](#); [panel record](#); [panel updates 1, 2 and 3](#); [findings](#); [roadmap](#); [handoff](#). These are planning and summary records and admit nothing.
- [7] Round33 skeptic, *Triage, control catalogue, pre-freeze reviews and repair records*, 2026: [triage](#); [prospective controls](#); [loop-2 review](#); [BB](#), [BC](#) and [BD](#) pre-freeze reviews; [Kotecký–Preiss repair record](#); [BB1 admission record for BB2](#). This is model-agent review with correlated ancestry, not human peer review.
- [8] Round33 lenses (Newton/Tesla, Jung/Pauli, Penrose/Feynman), *Memos, source ledgers, updates and assistant packages*, 2026: [Newton/Tesla](#), [Jung/Pauli](#) and [modern](#) ledgers; memos [experts/<lens>/memo.md](#); lens updates [experts/<lens>/update-1.md](#) to [update-4.md](#); and assistant packages [experts/<lens>/assistant-1/](#) to [assistant-4/](#). These are advisory. The assistant packages count zero research loops.
- [9] Round33 Jung/Pauli lens, *Occult, mystical and government-record reading ledger*, 24 September 2026, [reading ledger](#). Custody is provenance, not confirmation, and no entry became a premise.
- [10] Round33 advisor, *Contracts, gates and shared tools*, 2026: the eight contracts [contracts/<loop>.json](#), the eight gates [advisor/<loop>-gate.json](#) under [research/round33/](#), and the [tools README](#).
- [11] B. Nachtergaele and R. Sims, *On the dynamics of lattice systems with unbounded on-site terms in the Hamiltonian*, arXiv:1410.8174v1, 29 October 2014. <https://arxiv.org/abs/1410.8174>. Committed excerpt of Section 3 (Theorem 3.1) and Section 4 (Theorem 4.1), PDF sha256 501af040...4fba, [source excerpt](#). It is cited, not machine-checked.
- [12] D. Ueltschi, *Cluster expansions and correlation functions*, Moscow Math. J. 4 (2004) 511–522, arXiv:math-ph/0304003v3. <https://arxiv.org/abs/math-ph/0304003>. Committed excerpt of Theorems 1 and 3 and Section 4.2, PDF sha256 2cc3e036...bdfc, [source excerpt](#), added after the BB1 freezes as a repair source. It is cited, not machine-checked.
- [13] R. Kotecký and D. Preiss, *Cluster expansion for abstract polymer models*, Comm. Math. Phys. 103 (1986) 491–498. Cited by the BB1 forward producer in the form of [12]; the paper is not committed and was not re-read in Round33.

- [14] S. Gauvin, arXiv:2503.15539, physics.gen-ph; v3 of 14 September 2026, <https://arxiv.org/abs/2503.15539v3>, which the project read, with the fixed-cutoff material first appearing in v2 of 11 September 2026; its title, which this project neither adopts nor endorses, is *Pinched Multi Affine Geometry and Confinement: Describing the Yang–Mills Mass Gap*. Read: the abstract and theorem statements, with proofs unaudited; the version record is in the Round32 addendum [4].
- [15] J. Kogut and L. Susskind, *Hamiltonian formulation of Wilson’s lattice gauge theories*, Phys. Rev. D 11, 395–408 (1975), doi:10.1103/PhysRevD.11.395. <https://journals.aps.org/prd/abstract/10.1103/PhysRevD.11.395>. This is provenance of the Hamiltonian form only.
- [16] D. A. Yarotsky, *Quasi-particles in weak perturbations of non-interacting quantum lattice systems*, arXiv:math-ph/0411042v1, 11 November 2004. <https://arxiv.org/pdf/math-ph/0411042>. Named in the obligations table; its constants are not evaluated for this model.
- [17] J. Henheik, S. Teufel and T. Wessel, *Local stability of ground states in locally gapped and weakly interacting quantum spin systems*, Lett. Math. Phys. 112 (2022), arXiv:2106.13780v3, 3 March 2022. <https://arxiv.org/abs/2106.13780>. Named in the obligations table as a candidate route; its smallness and decay constants are not evaluated at $|\tau| = 10^{-8}$.
- [18] A. Jaffe and E. Witten, *Quantum Yang–Mills Theory*, official Clay Mathematics Institute problem description. <https://www.claymath.org/wp-content/uploads/2022/06/yangmills.pdf>. The continuum problem is distinct from the fixed-spacing models studied here.